

Christian Schmidt

Interleaving Concepts for Digital-to-Analog Converters

Algorithms, Models,
Simulations and Experiments



Springer Vieweg

Interleaving Concepts for Digital-to-Analog Converters

Christian Schmidt

Interleaving Concepts for Digital-to-Analog Converters

Algorithms, Models, Simulations and
Experiments

With a foreword by Prof. Dr.-Ing. Friedel Gerfers



Springer Vieweg

Christian Schmidt
Fraunhofer-Institut für Nachrichtentechnik
Heinrich-Hertz-Institut, HHI
Berlin, Germany

Also: Technische Universität Berlin, Dissertation, 2019 entitled
„Interleaving Concepts for Performance Enhancement of High-Speed Digital-to-Analog
Converters“

ISBN 978-3-658-27263-0 ISBN 978-3-658-27264-7 (eBook)
<https://doi.org/10.1007/978-3-658-27264-7>

Springer Vieweg

© Springer Fachmedien Wiesbaden GmbH, part of Springer Nature 2020

This work is subject to copyright. All rights are reserved by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.

The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer Vieweg imprint is published by the registered company Springer Fachmedien Wiesbaden GmbH part of Springer Nature

The registered company address is: Abraham-Lincoln-Str. 46, 65189 Wiesbaden, Germany

Foreword

The internet data traffic increases between 30 and 60 % per year. This demand for high data rates is largely driven by new wireless applications, cloud storage, cloud computing, self-driving cars and smart-city technologies. Thus, the infrastructure for tomorrow's information and communication technology (ICT) requires not only higher bandwidth but also more energy-efficient data communications to accommodate the increasing amount of data traffic.

For this, optical data transmission is the key technology. Important components are the data converters performing the transformation between the digital and the analog domain and vice versa. Digital-to-analog converters (DACs) and analog-to-digital converters (ADCs) enable the utilization of real-time digital signal processing (DSP) for software-defined waveforms with both flexible modulation bandwidth and modulation formats, improved coding techniques as well as pre- and post-equalization possibilities.

As the energy-efficiency is becoming more and more important for optical networks, low-power complementary metal oxide semiconductor (CMOS) DACs are highly desired to generate the transmit signal. The CMOS DACs can be integrated with the CMOS DSP on a single chip. With an increased DAC bandwidth, less optical wavelengths are required to achieve the same capacity over the fiber. However, the analog bandwidth of CMOS DACs is fundamentally limited by technological constraints.

Mr. Schmidt addresses in his book two innovative DAC interleaving concepts to overcome the limitations of CMOS DACs enabling a new generation of high-speed DACs. A key question that this book tries to answer is how to enhance the performance of these state-of-the-art DACs. The DAC interleaving concepts enhancing both the bandwidth and the sampling rate are the analog multiplexer (AMUX)-DAC and the frequency interleaving (FI)-DAC concept. Both techniques are analyzed from a system-level perspective to overcome the bandwidth limitations of a single DAC.

The overall objective of this work is to analyze and examine the limitations and the potential of both the AMUX-DAC and the FI-DAC concept to gain a profound understanding of the approaches. The in-depth scientific analysis shows for the first time the possibilities and limitations of both techniques, which are primarily in the

analog components that are necessary for both methods. Methodically, analytical system models are created for both techniques, which are calibrated and verified by means of measurements. In this way, the effects of various disturbing effects are investigated. Furthermore, discrete-time system models are developed enabling the application of compensation techniques by a suitably adapted DSP.

This way, a reasonable decision can be made on the best approach to increase the network capacity by means of transmitter performance enhancement. Since, analog components are employed for the DAC interleaving, there is a trade-off between an enhanced analog bandwidth and a reduced effective number of bits (ENOB).

Overall, this book has an excellent combination of scientific improvements and practical advices in the area of interleaved DACs. Further, it provides valuable suggestions for additional investigations.

Prof. Dr.-Ing. Friedel Gerfers

Acknowledgement

Für die Betreuung meiner Dissertation danke ich Prof. Dr.-Ing. Friedel Gerfers und Priv.-Doz. Dr. Volker Jungnickel. Die Unterstützung während der täglichen Arbeit durch Dr. Jungnickel, sowie die regelmäßigen Treffen mit Prof. Gerfers haben maßgeblich zum erfolgreichen Abschluss der Arbeit beigetragen.

Für die Begutachtung der Arbeit danke ich Prof. Dr.-Ing. Sebastian Randel und für die Übernahme des Vorsitzes des Promotionsausschusses geht mein Dank an Prof. Dr.-Ing. Lars Zimmermann.

Überdies bedanke ich mich bei Prof. Dr.-Ing. Ronald Freund, Prof. Dr.-Ing. Klaus Petermann und Prof. Dr.-Ing. Hans-Joachim Grallert für die Ermöglichung der dieser Dissertation zugrunde liegenden Projekte.

Mein ausdrücklicher Dank gilt Christoph Kottke für die gemeinsamen Diskussionen, Experimente, Veröffentlichungen und Patente. Ebenso danke ich Christoph Caspar für die gemeinsame Verfolgung des AMUX Projekts.

Weiter danke ich den Kollegen Dominic Schulz, Julian Hohmann, Kai Habel, Stefan Weide, Jasper Rödiger, Lutz Molle, Peter Hellwig, Jonas Hilt, Felix Frey, Pablo Wilke Berenguer und Johannes Fischer für die fachliche und moralische Unterstützung während der Arbeit. Überdies danke ich meinen beiden Master-Studenten Hamzeh Kouider und Patrick Zielonka für die gute Zusammenarbeit. Bedanken möchte ich mich ebenfalls bei Olivia Tex, Kristina Hepting und Henrietta Schiffer für die Unterstützung bei Fragestellungen administrativer Natur.

Weiterhin danke ich meinen Freunden und insbesondere meiner Familie für die Unterstützung während des Studiums und während der Erarbeitung der Dissertation. Schließlich danke ich allen anderen, die mich während der Dissertation unterstützten und die ich an dieser Stelle namentlich nicht nennen konnte.

Christian Schmidt

Table of Contents

List of Figures XIII

List of Tables XVII

List of Symbols XIX

List of Acronyms XXIX

Zusammenfassung XXXIV

Abstract XXXVI

1 Introduction 1

1.1 Motivation 1

1.2 State of the Art 1

1.3 Objectives 3

1.4 Structure of the Thesis and Main Contributions 4

2 DAC: Fundamentals and Interleaving Concepts 7

2.1 State of the Art DAC Realizations 8

2.2 Manufacturing Technologies 10

2.3 DAC Architectures 13

2.4 Nyquist Rate DAC ZOH Operation 14

2.5 DAC Performance Impairments 18

2.5.1 Static Performance 19

2.5.2 Dynamic Performance 20

2.5.3 Quantization Noise 21

2.5.4 Phase Noise and Jitter 22

2.5.5 Thermal Noise 25

2.6 DAC Interleaving Concepts 25

2.7 Analog Components for DAC Interleaving Concepts 27

2.7.1 Mixer 27

2.7.2 Frequency Multiplexer 31

2.7.3 Analog Multiplexer 33

3	Time Interleaving DAC (TI-DAC)	37
3.1	Concept	37
3.2	Mathematical Model	38
3.2.1	TI-DAC with RZ Pulses	38
3.2.2	TI-DAC with NRZ Pulses	40
3.3	Pre-Processing Filter for TI-DAC with NRZ Pulses	41
3.4	Characteristics and Limitations	43
3.5	Conclusion	44
4	Analog Multiplexing DAC (AMUX-DAC)	45
4.1	Introduction	45
4.2	Concept	46
4.3	Analytical System Model	47
4.3.1	General Model	48
4.3.2	Special Case for Two DACs	52
4.4	Signal Processing	58
4.4.1	Synchronization and Clocking	59
4.4.2	Digital Signal Processing	60
4.5	AMUX-DAC Impairment Model	65
4.6	Behavioral 2:1 AMUX Model	68
4.6.1	2:1 AMUX IC	68
4.6.2	Block Diagram	70
4.6.3	Parameter Fitting	71
4.7	Simulation Setup	74
4.8	Impact of AMUX Parameter Variations	76
4.8.1	Data LPF In Bandwidth	76
4.8.2	Clock LPF Bandwidth	77
4.8.3	Clock Duty Cycle Mismatch	78
4.9	Impact of AMUX-DAC Parameter Variations	78
4.9.1	Reference Case	78
4.9.2	DAC Bandwidth	80
4.9.3	Combined vs. MIMO Pre-Equalization	81
4.9.4	Phase Noise	82
4.10	2:1 AMUX Experimental Results	83
4.10.1	On-Chip S-Parameter Measurement	83
4.10.2	On-Chip Time Domain Measurement	86
4.10.3	Discussion of Results	91
4.11	Scaling Model	92
4.12	Limitations and Challenges	95
4.13	Summary	96

5	Frequency Interleaving DAC (FI-DAC)	99
5.1	Introduction	100
5.2	Block Diagram and Spectral Illustration	101
5.3	Brief Analytical System Description	105
5.4	Mathematical Framework	106
5.4.1	Definitions	107
5.4.2	Distribution of Data Samples	110
5.4.3	Local Oscillator Frequencies	113
5.4.4	Feasible Symbol Rates	114
5.5	Parameter Balancing	116
5.5.1	Standard FI-DAC Optimization Program	117
5.5.2	Extended FI-DAC Optimization Program	118
5.6	Parameter Balancing Examples	119
5.6.1	Standard FI-DAC Optimization Program Example	119
5.6.2	Extended FI-DAC Optimization Program Example	121
5.7	Digital Signal Processing for FI-DAC	122
5.7.1	Block Diagram	123
5.7.2	MIMO System Model for two DACs	124
5.7.3	Algorithms	128
5.7.4	MIMO Algorithm	134
5.7.5	Calibration	136
5.7.6	Sub-Signal Characteristics	139
5.8	Initial Experiments with MHz-DACs	141
5.9	FI-DAC Impairment Model	144
5.10	Experiments with GHz-DACs	147
5.10.1	Overview on Signal Processing for Two DACs	148
5.10.2	Digital Signal Processing for Two DACs	149
5.10.3	Initial Experiment	151
5.10.4	Improved Experiment	154
5.11	System Parameter Impact on Performance	160
5.11.1	Simulation Setup	160
5.11.2	SNR Estimations Based on DMT	161
5.11.3	Power Spectral Density Difference	163
5.11.4	LO Phase Variation	164
5.11.5	Overlap Variation	164
5.11.6	Clipping	165
5.11.7	Sub-Sample Shifting	166
5.11.8	DAC Resolution Variation	167
5.11.9	Mixer Input Signal Power	168
5.11.10	Phase Noise Variation	168
5.12	Scaling Model	169

5.13	Limitations and Challenges	173
5.14	Summary	175
6	Summary and Outlook	179
6.1	Summary	179
6.2	Outlook	183
	References	215
	List of Publications	217
A	DAC Behavioral Model	221
B	RF Mixer Behavioral Model	227
C	FI-DAC with I/Q Mixer	233
D	FI-DAC Distribution of Data Samples	237
E	FI-DAC Digital Frequency Demultiplexer	239
F	FI-DAC MIMO Model and Pre-Equalizer	241

List of Figures

2.1	High-speed DAC current-steering architectures.	13
2.2	DAC block diagram.	15
2.3	ZOH-DAC operation in time and frequency domain.	17
2.4	ZOH-DAC frequency response.	18
2.5	ENOB and SNR vs frequency for different RMS jitter values.	24
2.6	DAC concepts: single vs. interleaved DAC.	25
2.7	Simplified block diagrams of DAC interleaving concepts.	26
2.8	RF upconversion mixer block diagram.	28
2.9	Topologies for diode-based mixers.	29
2.10	Diplexer equivalent circuit and triplexer.	32
2.11	$N:1$ AMUX block diagram with clock signals.	33
3.1	TI-DAC block diagram and operation.	38
3.2	TI-DAC frequency response for RZ and NRZ.	40
3.3	TI-DAC with NRZ pulses: DSP block diagram.	42
3.4	TI-DAC pre-processing filter frequency response.	43
4.1	$2:1$ AMUX-DAC block diagram and NRZ operation.	47
4.2	Block diagram of the AMUX-DAC system model with clock signals.	48
4.3	AMUX clock signals for both $2:1$ and $4:1$ AMUX.	51
4.4	AMUX-DAC signal processing in the frequency domain.	53
4.5	Block diagram of AMUX-DAC clocking.	59
4.6	AMUX DSP block diagram.	61
4.7	Block diagram of the AMUX-DAC impairment model.	66
4.8	AMUX IC: block diagram and photograph.	69
4.9	$2:1$ AMUX core schematic.	70
4.10	Behavioral $2:1$ AMUX model block diagram.	71
4.11	AMUX fitting results: comparison with EDA tool.	72
4.12	Block diagram of the AMUX-DAC simulation setup.	75
4.13	AMUX model simulation results: frequency responses, EVM, eye diagrams.	77
4.14	AMUX-DAC simulation results: frequency response, eye diagram, spectrum.	79
4.15	AMUX-DAC frequency response magnitude for varying DAC bandwidth.	80

4.16	AMUX-DAC simulation results: EVM for different pre-equalizers and modulation formats.	81
4.17	AMUX-DAC simulation results: EVM for varying clock jitter.	82
4.18	AMUX on-chip S-parameter measurement setup.	84
4.19	Photographs of the AMUX S-parameter measurement setup.	85
4.20	Measured AMUX S-parameters.	86
4.21	Time-domain AMUX on-chip measurement setup.	87
4.22	Photographs of the time domain AMUX measurement.	88
4.23	Time domain AMUX measurement eye diagrams.	89
4.24	Measurement results: spectrum, impulse response, frequency response.	91
4.25	AMUX-DAC scaling model for N DACs.	93
4.26	SNR values for AMUX-DAC scaling model examples.	94
5.1	Block diagram of the FI-DAC concept.	101
5.2	Spectral illustration of the FI-DAC's digital and analog signal processing steps.	103
5.3	FI-DAC preprocessed digital data spectrum.	108
5.4	Feasible integer GBd-symbol rates for different integer GS/s-sampling rates.	115
5.5	FI-DAC parameter balancing examples solutions.	120
5.6	FI-DAC DSP block diagram.	123
5.7	FI-DAC block diagram for two DACs with spectrum insets without guard bands.	125
5.8	FI-DAC MIMO system model: spectral analysis.	127
5.9	FI-DAC MIMO system model block diagram for two DACs.	128
5.10	FI-DAC MIMO equalizer block diagram for two DACs.	135
5.11	Setup for the calibration of the FI-DAC.	137
5.12	Different realizations of the system identification for calibration.	138
5.13	Relative frequency distribution of the FI-DAC's digital sub-signals.	139
5.14	FI-DAC experiments with MHz-DACs: setup.	142
5.15	FI-DAC experiments with MHz-DACs: frequency response, eye diagram and spectrum.	143
5.16	FI-DAC impairment model block diagram.	145
5.17	FI-DAC block diagram for two DACs with spectrum insets with guard bands.	149
5.18	FI-DAC DSP block diagram for two DACs.	150
5.19	FI-DAC experiments with GHz-DACs: initial experimental setup.	151
5.20	FI-DAC experiments with GHz-DACs: initial experimental results I.	153
5.21	FI-DAC experiments with GHz-DACs: initial experimental results II.	154
5.22	FI-DAC experiments with GHz-DACs: improved experimental setup.	156
5.23	FI-DAC experiments with GHz-DACs: improved measurement results.	158

5.24	FI-DAC SNR estimations based on DMT.	162
5.25	FI-DAC performance impact of power and phase mismatches.	163
5.26	FI-DAC performance impact of spectral overlap.	165
5.27	FI-DAC performance impact of clipping and sub-sample shifting.	166
5.28	FI-DAC performance impact of DAC resolution and mixer input power.	167
5.29	FI-DAC performance impact of clock phase noise.	169
5.30	Block diagrams for the FI-DAC scaling model examples.	170
5.31	FI-DAC scaling model examples results.	172
6.1	Qualitative comparison of DAC interleaving concepts.	180
A.1	Behavioral DAC model block diagram.	222
A.2	Behavioral DAC model fitting: SINAD, ENOB, frequency response.	223
A.3	Behavioral DAC model fitting: power levels of harmonics.	224
A.4	Behavioral DAC model fitting: THD and SFDR.	225
A.5	Behavioral DAC model fitting: SINAD and RMS jitter.	226
B.1	Block diagram for the behavioral RF mixer model.	229
C.1	Block diagram of the FI-DAC with an I/Q mixer.	234
C.2	Spectrum partitioning for the FI-DAC consisting of three DACs and an I/Q mixer.	235
E.1	Exemplary FI-DAC digital frequency demultiplexer frequency response.	240

List of Tables

2.1	Overview on commercially available high-speed DACs and AWGs. . .	9
2.2	Overview on recent research results in the field of high-speed DAC ICs.	10
2.3	Overview on current and upcoming SiGe HBT BiCMOS processes. .	11
2.4	Overview on current and upcoming InP HBT processes.	12
2.5	Comparison of DAC architectures.	14
2.6	SNR values for different vertical resolution values.	21
2.7	State of the art AMUX research results.	34
4.1	Extracted parameters for the SiGe-HBT 2:1 AMUX behavioral model.	73
5.1	Variables for the FI-DAC mathematical framework.	109
5.2	PAPR values of the FI-DAC sub-signals for different PAM orders. .	140
5.3	Parameters for the FI-DAC scaling model examples.	171
5.4	Performance results for the FI-DAC scaling model examples. . . .	173
6.1	Qualitative comparison of the DAC interleaving concepts.	182

List of Symbols

Symbols

$a_{m,n}$	nonlinear polynomial coefficient for LO harmonic n
B	(signal) bandwidth
b	number of bits
$\underline{\mathbf{C}}(\mu)$	MIMO channel matrix
$\underline{\mathbf{C}}(f)$	MIMO channel matrix
C	Shannon capacity
CL	mixer conversion loss
$C_{ln}(\mu)$	MIMO channel frequency response
$C_n(f)$	Fourier transform of $c_n(t)$
$c_n(t)$	clock signal for AMUX input n
C_L	load capacitance
CL	mixer conversion loss
$\mathbf{d}$	digital signal vector
$\mathbf{D}(\mu), \mathbf{D}_n(\mu)$	digital spectrum vector
$D(\mu), D_n(\mu)$	digital spectrum
$d(k)$	digital signal
$d_n(k)$	digital signal of the n th transmitter/DAC
e_Q	quantization error
ENOB	effective number of bits
ENOB _{TI}	effective number of bits for TI-DAC
f	frequency
f_A, f_B	frequency interval
$f_{\text{AMUX},m}$	AMUX clock frequency
f_c	(AMUX) switching frequency
$f_{co}, f_{co,n}$	filter cutoff frequency, frequency demultiplexer split frequency
f_{DAC}	DAC clock frequency
f_{LO}	local oscillator frequency

$f_{\text{LO,IQ},n}$	LO frequency for mixer n (I/Q mixer)
$f_{\text{LO,LSB},n}$	LO frequency for mixer n (RF mixer LSB)
$f_{\text{LO,USB},n}$	LO frequency for mixer n (RF mixer USB)
$f_{\text{LO},M_n,n}$	LO frequency for mixer n of type M_n
$f_{\text{LO},n}$	LO frequency for mixer n
$f_{\text{min},n}$	minimum frequency of sub-signal n
f_{max}	transistor maximum oscillation frequency
$f_{\text{max},n}$	maximum frequency of sub-signal n
$f_{\text{PN},1}, f_{\text{PN},2}$	lower and upper phase noise integration boundary
f_s	sampling rate
$f_{s,\text{analog}}$	sampling rate for analog simulation
$f_{s,n}$	sampling rate of DAC n
$f_{s,\text{tot}}$	total sampling rate
$f_{\text{sym,tot,target}}$	target signal symbol rate
$f_{\text{sym,tot}}$	symbol rate
f_T	transistor current gain cutoff frequency
$G(f)$	DAC frequency response
$g(t)$	DAC impulse response
$\hat{\mathbf{H}}_{\text{LS}}$	LS CE impulse response matrix
$H_{\text{B},n}^{\text{RC}}(f)$	raised cosine filter band-pass characteristic
$H_{\text{H},n}^{\text{RC}}(f)$	raised cosine filter high-pass characteristic
$H_{\text{L},n}^{\text{RC}}(f)$	raised cosine filter low-pass characteristic
$H_{\text{In}}(\mu)$	MIMO system frequency response
$H_{\text{SF},n}(f)$	digital demultiplexer frequency response
$\mathbf{I}$	identity matrix
I_{IN}	switch pair input current
I_{OUT}	DAC core output current
k	discrete time
$\mathbf{K}_{\text{D}}$	# of data samples vector
$\mathbf{K}_{\text{D,tot}}$	# of total data samples vector
$K_{\text{D,tot}}$	# of total data samples
$K_{\text{D},n}$	# of data samples for DAC n
K_{H}	length of channel impulse response
$K_{\text{O},n}$	# of additional samples for guard bands for DAC n

$K_{OH,n}$	# of additional samples for guard bands at high frequencies for DAC n
$K_{OL,n}$	# of additional samples for guard bands at low frequencies for DAC n
K_S	# of samples
$K_{S,n}$	# of samples for DAC n
$K_{S,tot}$	# of total samples
L	# of receivers for MIMO system
M	# of AMUX stages
$M_{AMUX-DAC}^{(a)}$	# of multiplications for AMUX-DAC DSP variant (a)
$M_{AMUX-DAC}^{(b)}$	# of multiplications for AMUX-DAC DSP variant (b)
$M_{CE,MIMO,f}$	# of multiplications for MIMO frequency domain CE
$M_{CE,MIMO,t}$	# of multiplications for MIMO time domain CE
$M_{Delay,f}$	# of multiplications for delay filter
$M_{FFT,c}^{sr}, M_{FFT,r}^{sr}$	# of multiplications for complex- and real-valued split-radix FFT
$M_{FFT}^{r2,m}$	# of multiplications for radix-2 ^m FFT
M_{FI-DAC}	# of multiplications for FI-DAC DSP
$M_{FI,n}$	# of elements in the n th FI-DAC signal path
$M_{IFFT,r}^{sr}$	# of multiplications for real-valued split-radix IFFT
M_n	mixer type for signal path n
M_P	order of nonlinear polynomial
$M_{W,f}$	# of multiplications for frequency domain filter
$M_{W,MIMO,f,calc}$	# of multiplications for MIMO equalizer calculation
$M_{W,MIMO,f,diag}$	# of multiplications for MIMO equalizer (only diagonal elements)
$M_{W,MIMO,f}$	# of multiplications for MIMO equalizer
N	# of DACs, # of AMUX inputs, # of clock signals, # of sub-signals, # of transmitters
N_A	# of amplifiers in AMUX clock buffer
N_D	# of data samples for CE
N_{FFT}	FFT length
N_H	# of frequency bins for channel estimation
N_{IFFT}	IFFT length
N_{LO}	# of LO harmonics
N_P	degree of parallelization

N_W	# of frequency bins for frequency domain filter
NMSE	normalized mean squared error
$\underline{\hat{\mathbf{P}}}$	inverse matrix of oversampling ratios
$\underline{\mathbf{P}}$	matrix of oversampling ratios
$\underline{\tilde{\mathbf{P}}}$	cofactor matrix of oversampling ratios
$\tilde{p}_{1n}$	first row of the cofactor matrix $\underline{\tilde{\mathbf{P}}}$
P_{avg}	average constellation power
$P_{\text{AMP},n}$	power consumption of the n th amplifier
P_{DAC}	DAC's power consumption
$p_{H,n}$	oversampling ratio DAC n for high frequencies
$p_{H,n}^{\min}, p_{H,n}^{\max}$	lower and upper bound for oversampling ratio DAC n for high frequencies
P_{IN}	input power
$P_{\text{IN,ref}}$	input reference power
$P_{\text{LO},n}$	power consumption for the generation of the n th LO
$p_{L,n}$	oversampling ratio DAC n for low frequencies
$p_{L,n}^{\min}, p_{L,n}^{\max}$	lower and upper bound for oversampling ratio DAC n for low frequencies
$P_{m,n}$	spurious power
$\hat{p}_{n1}$	first column of the inverse matrix $\underline{\hat{\mathbf{P}}}$
p_n	oversampling ratio DAC n
P_{TOT}	total power consumption
$\mathbf{R}$	received samples matrix for CE
R	resistance
$r(k)$	received discrete-time signal
R_C	collector resistance
R_E	emitter degeneration resistance
R_L	load resistance
$\underline{\mathbf{S}}$	transmit data matrix for CE
$\underline{\mathbf{S}}_n$	transmit data sub-matrix for CE of the n th transmitter
$s_{\text{CLK}}(t)$	clock signal
$s_{\text{IF}}(t)$	mixer IF input signal
s_{IN}	DAC input signal
$s_{\text{LO},n}(t)$	mixer LO signal's n th harmonic
$s_{\text{OUT}}(t)$	DAC behavioral model output signal, mixer behavioral model output signal

$s_{\text{OUT},n}(t)$	sub-signal n of behavioral mixer model
$s_{\text{RF}}(t)$	mixer RF output signal
SINAD	signal to interference, noise and distortion ratio
SNR	signal to noise ratio
$\text{SNR}_{\text{in},n}$	input SNR of the FI-DAC's analog processing system for the n th signal path
$\text{SNR}_{\text{out},n}$	output SNR of the FI-DAC's analog processing system for the n th signal path
$\text{SST}_{m,n,\text{ref}}$	spurious suppression reference table entry
$\text{SST}_{m,n}$	spurious suppression table entry
T	temperature
t	time
T_c	AMUX clock switching period
T_{IN}	time period of the input signal
T_s	sampling period
$\mathbf{V}(\mu)$	noise samples vector
$V(f), V_n(f)$	analog filter's frequency response
V_0	amplitude
V_{CC}	positive supply voltage
V_{CLK}	AMUX clock voltage
$V_{\text{clk},n}(f), V_{\text{clk}}(f)$	AMUX clock path frequency response
$v_{\text{clk},n}(t)$	AMUX clock path impulse response
V_{EE}	negative supply voltage
$v_{\text{FM},n}(t)$	frequency multiplexer impulse response
$V_{\text{IN1}}, V_{\text{IN2}}$	AMUX input voltages
$v_n(t)$	analog filter's impulse response
V_{OUT}	AMUX output voltage
$V_{\text{out,A}}(f), V_{\text{out,B}}(f)$	2:1 AMUX output filter frequency response sub-bands
$V_{\text{out}}(f)$	AMUX data output frequency response
$v_{\text{out}}(t)$	AMUX data output impulse response
V_{pp}	peak-to-peak amplitude
V_{T}	temperature voltage
$\mathbf{W}(\mu)$	pre-equalizer frequency response matrix
$\mathbf{W}_{\text{LMS},i}(\mu)$	LMS solution for pre-equalizer matrix
$\mathbf{W}_{\text{MMSE}}(\mu), \mathbf{W}_{\text{ZF}}(\mu)$	MMSE and ZF solution for pre-equalizer matrix
$W(e^{j\Omega})$	pre-equalizer frequency response

w, w_R, w_I	complex number, its real part and its imaginary part
$W_{ln}(\mu)$	MIMO equalizer coefficients
$\mathbf{X}(\mu)$	digital signal spectrum vector for multiple sub-bands
$\mathbf{X}(f)$	vector of DAC input spectra
$X(e^{j\Omega})$	DTFT of $x(k)$
$X(\mu), X_n(\mu)$	DFT of digital signal $x(k), x_n(k)$
$X(f)$	Fourier transform of $x(t)$
$x(k)$	digital (DAC input) signal
$X_A(f), X_B(f), X_Z(f)$	2:1 AMUX input sub-bands
$X_n(e^{j\Omega})$	DTFT of $x_n(k)$
$X_n(f)$	band-limited Fourier transform of $x_n(k)$
$x_n(k)$	digital input signal of the n th DAC
$\mathbf{Y}(\mu), Y(\mu)$	DFT of $y(k)$
$\mathbf{Y}(f)$	vector of AMUX output spectra
$Y(f)$	Fourier transform of $y(t)$
$y(k)$	sampled DAC output signal
$y(t)$	(combined) analog DAC output signal
$y_0(t)$	combined analog DAC output reference signal
$Y_A(f), Y_B(f), Y_Z(f)$	2:1 AMUX output sub-bands
$\mathbf{Y}_n(\mu), Y_n(\mu)$	part of sampled DAC output signal's spectrum
$\tilde{y}_n(t)$	upconverted analog signal
$Y_n(f)$	Fourier transform of $y_n(t)$
$y_n(t)$	analog output signal of DAC n
z, z_R, z_I	complex number, its real part and its imaginary part
$Z_{IN,LP}, Z_{IN,HP}$	input admittance of LPF and HPF
Z_{IN}	input admittance
Z_L	load admittance
α_{DAC}	DAC clock/sampling rate ratio
$\alpha_{LO,M_n,n}$	DAC clock/LO ratio
β_n	raised cosine roll-off factor
$\chi(e_Q)$	quantization error probability distribution
$\Delta V(t)$	time-variant amplitude deviations
Δ_Q	quantization interval amplitude
$\delta_\theta(t)$	rectangular function for Dirac distribution definition

$\Delta_{\text{SNR},m,n}$	SNR penalty of the m th element in the n th signal path
ε	2:1 AMUX clock line attenuation factor
Γ	set of mixer types for the FI-DAC
γ	tanh argument in AMUX clock buffer
κ	switch pair correction coefficient
Λ	set of DACs
λ	LMS update coefficient
Λ_n	set of DACs without the n th DAC
μ	discrete frequency of DFT
Ω	normalized angular frequency of the DTFT
ω_{IN}	angular input frequency
ϕ	phase shift
$\phi(t)$	time-variant phase
$\Phi_{\text{PN}}(f)$	one-sided phase noise spectrum
Ψ	constellation given as a set of symbols
ψ_i	symbols in the constellation
σ_{J}^2	average jitter power
σ_{N}^2	noise variance
σ_{Q}^2	average quantization noise power
σ_{S}^2	average signal power
τ	RMS jitter
$\tau_{\text{AMUX}}, \tau_{\text{AMUX},m}$	AMUX clock RMS jitter
τ_{DAC}	DAC clock RMS jitter
τ_{tot}	total AMUX-DAC RMS jitter
θ	height of rectangular function for Dirac definition
$\zeta_{\text{TC}}(f)$	thermal noise PSD for sampling switch ($k_{\text{B}}T/C$)
$\zeta_{\text{T}}(f)$	thermal noise PSD

Indices

i	index variable
l	index variable
m	index variable
n	index variable

Constants and imaginary unit

j	imaginary unit $\sqrt{-1}$
k_B	Boltzmann constant

Set symbols

$\mathbb{N}_0$	set of natural numbers including zero
$\mathbb{N}_{>0}$	set of natural numbers excluding zero
$\mathbb{R}_{>0}$	set of non-negative rational numbers
$\mathbb{Z}$	set of integers

Operators

$*$	convolution operator
$\langle \cdot \rangle$	expectation operator
$\lceil \cdot \rceil$	ceiling operator
$\prod$	product operator
$\sum$	summation operator
$\{\cdot\}^+$	operator selecting the positive frequencies of the spectrum
$\{\cdot\}^-$	operator selecting the negative frequencies of the spectrum
$\{\cdot\}^*$	complex conjugated operator
$\{\cdot\}^\dagger$	reversed frequency axis + complex conjugate operator
$\{\cdot\}^{-1}$	scalar/vector/matrix inverse operator
$\{\cdot\}^\ddagger$	vector/matrix conjugate transpose operator
$\{\cdot\}^T$	vector/matrix transpose operator
$\text{adj}(\cdot)$	adjugate matrix
$\text{cof}(\cdot)$	cofactor matrix
$\det(\cdot)$	matrix determinant
$\mathfrak{F}\{\cdot\}$	Fourier transform
$\mathfrak{F}_{\text{DFT}}\{\cdot\}$	discrete Fourier transform

Functions

$\delta(t)$	Dirac delta distribution
$\text{rect}(t)$	rectangle function
$\text{sinc}(t)$	sinc function

Notation aspects

x, X	scalar in time and frequency domain
$\mathbf{x}, \mathbf{X}$	vector in time and frequency domain
$\underline{\mathbf{x}}, \underline{\mathbf{X}}$	matrix in time and frequency domain

List of Acronyms

ADC	analog-to-digital converter
AM	amplitude modulation
AMUX	analog multiplexer
AMUX-DAC	analog multiplexing DAC
ASIC	application-specific IC
AWG	arbitrary waveform generator
AWGN	additive white Gaussian noise
b2b	back-to-back
BER	bit error rate
BiCMOS	bipolar CMOS
BPF	band-pass filter
BPG	bit pattern generator
BPSK	binary phase shift keying
CE	channel estimation
CL	conversion loss
CMOS	complementary metal oxide semiconductor
CPU	central processing unit
CSI	channel state information
D/A	digital-to-analog
DAC	digital-to-analog converter
DBBS	De Bruijn binary sequence
DBI	digital bandwidth interleaving
DC	direct current
DEMUX	demultiplexer
DFT	discrete Fourier transform
DMT	discrete multi-tone
DMUX	digital multiplexer
DNL	differential nonlinearity
DSP	digital signal processing
DTFT	discrete-time Fourier transform
EDA	electronic design automation
ENOB	effective number of bits
EVM	error vector magnitude

FDE	frequency domain equalization
FEC	forward error connection
FET	field-effect transistor
FFT	fast Fourier transform
FI	frequency interleaving
FI-ADC	frequency interleaving ADC
FI-DAC	frequency interleaving DAC
FIR	finite impulse response
FoM	figure of merit
FPGA	field programmable gate array
GaAs	gallium arsenide
HBT	heterojunction bipolar transistor
HEMT	high electron mobility transistors
HHI	Fraunhofer Heinrich-Hertz-Institute
HPF	high-pass filter
IC	integrated circuit
ICT	information and communication technology
IDFT	inverse discrete Fourier transform
IF	intermediate frequency
IFFT	inverse fast Fourier transform
IIR	infinite impulse response
IMD	intermodulation distortion
IM/DD	intensity modulation and direct detection
IMP	intermodulation product
INL	integral nonlinearity
InP	indium phosphide
IP	internet protocol
I/Q	in-phase and quadrature
ISI	inter-symbol interference
LMS	least mean squares
LO	local oscillator
LPF	low-pass filter
LS	least squares
LSB	lower sideband
MIMO	multiple-input multiple-output
MINLP	mixed-integer nonlinear optimization program
MISO	multiple-input single-output
MMSE	minimum mean square error
MSE	mean square error
MZM	Mach-Zehnder modulator
NF	noise figure

NMSE	normalized mean squared error
NRZ	non-return-to-zero
OAWG	optical arbitrary waveform generation
OAWM	optical arbitrary waveform measurement
OFDM	orthogonal frequency division multiplexing
PA	power amplifier
PAM	pulse amplitude modulation
PAPR	peak-to-average power ratio
PDF	probability density function
PLL	phase-locked loop
PRBS	pseudo-random binary sequence
PSD	power spectral density
QAM	quadrature amplitude modulation
RF	radio frequency
RFD	relative frequency distribution
RLS	recursive least squares
RMS	root mean square
Rx	receiver
RZ	return-to-zero
SCIP	Solving Constraint Integer Programs
SE	single-ended
SFDR	spurious free dynamic range
Si	silicon
SiGe	silicon-germanium
SINAD	signal to noise and distortion ratio
SISO	single-input single-output
SNR	signal-to-noise ratio
SSB	single sideband
SSL	suspended stripline
SST	spurious suppression table
THD	total harmonic distortion
TI	time interleaving
TI-DAC	time interleaving DAC
TUB	Technische Universität Berlin
Tx	transmitter
USB	upper sideband
VNA	vector network analyzer
ZF	zero-forcing
ZIMPL	Zuse Institut Mathematical Programming Language
ZOH	zero-order-hold

Zusammenfassung

Die Nachfrage nach Hochgeschwindigkeits-Datenverbindungen ist in den letzten Jahrzehnten stetig gestiegen. Flexible optische Transceiver auf Basis von Digital-Analog-Wandlern (DACs) und Analog-Digital-Wandlern sind gefragt, um software-definierte Kommunikationssysteme zu ermöglichen. Die Performanz von Hochgeschwindigkeits-DACs wird durch drei zentrale Parameter bestimmt: Abtast-rate, analoge Bandbreite und effektive Bitanzahl (ENOB). Moderne Complementary Metal Oxide Semiconductor (CMOS) DACs sind aufgrund technologischer Beschränkungen in ihrer Bandbreite begrenzt. Zur Überwindung dieser Beschränkungen können parallele DAC-Architekturen genutzt werden, die als Interleaving Konzepte bezeichnet werden.

In dieser Arbeit werden zwei DAC Interleaving Konzepte im Detail untersucht: Analog Multiplexing DAC (AMUX-DAC) und Frequency Interleaving DAC (FI-DAC). Der AMUX-DAC enthält einen analogen Multiplexer mit hoher Bandbreite, um die Ausgangssignale mehrerer DACs sequentiell auf einen kombinierten Ausgang umzuschalten. Der FI-DAC enthält ein analoges Signalverarbeitungssystem bestehend aus Mischen, Filtern und einem Kombiniierer, um das Ausgangsspektrum aus mehreren Frequenzbändern zu synthetisieren. Bei beiden Konzepten weist der kombinierte DAC sowohl eine höhere Abtast-rate als auch eine höhere analoge Bandbreite im Vergleich zu einem einzelnen DAC auf.

Diese Arbeit analysiert und untersucht die Grenzen und Möglichkeiten beider Konzepte, um die Grundlage für eine Umsetzung in der Praxis zu schaffen. Hierfür werden Algorithmen der digitalen Signalverarbeitung (DSP), analytische und zeitdiskrete Modelle, numerische Simulationen und Laborexperimente präsentiert. Für die numerischen Simulationen werden geeignete Verhaltensmodelle entwickelt und anschließend angewendet, um die wesentlichen Limitierungen der Performanz zu untersuchen. Die Experimente validieren beide Konzepte in einer Laborumgebung. Darüber hinaus werden analytische Systemmodelle abgeleitet und analysiert. Basierend auf zeitdiskreten Systemmodellen werden geeignete Algorithmen zur Kompensation der analogen Beeinträchtigungen entwickelt. Ihre Funktionsweise wird verifiziert und der Beitrag zur Verbesserung der Performanz ausgewertet. Durch die Analyse und Bewertung verschiedener Einflussfaktoren auf die Performanz werden sowohl die Anforderungen für eine spätere Integration als auch Skalierungs-limitierungen ermittelt.

Abstract

The demand for high-speed data interconnects has been rising over the last decades. Flexible optical transceivers based on digital-to-analog converters (DACs) and analog-to-digital converters are desired to enable software-defined communication systems. The performance of high-speed DACs is measured with three main parameters: sampling rate, analog bandwidth, and effective number of bits (ENOB). Modern complementary metal oxide semiconductor (CMOS) DAC performance is bounded by the DAC's analog bandwidth due to technological constraints; hence, parallel DAC architectures are proposed to overcome these limitations, i.e., interleaving concepts.

In this thesis, two interleaving concepts are investigated in detail: analog multiplexing DAC (AMUX-DAC) and frequency interleaving DAC (FI-DAC). The AMUX-DAC comprises a high-bandwidth analog multiplexer (AMUX) to switch the output signals of multiple DACs to a combined output sequentially. The FI-DAC comprises an analog signal processing system consisting of mixers, filters, and a combiner to synthesize the output spectrum from multiple frequency bands. For both interleaving concepts, the combined DAC has a higher sampling rate and an increased analog bandwidth compared to a single DAC.

In order to provide the basis for a practical implementation, this work analyzes and examines the limitations and the potential of both concepts. For this, digital signal processing (DSP) algorithms, both analytical and discrete-time models, numerical simulations, and laboratory experiments are presented. For the numerical simulations, adequate behavioral models are developed and applied to investigate the major performance impairments. The experiments validate the proposed concepts in laboratory environments. Moreover, analytical system models are derived and analyzed. Discrete-time system models allow for formulating appropriate algorithms pre-compensating the analog impairments; they are verified and evaluated regarding performance improvement. By analyzing and evaluating the influence of various impairments onto the overall performance, the requirements for a later integration are deduced and practical solutions are developed. Further, conceptual scaling bottlenecks are identified and analyzed.



1 Introduction

1.1 Motivation

Over the last decades, the annual global internet protocol (IP) traffic has been increasing at an enormous rate and is going to further increase. Overall IP traffic will grow at a compound annual growth rate of 24 % from 2016–2021 [1]. The total annual global IP traffic has already surpassed the zettabyte (1 000 exabytes) threshold in 2016 [1].

This ongoing traffic expansion can be attributed to several trends. In the internet of things, more and more devices are connected in networks, e.g., smart phones, TVs, machines, computers, etc. [2]. In addition, new concepts such as smart factory or smart home accelerate this trend even more [3, 4]. Another key factor for the increased traffic is video content, which is expected to drive 80 % of all internet traffic in 2021 [1]. Next to end user data traffic, both inter and intra data center traffic are market drivers for high data rate products, especially for cloud computing applications [5–8]. Another upcoming traffic driver is the 5th generation of mobile communications (5G), which aims at a coverage of up to 1 Tbit/s/km² [9, 10].

The data traffic growth caused by these applications requires high-speed communication systems. Optical transport networks are required, since they offer the highest data rates at a minimum energy consumption [11, 12]. The demand for high data rate transceivers is present in all network segments from long-haul over metro to short-range optical fiber communication links [1].

1.2 State of the Art

For nowadays communications systems, flexible transceivers based on digital-to-analog converters (DACs) and analog-to-digital converter (ADC) are desired to enable software-defined transmitters (Tx) and receivers (Rx) [13–15]. The utilization of real-time digital signal processing (DSP) enables software-defined waveforms with both flexible modulation bandwidth and modulation formats [16, 17], improved coding techniques [18, 19], as well as pre- and post-equalization possibilities [20, 21]. The application of DSP-supported transmission improving the transmission performance has been increasing over the last years and is expected

not to be a limiting factor in terms of costs in the future anymore, even for short links [22–25]. Integrated transceivers are becoming state of the art in optical communication systems [26, 27].

As the energy-efficiency is becoming more and more important for optical networks, low-power complementary metal oxide semiconductor (CMOS) DACs are highly desired key components to generate the actual Tx signal to be transmitted [28]. Furthermore, the CMOS DACs can be integrated with the DSP on a single chip to prevent the interface data rate bottleneck [28–30]. Principally, with an increased DAC bandwidth, less optical wavelengths are required to achieve the same capacity.

A key question to be answered in this thesis, is how to enhance the capacity of the DAC and in turn, the Tx. According to Shannon [31], the capacity C that can be transmitted over a channel with the bandwidth B and the signal-to-noise ratio SNR is given as

$$C = B \cdot \log_2(1 + \text{SNR}) . \quad (1.1)$$

Concerning DACs, the SNR relates to the effective vertical resolution, i.e., the effective number of bits (ENOB). An increase of both SNR and bandwidth B is desirable. However, for high-speed DACs there is a fundamental trade-off between bandwidth and ENOB [32–34]. Furthermore, the electrical bandwidth offered by optical modulators exceeds the bandwidth offered by modern high-speed CMOS DACs [29, 35–37]; thus, the modulators cannot be fully utilized. As the bandwidth B is linearly related to the capacity C , whereas the SNR is related logarithmically to it, increasing the bandwidth is more effective than increasing the SNR.

Current fabrication technologies and materials, respectively, which can be used for high-speed DACs, are CMOS, silicon-germanium (SiGe) and indium phosphide (InP). For these technologies there is a trade-off between energy-efficiency and bandwidth, e.g., CMOS is highly energy efficient, but the maximum achievable bandwidth is not as high as with SiGe or InP and vice versa.

Progress in CMOS technology has led to reduced transistor dimensions, which made them faster, reduced the power consumption, and implicitly enhanced the ability for denser integration [30, 38, 39]. However, subsequent silicon process nodes have become exponentially more expensive [40]. While the advantages are true for digital circuits, the reduction of power consumption and the performance enhancement are not necessarily true for analog silicon circuits [30, 41]. Hence, for high-speed and ultra-high-speed analog applications, viable alternatives are bipolar CMOS (BiCMOS) processes and compound semiconductor technologies such as InP, SiGe and gallium arsenide (GaAs) [30]. However, due to the higher supply voltages, the circuits manufactured in these materials consume more power. Moreover, integration with CMOS technology is challenging.

In order to increase the capacity, the bandwidth limitations need to be overcome by new concepts. These concepts use multiple parallel DACs with additional circuitry to generate a combined electrical signal having both an increased sampling rate and analog bandwidth compared to a single DAC: this is called DAC interleaving.

For interleaved DACs, three concepts are viable: time interleaving DAC (TI-DAC), analog multiplexing DAC (AMUX-DAC), and frequency interleaving DAC (FI-DAC). For the TI-DAC concept, the outputs of multiple DACs are added to generate a higher overall sampling rate. For the AMUX-DAC concept, an analog multiplexer (AMUX) switches between the outputs of multiple DACs to generate both a higher overall sampling rate and analog bandwidth. For the FI-DAC concept, multiple DAC output signals are combined in the frequency domain using an analog processing system consisting of upconversion mixers, filters and a combiner to achieve both a higher overall sampling rate and analog bandwidth.

1.3 Objectives

In this thesis, the two innovative DAC interleaving concepts enhancing both the bandwidth and the sampling rate, i.e., the AMUX-DAC and the FI-DAC concept, are investigated from a system-level perspective to overcome the bandwidth limitations of a single DAC.

The overall objective of this work consists in analyzing and examining the limitations and the possibilities of both the AMUX-DAC and the FI-DAC concept to obtain a profound understanding of both approaches. This way, a reasonable decision can be made on the best approach to increase the network capacity by means of transmitter performance enhancement. Since, analog components are employed for the interleaving, there is a trade-off between an enhanced analog bandwidth and a reduced ENOB. The trade-off depends on the selected technology and it is not immediately intuitive, what the best choice will be.

Both, numerical simulations and experiments are conducted to pursue the overall objective. The experiments are conducted to validate both concepts in laboratory environments. For the numerical simulations, adequate behavioral models are developed and applied to determine and investigate the major performance impairments.

Next to simulations and experiments, analytical system models are developed and analyzed. Discrete-time system models allow for formulating appropriate mitigation algorithms to compensate the analog impairments. By using these mitigation algorithms for pre-equalization, the performance of the FI-DAC and the AMUX-DAC is improved.

Further, the influence of various impairments onto the overall DAC performance is analyzed to deduce the requirements for the integration and to develop practical solutions. Finally, conceptual scaling bottlenecks are identified and analyzed.

1.4 Structure of the Thesis and Main Contributions

In this section, an outline of this thesis is provided, whereby the main contributions are briefly summarized for each chapter. Moreover, the previously published works of the author are related to the corresponding chapters of the thesis.

In the *second chapter*, an overview of state of the art DACs and manufacturing technologies is provided. Current-steering DAC architectures are introduced and the DAC operation is described analytically. DAC performance impairments as well as relevant DAC measures are described and explained. Thereafter, the three interleaving concepts are briefly introduced, followed by a description of the required analog components and their effects.

In the *third chapter*, the TI-DAC concept is briefly covered to address all interleaving concepts for high-speed DACs in this thesis. The TI-DAC operation for both return-to-zero (RZ) and non-return-to-zero (NRZ) pulses is described analytically. For NRZ, the required pre-processing filter is introduced and its implications are reviewed. Finally, the principal shortcomings of the TI-DAC concept are discussed.

In the *fourth chapter*, the AMUX-DAC concept is investigated. The main contributions and findings are as follows:

- Development of a behavioral 2:1 AMUX model to reduce the simulation time for system-level simulation
- Measurements and experiments with an AMUX having the highest bandwidth worldwide at that point in time
- Development of an analytical N :1 AMUX system model
- Development of a multiple-input multiple-output (MIMO) system model for the 2:1 AMUX
- Derivation of a novel MIMO equalizer compensating the impairments for unequal AMUX signals paths
- Development of an AMUX-DAC scaling model to evaluate jitter requirements for an N :1 AMUX-DAC
- Development of an AMUX-DAC impairment model

Part of the work included in Ch. 4 has been previously published in [42, 43].

In the *fifth chapter*, the FI-DAC concept is investigated. The main contributions and findings are as follows:

- Development of a comprehensive mathematical FI-DAC framework
- Solution for the mathematical problem of distributing the data samples among the DACs for the FI-DAC
- Demonstration of the limited number of feasible symbol rates for the FI-DAC
- Mathematical optimization programs for balancing the interdependent FI-DAC parameters
- Development of a MIMO system model for FI-DACs without guard bands
- Development of a MIMO pre-equalizer for FI-DACs without guard bands
- Development of FI-DAC calibration routines
- Development of a general FI-DAC impairment model
- Experimental demonstration of the first reported open eye diagram for a FI-DAC without using a post-equalizer
- The first reported short-range optical intensity modulation and direct detection (IM/DD) transmission using a FI-DAC generated signal
- The first proposal of the FI-DAC concept with an in-phase and quadrature (I/Q) mixer
- Investigation of performance limitations by both experiments and simulations
- Development of a FI-DAC scaling model

Part of the work included in Ch. 5 has been previously published in [44–52].

In the *sixth chapter*, the thesis is summarized. Furthermore, an outlook is provided on present and upcoming issues that could be investigated in the future.

The *appendices* follow at the end of the thesis. The behavioral DAC model and the behavioral radio frequency (RF) mixer model are introduced. Furthermore, the FI-DAC concept with I/Q mixers is described. Then, the FI-DAC's distribution of data samples among the DACs is derived, which is followed by the description of the FI-DAC's digital frequency demultiplexer. Lastly, the MIMO system model and the pre-equalizer are derived for a FI-DAC operated without guard bands.



2 DAC: Fundamentals and Interleaving Concepts

A data converter is a device that transforms a signal between the digital and the analog domain and vice versa, whereby the two manifestations are DACs and ADCs [53]. The data converter's flexibility enables the application in multiple communication systems, i.e., for optical, wireless, wireline, etc.

For wireless transmission, DACs and ADCs are key components of software defined radios [54] and cognitive radios [55]. They provide the interface between the analog domain and the DSP to achieve an ultimate configurability in terms of modulation format, coding, bandwidth, etc. [56]. However, compared to the bandwidths required for optical communications, wireless applications can be characterized as narrow-band.

For optical communication systems, high-speed DACs, ADCs and DSP have attained rising interest in the last decade. Today, optical transmission systems use DACs in combination with Tx DSP for higher-order modulation formats, compensation of channel impairments, forward error connection (FEC), etc. [57]. DACs offer a great deal of control and interface with other electronics very well [58].

DACs can be segmented into two categories: oversampling converters and Nyquist rate converters. Oversampling converters operate at sampling rates of multiples of their output frequencies; they are usually implemented as $\Sigma\Delta$ -modulators. Due to the high oversampling in combination with digital noise shaping, the accuracy is enhanced [59]. More information on oversampling converters can be found in [59–61].

Nyquist rate converters operate at sampling rates close to their output frequencies. For high-speed DACs, the sampling rate is usually close to technologically maximum feasible values. This thesis focuses on Nyquist rate DACs.

In this chapter, an overview on state of the art DACs and manufacturing technologies is given. Further, DAC interleaving concepts and the required analog components for DAC interleaving are introduced. For the detailed treatment of DAC interleaving concepts please refer to Chs. 3 to 5.

This chapter is structured as follows: first, an overview on the state of the art high-speed DACs is given. Then, different manufacturing technologies for DACs

are discussed. After introducing current-steering DAC architectures, the zero-order-hold (ZOH) operation of the Nyquist rate DAC is described. DAC performance impairments are stated, including static, dynamic and noise limitations. Then, the three DAC interleaving concepts are shortly introduced; they are later analyzed and discussed in detail in Chs. 3 to 5. Finally, analog components for DAC interleaving are presented.

2.1 State of the Art DAC Realizations

Today's fastest high-speed DACs have sampling rates of around 100 GS/s. For DACs, there is usually a trade-off between high bandwidth and high resolution. DACs are manufactured using different materials and processes, respectively. The most common ones are CMOS, SiGe and InP, which are covered in the next section. Good insights into CMOS data converters can be obtained from [62, 63]. Information on advanced data converters can be found in [30]. A general overview on fast signal generation solutions is given in [64]. The most recent results regarding interleaving concepts, which originated during the time this thesis has been authored, are mentioned later in Chs. 4 and 5. Note that parts of this section have been previously published in [47].

Table 2.1 lists commercially available DAC solutions and arbitrary waveform generators (AWGs). The entries are sorted by sampling rate. Furthermore, the bandwidth, the vertical resolution, the ENOB, the single-ended (SE) output amplitude, and the manufacturing technology are stated. The highest bandwidth reported so far, is provided by Micram's DAC4, which has 40 GHz analog bandwidth [65]. In [66], insights are provided into Micram's DACII. Keysight's AWG provides two more bits of vertical resolution, although the ENOB is only 0.5 bit higher. The analog bandwidth is 20 % lower.

There are further commercial DACs, but the knowledge about them is kept proprietary, e.g., Broadcom's 64 GS/s 8-bit DAC in 20 nm CMOS [67]; ENOB and bandwidth specifications have not been published.

Recent research results of DAC integrated circuits (ICs) are listed in Tab. 2.2. The DAC with the highest sampling rate of 107 GS/s is a power DAC with only three bits, whereby digital-to-analog (D/A) conversion and amplification are realized in a single device. Furthermore, there is virtually no difference concerning the performance between the results of research and commercial products. This underlines the importance of advances in this area, as they would be adopted by the industry immediately. For the industry, CMOS DACs are most attractive due to their low power consumption and the ability for mass production. A table with DAC performances from 2013 and before can be found in [68].

Table 2.1 Overview on commercially available high-speed DACs and AWGs (sorted by sampling rate); sources: [47] & open literature.

	Micram DAC4 [65]	Keysight M8196A [29, 69]	Socionext [28]	Anritsu G0374A [70]	SHF 614B [71]	Tektronix AWG70001A [72]
Sampl. rate (GS/s)	100	92	92	64	60	50
3 dB Bandw. (GHz)	40	32	> 26 GHz	n/a	27	15
Resolution (bit)	6	8	8	2	6	10
ENOB (bit)	4.6 (@48 GHz)	5.3 (@31.9 GHz)	> 6	n/a	n/a	4.6 (@14.99 GHz)
Ampl. (mVpp, SE)	500	1000	n/a	900	1500	500
Techn.	SiGe BiCMOS	28 nm CMOS	28 nm CMOS	n/a	n/a	n/a

Table 2.2 Overview on recent research results in the field of high-speed DAC ICs (sorted by sampling rate); sources: [47] & open literature.

	III-V Labs [77, 78]	University of Stuttgart [79]	NTT [80]	Anritsu [81]
Sampling rate (GS/s)	107	100	90	80
3 dB Bandwidth (GHz)	n/a	13	>40	n/a
Resolution (bit)	3	8	6	6
ENOB (bit)	n/a	3.2 (@24.9 GHz)	n/a	n/a
Amplitude (mVpp, SE)	ca. 4000	ca. 400	500	360
Technology	InP DHBT	28 nm CMOS	InP HBT	InP HBT
Interl. Concept	none	time-interleaved	none	none

Besides the high-speed DACs presented in this section, there are also DACs in the low GS/s range with a higher vertical resolution, e.g., a 12 bit 10 GS/s DAC based on InP heterojunction bipolar transistor (HBT) [73] or a 65 nm CMOS 10 GS/s 16 bit RF-DAC [74]. More information on RF-DACs can be found in [75, 76].

2.2 Manufacturing Technologies

ICs are manufactured using different materials and technologies dependent on the targeted application. The emergence of fabless design companies and foundries has altered the semiconductor industry's business models, since system integrators dictate the development of both ICs and processes that are required to integrate multiple functionalities in a single chip or in a single package [82].

The international roadmap for devices and systems [82], which is the successor of the international roadmap for semiconductors, describes both the evolution of the industry and upcoming trends with a focus on CMOS technology. CMOS technology has been continuously improved by reducing both the transistor size and the power per transistor. This enables a denser integration, while at the same time, increases the switching speed [38, 82]. However, the technology advancement is not as fast as Moore's law observed and predicted in the 1960's anymore, but follows a constant path [39, 40]. The main issue for scaling circuits by going to a smaller process node is, that the cost per transistor rises for smaller process nodes, since the lithography costs rise and the yield declines [28, 30, 40].

Although, smaller process nodes become more expensive, there is still an advantage for digital circuitry, due to the smaller footprint, reduced power consumption, etc.

Table 2.3 Overview on current and upcoming SiGe HBTs BiCMOS processes; sorted in alphabetical order.

Foundry	Process	Node	$f_T/f_{\max}$ (GHz)	Ref.
GlobalFoundries	SiGe 9HP	90 nm	310/370	[83, 84]
GlobalFoundries	SiGe 8XP	130 nm	250/340	[85, 86]
IHP	SG13G2	130 nm	300/500	[87]
IHP	n/a*	130 nm	505/720	[88]
Infineon	B11HFC	130 nm	250/370	[89]
STM	BiCMOS55	55 nm	$> 300/\gg 300$	[90]
STM	BiCMOS9MW	130 nm	230/280	[91]

* recent research result

However, for analog circuitry, the power consumption does not necessarily decrease, since, higher dynamic performance can normally be obtained by utilizing more power [30, 41]. For high-speed CMOS DACs the design challenges are becoming more critical with every technology generation, since smaller geometries require lower voltages and smaller devices have both larger variability and less electrostatic discharge immunity. Furthermore, interconnects are closer spaced, which increases cross talk and noise coupling [28].

For the manufacturing of high-speed DACs, multiple materials and processes are viable: Si, SiGe, InP, etc. In communication systems, CMOS DACs are desired, since they are energy-efficient and can be integrated with digital functionality on the same chip [30].

A typically used rule of thumb in CMOS high-speed current switching ICs is to limit the switching speed to 10 % of the technologies f_T [92, 93].¹ The performance in deep sub-micron processes is primarily limited due to the large parasitic effects associated with the backend-of-line interconnects, i.e., resistance and capacitance [95, 96]. A 28 nm fully depleted silicon on insulator CMOS process reaches a transition frequency f_T and a maximum oscillation frequency $f_{\max}$ of about 300/330 GHz [38, 97].² A more recent 22 nm Fin field-effect transistor (FET) process achieves an $f_T/f_{\max}$ of 300/450 GHz [98]. However, subsequent Fin FET process nodes have a lower performance due to higher parasitics [98].

An overview on the evolution of analog CMOS is found in [99]. Both the DAC's analog bandwidth and the dynamic linearity are affected by the switch pair's speed.

¹ Transition frequency f_T : the frequency at which the current gain of the transistor is 0 dB [94].

² Maximum oscillation frequency $f_{\max}$: the frequency at which the maximum available power gain of the transistor is 0 dB [94].

Table 2.4 Overview on current and upcoming InP HBT processes; sources: [107] & open literature.

Foundry	Process	$f_T/f_{\max}$ (GHz)	Ref.
III/V Labs	n/a*	> 200/ > 400	[108, 109]
ETHZ	n/a*	495/882	[110]
FBH	n/a*	410/480	[111, 112]
FhG IAF	n/a*	> 350/ > 450	[113]
GCS	DHBT3	300/250	[114, 115]
NGAS	TF4 (4Met, HBT)	> 250/> 300	[116]
NGAS	n/a*	410/850	[117]
Teledyne	n/a*	> 520/> 1100	[118]
UCSB	n/a*	> 480/> 1000	[107, 119]

* recent research result

Further, the analog bandwidth depends on the circuit complexity and the desired output current. New manufacturing technologies can be considered to overcome the present limitations.

Two other common technologies for DACs are evaluated by means of f_T and $f_{\max}$, i.e., InP and SiGe HBTs [100]. For analog and mixed-signal applications, it is important to have both a high f_T and $f_{\max}$ [78].

SiGe HBTs are promising for low-noise applications due to their low noise figures [101]. With BiCMOS processes, both bipolar and n- and p-type CMOS transistors can be utilized on the same IC [30, 102]. This way, the advantages of both technologies are synergized. In Tab. 2.3, an overview on current SiGe HBT BiCMOS processes from multiple foundries is given: $f_T/f_{\max}$ values of up to 505/720 GHz have been achieved in SiGe.

For ultra-high-speed, purely compound semiconductors are the best choice, i.e., III-V semiconductors. InP-based HBTs have superior material properties, i.e., an extraordinary electron mobility and an outstanding frequency response [101, 103, 104]. However, higher supply voltages are required, which translate to higher power consumption. Moreover, the size of InP transistors is large compared to SiGe transistors. The emitter width is in the range of 0.1 to 1 μm [78]. In order to combine different materials, i.e., InP and CMOS, multiple ICs need to be integrated into a single package nowadays. However, packaging and interconnect constraints limit the overall performance [105]. Substrate transfer processes may overcome these limitations in future commercial applications [106].

In Tab. 2.4, an overview on InP HBTs is given, whereby $f_{\max}$ values of over 1.1 THz and f_T values around 500 GHz have been demonstrated; a node size definition, as for CMOS, is not established for InP. Next to InP HBTs, there are investigations on high electron mobility transistorss (HEMTs) with $f_{\max}$ around 1 THz [120], above 1 THz [121] and even around 1.5 THz [122].

In the future, horizontal features are going to approach the range of a few nanometers shortly beyond 2020. Hence, the transistors need to evolve from 2D to 3D structures, while reducing the power consumption [82]. It is highly desired to integrate new materials in present CMOS processes in order to profit from the high reliability and the mass production abilities. Further approaches may comprise tunnel transistors, carbon nanotubes, III-V materials [103], Ge [123], 2D materials, etc. [82, 99].

For optical communications, integration of both electronics and optics in a single opto-electronic IC is desired [124, 125]. Therefore, many research activities address silicon photonics [38, 58, 124]. Next to silicon photonics, approaches towards a generic photonic integration technology are pursued in order to facilitate photonic circuit design [126].

2.3 DAC Architectures

In general, DACs consist of four building blocks: digital interface, digital logic, converter core, and output node. The most critical block is the converter core [66], which can be built using different architectures, i.e., resistor-based, capacitor-based and current-source-based [59]. High-speed DACs are usually designed with a current-source architecture, since the current-steering nature of the circuitry is inherently fast and the demanding output buffer can be replaced with a simple resistor [66, 68, 127]. For information on the two other architectures, please refer to [59].

In Fig. 2.1, the two prominent architectures for current-source-based DACs with b bits are depicted, which are also called current-steering DACs. The unary-weighted architecture depicted in Fig. 2.1(a) consists of $2^b - 1$ switchable equal current

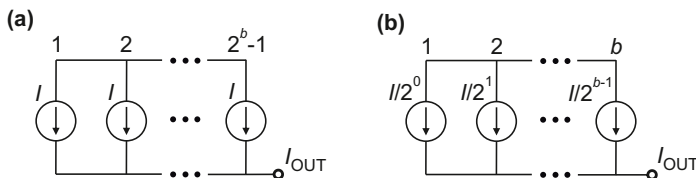


Figure 2.1 High-speed DAC current-steering architectures: unary weighted (a), binary weighted (b).

Table 2.5 Comparison of DAC architectures; source: [68].

Architecture	Advantages	Disadvantages
Unary weighted	+ identical stages + monotonicity	- 2^{b-1} current sources - decoding logic - large area
Binary weighted	+ b current sources + no decoding logic	- large current ratios
Segmented	+ strong glitch reduction	- additional logic

sources [59, 128]. By increasing the digital input code by one, one more current is switched on, thus enabling inherent monotonicity, but for the expense of a rather large circuit size as $2^b - 1$ DAC cells are required. The individual currents are combined to form the output current I_{OUT} . This architecture requires an additional decoding logic before the converter core to convert the binary-coded bit stream to a unary-coded bit stream.

The binary weighted architecture consists of b current sources, which are weighted in the ratio $2^0 : 2^1 : \dots : 2^{b-1}$ as depicted in Fig. 2.1(b) [59, 128]. It is a straightforward architecture. However, the manufacturing requirements for current ratios are significant, e.g., 128:1 for an 8-bit DAC [128]. In Tab. 2.5, both the benefits and drawbacks of the different architectures are summarized.

For the design of high-speed DACs with high resolution, none of the aforementioned architectures might be suitable and hence, a hybrid alternative is favored [59, 128]. In this case, the DAC is segmented into two sub-DACs with different architectures, whereby one sub-DAC takes care of the most significant bits and the other handles the least significant bits and their output currents are summed. Usually, the most and least significant bits are covered by the unary- and the binary-weighted architecture, respectively. The segmented architecture allows for higher-resolution implementations and reduces undesired transients like glitches. However, the increase in DAC performance is traded for a significantly higher circuit complexity due to the decoding logic [59, 68].

2.4 Nyquist Rate DAC ZOH Operation

A DAC is a device that transforms a discrete-time signal $x(k)$ into an analog signal $y(t)$ as depicted in Fig. 2.2; k and t denote the discrete and the continuous time, respectively.

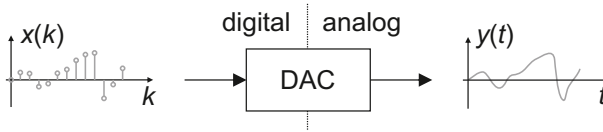


Figure 2.2 DAC block diagram with input and output signal.

The DAC generates an output stream of samples, which are spaced by the sampling period T_s . The sampling rate f_s is equal to the inverse of the sampling period $f_s = 1/T_s$ and the real-valued output of the DAC is given as [129, 130]:

$$y(t) = T_s \cdot \left(\sum_{k=-\infty}^{+\infty} x(k) \cdot \delta(t - kT_s) \right) * g(t), \quad (2.1)$$

whereby $\delta(t)$, $*$, and $g(t)$ denote the Dirac distribution,³ the convolution operation,⁴ and the DAC impulse response, respectively.

The DAC output can be either in the unit volt (V) or ampere (A). In this thesis, it is assumed to be in volt. Since the convolution operation has the unit second (s), the Dirac distribution $\delta(t)$ and the DAC impulse response $g(t)$ are given in Hertz (1/s) and volt per second (V/s), respectively. The discrete-time signal $x(k)$ is assumed dimensionless [130].

Ideally, the DAC generates an output stream of infinitely sharp pulses [53] as described by $g(t) = \frac{1}{T_s} \frac{V_{pp}}{2}$:

$$y(t) = \frac{V_{pp}}{2} \sum_{k=-\infty}^{+\infty} x(k) \cdot \delta(t - kT_s). \quad (2.2)$$

³ The Dirac distribution can be approximated by means of the rectangular function [131]:

$$\delta_\theta(t) = \begin{cases} \frac{1}{\theta} & |t| \leq \frac{\theta}{2} \\ 0 & \text{else} \end{cases}.$$

For the case $\theta \rightarrow 0$ the rectangular function converges to the Dirac distribution

$$\delta(t) = \lim_{\theta \rightarrow 0} \delta_\theta(t).$$

⁴ The convolution for two functions $x(t)$ and $g(t)$ is defined as [131]:

$$g(t) * x(t) = \int_{-\infty}^{\infty} g(\tau) x(t - \tau) d\tau.$$

However, real-world DACs cannot generate infinitely sharp pulses. Instead, the output value is kept constant for the sampling period T_s , which is called either zero-order-hold (ZOH) or NRZ mode. The operation can be described with a rectangular DAC impulse response $g(t)$ as:

$$y(t) = T_s \cdot \left(\sum_{k=-\infty}^{+\infty} x(k) \cdot \delta(t - kT_s) \right) * \frac{V_{pp}}{2T_s} \text{rect} \left(\frac{t}{T_s} \right), \quad (2.3)$$

whereby $\text{rect}(t)$ denotes the rectangular function, which is defined as

$$\text{rect} \left(\frac{t}{T_s} \right) = \begin{cases} 1 & |t| < T_s/2 \\ \frac{1}{2} & |t| = T_s/2 \\ 0 & \text{else} \end{cases}. \quad (2.4)$$

The DAC operation is depicted in Fig. 2.3, whereby the graphs are normalized to a maximum value of one. The discrete-time signal $x(k)$ in Fig. 2.3(a) is convolved with the rectangular DAC impulse response $g(t)$ in Fig. 2.3(b). The resulting DAC output signal is given by $y(t)$ in Fig. 2.3(c).

The operation in (2.3) can be described equivalently in the frequency domain by applying a Fourier transform.⁵ The operation is depicted in Figs. 2.3(d)-2.3(f). The spectrum $X(e^{j2\pi f})$ of the discrete-time signal $x(k)$ is visualized in Fig. 2.3(d) and obtained by means of the discrete-time Fourier transform (DTFT).⁶ The spectrum is periodic, and thus, infinitely broad with respect to continuous time [130].

The Fourier transform $G(f)$ of the impulse response $g(t)$ is denoted as transfer function or frequency response; it is depicted in Fig. 2.3(e) and given as:

$$G(f) = \mathfrak{F} \{g(t)\} = \mathfrak{F} \left\{ \frac{V_{pp}}{2T_s} \cdot \text{rect} \left(\frac{t}{T_s} \right) \right\} = \frac{V_{pp}}{2} \cdot \text{sinc}(\pi f T_s), \quad (2.5)$$

whereby $\mathfrak{F} \{ \cdot \}$ and f denote the continuous Fourier transform and the continuous frequency, respectively. The spectrum $Y(f)$ of the DAC output signal $y(t)$ is depicted in Fig. 2.3(f).

⁵ The Fourier transform $G(f)$ of the function $g(t)$ is defined as [131]:

$$G(f) = \mathfrak{F} \{g(t)\} = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt.$$

⁶ The DTFT $X(e^{j2\pi f})$ of the discrete-time signal $x(k)$ is defined as [53]:

$$\mathfrak{F}_{\text{DTFT}} \{x(k)\} = X(e^{j2\pi f}) = \sum_{k=-\infty}^{\infty} x(k) e^{-j2\pi f k}.$$

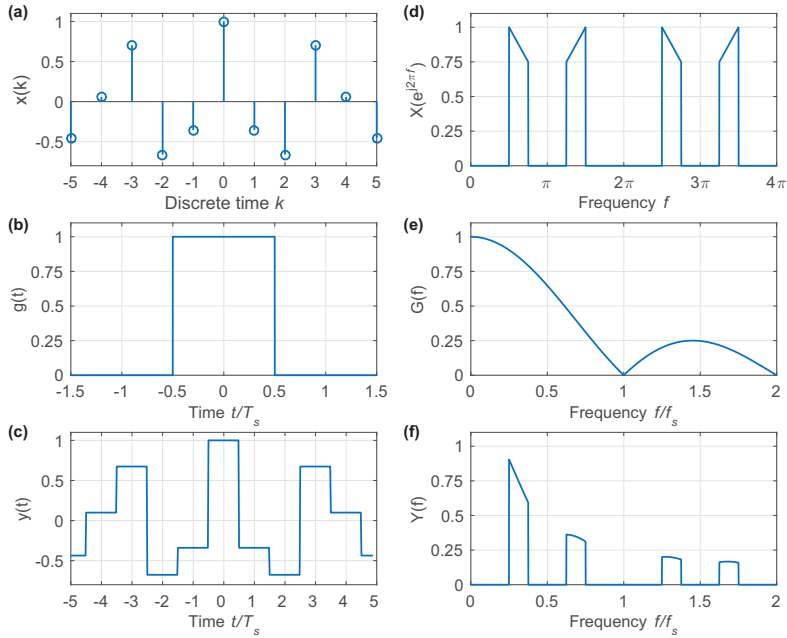


Figure 2.3 ZOH-DAC operation in time and frequency domain: the discrete-time signal (a) is convolved with the DAC impulse response (b) resulting in the DAC output signal (c); the spectrum (d) is multiplied with the DAC transfer function (e) resulting in the output spectrum (f).

Additionally, the frequency response of the ZOH-DAC is visualized in Fig. 2.4. The ZOH operation relates to a sinc-function in the frequency domain according to (2.5), which equals zero at multiples of the sampling frequency f_s . The spectrum of a discrete-time signal with a sampling frequency of f_s is periodic with f_s . Due to the periodic nature of the discrete spectrum, images are present. These are called Nyquist zones and are defined by the frequency bands $[lf_s/2, (1+l)f_s/2]$, $l \in \mathbb{N}_0$ [132]. The odd Nyquist zones contain regular images and the even Nyquist zones contain mirror images. The mirror images are complex conjugated and have a reversed frequency axis orientation relative to the regular images as depicted in Figs. 2.3(d) and 2.3(f).

The frequency response of a real-world DAC is composed of the sinc-function and the actual low-pass characteristic of the DAC itself. For high-speed DACs, the first Nyquist zone is of primary importance. The sinc-attenuation in the first Nyquist zone can be easily compensated by means of digital pre-distortion [133]. For some applications, the output signal can be low-pass filtered to suppress the

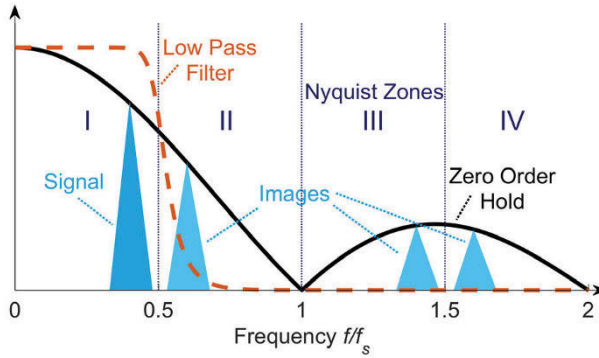


Figure 2.4 ZOH-DAC frequency response with Nyquist zones I-IV; Generated signal spectrum and generated image replicas of the original spectrum; low-pass filter (LPF) for filtering the images. Adapted, with permission, from [45].

images [53]. Due to the finite roll-off of the LPF, residual spectral components remain for frequencies $> f_s/2$, which may distort the signal quality. Therefore, an exemplary rule of thumb states, that a DAC can be utilized for band-limited baseband signals up to frequencies $\leq 0.4f_s$ by means of oversampling in order to sufficiently suppress the images [134].

Next to the ZOH mode, there are other modes of operation, e.g., RZ mode with a duty cycle of 25 % or 50 % [135], first order hold with a triangular pulse shape [136] or partial order hold (with a trapezoidal shape) [76]. These offer beneficial spectral properties for certain applications, but a high-speed implementation is difficult. Hence, they are not considered further in this thesis.

2.5 DAC Performance Impairments

The important parameters for high-speed DACs are the sampling rate, the analog output bandwidth, and the ENOB [137]. DACs are limited by both static and dynamic effects as well as noise. The main effects limiting high-speed DAC performance are image replica generation, bandwidth limitations, nonlinearity spurs and jitter [59]. Additional impairments need to be considered for the IC packaging [59].

In Secs. 2.5.1 and 2.5.2, the measures and limitations for both static and dynamic DAC performance are stated and explained. Furthermore, the main noise contributions, i.e., quantization noise, phase noise & jitter, and thermal noise are introduced in Secs. 2.5.3 to 2.5.5.

2.5.1 Static Performance

DACs are limited by static and dynamic effects. This section starts by stating the static performance measures, whereby differential nonlinearity (DNL) and integral nonlinearity (INL) are emphasized. Then, the origins of static performance limitations are described. For high-speed DACs with moderate resolution, the dynamic effects are dominant [59], which are described in Sec. 2.5.2.

Static Performance Measures

The static performance of a DAC is measured with multiple parameters, e.g., analog resolution, offset, gain error, DNL, monotonicity, hysteresis, INL, etc. [59, 138]. In the following, DNL and INL are shortly described.

Differential Nonlinearity The amplitude difference between the output signal levels of subsequent input codes is ideally equal. In reality, there are mismatches resulting in nonlinear distortions. The deviation in amplitude between subsequent output codes is called differential nonlinearity (DNL) and is usually measured in least significant bits [59]. The DNL is specified for each input code; however, often the maximum DNL is simply referred to as DNL [59].

Integral Nonlinearity Integral nonlinearity (INL) measures the deviation of the transfer function from the ideal characteristic [59]. It provides another measure for static nonlinearities and is calculated by summing the DNL values. Usually, the INL is stated in least significant bits and the maximum INL is referred to as INL. The correlated part of the INL is the main source for nonlinearity, whereas the uncorrelated part can be treated as noise [59].

Static Performance Limiters

The origins of the static performance limitations are briefly stated in the following.

A mismatch of the current sources results in a deviation from the actual to the nominal current value. This is based on manufacturing process variations and increases both DNL and INL [30].

Further, the finite output impedance of the current sources results in a code-dependent output impedance. This can be characterized with a static nonlinearity,

whereby the shape, i.e., bow-shape, or S-shape, depends on whether a single-ended or a differential architecture is used [139, 140]. Using a differential architecture with two complementary signals can improve the INL by one order of magnitude [30].

2.5.2 Dynamic Performance

In this section, the dynamic performance measures are summarized. Then, the root causes of the dynamic performance limitations are briefly explained.

Dynamic Performance Measures

The DAC's dynamic performance can be evaluated with different measures, e.g., sampling rate, frequency response, analog bandwidth, eye diagram, step response, glitch impulse area, SNR, signal to noise and distortion ratio (SINAD), ENOB, dynamic range, total harmonic distortion (THD), spurious free dynamic range (SFDR), intermodulation distortion (IMD), figure of merit (FoM), etc. [59, 141]. For communication applications, the most common measures are sampling rate, analog bandwidth, and ENOB, which are introduced in Secs. 2.4 and 2.5.3.

Although, FoMs characterizing the performance of DACs are available, their usage is limited. There is a wide variety of FoMs and not an one-fits-all solution due to the wide variety of DAC applications and the frequency-dependent DAC performance [30, 59, 142].

Dynamic Performance Limiters

The timing accuracy is essential for a good dynamic performance. Timing mismatch can be present between the digital logic and the converter core as well as in between the converter core's current sources. Timing imprecision is caused by clock skew, process mismatch in latches, drivers and switches, unequal loading or interconnect length, transmission line effects in the clock network, etc. [30, 127, 143, 144].

Furthermore, both the output resistance and capacitance are not constant with respect to frequency, and hence, there is a signal dependent output impedance, which results in nonlinear distortions [30, 127].

The clock phase noise translates to DAC jitter, as covered later in Sec. 2.5.4. Moreover, a fraction of the clock signal, that switches the current cells, is fed through to the output [59, 127]. The clock feedthrough can be reduced with a differential design.

Table 2.6 SNR values according to (2.7) for different number of bits b .

b in bit	1	2	3	4	5	6	7	8
SNR in dB	7.8	13.8	19.8	25.8	31.9	37.9	43.9	49.9

Furthermore, the DAC draws large current pulses from the power sources that are used to charge and discharge capacitances [30,59]. These dynamic currents result in additional data-dependent supply noise due to the inductive behavior of the power supply circuit. Their spectral level must stay well below the quantization noise floor [59].

2.5.3 Quantization Noise

A fundamental limitation on the quality of the converter's output signal is set by the resolution of the DAC, i.e., the number of bits b [59, 129]. The digital signal is subject to noise associated with the finite amplitude resolution in order to represent the signal. Actually, quantization is a nonlinear operation, but is commonly referred to as noise.

The DAC output is either a current or a voltage: it is assumed that the full-range DAC output is equivalent to the peak-to-peak voltage V_{pp} and this full range can be divided into 2^b quantization intervals, each with an amplitude of $\Delta_Q = \frac{V_{pp}}{2^b}$. Each amplitude value of the signal is mapped to a discrete amplitude level, which results in a quantization error e_Q . Assuming an equal probability distribution of the quantization error $\chi(e_Q) = 1/\Delta_Q$ in each interval $[-\Delta_Q/2, +\Delta_Q/2]$, the resulting average quantization noise power σ_Q^2 is given as [129, 145]:

$$\sigma_Q^2 = \int_{-\Delta_Q/2}^{\Delta_Q/2} \chi(e_Q) e_Q^2 de_Q = \frac{1}{\Delta_Q} \int_{-\Delta_Q/2}^{\Delta_Q/2} e_Q^2 de_Q = \frac{\Delta_Q^2}{12} = \frac{V_{pp}^2}{12 \cdot 2^{2b}}. \quad (2.6)$$

For an infinite resolution, the quantization noise reduces to zero according to $\lim_{b \rightarrow \infty} \sigma_Q^2 = 0$. The interested reader finds information on quantization noise for signals with an unequal probability distribution in [146].

In order to calculate the SNR according to the quantization noise power, a full swing sine signal is assumed to be the input signal of the quantizer with the average power $\sigma_S^2 = V_{pp}^2/8 = (\Delta_Q 2^b)^2/8$ [59]. The resulting SNR is given in dB as

$$\text{SNR}|_{\text{dB}} = 10 \cdot \log_{10} \left(\frac{\sigma_S^2}{\sigma_Q^2} \right) \approx 6.02 \cdot b + 1.76 . \quad (2.7)$$

Table 2.6 lists the SNR values for different resolution values b . Every additional bit enhances the SNR by 6.02 dB, e.g., for an SNR > 30 dB, five bits are required.

The SNR calculation only considers quantization noise. However, in real-world circuits additional noise and distortion are present, which are based on both static and dynamic effects as noted in Secs. 2.5.1 and 2.5.2. By considering b as an effective resolution, i.e., the ENOB, and by considering all of noise and distortion as SNR, (2.7) can be rewritten as

$$\text{ENOB} = \frac{\text{SNR}|_{\text{dB}} - 1.76}{6.02} . \quad (2.8)$$

For signals other than full-range sine signals, the resulting signal power is different, and hence, (2.7) and (2.8) may change [146].

2.5.4 Phase Noise and Jitter

Phase noise and jitter are often subject to misunderstanding: phase noise denotes the random phase deviations of a periodic signal from the ideal phase; it is usually specified in the frequency domain in dBc/Hz. Jitter denotes the phase deviations of a data or clock signal from the ideal phase; it is a time-domain term and is usually stated in s.

Phase Noise

The clock signal is given as a sine wave according to

$$s_{\text{CLK}}(t) = (V_0 + \Delta V(t)) \cdot \sin(2\pi f_c t + \phi(t)) , \quad (2.9)$$

whereby V_0 , $\Delta V(t)$, f_c , and $\phi(t)$ denote the amplitude, the time-variant amplitude deviations, the clock frequency, and the time-variant phase, respectively. The contributions to the clock signal of $\Delta V(t)$ and $\phi(t)$ are also known as amplitude and phase noise, respectively [147]. Usually, the phase noise contributions dominate the resulting jitter; hence, amplitude noise can be neglected [59].

A typical phase noise spectrum from a free-running oscillator has different regions, i.e., a $1/f^3$, a $1/f^2$ and a white noise region [148–150]. For communication systems, a phase-locked loop (PLL) is normally used, which stabilizes the oscillator at lower frequencies and alters the phase noise spectrum [151, 152]. In this thesis, phase noise spectra from PLLs rather than phase noise spectra from free-running oscillators are considered.

Jitter

Jitter can be decomposed into random and deterministic jitter [153]. Random jitter originates from noise, i.e., phase noise of the clock signal.

The deterministic jitter results from spurious tones in the clock signal (periodic jitter), the frequency response of the DAC (data-dependent jitter) and timing inaccuracies of the digital logic (duty cycle distortion) [154, 155].

Root mean square (RMS) jitter τ can be calculated from the phase noise spectrum according to [156]:

$$\tau = \frac{1}{2\pi f_c} \sqrt{2 \int_{f_{PN,1}}^{f_{PN,2}} 10^{\Phi_{PN}(f)/10} df}, \quad (2.10)$$

whereby $f_{PN,1}$, $f_{PN,2}$, $\Phi_{PN}(f)$ denote the integration boundaries and the one-sided phase noise spectrum in dBc/Hz, respectively. The lower integration boundary should be as low as possible to calculate the RMS jitter; however, oscillator specifications are usually not given for frequencies < 10 Hz [157]. The choice of the upper boundary depends on the application: for example, if a band-pass filter (BPF) is employed at the clock input, $f_{PN,2}$ should account for its cutoff frequency. Usually, the DAC current sources' jitter contributions differ for each current cell, i.a., due to transmission line effects as described in Sec. 2.5.2.

In order to estimate the jitter's impact on the DAC output signal, a sine wave signal is assumed. The average power of the jitter error is given as $\sigma_J^2 = V_{pp}^2/8 \cdot (2\pi f)^2 \tau^2$ [59] and the resulting jitter-based SNR is given according to [59]:

$$\begin{aligned} \text{SNR}|_{\text{dB}} &= 10 \cdot \log_{10} \left(\frac{\sigma_S^2}{\sigma_J^2} \right) \\ &= 10 \cdot \log_{10} \left(\frac{\frac{V_{pp}^2}{8}}{\frac{V_{pp}^2}{8} (2\pi f)^2 \tau^2} \right) = 20 \cdot \log_{10} \left(\frac{1}{2\pi f \tau} \right). \end{aligned} \quad (2.11)$$

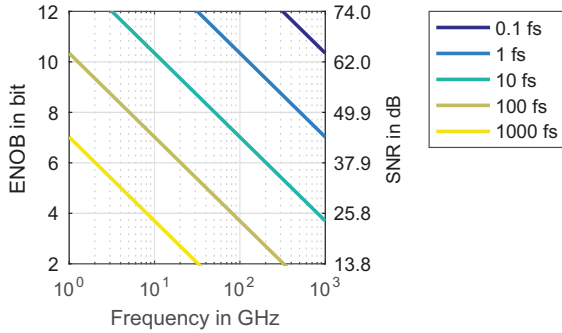


Figure 2.5 ENOB and SNR with respect to the frequency for different RMS jitter values.

By inserting (2.11) into (2.8), the ENOB can be stated with respect to the jitter to derive jitter requirements. The ENOB and the corresponding SNR values are visualized for different jitter values with respect to the generated frequency in Fig. 2.5. For an ENOB > 4 bit at a signal frequency of 100 GHz, an RMS jitter < 100 fs is required. For wideband signal synthesis, the jitter requirements can be relaxed, since a sine signal is assumed for this calculation.

Next to jitter, other impairments limit the maximum ENOB as mentioned in Secs. 2.5.1 to 2.5.3 and 2.5.5. Jitter in (2.11) and quantization noise in (2.7) can be interpreted as upper bounds for the performance. Both effects can be combined to a single formula stating the SNR for a sine signal as [59]:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{\sigma_S^2}{\sigma_J^2 + \sigma_Q^2} \right) = 10 \cdot \log_{10} \left(\frac{1}{(2\pi f \tau)^2 + \frac{2}{3} 2^{-2b}} \right). \quad (2.12)$$

A more detailed treatment on the relation between clock phase noise and DAC output signal quality is found in [158–161].

For many applications, frequency dividers and multipliers are required. Their impact on phase noise and jitter is investigated in [162–167].

2.5.5 Thermal Noise

Thermal noise or Johnson-Nyquist noise is a fundamental limitation of electronic circuits resulting from the thermal movement of charge carriers [168–170]. The white noise power spectral density (PSD) is given by

$$\zeta_T(f) = k_B T, \quad (2.13)$$

whereby k_B and T denote the Boltzmann constant and the temperature, respectively. The formula is valid up to frequencies in the THz range and equals to about -174 dBm/Hz at room temperature in the case of impedance matching.

For data converters, an important limit is set by the thermal noise associated with the current-sources, which is termed $k_B T/C$ noise [59]. The noise PSD is given by

$$\zeta_{TC}(f) = \frac{k_B T}{1 + (2\pi f R C_L)^2}, \quad (2.14)$$

whereby C_L and R denote the load capacitance and the impedance of the individual current source. Hence, $k_B T/C$ noise is thermal noise filtered with a low-pass RC-filter.

Besides the previously mentioned effects, other parasitic effects need to be considered for the DAC design, e.g., ground noise, supply noise, etc. [59, 62].

2.6 DAC Interleaving Concepts

The performance of today's high-speed DACs is limited in terms of analog bandwidth and vertical resolution as described in Ch. 1. In order to overcome these limitations, parallel DAC concepts have been introduced to aggregate the output signals of multiple DACs [47, 68]. The term “interleaving” is commonly used for these parallelization concepts. In this section, a short introduction on DAC

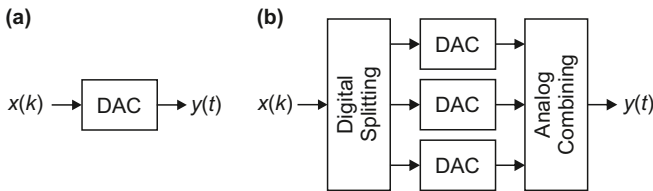


Figure 2.6 DAC concepts: single DAC (a) and interleaved DAC (b).

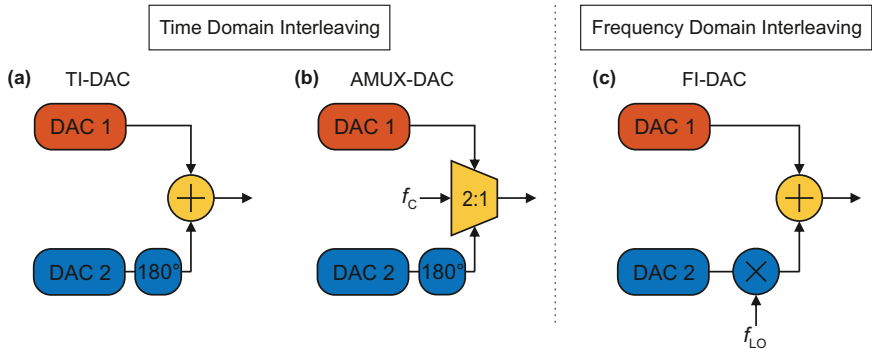


Figure 2.7 Block diagrams of DAC interleaving concepts with two DACs each (simplified): TI-DAC (a), AMUX-DAC (b), FI-DAC (c).

interleaving concepts is given. These concepts are covered in more detail in Chs. 3 to 5.

In Fig. 2.6, the basic interleaving idea is visualized. In Fig. 2.6(a), the classical single DAC topology is depicted. In Fig. 2.6(b), the generic interleaved DAC topology is visualized. For interleaving, the digital input signal is split into multiple sub-signals, which are each individually D/A converted. Finally, they are recombined by means of analog circuitry.

The interleaving can be performed in both the time and the frequency domain. For this thesis, three concepts are distinguished:

1. time interleaving DAC (TI-DAC) [79, 171],
2. analog multiplexing DAC (AMUX-DAC) [42, 43, 172–175], and
3. frequency interleaving DAC (FI-DAC) [44–52, 176].

Although, the first two concepts are both time domain concepts, a distinction is made, since the first and the second use a linear and a nonlinear operation for combining the analog sub-signals, respectively. Only the nonlinear operation enhances the analog bandwidth as explained later. In Fig. 2.7, simplified block diagrams of the three interleaving concepts are visualized with two DACs each.

In Fig. 2.7(a), a block diagram of the TI-DAC concept is visualized. The combined analog output signal is generated by adding the two phase-shifted analog sub-signals. Thereby, the sampling rate is doubled, which doubles the usable bandwidth. However, the analog bandwidth is still limited due to both the analog frequency response of the individual DACs and the sinc roll-off. The phase shift between the

analog sub-signals will be preferably realized by phase-shifting one of the DAC clocks.

In Fig. 2.7(b), a block diagram of the AMUX-DAC concept is shown. Thereby, an AMUX is utilized to switch between the two individual sub-signals. Thereby, the AMUX “samples” the analog sub-signals in the center of each hold period. Due to the switching operation, the sampling rate is doubled, which doubles the usable bandwidth; furthermore, the analog bandwidth is enhanced due to the nonlinear switching operation.

In Fig. 2.7(c), a block diagram of the FI-DAC concept is depicted. The digital signal is divided into two sub-bands in the frequency domain. The analog representations of these sub-bands are each generated with a separate DAC. The 2nd sub-band is upconverted to f_{LO} . Then, both sub-bands are seamlessly added to form the analog representation of the digital signal. With this concept, the analog bandwidth is enhanced. Since, two DACs are used the sampling rate is doubled. However, the sampling rate does not have the same meaning as it has for a single DAC, since the sub-signals are processed in the analog domain after D/A conversion.

It can be noted that all interleaving concepts are usually accompanied by a reduced ENOB due to the impairments of the additional analog components for DAC interleaving. Further, different interleaving concepts can be combined to take advantage from their effects.

2.7 Analog Components for DAC Interleaving Concepts

In this section multiple analog components are introduced, which are required for the DAC interleaving concepts: mixer, frequency multiplexer and analog multiplexer.

2.7.1 Mixer

In this section, the mixer is introduced. It is described analytically, and its topologies, characteristics and impairments are covered.

Ideally, an upconversion mixer is a three-port device that multiplies two signals. It is used to upconvert the input signal $s_{IF}(t)$ with the local oscillator (LO) signal

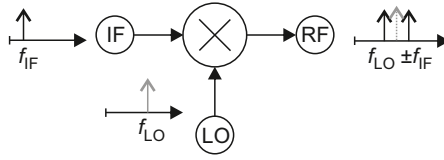


Figure 2.8 RF upconversion mixer block diagram: the input signal at the intermediate frequency (IF) port is upconverted with the LO signal to generate the output signal at the RF port.

$s_{LO}(t)$ at frequency f_{LO} . The mathematical relation between the input signal $s_{IF}(t)$ and the output signal $s_{RF}(t)$ is given for a sine LO signal with frequency f_{LO} as

$$s_{RF}(t) = s_{IF}(t) \cdot s_{LO}(t) \quad (2.15)$$

$$= s_{IF}(t) \cdot \sin(2\pi f_{LO}t) . \quad (2.16)$$

In Fig. 2.8, an upconversion mixer is depicted: the two input ports and the output port are termed IF port, LO port and RF port, respectively. For a cosine input signal, the output signal has frequency lines at the sum and the difference frequency of the input signals [177]. Due to the focus of the thesis on Tx, this section is limited to upconversion mixers.

In general, two RF mixers can be combined to form an I/Q mixer. Then, the LO is split and applied to the in-phase and the quadrature mixer with a 90° phase shift [178].

Mixers are available for many frequency bands, i.e., up to 100 GHz [179–182] and have been demonstrated even up to 300 GHz in research experiments [183, 184].

In general, any nonlinear device can be used as a mixer. Modern mixers are made of Schottky diodes, GaAs FETs, and CMOS transistors. For high volume markets, FET and CMOS mixers are typically used [185–187]; however, for broadband high-performance applications passive diode-based mixers are commonly preferred, e.g., GaAs Schottky diode mixers [185, 188, 189]. For the experiments in this thesis, diode-based mixers are used. The relevant topologies for this mixer type are introduced in the next section.

Topologies

In Fig. 2.9, three diode-based mixer topologies are visualized: unbalanced, single-balanced, and double-balanced. Dependent on the application, one topology can have advantages over the other [190].

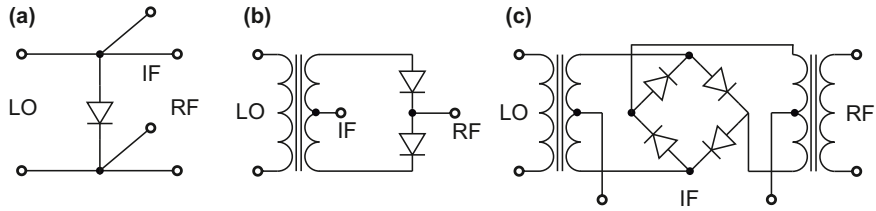


Figure 2.9 Topologies for diode-based mixers: unbalanced (a), single-balanced (b), double-balanced (c).

The unbalanced mixer is depicted in Fig. 2.9(a) [191]. Although, it has a low complexity it is rarely used, since the suppression of even order spurious components is not achieved as for one of the balanced topologies [192]. Furthermore, additional circuitry is necessary to separate the LO, IF and RF ports, i.e., filters.

In Fig. 2.9(b), the single-balanced mixer is depicted. A single-balanced mixer is composed of two unbalanced mixers and a balun [192]. Due to the balun, two copies of the LO signal are combined at the output port with opposite phases. Hence, the LO is significantly suppressed, while the IF signal is not. Alternatively, the IF signal can be fed into the balun in order to suppress it rather than the LO signal at the RF port [191].

In Fig. 2.9(c), a double-balanced mixer is depicted, which is also known as ring modulator [177]. It is composed of a diode ring and two baluns; essentially, two single-balanced mixers are combined [185]. Due to the balancing, both input signals are suppressed at the output yielding an improved performance in terms of both isolation and spurious components [190, 193].

Next to these topologies, there is an advanced type of the double-balanced mixer, i.e., the triple-balanced or double-doubly balanced mixer [192]. Contrary to the previous topologies, this mixer type can have overlapping RF/LO and IF frequency bands [185]; however, the IF port is not DC coupled anymore and the conversion loss is higher [185].

Characteristics and Impairments

The performance of mixers is commonly evaluated with multiple parameters; among them are conversion loss, noise figure, 1 dB and 3 dB compression points, intermodulation products (IMPs), port-to-port isolation, as well as IF and RF frequency responses.

Conversion Loss The conversion loss (CL) specifies the difference in power between the IF signal power level and the power level of the desired one-sided RF signal [193]. The theoretically minimum CL for a passive double-balanced diode mixer is 3.92 dB [194, 195];⁷ typical values range between 4.5 and 9 dB. The additional losses can be attributed to IF and RF mismatch loss, loss in the diode spreading resistance, and loss in the diode junction due to junction resistance as well as generation of IMD products [195].

Noise Figure The noise figure (NF) for passive devices can be approximated solely by the gain, i.e., the CL [185, 195]. However, the actual mixer NF is higher than the CL due to additional noise contributions from the diode(s). The major noise sources in Schottky diodes are thermal noise, shot noise and flicker noise [195]. In addition to these effects, LO phase noise will translate to the RF output noise as well [195].

Nonlinearity The mixer's nonlinear distortions are usually stated by a spurious suppression table (SST) or by the 1 dB and 3 dB compression points [196]. The mixer's nonlinear behavior is twofold: First, there are IMPs of the IF signal and second, there are harmonics of the LO due to saturating the LO input. Hence, for static nonlinear effects the RF output signal is given as

$$s_{\text{RF}}(t) = \sum_{n=1}^{\infty} \left(\sum_{m=1}^{\infty} a_{m,n} \cdot s_{\text{IF}}^m(t) \right) \cdot \sin(2\pi n f_{\text{LO}} t) , \quad (2.17)$$

⁷ For a passive double-balanced diode mixer, ideal diodes are assumed acting as linear rectifiers with a zero forward and an infinite backward resistance [195]. The Fourier series for the square-wave switching function $s_{\text{CLK}}(t)$ with an amplitude of 1, i.e., amplitude levels of ± 1 , is given by

$$s_{\text{CLK}}(t) = \frac{4}{\pi} \sum_{l=1}^{\infty} \frac{\sin((2l-1)2\pi f_{\text{LO}} t)}{2l-1} .$$

Then, the RF output of the mixer $s_{\text{RF}}(t)$ is given for a sine IF signal $s_{\text{IF}}(t)$ by:

$$\begin{aligned} s_{\text{RF}}(t) &= s_{\text{IF}}(t) \cdot s_{\text{CLK}}(t) \\ &= \sin(2\pi f_{\text{IF}} t) \cdot s_{\text{CLK}}(t) \\ &= \frac{4}{\pi} \left(\frac{1}{2} [\cos(2\pi(f_{\text{LO}} - f_{\text{IF}})t) - \cos(2\pi(f_{\text{LO}} + f_{\text{IF}})t)] + \dots \right) . \end{aligned}$$

Thus, the conversion loss CL is given by the factor $2/\pi$ as:

$$\text{CL} = 20 \cdot \log_{10} \left(\frac{\pi}{2} \right) \approx 3.92 \text{ dB} .$$

whereby the nonlinear polynomial coefficients $a_{m,n}$ relate to the IMPs of the IF signal for each LO harmonic nf_{LO} .

These nonlinear effects are determined by diode nonlinearity according to the voltage-current characteristic and overloading of the diode [193]: if the IF signal power approaches the LO power level, the overloading is increased. Hence, the switching time of the diode depends on both the IF and the LO signal [195]. Theoretically, only odd LO harmonics are generated due to the square-wave LO characteristic [197]; however, real mixers generate both even and odd LO harmonics due to imperfections.

Port-to-Port Isolation The port-to-port isolation, characterizing the power that leaks from one port to the other, depends primarily on the chosen mixer topology [185]. For upconversion mixers, both IF-RF and LO-RF isolation are important.

Passive mixers are driven with high LO power levels, i.e., 10-23 dBm [188, 189, 198]. Hence, insufficient LO-RF isolation will result in a significantly present LO at the output. Furthermore, if the IF signal has components close to DC, the LO signal cannot be suppressed by analog filtering without distorting the signal [185].

Normally, IF-RF isolation is not a major concern for system designers, since the upconverted frequency band is far away from the IF band. However, for the FI-DAC application the first upconverted sub-signal at the RF port is spectrally close to the baseband signal at the IF port, if the first and the second sub-signal have similar bandwidths. Hence, a good IF-RF isolation is required, which could be achieved by means of balancing or by using a diplexer with a steep roll-off at the mixer output.

Further, port-to-port isolation is usually reciprocal for passive mixers. It can be enhanced by means of balancing and the use of hybrid junctions as explained in the topologies section [185].

Frequency Response For broadband mixers, both the IF and RF frequency response need to support the bandwidths of the IF and the upconverted RF signal. The frequency response depends on the circuitry, i.e., the diodes, the baluns and the transmission lines.

2.7.2 Frequency Multiplexer

A frequency multiplexer or multiplexer filter is a group of filters, each of which has a different frequency range, that are interconnected to provide multiple output ports

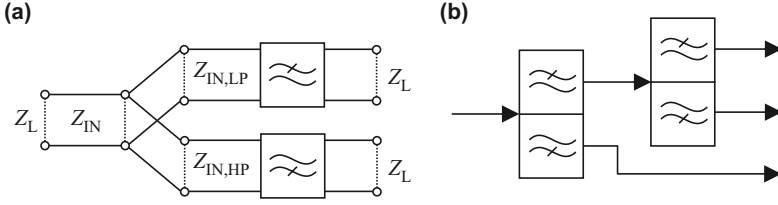


Figure 2.10 Diplexer equivalent circuit (a); triplexer consisting of two diplexers (b).

and a common input port [199]. Frequency multiplexers are usually reciprocal, such that input(s) and output(s) can be interchanged, which is required for the FI-DAC concept.

Diplexers are well known for the application with mixers in wireless communications. They enable a wideband impedance matching for the mixer output. Furthermore, the mixer's output signal is filtered, while at the same time preventing spectral components in other frequency ranges from reflecting back into the mixer.

Standard filters such as LPFs, high-pass filters (HPFs), and BPFs are characterized by the following parameters: cut-off frequencies, center frequency, roll-off, reflection characteristics, etc. Frequency multiplexers are additionally characterized by the crossover insertion loss and the crossover frequency [200]. The overall performance is usually close to single filters in the respective manufacturing technology [200].

There are continuous and non-continuous frequency multiplexers. The continuous multiplexer has a minimum insertion loss at the crossover frequency of 3 dB [201]. A non-continuous multiplexer trades a worse impedance matching at the crossover frequency for an improved insertion loss.

In Fig. 2.10(a), a diplexer is depicted, which is realized by an LPF and an HPF. The complex input admittances of both filters $Z_{IN,LP}$ and $Z_{IN,HP}$ need to satisfy the following equation over the entire frequency band in order to obtain a matched input port [201]:

$$Z_{IN} = Z_{IN,LP} + Z_{IN,HP} = Z_L, \quad (2.18)$$

whereby Z_{IN} and Z_L denote the input and the load admittances, respectively. In order to satisfy this equation, complementary filters are needed [201–203]. Fully complementary filters can be designed only with Butterworth filters. For the desired steep roll-off Chebyshev filters, only pseudo-complementary filters could be designed, which do not fulfill condition (2.18) over the entire frequency band [201, 203].

Higher order frequency multiplexers, i.e., triplexer or quadruplexer, are usually built using multiple diplexers. In Fig. 2.10(b), a triplexer is shown being built from two diplexers. The second diplexer is located in the low-frequency path of the first diplexer [200, 201]. A single-stage design with multiple filters is challenging, since impedance matching is required over the entire frequency band [202]; hence, advanced filter synthesis algorithms are required. Information on designing complementary triplexer filters can be found in [202]. Furthermore, interconnection problems may arise in a bulky design [204].

Diplexers are fabricated using lumped elements, combines, interdigital or suspended stripline (SSL) designs [200]. Recently, a diplexer in SSL technology on a RO5880 dielectric substrate has been developed for the application in oscilloscopes, which is suitable for signals up to 65 GHz [205]. A similar performance is achieved with a hybrid diplexer consisting of an SSL and a waveguide filter [206]. In a more recent paper, a diplexer for signals up to 100 GHz is presented by the same authors based on SSL technology using an organic liquid crystal polymer [207].

In [208], a wideband BPF from 40 to 80 GHz has been demonstrated in a 90 nm BiCMOS process paving the way for the integration of all FI-DAC components in a single IC.

2.7.3 Analog Multiplexer

An analog multiplexer (AMUX) is a device that switches multiple input ports in a sequential fashion to a common output port. There has been only little research on AMUXs up to now.

High-speed digital multiplexers (DMUXs) are widely used, e.g., for aggregating multiple digital input signals to a single output signal, which is known as serializing [209]. This way, a two-level pulse amplitude modulation (PAM) signal can be efficiently generated for high-speed communication applications [210]. In [211],

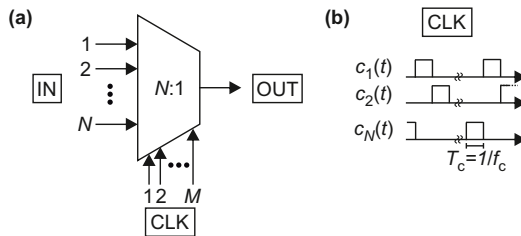


Figure 2.11 $N:1$ AMUX block diagram (a); AMUX clock signals (b).

Table 2.7 State of the art AMUX research results; sources: [42] & open literature.

Techn.	Power	Bandwidth	Sampling rate	Ref.
InP	1.35 W	> 40 GHz	50 GS/s ¹	[174]
InP	0.54 W	> 50 GHz	100 GS/s	[217]
InP	0.54 W	63 GHz	128 GS/s	[218]
SiGe	1.06 W	> 67 GHz	> 56 GS/s, 128 GS/s ¹	[42]
InP	0.9 W	> 110 GHz	112 GS/s	[175]

¹ simulation

a power DMUX is presented, which can generate a 100 Gbit/s 2 V peak-to-peak driving signal for a Mach-Zehnder modulator (MZM). For the DAC design, DMUXs commonly aggregate the digital input signals, that arrive at a reduced rate in parallel at the DAC input interface, to the DAC sampling rate [65,80,212]. More information on DMUXs can be found in [209,213–216]

An AMUX is different from a DMUX, since its transfer characteristic needs to be linear and furthermore, there are no logic levels to be regenerated. In Fig. 2.11(a), a block diagram of an $N:1$ AMUX is depicted with N input ports, M clock signals and a single output port. In Fig. 2.11(b), the corresponding multi-phase clock signal for a single-stage AMUX is depicted. The clock signals $c_1(t) \dots c_N(t)$ are orthogonal and each consists ideally of a rectangular pulse train, whereby the pulse width is equal to the switching period T_c , which is the inverse of the switching frequency f_c . The pulses are spaced in the interval NT_c and the n th clock signal is delayed by $(n-1)T_c$. A complete system model is provided later in Sec. 4.4. Furthermore, the schematic of a recent 2:1 AMUX IC is presented in Sec. 4.6.1.

The AMUX is subject to multiple impairments: bandwidth limitations of both data and clock path, nonlinearity in the data path, clock feed-through, and duty cycle noise due to unequal switching periods. Furthermore, the AMUX performance is limited by both thermal noise and clock phase noise. More details are provided in Sec. 4.6.

An overview on recent research results is given in Tab. 2.7. The listed AMUXs are designed by two research groups, i.e., from NTT and from the University of Stuttgart, whereby different manufacturing processes are utilized, i.e., InP and SiGe. The AMUX bandwidth exceeds 110 GHz for on-chip and 63 GHz for module measurements, respectively. The direction of the data flow on the ICs varies for the presented implementations: there is a straight data flow from left→right for [42] and an angular data flow configuration for the others, in which the data inputs are supplied from the sides (up & down) and the output is to the right. The first data

flow configuration eases alignment of input signals for packaging; however, the clock transmission line crosses a data transmission line on the IC.

Next to the results presented in the table, an AMUX was integrated with two DACs on a single IC in 28 nm CMOS achieving a combined DAC with 9 bit resolution operating at 11 GS/s [219]. As it is integrated, the specifications for the AMUX itself are not available.



3 Time Interleaving DAC (TI-DAC)

Two concepts are viable in the time domain to achieve a performance enhancement for DACs: TI-DAC and AMUX-DAC. In this chapter, the TI-DAC concept is covered, in which the output signals of multiple DACs are delayed relative to each other and finally combined. Either a simple analog power combiner or a summing amplifier is used to combine the sub-signals.

Note that this chapter is rather short, since the TI-DAC is not the main topic of this thesis. It is included in the thesis in order to cover all relevant interleaving concepts for high-speed DACs.

This chapter is structured as follows: first, the TI-DAC concept is introduced. It is followed by an analytical description of the TI-DAC operation for both RZ and NRZ pulses. Then, the required pre-processing filter for NRZ pulses is introduced and its implications are discussed. Characteristics and limitations of the concept are stated thereafter. Finally, the chapter is summarized.

3.1 Concept

The TI-DAC concept is based on summing the delayed output signals of multiple DACs in the time domain, such that a higher overall sampling rate is achieved. Note that parts of this section have been previously published in [45, 47, 49].

A block diagram of a TI-DAC with two DACs is depicted in Fig. 3.1(a). Both DACs are provided with the same clock frequency. The clock signals are phase shifted relative to each other to ensure a correct summation of the DAC output signals. The output signals are added either passively, e.g., with a power combiner, or actively, e.g., with a summing amplifier.

Furthermore, there are two basic variants for the TI-DAC concept to add the sub-signals as depicted in Fig. 3.1(b): NRZ and RZ pulses.

First, NRZ pulses are utilized [220], whereby the sub-signals overlap in the analog domain. They need to add up correctly to form the combined output signal. A digital pre-distortion algorithm is required to calculate the correct digital sub-signals [221]; the algorithm will be presented in Sec. 3.3. This variant is also referred to as pseudo-interleaving, and is used, e.g., in the Fraunhofer HHI AWG with 70 GS/s,

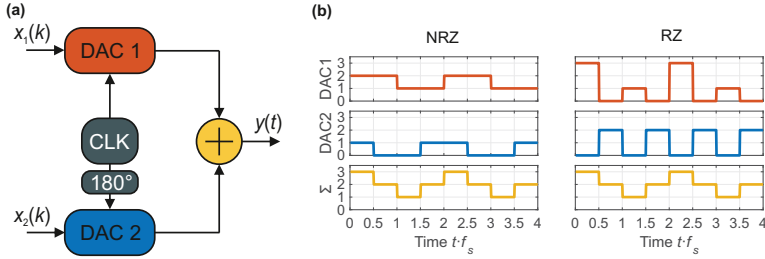


Figure 3.1 TI-DAC block diagram (a); operation with NRZ and RZ pulses (b).

18 GHz bandwidth and 6 bit vertical resolution [171] and in the Tektronix AWG 70000 with 50 GS/s, 15 GHz bandwidth and 10 bit resolution [72].

Second, RZ pulses are used. This variant does not require a pre-distortion algorithm per se, since the sub-signals are orthogonal in the time domain, i.e., the pulses do not overlap. This variant is also referred to as true interleaving. However, the requirements on the DACs' switching speed and thus, on the analog bandwidth, are increased compared to NRZ pulses. Both variants, NRZ and RZ, have been combined by Huang [79] to design an 8 bit 100 GS/s CMOS DAC consisting of two 50 GS/s NRZ-sub-DACs, which each consist of two 25 GS/s RZ-sub-DACs. However, the combined DAC's bandwidth is 13 GHz, limiting the usage of the DAC at high symbol rates.

A good overview on TI-DAC concepts can be found in [76, 132]. The previously mentioned references are focused on ZOH-DACs, however, other pulse shaping approaches are viable as well, such as partial order hold, which is demonstrated for a TI-DAC in [222]. The interested reader can find information on TI-ADCs for the receiver side, e.g., in [68, 223–229]. Moreover, TI concepts have been studied in the optical domain as well, e.g., in [230].

3.2 Mathematical Model

In this section, the operation of the TI-DAC is described analytically for both RZ and NRZ pulses.

3.2.1 TI-DAC with RZ Pulses

The operation of the TI-DAC with RZ pulses does not require a special pre-processing algorithm, since the samples are orthogonal in the time domain. A

simple digital demultiplexer (DEMUX) splits the combined digital signal $x(k)$ into N sub-signals $x_n(k)$ according to

$$x_n(k) = \sum_{k \in \mathbb{Z}} x(Nk + (n-1)) , \quad (3.1)$$

with $n \in \Lambda$, whereby Λ denotes the set of DACs according to $\Lambda = \{1, \dots, N\}$.

The output of the n th DAC $y_n(t)$ is given as

$$y_n(t) = \underbrace{\sum_{k \in \mathbb{Z}} x_n(k) \delta(t - kT_s)}_{\text{Ideal D/A Conversion}} * \underbrace{\text{rect}\left(\frac{N}{T_s}t\right)}_{\text{ZOH}} * \underbrace{v_n(t)}_{\text{Analog Filter}} * \underbrace{\delta\left(t - (n-1)\frac{T_s}{N}\right)}_{\text{Delay}} . \quad (3.2)$$

The D/A conversion is described by ideal D/A conversion, ZOH operation at the sampling rate f_s as described in Sec. 2.4, an analog LPF $v_n(t)$ accounting for the n th DAC's bandwidth limitations, and a delay $(n-1)\frac{T_s}{N}$ that accounts for the n th DAC clock phase shift. $\text{rect}(\cdot)$, $*$, and $\delta(t)$ denote the rectangular function, the convolution operator, and the Dirac distribution, respectively. The usage of additional variables for unit consistency, as in (2.3), is omitted to improve clarity and readability.

The output of the TI-DAC $y(t)$ is given as the sum of the sub-signals:

$$y(t) = \sum_{n \in \Lambda} y_n(t) . \quad (3.3)$$

In Figs. 3.2(a) and 3.2(b), the frequency responses of the DACs and the combined TI-DAC are depicted for the case of two DACs generating RZ pulses. The DACs are assumed free of bandwidth limitations; they are only limited by the sinc roll-off due to the hold operation. The frequency responses' magnitudes are each normalized to 0 dB at DC.

The magnitude responses for DAC1 and DAC2 are identical. The phase responses differ, since the DAC clocks are phase shifted relative to each other. Although, the first null of the sinc roll-off is at $2f_s$ due to the RZ pulses, the first Nyquist zone is defined by the frequency interval $[0, f_s/2]$. The phase responses are linear and show phase steps of π at the frequencies, where the roll-off due to the sinc is equal to zero.

The combined magnitude response is equal to the magnitude response of DAC1 and DAC2. However, the previously empty RZ time slots are now filled with samples from the other DAC; hence, the TI-DAC's first Nyquist zone is defined by the

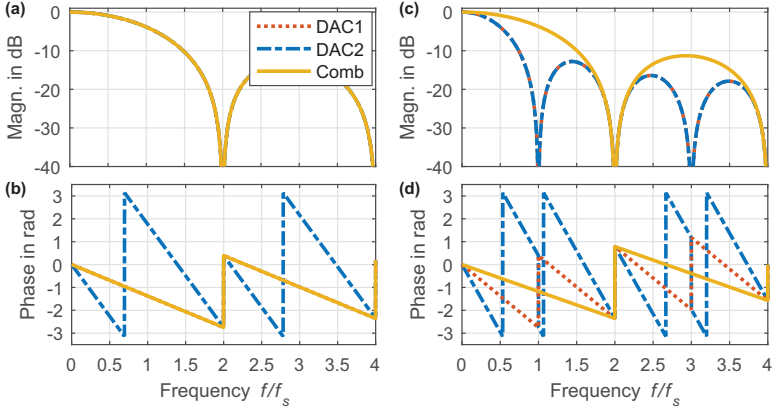


Figure 3.2 TI-DAC frequency response: magnitude (a) and phase (b) for RZ operation; magnitude (c) and phase (d) for NRZ operation.

frequency interval $[0, f_s]$. The combined phase response shows steps of π at the sinc roll-off's zeros.

For high-speed CMOS DACs, RZ pulses with a 50 % duty cycle can be generated with a clock switch, which is placed in front of the current cells. It switches the current to the DAC's analog output at the clock's "high" level and to a dummy load at the clock's "low" level in an alternating manner [79].

Concluding, the combined DAC operates like a single DAC with a cumulative sampling rate of $2f_s$. It can be noted that the bandwidth of a DAC with RZ pulses need to be higher than for NRZ pulses, which is difficult to achieve for high-speed DACs and constitutes a fundamental drawback.

3.2.2 TI-DAC with NRZ Pulses

The TI-DAC with NRZ pulses requires an additional digital pre-processing filter to mitigate the effect of overlapping analog samples. In the DSP, the pre-processing filter is applied first and the digital DEMUX is applied second. The pre-processing filter will be described in Sec. 3.3.

For NRZ operation the output of the n th DAC is given as

$$y_n(t) = \underbrace{\sum_{k \in \mathbb{Z}} x_n(k) \delta(t - kT_s)}_{\text{Ideal D/A Conversion}} * \underbrace{\text{rect}\left(\frac{t}{T_s}\right)}_{\text{ZOH}} * \underbrace{v_n(t)}_{\text{Analog Filter}} * \underbrace{\delta\left(t - (n-1)\frac{T_s}{N}\right)}_{\text{Delay}}. \quad (3.4)$$

The difference between NRZ and RZ operation is the width of the rectangle filter describing the ZOH operation. The output of the TI-DAC is given by the sum of the sub-signals according to (3.3).

In Figs. 3.2(c) and 3.2(d), the frequency responses of the DACs and the combined TI-DAC output are depicted for the case of two DACs, which generate NRZ pulses. The DACs are assumed free of bandwidth limitations; they are only limited by the sinc roll-off due to the hold operation. The frequency responses' magnitudes are each normalized to 0 dB at DC.

The magnitude responses for DAC1 and DAC2 are identical. The phase responses differ, since the sub-signals are phase shifted relative to each other. The phase responses are linear and have phase steps of π at the frequencies, where the sinc roll-off is equal to zero.

By applying the pre-processing filter, the TI-DAC operates like a single DAC at the doubled sampling rate [221]. The digital pre-processing filter will be introduced in Sec. 3.3. The combined phase response has steps of π at the new sinc roll-off's zeros.

3.3 Pre-Processing Filter for TI-DAC with NRZ Pulses

For the NRZ operation, a digital-preprocessing algorithm is required to pre-compensate the impact of the overlapping output signals of the DACs.

The digital pre-processing algorithm is applied on the combined digital signal, which is denoted as $d(k)$ in this section. Since, the samples overlap in the time domain, the contribution of the previous samples on the current sample need to be subtracted. This is performed according to [221]:

$$x(k) = d(k) - \sum_{n=1}^{N-1} x(k-n) . \quad (3.5)$$

This operation corresponds to an infinite impulse response (IIR) filter and can be described in the frequency domain as [221]:

$$W(e^{j\Omega}) = \frac{1}{1 + \sum_{n=1}^{N-1} e^{-j\Omega n}} , \quad (3.6)$$

whereby Ω denotes the normalized angular frequency of the DTFT.

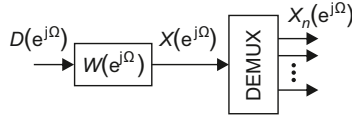


Figure 3.3 TI-DAC with NRZ pulses: DSP block diagram.

A block diagram of the DSP is depicted in Fig. 3.3. The spectrum $D(e^{j\Omega})$ of the combined discrete-time signal $d(k)$ is pre-processed with the filter $W(e^{j\Omega})$ according to:

$$X(e^{j\Omega}) = W(e^{j\Omega}) \cdot D(e^{j\Omega}) . \quad (3.7)$$

After applying the pre-processing filter, the signal $x(k)$ corresponding to $X(e^{j\Omega})$ is demultiplexed into the sub-signals $x_n(k)$ according to (3.1).

By using the pre-processing filter in (3.6), the TI-DAC consisting of N DACs, each operating at a sampling rate of f_s , behaves like a single DAC operating at a sampling rate of Nf_s [221]. By neglecting the analog filter characteristic in (3.4) the output of the TI-DAC in (3.3) is given by [221]:

$$y(t) = \sum_{k \in \mathbb{Z}} x(k) \delta\left(t - k \frac{T_s}{N}\right) * \text{rect}\left(\frac{N}{T_s} t\right) . \quad (3.8)$$

By using ideal pre-processing for a TI-DAC, the first $N - 1$ image replica can be suppressed [132].

In Fig. 3.4, the pre-processor's frequency response is shown for a TI-DAC consisting of two and four DACs, respectively. For the case of two DACs, the filter has a high-pass characteristic as depicted in Fig. 3.4(a). It compensates for the sinc roll-off, which has a spectral zero at f_s . The phase is linear over the entire frequency band.

The filter for the case of four DACs is visualized in Fig. 3.4(b). Due to the increased number of DACs, there are two sinc roll-off's spectral zeros, that need to be compensated: the first at f_s and the second at $2f_s$. Furthermore, the reduced spectral height of the sinc roll-off's second side lobe between f_s and $2f_s$ is compensated by lifting the spectrum by > 11 dB. The phase response is linear and further subject to a phase step of π at f_s .

Although, the DACs operate like a single DAC by using the pre-processing filter as shown in (3.8), the pre-processing has an impact on the effective resolution. On the one hand, the effective resolution of the TI-DAC is reduced due to the pre-processing. On the other hand, the enhanced sampling rate is beneficial in

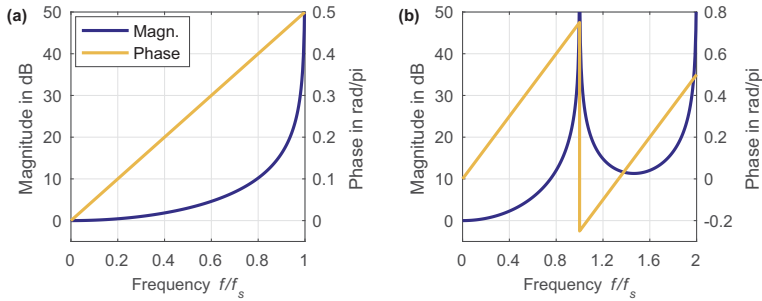


Figure 3.4 TI-DAC pre-processing filter frequency response: for two DACs (a); for four DACs (b).

terms of effective resolution. For two DACs, it is obvious that the closer the signal bandwidth is to f_s , the more of the sinc roll-off will be compensated. For $N > 2$, one of the spectral zeros being compensated is located in the center of the frequency band, which inhibits the generation of broadband signals with NRZ TI-DACs with more than two DACs.

Next to the sinc roll-off, the analog impulse response $v_n(t)$ need to be compensated as well. It is recommended for two DACs to generate signals with frequency components $\leq \frac{2}{3}f_s$ in order to limit the resolution loss to an acceptable degree [231]. As a consequence, the usable combined bandwidth is reduced from f_s to $\frac{2}{3}f_s$.

3.4 Characteristics and Limitations

Both the RZ and the NRZ variant increase the sampling rate by a factor equal to the number of DACs N . Thus, the usable bandwidth is increased by N , since a baseband signal with the bandwidth $Nf_s/2$ can be generated with a sampling rate of Nf_s . The sampling rate enhancement may be used either to cancel images above the first Nyquist zone [221, 232–235] or to generate signals with bandwidths exceeding the first Nyquist zone. Although, the usable bandwidth is extended, the analog bandwidth remains constant as shown in Secs. 3.2.1 and 3.2.2. It is still limited by both the DACs' low-pass characteristic and sinc roll-off. The increased sampling rate cannot be used for higher symbol rates per se. However, adaptive modulation schemes in the frequency domain, such as discrete multi-tone (DMT) or orthogonal frequency division multiplexing (OFDM) with bit- and power-loading can efficiently utilize the higher usable bandwidth.

In general, the ENOB is increased due to the time-interleaved operation of the TI-DAC; the ENOB reduction for NRZ pulses is not considered here. The resulting ENOB of the TI-DAC ENOB_{TI} is given as [132]:

$$\text{ENOB}_{\text{TI}} = \text{ENOB} + \log_4 \left(\frac{N f_s}{2B} \right), \quad (3.9)$$

whereby ENOB and B denote the ENOB of the single DAC and the signal bandwidth, respectively. Although the ENOB is enhanced, the pre-processing filter for NRZ pulses will decrease the ENOB again. The high magnitude attenuation close to f_s due to the sinc roll-off can be avoided by using RZ instead of NRZ signals. However, NRZ interleaving improves the image replica cancellation compared to RZ interleaving, since the nonlinearity error is averaged out across the parallel DACs [132]. In conclusion, there is a trade-off between both variants.

For the TI-DAC operation, synchronization between the DACs is highly important. This can be achieved, e.g., with a power detector at the TI-DAC output or by controlling the phase delay between the DAC clock signals with a phase detector.

Furthermore, mismatches between the DACs limit the TI-DAC performance, i.e., offset mismatch, gain mismatch, timing mismatches or more complex frequency response differences [132]. These mismatches can be pre-compensated by means of DSP, which in turn increases the overall calibration complexity and therewith the TI-DAC's power consumption.

3.5 Conclusion

The TI-DAC concept is a straightforward technique to enhance both the sampling rate and the ENOB of a DAC. It has both low hardware complexity and low synchronization effort. However, the concept does not enhance the analog bandwidth. The increased sampling rate can be used to generate a higher signal bandwidth than with a single DAC. The combined output signal is still limited by the sinc roll-off and the low-pass characteristic of the individual DACs. The TI-DAC with NRZ pulses requires an additional pre-processing filter to subtract the effect of overlapping analog samples and the pre-processing filter reduces the TI-DAC's effective resolution.



4 Analog Multiplexing DAC (AMUX-DAC)

In the previous chapter, the TI-DAC concept was introduced, which enhances the DAC sampling rate. In this chapter, the AMUX-DAC concept is addressed, which enhances both the sampling rate and the analog bandwidth.

A short introductory literature overview is followed by the description of the AMUX-DAC concept. Then, an analytical system model for N DACs is derived, which has not been available in the literature up to now. Based on this, a MIMO system model is derived for the case of two DACs and extended towards N DACs. Signal processing aspects are discussed next, covering both calibration requirements and DSP algorithms including algorithmic complexity estimations. A novel MIMO algorithm is derived from the MIMO system model, which mitigates the impairments also for signal paths with unequal frequency responses.

Thereafter, a general impairment model for the AMUX-DAC is introduced. Based on this, a behavioral 2:1 AMUX model for system-level simulation is developed [43]. With this model, the impact of various parameters on the combined output signal is investigated both for the AMUX by itself and for the AMUX-DAC. Furthermore, the performance of the novel MIMO equalizer is compared to a single-input single-output (SISO) equalizer.

The performance of an AMUX IC with the highest bandwidth reported at that time [42] is evaluated by an on-chip S-parameter measurement and an on-chip time-domain measurement [42]. Finally, a scaling model is introduced, limitations and challenges of the AMUX-DAC concept are described and the chapter is summarized.

4.1 Introduction

For the AMUX-DAC concept, a high-speed AMUX switches between multiple DAC output signals to aggregate these signals. Due to the nonlinear switching operation, the combined output signal has both an increased sampling rate and a higher bandwidth compared to a single DAC while assuming that the AMUX bandwidth is greater than the DAC bandwidth [47, 172].

The concept was introduced in 2011 by Ferenci [174], who demonstrated an InP AMUX operating up to 50 GHz. A 2:1 AMUX integrated with two DACs on a single IC in 28 nm CMOS was presented with 9 bit and 11 GS/s in 2015 [219].

Recently, an 80 GBd signal with 40 GHz bandwidth was generated by multiplexing the outputs of two CMOS DACs. The DACs and the AMUX have analog 3 dB bandwidths of ~ 20 GHz and > 50 GHz, respectively [172, 217]. Two sub-signals with a signal bandwidth of 20 GHz each were combined with the AMUX operating at a clock frequency of 43.3 GHz. Later, the same authors showed the generation of a 300 Gb/s DMT signal with a halved clock frequency of 37.5 GHz and digital pre-processing in [173]. These authors later presented an AMUX module with an analog bandwidth of 63 GHz operating up to 128 GS/s [218] and an 2:1 AMUX IC with an analog bandwidth > 110 GHz [175].

In [42], a SiGe AMUX with an analog bandwidth > 67 GHz is presented, which was the highest bandwidth reported at that time. This AMUX is later used both as a basis for the behavioral model in Sec. 4.6 and for the measurement sections in Sec. 4.10.

By realizing the complex DAC architecture in energy-efficient CMOS technology and the analog front-end in high-bandwidth bipolar technology (SiGe, InP, etc.), all of high bandwidth, high sampling rate, and high resolution could be achieved.

As outlined both above and in Sec. 2.7.3, only a limited number of research results for high-speed switching have been published so far. Consequently, the high-speed operation of these devices requires additional investigation.

4.2 Concept

In Fig. 4.1, the AMUX-DAC concept is visualized. In Fig. 4.1(a), a conceptual block diagram is depicted for the case of two DACs: the outputs of the DACs are combined with the 2:1 AMUX to generate the AMUX output signal $y(t)$. Both DACs are operated at the same sampling rate and are clocked with an 180° phase shift. The DAC input signals are denoted as $x_1(k)$ and $x_2(k)$. The AMUX operates in a half-clock configuration, i.e., the input signals are switched through in an alternating fashion by the positive and the negative half of the alternating clock signal. A full-rate DAC clock signal is assumed for this exemplary configuration.

In Fig. 4.1(b), the idealized operation of the AMUX-DAC is depicted in the time domain for NRZ signals. The AMUX is transparent for the DAC output signals in the center of each sample, which is denoted with a drawn-through line. In this switching period, the other input signal is isolated from the output. The isolated

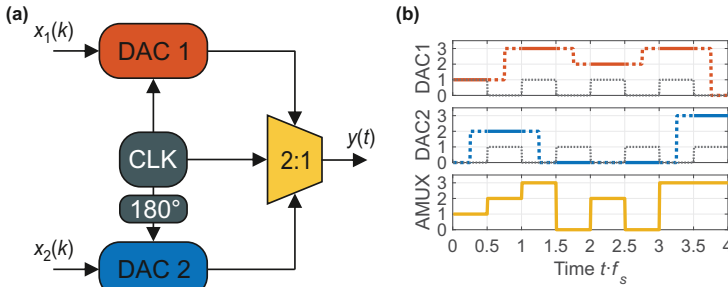


Figure 4.1 2:1 AMUX-DAC: block diagram (a); operation in the time domain (b).

DAC can perform a transition without affecting the AMUX output. The clock signal is visualized with a dotted line for DAC1 and DAC2.

The correct clocking of the AMUX-DAC is essential: the AMUX should “sample” the DAC output signals in the center of each hold period to ensure the maximum margin for timing inaccuracies, i.e., clock jitter.

4.3 Analytical System Model

In this section, an analytical system model is derived for the AMUX-DAC. In general, DAC interleaving in the time domain has been only sparsely addressed in the research literature as already described in Ch. 3. Although, there is plenty of literature on time-interleaved ADCs [226–229].

In [132], an analytical analysis of time-interleaved DACs is performed, in which the DACs are interleaved by simple addition. However, the nonlinear switching operation introduces additional effects that have not been covered by the mathematical analysis so far. For the AMUX-DAC, a spectral analysis for the case of two DACs was provided in [172] by means of a figure; however, no analytical insights were provided.

In this section, the AMUX operation is formulated analytically in the general case for N DACs. Then, a system model for the special case of two DACs is developed highlighting the implications for an AMUX-DAC with both equal and unequal sub-channels. Furthermore, the extension to N DACs is presented.

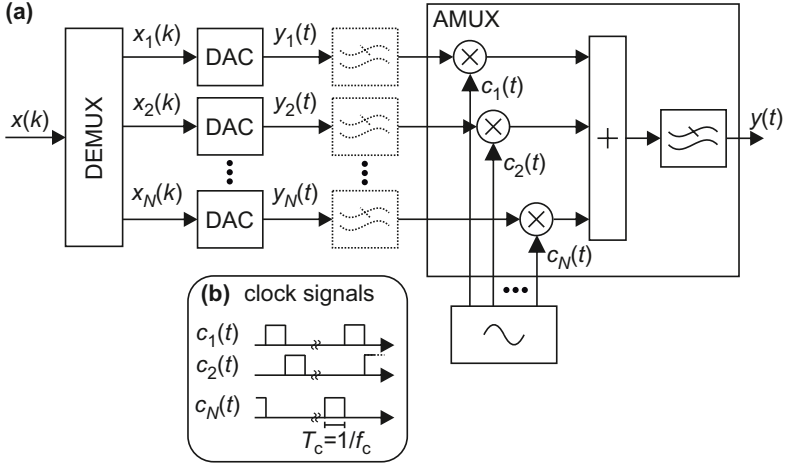


Figure 4.2 Block diagram of the AMUX-DAC system model (a); ideal clock signals (b).

4.3.1 General Model

In Fig. 4.2(a), a simplified block diagram of the AMUX-DAC operation is depicted covering the relevant aspects for the analytical system model. The digital input signal $x(k)$ at the sampling rate f_c is split among the N DACs with a digital DEMUX, whereby the set of DACs is given by $\Lambda = \{1, \dots, N\}$. The sub-signals $x_n(k)$, with $n \in \Lambda$, are obtained by decimating $x(k)$ in the time domain according to

$$x_n(k) = \sum_{k \in \mathbb{Z}} x(Nk + (n-1)). \quad (4.1)$$

The operation of decimation in the time domain is equivalent to summing shifted and frequency-scaled versions of the digital spectrum in the frequency domain [53]. Further, a delay in the time domain corresponds to a multiplication with an exponential phase term in the frequency domain [53]. Therefore, the frequency domain representation of (4.1) is given as

$$X_n(e^{j\Omega N}) = \frac{1}{N} \sum_{m=0}^{N-1} X(e^{j(\Omega - m\frac{2\pi}{N})}) \cdot e^{j(\Omega - m\frac{2\pi}{N})(n-1)}, \quad (4.2)$$

whereby $X(e^{j\Omega})$ and $X_n(e^{j\Omega N})$ are the DTFTs of $x(k)$ and $x_n(k)$, respectively; Ω denotes the normalized angular frequency of the DTFT with $\Omega = 2\pi f T_c$.⁸ Due to

⁸ Please see footnote 6 for the definition of the DTFT.

the decimation, the signal sampling rates before and after the digital DEMUX are f_c and f_c/N , respectively.

Assuming linear DACs with NRZ pulses without noise and jitter, the analog DAC output signals $y_n(t)$ are given by:

$$y_n(t) = \underbrace{\sum_{k \in \mathbb{Z}} x_n(k) \delta(t - kT_s)}_{\text{Ideal D/A Conversion}} * \underbrace{\text{rect}\left(\frac{t}{T_s}\right)}_{\text{ZOH}} * \underbrace{v_n(t)}_{\text{Analog Filter}} * \underbrace{\delta(t - (n-1)T_c)}_{\text{Delay}}. \quad (4.3)$$

The D/A conversion is described by ideal D/A conversion, ZOH operation at the sampling rate f_s as described in Sec. 2.4, an analog LPF accounting for bandwidth limitations $v_n(t)$, and a delay that accounts for the clock phase shift $(n-1)T_c$. In this model, $v_n(t)$ also accounts for the analog filters following the DACs that are depicted in Fig. 4.2(a) and are optionally utilized in order to suppress DAC hold images as well as the AMUX's low-pass characteristic that is associated with the input data paths as described in Sec. 4.5. $\delta(t)$ and $*$ denote the Dirac distribution and the convolution operator, respectively.⁹ The usage of additional variables for unit consistency, as in (2.3), is omitted to improve clarity and readability.

The sampling rate of the DACs f_s does not need to equal the sampling rate f_c/N of the digital sub-signals, although this fact simplifies the clock network. Prior to D/A conversion, the digital sub-signals could be upsampled to the DAC sampling rate as described in Secs. 4.4 and 4.7.

The corresponding frequency domain representation of (4.3) is given as

$$Y_n(f) = \underbrace{X_n(e^{j2\pi f T_s})}_{\text{Ideal D/A conversion}} \cdot \underbrace{T_s \text{sinc}(T_s f)}_{\text{ZOH}} \cdot \underbrace{V_n(f)}_{\text{Analog Filter}} \cdot \underbrace{e^{-j2\pi f (n-1)T_c}}_{\text{Delay}}, \quad (4.4)$$

whereby $X_n(e^{j2\pi f T_s})$ denotes the DTFT of $x_n(k)$.

The AMUX is operated with a multi-phase clock signal $c_n(t)$ to “sample” each input signal in the center of the hold period. The clock signals ideally have a rectangular shape with a pulse width of T_c as depicted in Fig. 4.2(b). The clock signal $c_n(t)$ for the AMUX input signal $y_n(t)$ with $n \in \Lambda$ is given by:

$$c_n(t) = \text{rect}\left(\frac{t}{T_c}\right) * \delta(t - (n-1)T_c) * \sum_{l \in \mathbb{Z}} \delta(t - lNT_c) * v_{\text{clk},n}(t), \quad (4.5)$$

⁹ The definition of the Dirac distribution and the convolution are given in the footnotes 3 and 4, respectively.

whereby $v_{\text{clk},n}(t)$ denotes the low-pass characteristic of the n th clock path. After minor transformations, the corresponding frequency domain representation is given as

$$C_n(f) = \underbrace{\frac{1}{N} \text{sinc}(T_c f)}_{\text{Amplitude}} \cdot \underbrace{e^{-j2\pi f(n-1)T_c}}_{\text{Phase}} \cdot \underbrace{V_{\text{clk},n}(f)}_{\text{Analog Filter}} \cdot \underbrace{\sum_{l \in \mathbb{Z}} \delta\left(f - l \frac{1}{NT_c}\right)}_{\text{Clock Lines}}. \quad (4.6)$$

Eq. (4.6) illustrates, that the clock lines are present at multiples of $l \frac{f_c}{N}$ for $l \in \{\mathbb{Z} \setminus \{N, 2N, 3N, \dots\}\}$. They are attenuated by the sinc-function with spectral nulls at lf_c with $l \in \mathbb{Z}$, are subject to a phase shift and are further attenuated by the low-pass characteristic of the clock path.

In Fig. 4.3(a), the time domain characteristics of the clock signals are depicted for $N = 2$. Note the DC level of $1/N = 0.5$ due to the rect-function. In Fig. 4.3(b), the clock signals' frequency domain representations are depicted for the positive half of the spectrum. Additionally, the sinc-function is plotted. For the case $N = 2$, the clock lines for sub-signal one and two have opposite phases and are only present at odd multiples of $f_c/2$.

In Fig. 4.3(c), the clock signals are visualized in the frequency domain for $N = 4$. The clock lines of the first and the third clock signal have a phase of zero. The second and the fourth clock signal are phase-shifted copies of the first and the third clock signal, respectively. The clock lines of the second and the fourth clock signal are phase-shifted compared to the first and the third clock signal by 0° , 270° , 180° , and 90° in a repetitive fashion.

Eventually, the analog output signal $y(t)$ is obtained by summing the multiplication of the DAC output signal $y_n(t)$ with the corresponding clock signal $c_n(t)$ as depicted in Fig. 4.2(a). Furthermore, the output has a low-pass characteristic denoted as $v_{\text{out}}(t)$. The output signal $y(t)$ is given as

$$y(t) = \left(\sum_{n \in \Lambda} c_n(t) \cdot y_n(t) \right) * v_{\text{out}}(t), \quad (4.7)$$

with the corresponding frequency domain representation

$$Y(f) = \left(\sum_{n \in \Lambda} C_n(f) * Y_n(f) \right) \cdot V_{\text{out}}(f). \quad (4.8)$$

Concluding, the AMUX operation consists of upconverting the input signals to multiples of the clock frequency and additional summing as stated by (4.6) and (4.8).

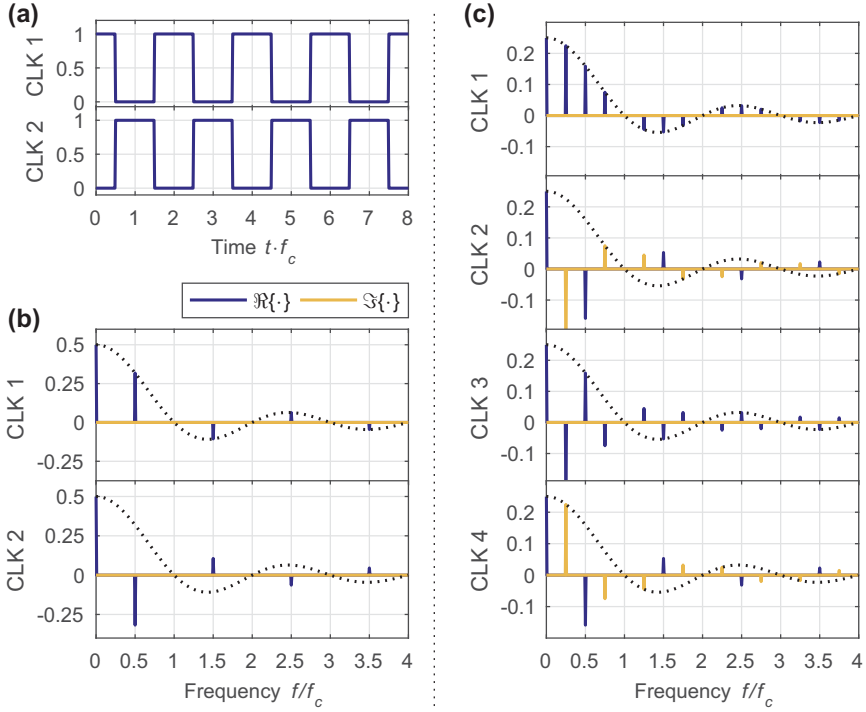


Figure 4.3 AMUX clock signals: for a 2:1 AMUX in the time domain (a) and in the frequency domain (b); for a 4:1 AMUX in the frequency domain (c).

The AMUX-DAC output signal is a superposition of multiple DAC NRZ signals, which each consists of multiple images in the frequency domain attenuated by the sinc-function and the DAC's frequency response as stated in (4.4). If oversampling is used in combination with an analog LPF, the DAC hold images can be suppressed prior to analog multiplexing.

From a mathematical perspective, the AMUX-DAC operation is similar to the FI-DAC operation, explained later in Sec. 5.3, due to the multiplication of the DAC output signals with clock signals. The main difference is that all sub-signals share the same clock frequency with multiple wanted harmonics, which are subject to different phases. Furthermore, each clock signal has a DC term that is also multiplied with the sub-signals.

4.3.2 Special Case for Two DACs

Within the scope of this thesis, bandwidth-limited DACs are used together with the AMUX. Thus, the DAC output signals are generated using oversampling, which enables the suppression of hold images either with an analog LPF following the DACs or with the bandwidth-limited frequency response of the DAC.

In this section, a frequency domain MIMO system model for the special case of two DACs is derived. It provides the basis for a pre-equalizer compensating the impairments, which is presented later in Sec. 4.4.

For this derivation, only the DAC part and the major clock line at $f_c/2$ are taken into account. The other clock lines are assumed negligible. They are suppressed either due to the limited bandwidth of the clock path or due to suppression with an analog LPF after the AMUX to save bandwidth on the following transmission channel. This section has been inspired by the AMUX-DAC spectral visualization in [172].

Sub-Signals

The sub-bands $X_1(e^{j2\Omega})$ and $X_2(e^{j2\Omega})$ correspond to the digital sub-signals $x_1(k)$ and $x_2(k)$, respectively. They are obtained according to (4.2) in the frequency domain, which corresponds to decimation with a factor of two in the time domain:

$$\begin{aligned} X_1(e^{j2\Omega}) &= \frac{1}{2} \sum_{m=0}^1 X(e^{j(\Omega-m\pi)}) \\ &= \frac{1}{2} \left[X(e^{j\Omega}) + X(e^{j(\Omega-\pi)}) \right], \end{aligned} \quad (4.9)$$

$$\begin{aligned} X_2(e^{j2\Omega}) &= \frac{1}{2} \sum_{m=0}^1 X(e^{j(\Omega-m\pi)}) \cdot e^{j(\Omega-m\pi)} \\ &= \frac{1}{2} \left[X(e^{j\Omega}) - X(e^{j(\Omega-\pi)}) \right] \cdot e^{j\Omega}. \end{aligned} \quad (4.10)$$

The spectra $X(e^{j\Omega})$, $X_1(e^{j2\Omega})$, and $X_2(e^{j2\Omega})$ are illustrated in Fig. 4.4(a).

Next to the decimation operation, the spectrum $X(e^{j\Omega})$ can be split into two frequency bands. This enables a simplified treatment of the 2:1 AMUX-DAC

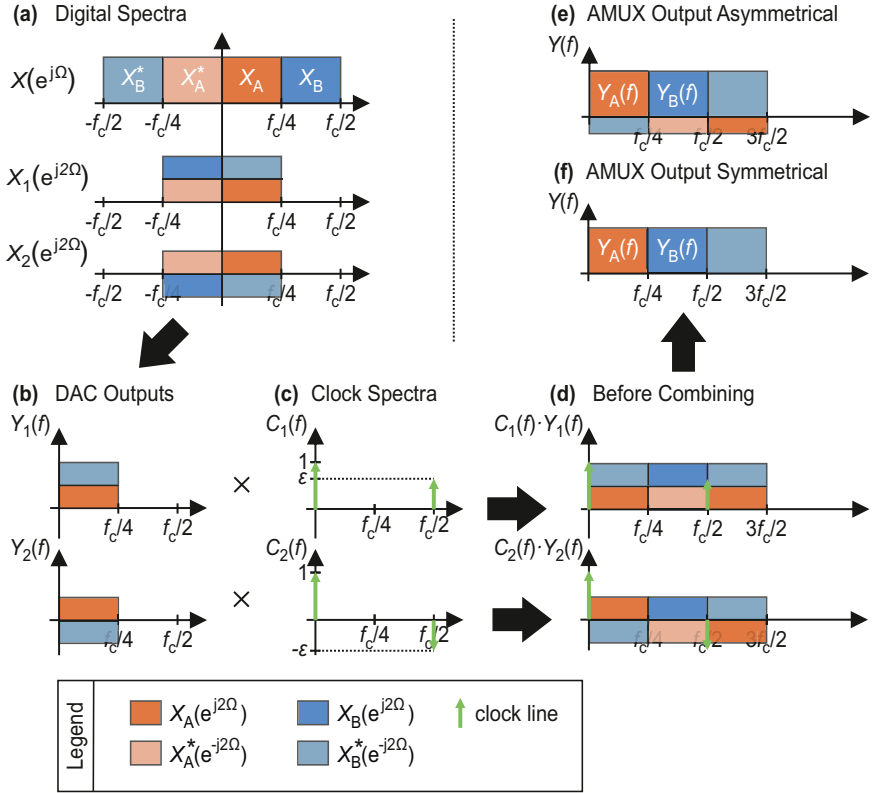


Figure 4.4 AMUX-DAC signal processing in the frequency domain (dark and light colors denote the spectrum in regular and in reversed frequency position, respectively): digital spectrum and sub-bands (a); DAC output spectra (b); clock spectra (c); spectra before combining in the AMUX (d); AMUX output spectrum with unequal sub-channels (e) and with equal sub-channels (f).

operation. The two sub-bands $X_A(e^{j2\Omega})$ and $X_B^\dagger(e^{j2\Omega})$ are depicted in Fig. 4.4(a) and are given in baseband as:

$$X_A(e^{j2\Omega}) = X(e^{j\Omega}) \cdot \text{rect}\left(\frac{\Omega}{f_c/2}\right), \quad (4.11)$$

$$X_B^\dagger(e^{j2\Omega}) = X(e^{j\Omega}) \cdot \left(1 - \text{rect}\left(\frac{\Omega}{f_c/2}\right)\right), \quad (4.12)$$

whereby the operator $(\cdot)^\dagger$ denotes both operations of reversing the frequency axis and applying the complex conjugate, i.e., $X^\dagger(e^{j\Omega}) = X^*(e^{-j\Omega})$.

By taking into account (4.11) and (4.12), the digital sub-bands in (4.9) and (4.10) are rewritten as

$$X_1(e^{j2\Omega}) = \frac{1}{2} \left[X_A(e^{j2\Omega}) + X_B^\dagger(e^{j2\Omega}) \right], \quad (4.13)$$

$$X_2(e^{j2\Omega}) = \frac{1}{2} \left[X_A(e^{j2\Omega}) - X_B^\dagger(e^{j2\Omega}) \right] \cdot e^{j\Omega}. \quad (4.14)$$

The analog representations of the digital sub-bands are given as

$$X_1(e^{j2\pi f 2T_c}) = \frac{1}{2} \left[X_A(e^{j2\pi f 2T_c}) + X_B^\dagger(e^{j2\pi f 2T_c}) \right], \quad (4.15)$$

$$X_2(e^{j2\pi f 2T_c}) = \frac{1}{2} \left[X_A(e^{j2\pi f 2T_c}) - X_B^\dagger(e^{j2\pi f 2T_c}) \right]. \quad (4.16)$$

The phase term $e^{j\Omega}$ in (4.10) in the digital domain will be canceled by the phase term in (4.4) in the analog domain; hence, it is neglected for this system model.

In order to improve the readability in the upcoming analysis in the analog domain, only the positive half of the spectrum is considered to shorten the equations. For the present case with real-valued signals, the spectrum has the complex-conjugated property; hence, the negative half of the spectrum can be disregarded without loss of generality.

The DAC output spectra are given according to (4.4):

$$Y_1(f) = X_1(e^{j2\pi f T_s}) \cdot V_1(f), \quad (4.17)$$

$$Y_2(f) = X_2(e^{j2\pi f T_s}) \cdot V_2(f), \quad (4.18)$$

whereby $X_1(e^{j2\pi f T_s})$ and $X_2(e^{j2\pi f T_s})$ denote the D/A converted digital sub-bands. The corresponding digital sub-signals have been upsampled from $f_c/2$ to f_s prior to D/A conversion to enable suppression of hold images with analog LPFs. All of the ZOH sinc roll-off, the frequency responses of the DACs and the AMUX inputs are denoted by $V_1(f)$ and $V_2(f)$. The DAC output spectra are depicted in Fig. 4.4(b) without the frequency response effects.

The clock signals in (4.6) can be simplified by summarizing terms and by assuming equal clock path LPFs $V_{\text{clk}}(f) = V_{\text{clk},1}(f) = V_{\text{clk},2}(f)$. Then, the clock spectra yield

$$C_1(f) = \delta(f) + \varepsilon \delta\left(f - \frac{f_c}{2}\right), \quad (4.19)$$

$$C_2(f) = \delta(f) - \varepsilon \delta\left(f - \frac{f_c}{2}\right), \quad (4.20)$$

whereby $\varepsilon = \text{sinc}\left(\frac{1}{2}\right) V_{\text{clk}}(f_c/2)$. Due to the 180° phase shift between the clock signals in the time domain, the second clock signal's second term has a negative sign. The clock spectra $C_1(f)$ and $C_2(f)$ are depicted in Fig. 4.4(c).

In Fig. 4.4(d), the analog sub-signals $Y_1(f) \cdot C_1(f)$ and $Y_2(f) \cdot C_2(f)$ are visualized before combining.

AMUX Output Signal

The combined AMUX output spectrum is given by inserting (4.17) to (4.20) into (4.8)

$$Y(f) = (C_1(f) * Y_1(f) + C_2(f) * Y_2(f)) \cdot V_{\text{out}}(f) \quad (4.21)$$

$$\begin{aligned} &= \left[X_1 \left(e^{j2\pi f T_s} \right) \cdot V_1(f) + X_2 \left(e^{j2\pi f T_s} \right) \cdot V_2(f) \right. \\ &\quad + \varepsilon \left(X_1 \left(e^{j2\pi \left(f - \frac{f_c}{2}\right) T_s} \right) \cdot V_1\left(f - \frac{f_c}{2}\right) \right) \\ &\quad \left. - \varepsilon \left(X_2 \left(e^{j2\pi \left(f - \frac{f_c}{2}\right) T_s} \right) \cdot V_2\left(f - \frac{f_c}{2}\right) \right) \right] \cdot V_{\text{out}}(f). \end{aligned} \quad (4.22)$$

Eq. (4.22) can be reformulated by using (4.15) and (4.16) to

$$\begin{aligned}
 Y(f) = & \left[\frac{1}{2} \left(X_A \left(e^{j2\pi f T_s} \right) + X_B^\dagger \left(e^{j2\pi f T_s} \right) \right) \cdot V_1(f) \right. \\
 & + \frac{1}{2} \left(X_A \left(e^{j2\pi f T_s} \right) - X_B^\dagger \left(e^{j2\pi f T_s} \right) \right) \cdot V_2(f) \\
 & + \frac{\varepsilon}{2} \left(X_A \left(e^{j2\pi \left(f - \frac{f_c}{2} \right) T_s} \right) + X_B^\dagger \left(e^{j2\pi \left(f - \frac{f_c}{2} \right) T_s} \right) \right) \cdot V_1 \left(f - \frac{f_c}{2} \right) \\
 & \left. - \frac{\varepsilon}{2} \left(X_A \left(e^{j2\pi \left(f - \frac{f_c}{2} \right) T_s} \right) - X_B^\dagger \left(e^{j2\pi \left(f - \frac{f_c}{2} \right) T_s} \right) \right) \cdot V_2 \left(f - \frac{f_c}{2} \right) \right] \cdot V_{\text{out}}(f).
 \end{aligned} \tag{4.23}$$

Three steps are performed to simplify the previous equation. First, only the frequency interval $[0, \frac{f_c}{2}]$ is considered. Second, the equation is split into two frequency intervals f_A and f_B , which are defined as $f_A = [0, f_c/4]$ and $f_B = [f_c/4, f_c/2]$. Accordingly, $Y(f)$ and $V_{\text{out}}(f)$ are each split into two parts $Y_A(f)$, $Y_B(f)$, $V_{\text{out},A}(f)$ and $V_{\text{out},B}(f)$, which correspond to the frequency intervals f_A and f_B . Furthermore, the spectra $X_A(e^{j2\pi f T_s})$ and $X_B(e^{j2\pi f T_s})$ are assumed bandwidth-limited to $f_c/4$. Third, the equation is reformulated in matrix notation as:

$$\begin{bmatrix} Y_A(f) \\ Y_B(f) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} (V_1(f) + V_2(f)) V_{\text{out},A}(f) & 0 \\ 0 & 0 \\ (V_1(f) - V_2(f)) V_{\text{out},A}(f) & 0 \\ \varepsilon \left(V_1^\dagger(f) - V_2^\dagger(f) \right) V_{\text{out},B}(f) & \varepsilon \left(V_1^\dagger(f) + V_2^\dagger(f) \right) V_{\text{out},B}(f) \\ 0 & 0 \end{bmatrix}^T \cdot \begin{bmatrix} X_A(e^{j2\pi f 2T_c}) \\ X_A^\dagger(e^{j2\pi f 2T_c}) \\ X_B(e^{j2\pi f 2T_c}) \\ X_B^\dagger(e^{j2\pi f 2T_c}) \end{bmatrix}. \tag{4.24}$$

Since, the AMUX output signal is real-valued, the spectra have the conjugate symmetry property and two equations are redundant. From (4.24), the first and the last column are considered (please note the transpose operator) to obtain an

equation system dependent on $X_A(e^{j2\pi f 2T_c})$ and $X_B^\dagger(e^{j2\pi f 2T_c})$, which are part of the sub-bands $X_1(e^{j2\pi f 2T_c})$ and $X_2(e^{j2\pi f 2T_c})$ as in (4.15) and (4.16):

$$\begin{bmatrix} Y_A(f) \\ Y_B^\dagger(f) \end{bmatrix} = \frac{1}{2} \underbrace{\begin{bmatrix} (V_1(f) + V_2(f)) V_{\text{out},A}(f) & (V_1(f) - V_2(f)) V_{\text{out},A}(f) \\ \varepsilon (V_1(f) - V_2(f)) V_{\text{out},B}^\dagger(f) & \varepsilon (V_1(f) + V_2(f)) V_{\text{out},B}^\dagger(f) \end{bmatrix}}_{\underline{\mathbf{C}}(f)} \begin{bmatrix} X_A(e^{j2\pi f 2T_c}) \\ X_B^\dagger(e^{j2\pi f 2T_c}) \end{bmatrix}. \quad (4.25)$$

The equation system in (4.24) can be easily generalized towards N DACs according to

$$\mathbf{Y}(f) = \underline{\mathbf{C}}(f) \cdot \mathbf{X}(e^{j2\pi f NT_c}), \quad (4.26)$$

whereby $\underline{\mathbf{C}}(f)$ is the channel matrix and the vectors $\mathbf{Y}(f)$ and $\mathbf{X}(e^{j2\pi f NT_c})$ are given as

$$\mathbf{Y}(f) = \begin{bmatrix} Y_A(f) & Y_A^\dagger(f) & Y_B(f) & Y_B^\dagger(f) & \dots & Y_Z(f) & Y_Z^\dagger(f) \end{bmatrix}^T, \quad (4.27)$$

and

$$\mathbf{X}(e^{j2\pi f NT_c}) = \begin{bmatrix} X_A(e^{j2\pi f NT_c}) & X_A^\dagger(e^{j2\pi f NT_c}) & X_B(e^{j2\pi f NT_c}) & X_B^\dagger(e^{j2\pi f NT_c}) \\ \dots & X_Z(e^{j2\pi f NT_c}) & X_Z^\dagger(e^{j2\pi f NT_c}) \end{bmatrix}^T. \quad (4.28)$$

In the following, the characteristics of the AMUX-DAC will be evaluated for both unequal and equal sub-channels.

Unequal Sub-Channels

For the case of unequal sub-channels, i.e., $V_1(f) \neq V_2(f)$, (4.25) does not change. The terms on the main diagonal of the matrix $\underline{\mathbf{C}}(f)$ are the sum of the sub-channels frequency responses multiplied with the frequency response of the AMUX output characteristic $V_{\text{out}}(f)$. The cross talk terms in the matrix are the difference of the sub-channels frequency responses multiplied with the frequency response of the AMUX output characteristic. Furthermore, the terms with $Y_B^\dagger(f)$ are subject to an attenuation with the factor ε .

In Fig. 4.4(e), the spectrum of the AMUX output is visualized for the asymmetrical case with unequal sub-channels resulting in residual distortions. $Y_A(f)$ and $Y_B(f)$ contain cross talk terms from $X_B^\dagger(e^{j2\pi f 2T_c})$ and $X_A^\dagger(e^{j2\pi f 2T_c})$, respectively. At first sight, the straightforward formulation of the analytical problem with respect to the AMUX input spectra $X_1(e^{j2\pi f 2T_c})$ and $X_2(e^{j2\pi f 2T_c})$ may seem appropriate. However, $X_1(e^{j2\pi f 2T_c})$ and $X_2(e^{j2\pi f 2T_c})$ are composed of a superposition of $X_A(e^{j2\pi f 2T_c})$ and $X_B(e^{j2\pi f 2T_c})$ with equal amplitudes. The present cross talk terms are composed of a superposition of $X_A(e^{j2\pi f 2T_c})$ and $X_B(e^{j2\pi f 2T_c})$ with possibly unequal amplitudes. Hence, the analytical formulation with respect to $X_A(e^{j2\pi f 2T_c})$ and $X_B(e^{j2\pi f 2T_c})$ represents the general case, whereas the straight-forward approach formulation with respect to $X_1(e^{j2\pi f 2T_c})$ and $X_2(e^{j2\pi f 2T_c})$ represents a special case.

These insights will be later used to compute MIMO equalizer coefficients in Sec. 4.4.

Equal Sub-Channels

If the frequency responses of the DACs and the AMUX inputs are equal, i.e., $V(f) = V_1(f) = V_2(f)$, the AMUX output signal is given as

$$\begin{bmatrix} Y_A(f) \\ Y_B^\dagger(f) \end{bmatrix} = \begin{bmatrix} V(f)V_{\text{out},A}(f) & 0 \\ 0 & \epsilon V(f)V_{\text{out},B}^\dagger(f) \end{bmatrix} \cdot \begin{bmatrix} X_A(e^{j2\pi f 2T_c}) \\ X_B^\dagger(e^{j2\pi f 2T_c}) \end{bmatrix}. \quad (4.29)$$

The cross talk terms are zero. Since, $X_A(e^{j2\pi f 2T_c})$ and $X_B^\dagger(e^{j2\pi f 2T_c})$ are non-overlapping in the frequency domain, a combined SISO equalization is sufficient to pre-equalize the impairments as shown later in Sec. 4.9. The upconverted spectrum is subject to a lower spectral amplitude due to the factor ϵ .

4.4 Signal Processing

Two aspects of the AMUX-DAC signal processing are discussed in this section. First, synchronization and clocking are discussed for both a single- and a multi-stage AMUX. Second, DSP aspects for the pre-equalization are addressed. The DSP chain is introduced and the MIMO equalizer is presented, which mitigates the AMUX-DAC impairments also for signal paths with different frequency responses. Moreover, algorithmic complexity estimations are provided for the DSP by analytical formulas.

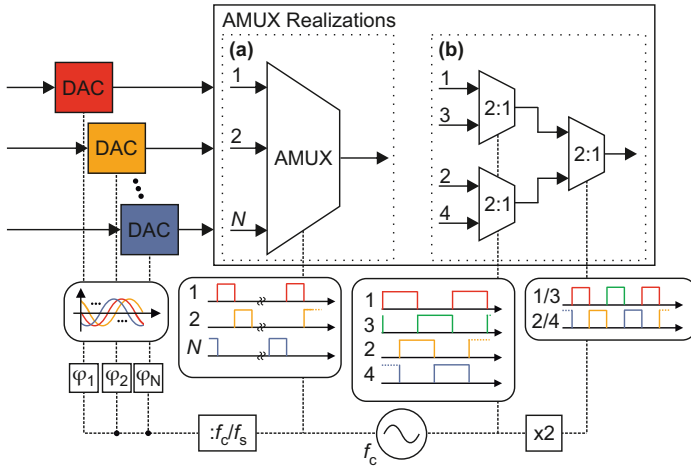


Figure 4.5 Block diagram of AMUX-DAC clocking for two different AMUX realizations: single stage (a) and multi-stage (b).

4.4.1 Synchronization and Clocking

For a correct operation of the AMUX-DAC, all components need to be synchronized: the DSP and the digital logic need to be synchronized to the DACs [236], the individual DACs need to be synchronized to each other [237], and the DACs need to be synchronized to the AMUX.

In order to synchronize the DACs with the AMUX, the AMUX output signal is measured by means of a power detector or by means of a sub-sampling oscilloscope. The measurement data is analyzed and a feedback signal is sent back to the clocking network and the DSP adjusting the clock phases. If a sub-sampling oscilloscope is utilized, the frequency response could be corrected, too. The synchronization will be performed at startup to obtain an initial clock phase configuration. During operation, a recalibration of the AMUX-DAC may be required, e.g., to account for heating effects.

Dependent on the design of the AMUX, different clocking concepts are feasible. In Fig. 4.5, two AMUX realizations are depicted, i.e., single-stage and multi-stage AMUX. Additionally, the clock signals for both the DACs and the AMUX are visualized.

The AMUX shares the clock source with the DACs to ensure phase-locking between the components. A clock divider may be used to generate the required DAC clock frequency. The DACs require phase-shifted clock signals, which either have a

sinusoidal or a rectangular shape. The phase-shift between the DAC output signals may also be realized by phase-shifting the digital sub-signals in the DSP. The required AMUX clock signals depend on the internal AMUX architecture.

In Fig. 4.5(a), a single stage AMUX is depicted. The AMUX requires rectangular RZ clock signals with a duty cycle $\leq 50\%$, which are phase-shifted relative to each other as in (4.6) to select one input at a time. If all inputs are connected to a single output node in one stage, a large capacitance will likely reduce the analog bandwidth [209].

In Fig. 4.5(b), an exemplary multi-stage 4:1 AMUX is depicted, consisting of three 2:1 AMUXs. For the first stage, a four-phase rectangular clock signal is required. For the second stage, a two-phase rectangular clock signal is required, that has the doubled frequency compared to the first stage [209]. In the figure, this is realized with a frequency multiplier. Alternatively, the highest frequency in the AMUX-DAC is provided by the clock source; then, only frequency dividers are required generating the AMUX and the DAC clock frequencies.

For both realizations, the input signals shall be well aligned in terms of timing to achieve a high-quality signal.

4.4.2 Digital Signal Processing

The block diagram for the AMUX DSP is visualized in Fig. 4.6. Two different schemes can be used for the pre-processing, depending on whether equal or unequal sub-channels are present. The bold arrows denote multiple parallel signals. The pre-processing is implemented either in the time domain or in the frequency domain, which is indicated with the dashed blocks for fast Fourier transform (FFT) and inverse fast Fourier transform (IFFT).

The first scheme in Fig. 4.6(a) uses a combined SISO equalizer and a time-domain DEMUX to split the signal at the sampling rate f_c into several sub-signals with the sampling rate f_c/N . Additionally, these signals may be individually pre-equalized after the DEMUX (not shown in the figure). Finally, they are upsampled to the sampling rate of the DACs f_s . For AMUX-DACs with equal sub-channels, this low-complexity scheme is well suited and a combined pre-equalizer can be utilized. However, this scheme is not well suited for AMUX-DACs with unequal sub-channels, since the cross talk between the signals cannot be fully compensated as shown in Sec. 4.3.

In the second scheme in Fig. 4.6(b), the time-domain DEMUX operation is split among multiple functional blocks to realize MIMO pre-equalization in between these blocks: frequency domain DEMUX, downsampling, crosswise adding, and

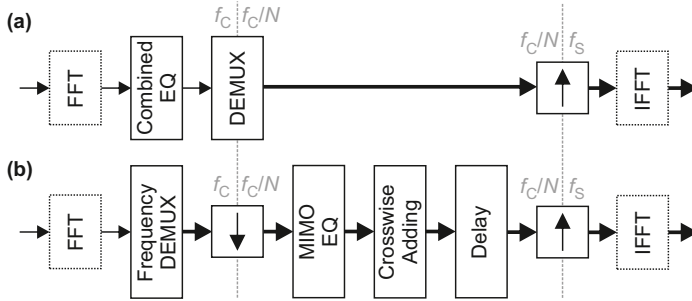


Figure 4.6 Block diagram for AMUX DSP with two possible schemes: time domain DEMUX (a); frequency domain DEMUX (b).

delay. The frequency domain DEMUX is a multi-band filter, i.e., a diplexer, a triplexer, or a quadruplexer, which splits the incoming signal into several frequency bands, whereby the cutoff frequencies are given by $f_c/(2n)$ with $n \in \Lambda$. The sub-signals are decimated to the sampling rate f_c/N , whereby the sub-signals in higher frequency bands are aliased to the baseband.

A MIMO pre-equalizer is utilized to mitigate the impairments of the analog components. The coefficients for the MIMO equalizer can be obtained by inverting the channel matrix $\underline{\mathbf{C}}(f)$ either by a zero-forcing (ZF) or by a minimum mean square error (MMSE) solution using the Moore–Penrose pseudoinverse [129]. For the case of two DACs, the channel matrix can be obtained by performing a 2×1 multiple-input single-output (MISO) least squares (LS) channel estimation (CE) in the time domain [238, 239] with the two times upsampled sub-signals corresponding to $X_A(e^{j2\pi f 2T_c})$ and $X_B^\dagger(e^{j2\pi f 2T_c})$ [44]. The two frequency responses are obtained by means of a discrete Fourier transform (DFT) from the MIMO impulse responses. They are each split in half in order to obtain the coefficients of $\underline{\mathbf{C}}(f)$. The same approach is used later for the FI-DAC in Sec. 5.7.3.

The ZF equalizer coefficients for the analytical case of two DACs are given by the inversion of the channel matrix in (4.25) according to

$$\underline{\mathbf{C}}^{-1}(f) = \frac{1}{2V_1(f)V_2(f)} \begin{pmatrix} \frac{V_1(f)+V_2(f)}{V_{\text{out},A}(f)} & \frac{V_2(f)-V_1(f)}{\mathcal{E}V_{\text{out},B}^\dagger(f)} \\ \frac{V_2(f)-V_1(f)}{V_{\text{out},A}(f)} & \frac{V_1(f)+V_2(f)}{\mathcal{E}V_{\text{out},B}^\dagger(f)} \end{pmatrix}. \quad (4.30)$$

After the MIMO equalizer, the sub-signals are cross-wise added as stated in (4.15) and (4.16). Further, the signals are delayed relative to each other as stated in (4.9) and (4.10). Finally, the sub-signals are upsampled to the DAC sampling rate f_s .

Algorithmic Complexity

The complexity of the algorithms is evaluated for a frequency-domain implementation by stating the number of real-valued multiplications for each processing block. The real-time processing is preferably implemented in the frequency domain to reduce the algorithmic complexity. The computation of the CE is not required in real-time and can be computed at a significantly reduced rate. It is assumed that a complex multiplication is equivalent to three real-valued multiplications.¹⁰ The complexity further depends on the accuracy for the implemented algorithms, which is not covered in this section.

In order to enable frequency domain processing, both FFT and IFFT algorithms need to be implemented. The complexity varies with the utilized algorithm [241–244]. For the split-radix complex FFT with length N_{FFT} , which needs to be a power of two, the number of real-valued multiplications is given as [244]:

$$M_{\text{FFT},c}^{\text{sr}} = N_P(N_{\text{FFT}}(\log_2(N_{\text{FFT}}) - 3) + 4) , \quad (4.31)$$

whereby N_P denotes the degree of parallelization. Parallel DSP architectures are required, if the desired signal processing rate is higher than the maximum clock frequency of the DSP. In the AMUX DSP, real-valued input signals are present, which require fewer multiplications. The number of multiplications is then given as [244]:

$$M_{\text{FFT},r}^{\text{sr}} = \frac{M_{\text{FFT},c}}{2} . \quad (4.32)$$

Although, the split-radix minimizes the arithmetic count of operations [245], it depends on the processor architecture, whether this algorithm is optimal in terms of hardware resources and latency [243, 246].

Both variants require resampling, if the DAC sampling rate is not equal to f_c/N .

Resampling is trivial in the frequency domain: it is performed by either insertion of zeros for upsampling or removing samples for downsampling prior to the IFFT.

Lastly, the IFFT is performed to convert the sub-bands to the time domain. For the IFFT, N sub-bands with conjugate symmetry are processed. Hence, the complexity is given for a split-radix IFFT as

¹⁰ The multiplication of two complex numbers $w = w_R + jw_I$ and $z = z_R + jz_I$ is given by [240]:

$$\begin{aligned} w \cdot z &= (w_R + jw_I)(z_R + jz_I) \\ &= w_R z_R - w_I z_I + j(w_R z_I + w_I z_R) \\ &= w_R z_R - w_I z_I + j((w_R + w_I)(z_R + z_I) - w_R z_R - w_I z_I) . \end{aligned}$$

$$M_{\text{IFFT},r}^{\text{sr}} = N \cdot \frac{N_P}{2} \cdot (N_{\text{IFFT}} (\log_2(N_{\text{IFFT}}) - 3) + 4) . \quad (4.33)$$

The degree of parallelization N_P may be subject to a change between the DSP blocks, depending on the overall DSP architecture.

For variant (a) in Fig. 4.6, a combined equalizer and a DEMUX are necessary. The combined equalizer is realized with a one-tap filter for each of the N_W frequency bins. By utilizing the conjugate symmetry property, the complexity is given as

$$M_{W,f} = N_P \cdot 3 \cdot \frac{N_W}{2} . \quad (4.34)$$

The DEMUX for variant (a) in Fig. 4.6 has a very low complexity, which is negligible. It consists of adding and shifting in the frequency domain. For a realization in the time domain, a simple rewiring is required.

The total complexity for variant (a) is hence given as

$$M_{\text{AMUX-DAC}}^{(a)} = M_{\text{FFT},r}^{\text{sr}} + M_{W,f} + M_{\text{IFFT},r}^{\text{sr}} . \quad (4.35)$$

For variant (b) in Fig. 4.6, the operations of frequency demultiplexing, and downsampling can be efficiently realized by selecting the appropriate samples from the FFT by rewiring. In addition, the cyclic shift transforming one of the sub-bands in reversed frequency position can be efficiently realized and is thus, negligible.

The number of real-valued multiplications required for the $N \times N$ MIMO equalizer with N_W taps is given as

$$M_{W,\text{MIMO},f} = N_P \cdot 3 \cdot \left(N^2 \cdot \frac{N_W}{2} \right) . \quad (4.36)$$

Since, the equalizer coefficients have the conjugate symmetry property, only half of the number of taps is required compared to an equalizer without the conjugate symmetry property.

The block crosswise adding does not require multiplications.

The delay operation corresponds to multiplication of each sub-band with a linear phase; it requires the following number of multiplications:

$$M_{\text{Delay},f} = N_P \cdot 3 \cdot \frac{N_{\text{FFT}}}{N} . \quad (4.37)$$

Hence, the total complexity for variant (b) is given as

$$M_{\text{AMUX-DAC}}^{(b)} = M_{\text{FFT},r}^{\text{sr}} + M_{\text{W,MIMO},f} + M_{\text{Delay},f} + M_{\text{IFFT},r}^{\text{sr}}. \quad (4.38)$$

Since, the system characteristics are expected to vary slowly over time, the coefficients of the MIMO equalizer are calculated in software, i.e., on a dedicated central processing unit (CPU) that handles the computation of the CE as well. FFT and IFFT operations are also necessary for the computations; their complexity has been stated before.

A simple MIMO CE in the frequency domain consists of dividing the matrix with received symbols by the matrix with the transmitted symbols for N_H frequency bins, which requires $\frac{N_H}{2} \cdot N^2$ complex-valued divisions. Additionally, averaging might be required to ensure a sufficient CE accuracy and N_H is preferably chosen equal to N_W . The associated complexity $M_{\text{CE,MIMO},f}$ in terms of multiplications is not trivial to determine, since multiple approaches exist for division.

Alternatively, the CE can be performed in the time domain. Thereby, the matrix with the transmitted symbols is inverted by means of the Moore-Penrose pseudoinverse and the complexity is given as [247]:

$$M_{\text{CE,MIMO},t} = \frac{3}{2} N_D N_H^2 + \frac{1}{2} (N_H^3 + N_H^2 + N_D N_H) - N_H, \quad (4.39)$$

whereby N_D denotes the number of data samples used for the CE. The coefficients N_D and N_H could be exchanged in order to minimize the total complexity [129] and N_H is preferably chosen equal to N_W . The inverse is calculated at startup once; then, a matrix multiplication is performed with the vector of received symbols to obtain the CE.

Finally, an inversion is performed for each frequency bin to calculate the equalizer coefficients. The number of multiplications with LU decomposition is given as [247]:

$$M_{\text{W,MIMO},f,\text{calc}} = 3 \cdot \frac{N_W}{2} \cdot \frac{N^3 - N}{3} = \frac{N_W}{2} \cdot (N^3 - N) \quad (4.40)$$

As mentioned at the beginning of this section, the processing can be divided into two domains: real-time and non-real-time. The data is processed in real-time in the “data plane” and both the CE and the calculation of the equalizer coefficients are performed in the “control plane” at a significantly reduced rate. The operations in the control plane can be performed in software, whereas the operations in the in the data plane need to be performed in hardware. For an implementation, the complexity in the data plane being stated in (4.35) and (4.38) is far more

critical than the complexity for the CE and the equalizer coefficients calculation. The algorithmic complexity for the real-time domain depends on the available hardware, i.e., if a field programmable gate array (FPGA) or an application-specific IC (ASIC) is utilized. The maximum DSP clock frequency sets a lower bound for the parallelization degree N_p and the FFT size depends on latency requirements, the number of available logic gates and the desired frequency resolution.

4.5 AMUX-DAC Impairment Model

In this section, an impairment model for the $N : 1$ AMUX-DAC is presented. The block diagram is depicted in Fig. 4.7. For this model, the characteristics of the following components are taken into account: DACs, AMUX, and clock generation network. Reflections between the components, thermal noise contributions from active components and supply coupling are present, but are not depicted in the figure.

The DACs are subject to multiple impairments as described in Sec. 2.5. Due to the finite vertical resolution, quantization noise is introduced as described in Sec. 2.5.3. Further, static and dynamic nonlinear distortions are present as described in Secs. 2.5.1 and 2.5.2. Moreover, the DAC is subject to clock phase noise, which translates to the DAC output as jitter, as described in Sec. 2.5.4. An LPF accounts for both the DACs' analog frequency response and the sinc roll-off due to the ZOH operation as described in Sec. 2.4.

The DAC clock signals are shifted relative to each other, such that the AMUX selects the DACs' output samples in the center of each hold period. This multi-phase clock architecture can be realized with transmission lines having different lengths, with an inverter cascade or with LC circuits.

Then, an optional LPF follows, which suppresses undesired hold images. Dependent on the filter design, it either absorbs the signal energy, which is not transmitted to the LPF's output, or reflects it back to the DAC. For NRZ operation, this filter can be omitted.

The AMUX consists of buffer amplifiers at both the data input and the data output to provide impedance matching. These buffer amplifiers introduce noise and small nonlinear distortions [248].

The data input signals are subject to a low-pass characteristic accounting for both a limited switching speed of the AMUX core and bandwidth limitations of the input buffers and transmission lines. The data signals are multiplied with the clock signals in the AMUX core and finally added in order to generate the output signal. The output is further subject to bandwidth limitations.

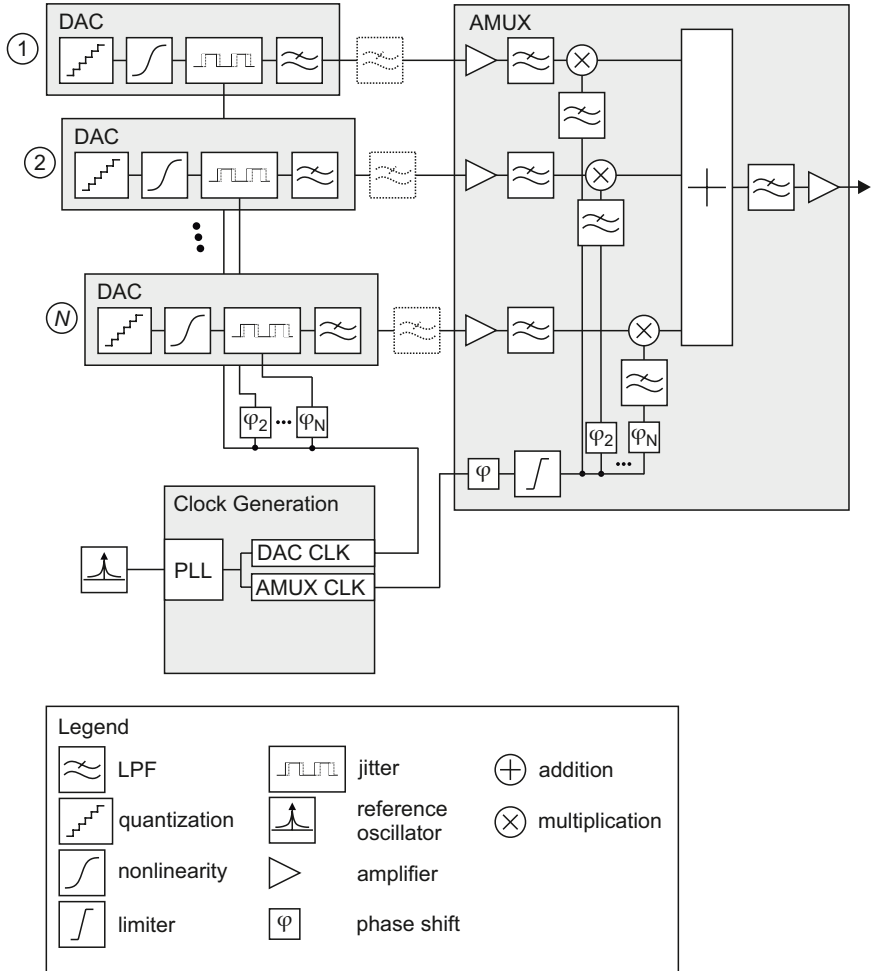


Figure 4.7 Block diagram of the AMUX-DAC impairment model consisting of DACs, AMUX, and clock network.

Nonlinear distortions in the AMUX data path, which are not depicted in the figure, result from both the non-ideal transmission characteristic of the transistors in the AMUX core and the buffer amplifiers at the input and the output. Moreover, there is signal feedthrough from the AMUX inputs to the AMUX output due to the finite isolation of the blocked data path.

For this model, the AMUX clock signals are rectangularly shaped and shifted relative to each other as in (4.6). The overall delay between the AMUX and the DACs is adjusted with the phase offset φ . In each clock path, an LPF accounts for the bandwidth limitations of the clock driver. Both, a DC offset of the clock signal and unequal clock signal amplitudes cause a switching offset. For an actual AMUX realization, the internal structure may differ, including the clock architecture.

The clock generation is a highly important sub-system, since the phase noise of both the AMUX clock and the DAC clock deteriorate the signal quality. Especially, for high-frequency operation, low-jitter clock signals are required as discussed in Sec. 2.5.4.

For the clock generation, an internal reference frequency is derived in a PLL from a reference oscillator, e.g., a crystal. This internal reference frequency is subject to phase noise. Dependent on the architecture of the AMUX and the required DAC clock frequency, the frequencies for the DACs and the AMUX may differ. However, they need to be phase-locked. The resulting phase noise and jitter will be determined by the overall clock architecture, which consists of multipliers, dividers or multiple PLLs. Moreover, the spectral purity of the clock source needs to be ensured, which may contain spurs from the PLL's reference frequency.

Power Consumption and Size

In general, a single DAC running at the doubled sampling rate consumes approximately twice the power. Hence, using multiple DACs in parallel does not enhance the power consumption for equal performance per se. The AMUX-DAC's power consumption, compared to a single DAC operating at twice the sampling rate is higher due to the AMUX IC and the more complex DSP.

The size of the AMUX-DAC depends on the size of both the DAC ICs and the AMUX IC. In general, the IC size is highly dependent on the underlying manufacturing process and hence, a definitive prognosis cannot be made. The AMUX IC presented in Sec. 4.6.1 was produced in IHP's 130 nm SG13G2 technology and the active area is $100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$. The AMUX IC could be integrated with the DACs, e.g., in a BiCMOS process. Due to reduced packaging impairments, the analog bandwidth is presumably higher. Furthermore, both the power consumption and the size are reduced.

4.6 Behavioral 2:1 AMUX Model

Simulating the AMUX-DAC architecture on transistor level in an electronic design automation (EDA) tool is very time-consuming, especially, if the impact of multiple parameters needs to be studied. In order to conduct simulations on system-level, a behavioral model is required. This way, the performance of AMUX-DACs in high-speed transceivers can be quantified and the IC design process can be supported with specifications for the AMUX's sub-blocks etc.

In this section, a generic behavioral model for the 2:1 AMUX is introduced; this model decreases the computation time by at least three orders of magnitude, which is of particular interest for system-level simulation. First, the investigated AMUX IC is introduced [42]. Then, the behavioral model is presented and explained in detail. Furthermore, the model is implemented in Matlab and fitted against simulation data obtained from an EDA tool.

In Secs. 4.8 and 4.9, the AMUX model will be embedded in an AMUX-DAC system to study the impact of multiple parameters on both system behavior and performance. The behavioral model was developed during the supervision of a master's thesis [249]. Parts of this section have been previously published in [42, 43].

4.6.1 2:1 AMUX IC

In this section, the AMUX IC is presented. The 2:1 AMUX IC is designed in IHP's SiGe-HBT BiCMOS technology SG13G2, which features high-speed bipolar transistors having a maximum transit frequency of $f_T = 300$ GHz and a maximum oscillation frequency of $f_{max} = 500$ GHz [87]. In Fig. 4.8, a block diagram of the IC and a photograph of the chip with bumps are depicted. The fully differential circuit design with a differential port impedance of 100 Ohm for both inputs and output is composed of the AMUX core, the clock buffer and a linear output amplifier [42]. The AMUX supports a maximum differential peak-to-peak input voltage of 1 V and is designed for an overall voltage gain of one.

In Fig. 4.9, the schematic of a 2:1 AMUX core is depicted. V_{CC} and V_{EE} denote the positive and the negative supply voltage, respectively. For the input signals V_{IN1} and V_{IN2} , differential transconductance amplifiers are put into place, which are linearized with the emitter degeneration resistors R_E . The input termination and the emitter followers at the signal inputs are not depicted in Fig. 4.9. The signal currents pass through clocked-current switch pairs that select one of the signal currents. If the clock voltage V_{CLK+} is in high state, the signal currents of Input 1 are switched to the output and the signal currents of Input 2 are switched to a

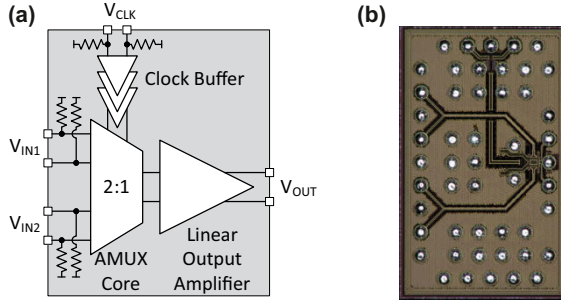


Figure 4.8 Block diagram of the 2:1 AMUX IC (a); photograph of the chip (b). (a): © 2017 IEEE. Adapted, with permission, from [42]; (b): © 2018 IEEE. Reprinted, with permission, from [43].

dummy load. If V_{CLK+} is in low state, the condition is vice versa. At the output, cascode transistors and load resistors R_L generate the output voltages V_{OUT+} and V_{OUT-} .

The differential output voltage V_{OUT} for each input current I_{IN} is given for a differential bipolar amplifier by [168]:

$$V_{OUT} = 2I_{IN}R_C \cdot \tanh\left(\frac{V_{CLK}}{2\kappa V_T}\right), \quad (4.41)$$

whereby R_C , V_{CLK} , κ , and V_T denote the collector resistance, the clock voltage, a correction coefficient, and the temperature voltage, respectively. The operating point is linearized with respect to the input current I_{IN} .

The clock buffer consists of multiple high-gain Cherry-Hooper amplifiers, which act as limiting amplifiers for the input clock signal to generate steep clock signal slopes, i.e., a quasi-rectangular clock signal for the AMUX core.

The output amplifier is implemented as a multi-stage amplifier consisting of two series connected emitter followers and a linearized differential transconductance cascode stage. It compensates the loss of the AMUX to achieve an overall gain of one and matches the AMUX core to the output impedance.

It is intended to operate the AMUX in an AMUX-DAC system consisting of two DACs, each running at a sampling rate of 64 GS/s and generating a symbol rate of 64 GBd. The outputs of the DACs are combined with the 2:1 AMUX to generate an 128 GBd signal. The AMUX operates at half clock rate, i.e., 64 GHz.

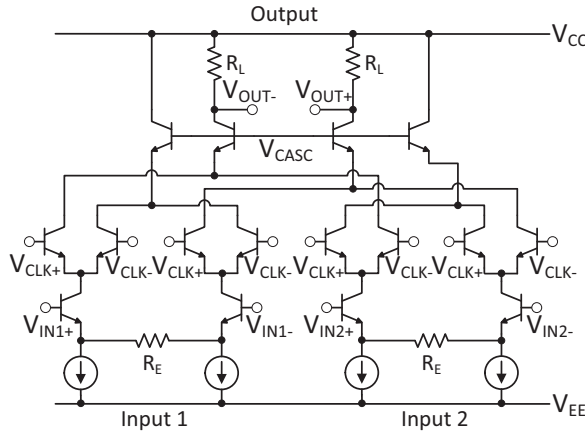


Figure 4.9 2:1 AMUX core schematic. © 2017 IEEE. Adapted, with permission, from [42].

4.6.2 Block Diagram

A behavioral model for the actual IC presented in the previous section is developed in this section, which can be easily adapted to other AMUX ICs. Therefore, a generic approach is taken and a quasi-linear model without nonlinear distortions in the data path is developed. The resulting model is depicted in Fig. 4.10. It consists of two data input signals *DataIn1* and *DataIn2*, a clock input signal *ClockIn* and a data output signal *DataOut*.

The data input signals *DataIn1* and *DataIn2* are low-pass filtered to account for the low-pass characteristic of the transmission lines including pads and bumps, as well as for the low-pass characteristic of the transconductance amplifiers. In the AMUX core, the data signals are multiplied with the clock signals and added. Since, the operating point of the core transistors is linearized as described in Sec. 4.6.1, the core operation is an ideal multiplication; the tanh-characteristic in (4.41) is included in the clock buffer.

The clock input signal *ClockIn* is processed in the clock buffer to generate a rectangular clock signal. First, *ClockIn* is fed N_A times through a combination of a tanh-characteristic and a LPF, which corresponds to a bandwidth-limited differential amplifier cascade. The argument γ defines the slope of the tanh-characteristic. The low-pass characteristic of the transmission lines, pads, and bumps has a minor impact on the sine clock input signal. Eventually, the clock signal is scaled and

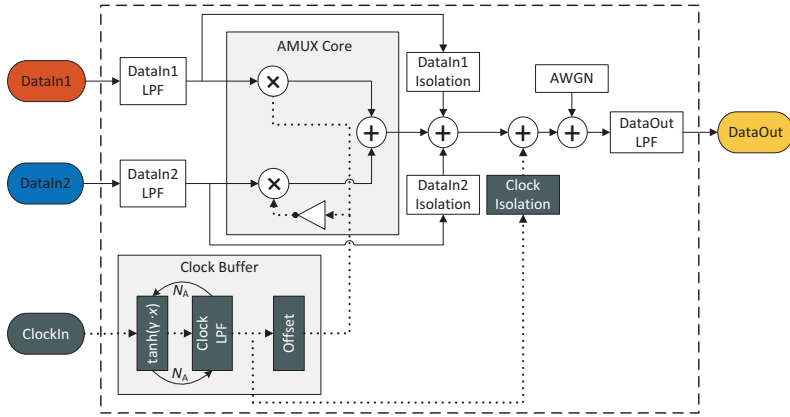


Figure 4.10 Behavioral 2:1 AMUX model block diagram: two data signals DataIn1, DataIn2 and one clock signal ClockIn are given as inputs and one data signal DataOut is given as output. The input data signals are multiplied with the clock signal in the AMUX core, combined and fed to the output port. Multiple parasitic effects are modeled. © 2018 IEEE. Adapted, with permission, from [43].

offset to convert the amplitude range of the tanh-characteristic $[-1, +1]$ to the required amplitude range $[0, +1]$. This amplitude range enables switching off one of the input signals during one half of the clock period. The clock signal is fed to the AMUX core, where it is duplicated; the clock signal for the second path is inverted, i.e., phase shifted by 180° .

Furthermore, feedthrough is accounted for by adding attenuated copies of the data signals DataIn1 and DataIn2 as well as the clock signal ClockIn to the combined data signal. Since, the clock signal has a DC term as shown in Sec. 4.3, the modeled data cross talk is additional cross talk that deviates from the intended value $1/N$.

Additive white Gaussian noise can be added to account for the thermal noise of the circuit. The combined data signal is eventually low-pass filtered to generate the output signal DataOut.

4.6.3 Parameter Fitting

In order to fit the generic model to the IC presented in Sec. 4.6.1, the parameters of the behavioral model are fitted with simulation data obtained with a SPICE-level

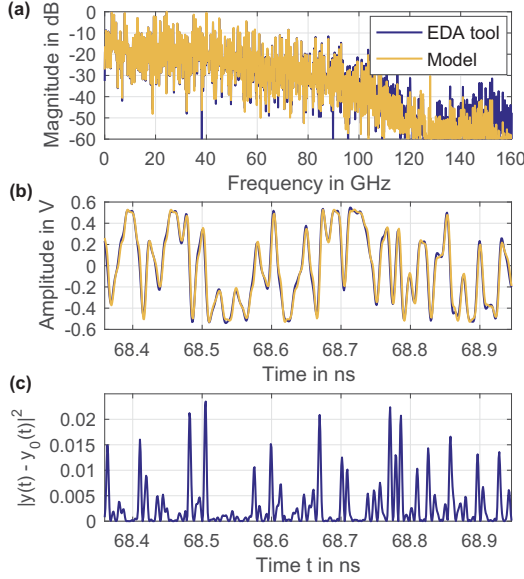


Figure 4.11 AMUX output difference between the EDA tool $y_0(t)$ and the behavioral model $y(t)$: spectrum (a), time domain signal excerpt (b), squared error between both time domain signals (c). © 2018 IEEE. Adapted, with permission, from [43].

circuit simulator, which is denoted as ‘reference data’ in the following. The fitting is performed on the differential signals and the low-pass characteristics are chosen from standard analog filter characteristics, i.e., Bessel, Butterworth, and Gaussian to easily vary their parameters in the following sections. The behavioral model is implemented in Matlab.

The reference data for the AMUX IC consists of PAM-8 input signals at 64 GBd, which are multiplexed to form a single 128 GBd PAM-8 output signal. The transient simulation reference data is interpolated on an evenly-spaced time vector with 16 samples per symbol relative to the output symbol rate [250].

The fitting of the behavioral model is performed in two steps. First, a LS channel estimation is performed on several sub-systems of the IC in order to roughly estimate the parameters of the standard filters [251, 252], e.g., from the bumps on the input pads to the AMUX core and from the AMUX core to the bumps on the output pads. Second, a multi-parameter optimization is performed, to both fit the parameters of the clock path and to optimize the roughly estimated parameters. Thereby, a brute force approach is chosen in order to check all parameter combinations. Note, that other approaches are viable as well, e.g., gradient-based. In order to evaluate the

Table 4.1 Extracted parameters for the SiGe-HBT 2:1 AMUX behavioral model. © 2018 IEEE.
Adapted, with permission, from [43].

Parameter	Value
DataIn1 LPF, DataIn2 LPF	4th order Butterworth filter $f_{co} = 120$ GHz
DataIn1 isolation, DataIn2 isolation	50 dB
DataOut LPF	4th order Butterworth filter $f_{co} = 125$ GHz
Clock offset	0
Tanh argument γ	2.5
Clock amplifier cascade N_A	1
Clock LPF	6th order Butterworth filter $f_{co} = 80$ GHz
Clock isolation	60 dB
AWGN	-

quality of the fitting, the normalized mean squared error (NMSE) is used, which can be calculated according to

$$\text{NMSE} = 10 \cdot \log_{10} \left(\frac{\langle (y(t) - y_0(t))^2 \rangle}{\langle y_0^2(t) \rangle} \right), \quad (4.42)$$

whereby, $y(t)$ and $y_0(t)$ denote the output signal of the behavioral model and the reference data, respectively. $\langle \cdot \rangle$ denotes the expected value operator. The optimization is stopped after an NMSE of < -20 dB is achieved, i.e., -20.8 dB, since no further improvements > 0.1 dB were possible.

In order to obtain a visual impression and to compare the results of the behavioral model simulation with the reference data, both the spectrum of the AMUX output signal as well as an excerpt of the corresponding time-domain signal are plotted in Figs. 4.11(a) and 4.11(b). The simulated spectrum in Fig. 4.11(a) fits well to the spectrum of the reference data for frequencies < 100 GHz as expected from the low NMSE. However, for frequencies > 100 GHz, the fitting quality degrades. The corresponding time-domain signal visualized in Fig. 4.11(b) appears to fit very well to the reference data. However, for fast signal transitions, which correspond to high frequencies, there is a non-negligible squared error as depicted in Fig. 4.11(c).

4.7 Simulation Setup

In the following two sections, several simulations are conducted in order to investigate the effects of multiple parameters on the AMUX-DAC system. A block diagram of the simulation setup is visualized in Fig. 4.12. The DSP is implemented as follows: The digital PAM input signal is fed through the combined pre-equalizer and a digital DEMUX splits the input signal into multiple sub-signals. Finally, the sub-signals are each upsampled to the DACs' sampling rate, which is denoted with the upward pointing arrow. If the MIMO pre-equalizer is utilized, the pre-processing is performed as in Sec. 4.4.2. The digital sub-signals are converted to the analog domain with the DACs and are optionally low-pass filtered, which is indicated in the respective section later. Then, the AMUX combines the analog sub-signals. The AMUX and the DACs share a common clock, whereby the DAC clock phases are adjusted, such that the AMUX samples the signals in the center of the hold period. In the simulation, this is realized by ideal analog phase shifting after the DACs.

The simulations are divided into two parts: first, the AMUX-DAC is investigated with ideal DACs and second, the AMUX-DAC is investigated with non-ideal DACs. For the simulations with ideal DACs in Sec. 4.8, a 128 GBd PAM-4 signal with 2^{20} random symbols is generated from a binary sequence with 2^{21} bits. The maximum lengths of equal bits and symbols in a row are 20 and 12 for the binary sequence and the PAM signal, respectively.

The PAM signal is split into two sub-signals, which are fed into the DACs. The DACs operate at 64 GS/s with one sample per symbol. The parameters of the DACs are ideal, i.e., the DACs generate ZOH pulses without bandwidth limitations. This way, the AMUX impairments can be studied in an isolated manner. The nominal parameters for the AMUX are chosen as in Tab. 4.1.

For the simulations with non-ideal DACs in Sec. 4.9, the DAC model and its parameters are chosen as in Appendix A. Again, a 128 GBd PAM signal with 2^{20} pseudo-random symbols is split into two sub-signals, which are fed into the DACs. The DACs operate at 90 GS/s, each generating a 64 GBd sub-signal. For a real-world application, the ratio between the DAC clock and the AMUX clock will be designed to allow for low-phase-noise clock signals.

In the DSP, the combined pre-equalizer is generally utilized. For the simulations with unequal sub-channels in Sec. 4.9.3, the MIMO pre-equalizer is chosen. A more than sufficient number of taps (1024) is selected to prevent a performance degradation. The combined SISO pre-equalizer's coefficients are obtained by auto- and cross-correlation matrices [129]. The MIMO pre-equalizer's coefficients are obtained by calculating the Moore–Penrose pseudoinverse [129]. A ZF solution

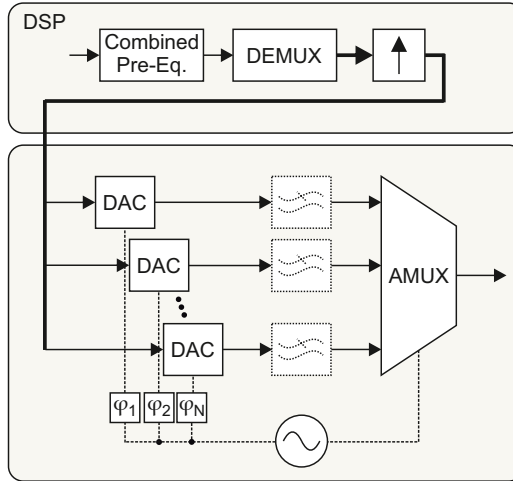


Figure 4.12 Block diagram of the AMUX-DAC simulation setup.

is obtained for both approaches and the results are equivalent in the present case [129].

The ratio between the AMUX's first clock line at 64 GHz and DC is denoted as ε in Sec. 4.3.2 and is pre-compensated. Due to the pre-processing, the DAC output signals' peak-to-peak amplitudes need to be matched to each other in a joint manner. The DACs' full-range peak-to-peak amplitude of $V_{pp} = 500$ mV is used. The difference in amplitude is compensated by means of ideal amplifiers following the DACs. In a real-world application, the output signal would be further distorted by the amplifiers' nonlinearities and noise. Alternatively, the output amplitude of one DAC could be limited by restricting the digital signal to a certain maximum code; however, this reduces the effective resolution of one of the DACs.

The LPFs after the DACs are 16th order Butterworth filters with a bandwidth of 42 GHz to suppress DAC hold images, while having a minor influence on the DAC bandwidth variation in Secs. 4.9.2 and 4.9.3. The parameters of the LPFs are consistent with commercially available filters [253]. The nominal parameters for the AMUX are chosen as in Tab. 4.1.

At the Rx, a rectangular frequency domain filter is utilized in order to suppress unwanted spectral components > 64 GHz. This could be represented in a real-world setup by an analog LPF before the ADC operating at a sampling rate > 128 GS/s with additional frequency domain DSP. The Rx DSP consists of synchronization,

clock phase recovery, error vector magnitude (EVM) estimation,¹¹ bit error counting, and system identification with 1024 taps at one sample per symbol.

The sampling rate for the analog simulation is 1024 GS/s. Please note, that simulating the AMUX-DAC system with 2^{20} pseudo-random symbols with the behavioral model, introduced in Sec. 4.6, took less than a minute, whereas simulation of 2^{17} symbols with the EDA tool took almost two days. This is a complexity reduction by a factor of multiple thousand times; a comparison of results was presented in Sec. 4.6.3.

4.8 Impact of AMUX Parameter Variations

In this section, the impact of multiple AMUX parameters is simulated in an AMUX-DAC system with ideal DACs. This way, new insights on the AMUX's behavior and the resulting performance are obtained. Note that parts of this section have been previously published in [43].

4.8.1 Data LPF In Bandwidth

In Fig. 4.13(a), the magnitude of the combined frequency response is depicted for varying the 3 dB bandwidth f_{co} of the data path input LPF. For both data paths, the filter parameters are chosen equally. The frequency responses are obtained with a SISO LS CE at two samples per symbol at the output [251]. The DACs' sinc-roll-off is not compensated. For low bandwidths, a dip of the frequency responses at 32 GHz, i.e., at the half clock frequency, as well as a plateau around 64 GHz is observed. This effect can be attributed to the AMUX behaving as a mixer, since the data signals are each multiplied with the clock signal.

Varying the AMUX's output filter bandwidth results in a common LPF bandwidth variation as in a SISO system, and thus, does not provide new insights.

¹¹ The EVM is normalized to the average constellation power and is calculated according to [254]:

$$\text{EVM} = \frac{1}{\sqrt{P_{\text{avg}}}} \sqrt{\langle |r(k) - d(k)|^2 \rangle},$$

whereby $r(k)$, $d(k)$, and $\langle \cdot \rangle$ denote the received symbols, the transmitted symbols and the expectation operator. The average constellation power P_{avg} is defined as

$$P_{\text{avg}} = \frac{1}{|\Psi|} \sum_{i \in \Psi} |\psi_i|^2,$$

whereby the constellation is given as a set by $\Psi = \{\psi_1, \psi_2, \dots\}$ with the symbols ψ_i .

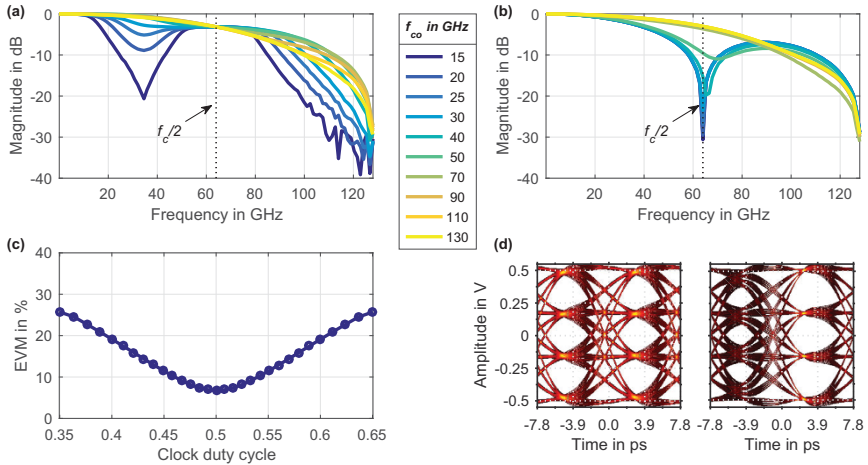


Figure 4.13 Simulation results for the behavioral 2:1 AMUX model: channel frequency response for variation of the 3 dB bandwidth f_{co} of the input low pass filter (a); channel frequency response for variation of the 3 dB bandwidth f_{co} of the clock low pass filter (b); EVM with respect to the clock duty cycle (c); PAM4 eye diagrams without a clock offset (left) and with a normalized clock offset of 0.56 (right) (d). © 2018 IEEE. Adapted, with permission, from [43].

4.8.2 Clock LPF Bandwidth

In Fig. 4.13(b), the combined frequency response is depicted for varying the 3 dB bandwidth f_{co} of the clock LPF. For low bandwidths, there is a clear dip of the frequency response at 64 GHz, which shifts to higher frequencies, if the bandwidth of the filter is increased. This can be attributed to the fact that both the AMUX clock frequency at 64 GHz and the clock harmonics are attenuated by the LPF, such that only the DC term is substantially present for low values of the clock LPF's bandwidth.

If the bandwidth of the LPF is decreased evermore, the clock signals converge to $c_n(t) = 0.5$. Since, the clock signals do not have frequency components $>DC$ anymore, the AMUX becomes equivalent to a simple addition with 6 dB insertion loss. Thus, a very low bandwidth clock corresponds to the NRZ TI-DAC operation described in Ch. 3. Then, the DACs' sinc roll-off is the primary limitation of the AMUX-DAC's combined bandwidth. Hence, only by applying the AMUX clock, the analog bandwidth can be enhanced.

4.8.3 Clock Duty Cycle Mismatch

In Fig. 4.13(c), the EVM of the received PAM-4 signal is visualized with respect to the clock duty cycle. During the clock's positive half-wave, the first input signal is switched to the output and during the clock's negative half-wave, the second input signal is switched to the output. By applying a DC offset to the clock, a clock duty cycle mismatch is present. The clock duty cycle is defined as the ratio between the "on"-time of the clock for the first signal path and the clock period $1/(f_c/2)$. A clock duty cycle mismatch is present, if the duty cycle deviates from the ideal value 0.5.

For small clock duty cycle mismatches 0.5 ± 0.04 , the EVM is below 10 %. In Fig. 5(d), eye diagrams for 128 GBd PAM-4 are shown for the basic configuration with a clock duty cycle of 0.5 (left) and 0.56 (right). Due to the clock duty cycle mismatch, the quality of the odd and the even samples is directly affected such that one becomes better, while the other becomes worse. For a practical implementation, an additional control voltage is recommended to minimize the duty cycle mismatch during operation.

4.9 Impact of AMUX-DAC Parameter Variations

In this section, the impact of several parameters on both the AMUX-DAC's behavior and performance is evaluated on the basis of simulation results. For the simulations, the 2:1 AMUX model presented in Sec. 4.6 and the DAC model presented in Appendix A are utilized.

First, the results with nominal parameters are presented, which represents the reference case. Then, the DAC bandwidth is varied to study the effects of limited bandwidth both for a system with equal sub-channels and unequal sub-channels. Furthermore, the performance of the MIMO pre-equalizer is compared with the combined pre-equalizer. Then, simulations with phase noise and jitter are performed.

4.9.1 Reference Case

For the reference case, all simulation parameters are set to their nominal values. As described in Sec. 4.7, the DACs operate at 90 GS/s and are parametrized as in Appendix A. Each DAC generates a 64 GBd sub-signal and the combined AMUX output signal is at 128 GBd.

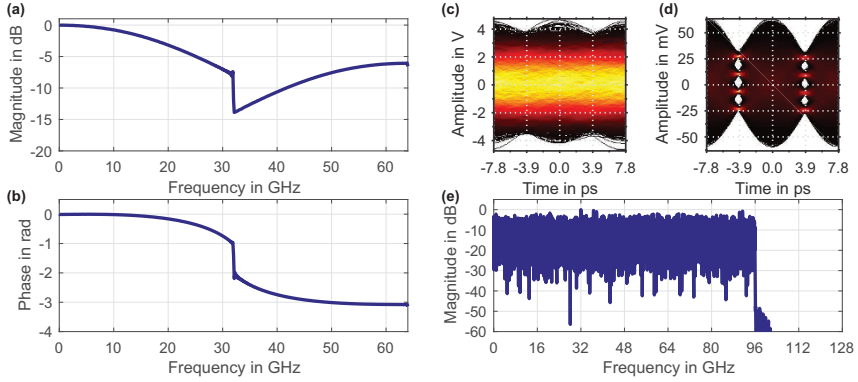


Figure 4.14 Simulated 2:1 AMUX-DAC results for the reference case: frequency response's magnitude (a) and phase (b); eye diagram without pre-equalizer (c); eye diagram with pre-equalizer (d); pre-equalized output spectrum (e).

In Fig. 4.14, the results for the reference case are depicted. In Fig. 4.14(a), the frequency response's magnitude is visualized. The magnitude is divided into two equally broad frequency intervals, which are split by a drop of about 6 dB at 32 GHz. The drop is based on the Nyquist-shaped DAC output sub-signals; the height can be attributed to the difference in spectral power of the clock line at 64 GHz and DC, which is denoted ϵ in Sec. 4.3.2. Furthermore, the frequency response's shape is reversed in the interval 32 to 64 GHz relative to the interval 0 to 32 GHz.

In Fig. 4.14(b), the frequency response's phase is depicted. It has a phase step at 32 GHz. Furthermore, the phase characteristic from 32 to 64 GHz is both the reversed and negated characteristic from 0 to 32 GHz as shown analytically in Sec. 4.3.2.

In Fig. 4.14(c), the AMUX output eye diagram is depicted with 2^{20} symbols for the AMUX-DAC without pre-equalization: the eye is severely distorted. Due to the non-NRZ operation, an open eye diagram is not visible at the AMUX output natively. By applying combined pre-equalization, the resulting eye diagram's appearance is considerably improved as depicted in Fig. 4.14(d): the EVM is 6.1 %. As the eye diagram is error free, the bit error rate (BER) is derived from the EVM as 1.13×10^{-13} .¹² The eye diagrams are obtained by upsampling the AMUX output signal prior to plotting.

¹² The SNR can be estimated from the EVM according to [255]:

$$\text{SNR} \approx \frac{1}{(\text{EVM})^2}.$$

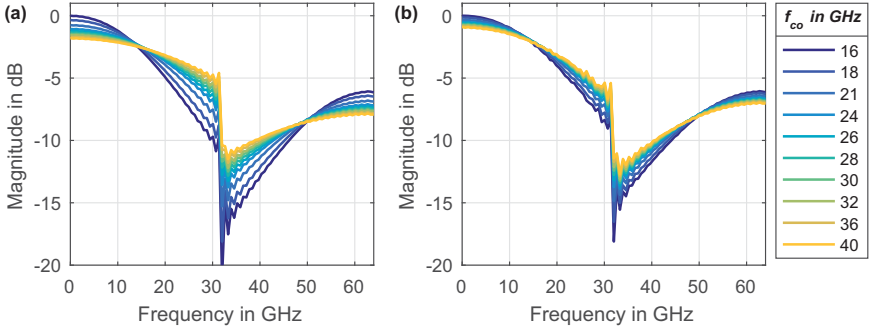


Figure 4.15 AMUX-DAC frequency response magnitude: varying the bandwidth of both DACs (a); varying the bandwidth of DAC two (b).

In Fig. 4.14(e), the pre-equalized AMUX output spectrum is depicted. It is clearly flat in the desired frequency range from 0 to 64 GHz. In the frequency range from 64 to 96 GHz, the upper sideband (USB) of the upconversion to $f_c/2 = 64$ GHz is visible. The spectrum seems to be flat as well, although the pre-equalization is not intended to compensate the frequency response in that frequency interval by purpose. Beyond 96 GHz the spectrum drops instantaneously due to the sub-signals' rectangular spectral shape. Some residual distortion is present ~ 50 dB below the signal level.

4.9.2 DAC Bandwidth

In order to investigate the impact of the DAC bandwidth onto the overall frequency response of the AMUX-DAC system, the bandwidth of the DACs is varied twofold: first, the bandwidth is varied for both DACs simultaneously, obtaining equal sub-channels. Second, the bandwidth of the first DAC is kept at its nominal value of 21 GHz and the bandwidth of the second DAC is varied. For both investigations, the pre-equalizer is disabled. The results are depicted in Fig. 4.15.

The symbol error rate (SER) is given for a PAM signal with the symbols chosen from the set Ψ of order $|\Psi|$ for additive white Gaussian noise as [256]:

$$\text{SER} = \frac{|\Psi| - 1}{|\Psi|} \operatorname{erfc} \left(\sqrt{\frac{3}{|\Psi|^2 - 1}} \text{SNR} \right).$$

For a signal with high SNR and a Gray-coded constellation, the BER is given accordingly as [256]:

$$\text{BER} = \frac{\text{SER}}{\log_2(|\Psi|)}.$$

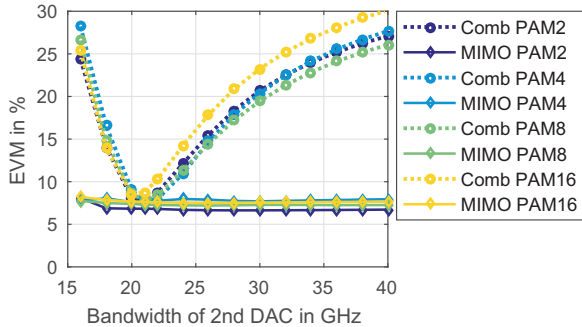


Figure 4.16 2:1 AMUX-DAC EVM performance comparison between the combined and the MIMO pre-equalizer by varying the 2nd DAC's bandwidth for different modulation formats.

In Fig. 4.15(a), the frequency response's magnitude is visualized for varying the bandwidth of both DACs simultaneously. The higher the bandwidth of the DACs, the more the frequency response of each sub-band flattens; this reduces the requirements for pre-equalization. A magnitude step of ~ 6 dB is visible at 32 GHz due to the power difference of the clock line at 64 GHz relative to DC, which is denoted ε in Sec. 4.3.2.

The ringing of the estimated frequency responses close to the step is caused by the step itself, since the estimation is performed in the time domain. More taps are required to represent the step more appropriately in the time domain.

In Fig. 4.15(b), the frequency response's magnitude is visualized for varying the bandwidth of DAC two only. The observations are similar to the first case: the higher the bandwidth of the DAC, the more the frequency response of each sub-band flattens. The observable impact is less compared to the first case, since the combined frequency response is an addition of both sub-channels' frequency responses as shown in Sec. 4.3.2.

4.9.3 Combined vs. MIMO Pre-Equalization

As mentioned in Sec. 4.3.2, the impact of unequal sub-channels can only be mitigated with a MIMO pre-equalizer rather than with a combined pre-equalizer.

In order to quantify the performance difference between a combined pre-equalizer and a MIMO pre-equalizer for unequal sub-channels, the bandwidth of the 2nd DAC is varied and the bandwidth of the first DAC is kept at its nominal value of 21 GHz. The resulting EVM is depicted in Fig. 4.16 for different modulation formats.

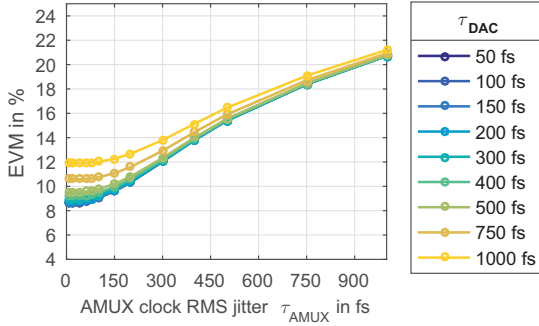


Figure 4.17 AMUX-DAC EVM performance with respect to different AMUX clock RMS jitter levels for multiple DAC clock RMS jitter levels.

The results are similar for the investigated modulation orders from PAM-2 to PAM-16. By using the MIMO pre-equalizer, an almost constant EVM is achieved between 6 and 8 %. If the difference between both sub-channels increases, the performance degrades for the combined pre-equalizer. The EVM increases from 7 % at 21 GHz to above 25 % at 38 GHz. For the nominal value of 21 GHz, the results for the combined and the MIMO pre-equalizer are equal in theory. The small difference can be attributed to different CE and equalizer coefficient estimation algorithms that are applied for the combined and the MIMO pre-equalizer as described in Sec. 4.7.

4.9.4 Phase Noise

A simulation is conducted to evaluate the phase noise influence of both the DAC clock and the AMUX clock on the combined output signal.

The parameters for the phase noise spectrum are obtained from the data sheet of the frequency synthesizer Agilent E8257D [257]; this device is used for the experiments in Sec. 4.10. The phase noise time-domain samples are generated by filtering white Gaussian noise in the frequency domain and a following IFFT. The clock signals for both DAC and AMUX are DC-free and clock duty cycle errors are not present. According to the DAC parameter fitting for the Socionext DAC in Appendix A, the DAC's RMS jitter is 150 fs; the DAC's static nonlinear distortions are reduced to retain the SINAD fitting with both static nonlinearities and phase noise. Please refer to Appendix A for more details. The length of the digital signal is set to 2^{21} samples.

In Fig. 4.17, the estimated EVM values are depicted for the AMUX-DAC output signal with respect to the AMUX clock RMS jitter τ_{AMUX} for multiple DAC clock RMS jitter values τ_{DAC} . It can be observed that the EVM is limited to $\sim 8.5\%$ by the DAC's nominal parameters; for low AMUX clock RMS jitter values < 100 fs, the DAC impairments dominate. By increasing the AMUX clock RMS jitter, its effect dominates, since the higher clock frequency is more susceptible to jitter. The EVM rises to $\sim 16\%$ for an AMUX clock RMS jitter of 500 fs. The difference in EVM between the DAC jitter values decreases by increasing the AMUX clock RMS jitter.

Concluding, the overall AMUX-DAC performance in terms of jitter is limited by both the DAC and the AMUX's clock jitter.

4.10 2:1 AMUX Experimental Results

In order to evaluate the 2:1 SiGe AMUX IC presented in Sec. 4.6.1 with the highest bandwidth reported at that time [42], on-chip measurements are conducted. In this section, the results of the S-parameter and the time-domain measurement are presented and discussed. Parts of this section have been previously published in [42].

4.10.1 On-Chip S-Parameter Measurement

In order to evaluate the frequency domain performance of the AMUX IC, an S-parameter measurement is conducted. In this section, the measurement setup is described and the results are presented.

Setup

The S-parameter measurements are performed with the vector network analyzer (VNA) Rohde&Schwarz ZVA67 in the frequency range 690 MHz - 67 GHz with microwave on-chip probes from Cascade Microtech, i.e., an InfinityQuad probe (DC-67 GHz) for the data input pads, an Infinity probe (DC-67 GHz) for the clock input pads, an Infinity probe (DC-110 GHz) for the data output pads, and a DCQ probe for the DC pads. The VNA is calibrated with calibration substrates from Cascade Microtech prior to conducting the measurements.

The differential measurement setup is depicted exemplarily for the signal path from input 1 to the output in Fig. 4.18. Port 1 of the VNA is connected with input 1 of the AMUX IC with additional DC blocks. The AMUX output is connected to the

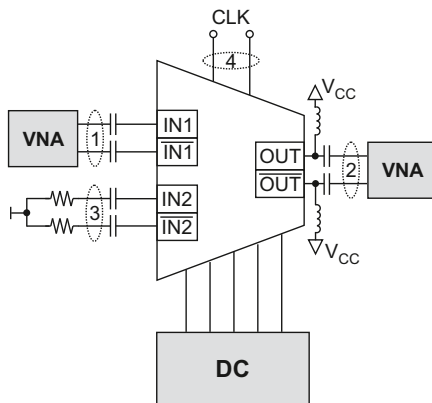


Figure 4.18 AMUX on-chip S-parameter measurement setup for measuring signal path 1.

VNA's port 2, whereby the internal bias tees of the VNA are utilized. Bias tees are required to provide a DC path for the output port, which reduces the current on the IC output resistors. The second AMUX input, which is denoted as port 3, is terminated with a differential impedance of 100Ω . The clock input, which is denoted as port 4, is set to a proper DC level, such that the signal path under test is either open or closed. Measuring of the second signal path is performed vice versa.

The DC supply sources set the operating voltages to $V_{CC} = 0\text{ V}$ and $V_{EE} = -4.5\text{ V}$ for both the AMUX core and the output amplifier, and $V_{EE,CLK} = -3.4\text{ V}$ for the clock buffer. The total power consumption of the AMUX IC is 1.06 W . Additional control currents and control voltages on the AMUX IC are used to optimize the performance.

For the measurement of the clock path, the VNA's port 4 is connected to the clock input and the VNA's port 2 to the AMUX output. One of the signal inputs is set to the "high" level with the differential voltage $+600\text{ mV}$ and the other is set to the "low" level with the differential voltage -600 mV by means of bias tees.

In Fig. 4.19, photos of the measurement setup are shown. The microwave probes and the chip with the contacted probes are presented in Fig. 4.19(a) and Fig. 4.19(b), respectively. The chip size is $970\mu\text{m} \times 1650\mu\text{m}$. In Fig. 4.19(c), the probe station is shown with the VNA in the background.

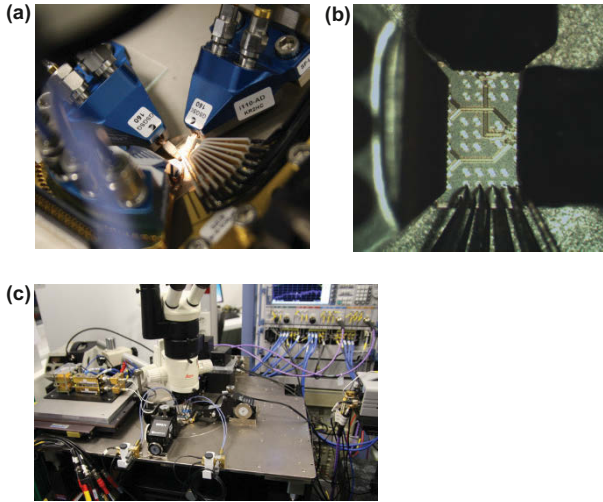


Figure 4.19 Photographs of the AMUX S-parameter measurement setup: microwave probes and DC probe (a); chip with contacted probes (b); probe station with VNA (c).

Results

In Fig. 4.20, the measured S-parameters are visualized. In Fig. 4.20(a), the transparent signal path's transmission coefficients are depicted for both AMUX inputs. The analog 3 dB bandwidth of the AMUX signal path exceeds 67 GHz and is approximately between 70 and 80 GHz, if the trend of the $|S_{21}|$ and $|S_{23}|$ is extrapolated linearly in the logarithmic scale. For various differential input power levels from -17 to $+1$ dBm the curves are very similar due to the signal path's high linearity. The curves are offset from 0 dB by -1.2 dB, which can be attributed to the output amplifier's reduced bias current compared to the nominal current in the measurement.

The attenuation for the isolated signal path is visualized in Fig. 4.20(b). The attenuation is between -35 and -29 dB in the measured frequency range for both data paths.

In Fig. 4.20(c), the reflection coefficients are depicted for both inputs and the output. The reflection coefficients for the inputs increase from < -49 dB at low frequencies to -13 dB at 67 GHz. The output reflection coefficient ranges from -31 dB at low frequencies to -11 dB at 67 GHz.

In Fig. 4.20(d), the clock path's transmission coefficient is depicted for a differential input power of 0 dBm. The clock path's 3 dB bandwidth is ~ 60 GHz.

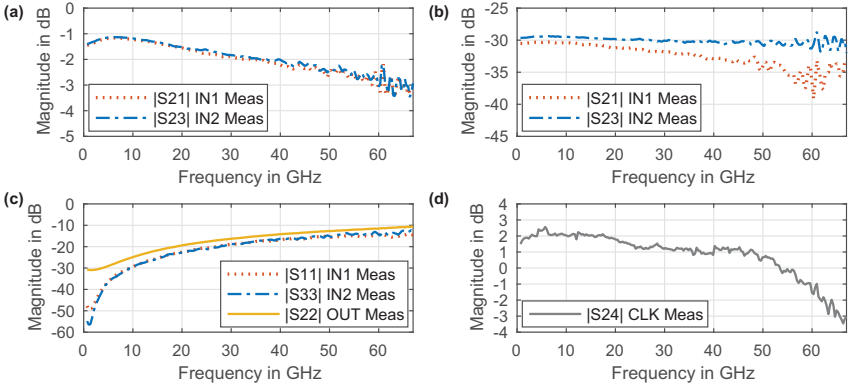


Figure 4.20 Measured AMUX S-parameters: signal path transmission (a); signal path isolation (b); signal path reflection (c); clock path transmission (d).

4.10.2 On-Chip Time Domain Measurement

In order to evaluate the performance of the AMUX IC in the time domain, an additional on-chip experiment is set up. This way, the AMUX is assessed for the AMUX-DAC application. In this section, the measurement setup is described and the results are presented.

Setup

The measurement setup is depicted in Fig. 4.21. It consists of a bit pattern generator (BPG), two DACs, the AMUX IC, a clock network, AMUX DC supplies and two oscilloscopes. In the setup, many phase shifters are used to provide enough setting options for the measurement. For a later real-world application, these can be mostly omitted.

The digital input signals for the DACs are provided by the BPG SHF 12103 with two high-speed digital output ports operating at up to 56 Gbit/s. Due to the lack of a BPG with four high-speed ports, the two output ports are used to drive both DACs. Hence, both the regular and the inverted output port are used. In order to decorrelate the digital signals, they are further interchanged and delayed. The BPG outputs generate $2^{15} - 1$ PRBS sequences. At the AMUX output, the underlying sequence of the combined PAM-2 signal has a maximum length of 22 zeros and 28 ones.

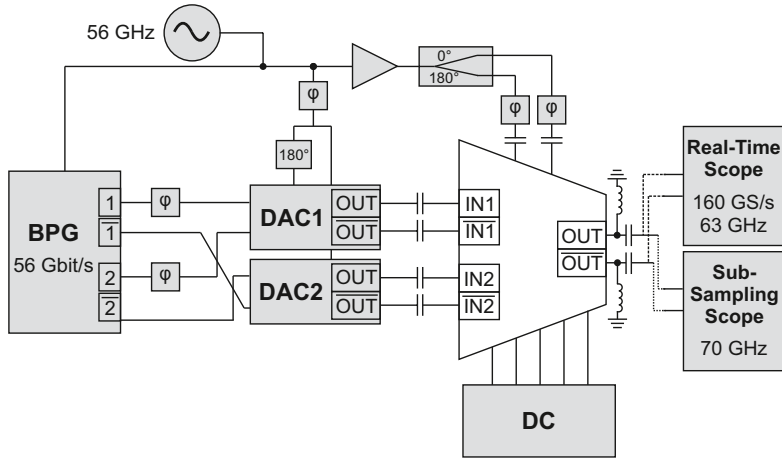


Figure 4.21 Time-domain AMUX on-chip measurement setup.

The differential AMUX input signals are generated with the SHF DACs 613A [258] with an analog output bandwidth of 35 GHz. Both the clock delay and the digital signal delays are adjusted such that the DACs operate with a phase shift of 180° relative to each other. DC blocks are inserted between the DACs and the AMUX.

The 56 GHz clock signal is provided by the frequency generator Agilent E8257D. It is split to drive all of the BPG, the DACs, and the AMUX. The balun Marki BAL-0067 generates the differential clock signal for the AMUX. In order to compensate the losses of both the cabling and the balun for the AMUX clock, the amplifier Spacek Labs SP547-25-19V is utilized prior to the balun. The amplifier operates in the frequency range from 49 to 56 GHz with a gain of 25 dB. Phase shifters are required both to optimize the phase of the differential signal and to synchronize the AMUX clock signal with the DACs' output signals to ensure an optimum operation of the AMUX. Moreover, DC blocks are utilized prior to the AMUX clock input port. The differential clock level is set to -0.4 dBm.

The differential AMUX output signal is acquired with two oscilloscopes one after the other: with the sub-sampling oscilloscope Agilent 86100 DCA with 70 GHz analog bandwidth and with the real-time oscilloscope Keysight DSA-Z 634A with 63 GHz analog bandwidth and 160 GS/s. Bias tees are required before the oscilloscopes to provide a DC path for the output port, which reduces the current on the IC output resistors.

Furthermore, multiple DC supplies are provided to power the AMUX and to optimize control voltages and currents on the chip. The chip is contacted with

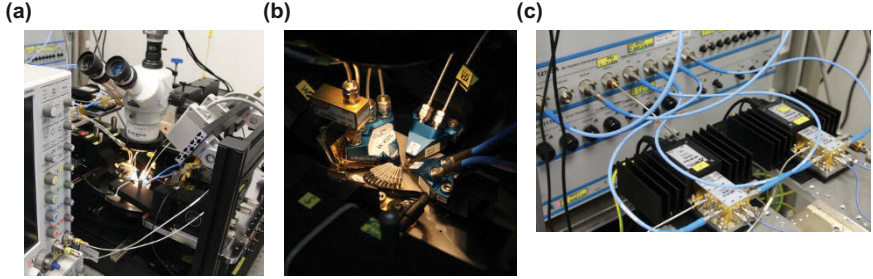


Figure 4.22 Photographs of the time domain AMUX measurement: probe station with oscilloscope (a); microwave probes (b); BPG and DACs (c).

Microwave probes from Cascade Microtech, i.e., an InfinityQuad probe (DC-67 GHz) for the data input pads, an Infinity probe (DC-67 GHz) for the clock input pads, an Infinity probe (DC-110 GHz) for the data output pads, and a DCQ probe for the DC pads.

In Fig. 4.22, photos of the time-domain measurement setup are shown. The probe station with the microscope and the real-time oscilloscope are depicted in Fig. 4.22(a) and the microwave probes are shown in Fig. 4.22(b). The DACs and the BPG are depicted in Fig. 4.22(c).

Results

For the AMUX-DAC, two different clock frequencies are investigated: 28 and 56 GHz. For the first experiment, the input signals are 28 Gb/s PAM-4 NRZ signals and the AMUX is operated at a clock frequency of 28 GHz. The eye diagrams of the DAC output signals are visualized in Fig. 4.23(a) and Fig. 4.23(b). Both are clearly open with average eye openings of ~ 60 mV and ~ 50 mV for the first and the second DAC, respectively. The eye opening of the second DAC is slightly worse due to a minor frequency response difference. The eye levels are at $[-126, -42, +42, +126]$ mV and $[-130, -43, +43, +130]$ mV for the first and the second DAC, respectively.

In Fig. 4.23(c), the resulting 56 Gb/s PAM-4 eye diagrams of the differential AMUX output signal are depicted. The eye diagrams are clearly open with average eye openings of ~ 50 and ~ 40 mV for the odd and even symbols, respectively. The odd and even symbols are generated by the first and the second DAC, respectively. In general, frequency response effects can be observed for the top and the bottom eye level for both single-ended signals. The eye levels for the upper single-ended eye diagram are at $[-170, -65, +45, +150]$ mV and $[-175, -65, +50, +165]$ mV for

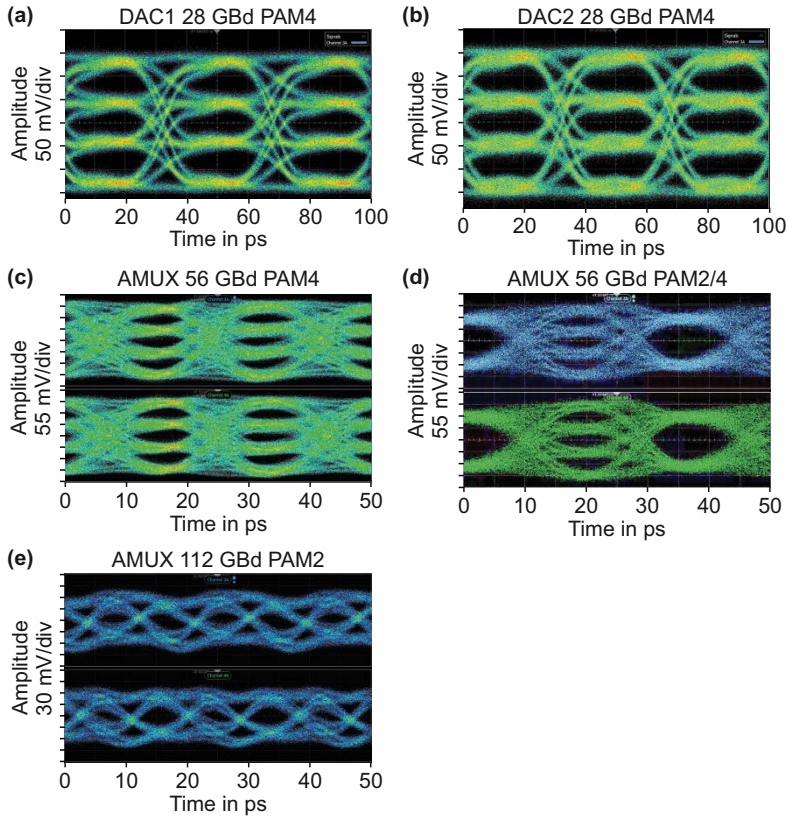


Figure 4.23 Time domain AMUX measurement eye diagrams: DAC1 28 GBd (a); DAC2 28 GBd (b); AMUX 56 GBd (c); AMUX 56 GBd (d); AMUX 112 GBd (e). (c): © 2017 IEEE. Reprinted, with permission, from [42].

the odd and even symbols, respectively. The amplitude noise of the output eyes is increased compared to the input eyes resulting in an eye opening penalty of 66 % and 61 % for the odd and the even symbols, respectively. The eye opening penalty is defined as the ratio of the average eye opening of the output to the average eye opening of the input, whereby both input and output are normalized to the average eye amplitude.

The quality of the eyes is degraded due to the frequency responses of DACs, DC blocks at the input, cabling, microwave probes at the input, the AMUX's data and clock path, microwave probes at the output, cabling at the output, and bias Ts at

the output. Furthermore, additional noise contributions are present both from the AMUX's buffer amplifiers and from the AMUX clock's phase noise.

In Fig. 4.23(d), the output of the AMUX is depicted for the case of DAC1 using 28 GBd PAM-2 and DAC2 using 28 GBd PAM-4 resulting in an interleaved PAM-2/PAM-4 modulation with 84 Gbit/s. By using modulation formats in this interleaved fashion, i.e., time domain hybrid modulation, an additional degree of freedom is enabled without changing the clock frequency [259]. Alternatively, a change in data rate can be easily obtained with the appropriate channel coding.

For the second experiment, the AMUX input is given by two 56 GBd PAM-2 NRZ signals and the AMUX is operated at 56 GHz.

In Fig. 4.23(e), the resulting 112 GBd PAM-2 eye diagram of the differential output signal is shown. The clock feedthrough is clearly visible by the envelope in the upper single-ended signal. The upper single-ended PAM-2 eye diagram has an eye opening of ~ 45 and ~ 30 mV for the odd and even symbols, respectively. The difference between odd and even symbols can be attributed either to different frequency responses of the input paths or to an unequal switching of the AMUX between both input signals, i.e., clock duty cycle mismatch. With PAM-4 an open eye diagram is not achieved and hence, not depicted.

Further measurement results are depicted in Fig. 4.24. The spectrum of the single-ended AMUX output signal is visualized in Fig. 4.24(a). The spectrum is attenuated linearly in the logarithmic scale up to -13 dB at the clock frequency 56 GHz, which is clearly visible. There is a strong roll-off at 63 GHz due to the oscilloscope's bandwidth limitation. The ripple in the spectrum results from the non-independent BPG data sequences.

The high attenuation is based on a superposition of the frequency responses of the employed components. The DACs have an analog 3 dB bandwidth of 35 GHz. Additionally, the DACs' sinc roll-off attenuates the input signals by 3.92 dB at $f_s/2 = 28$ GHz. The microwave on-chip probes used at the input have an attenuation of 2.2 dB at 30 GHz. The microwave on-chip probes at the output have an attenuation of 3.8 dB at 60 GHz. Furthermore, RF coaxial cables, DC blocks at the AMUX input and bias Ts at the AMUX output contribute to the overall attenuation. Besides, the AMUX impairments, i.e., the frequency responses of both data path and clock path need to be considered for the overall result.

In Fig. 4.24(b), the centered half-symbol spaced impulse response is depicted, which is obtained with a LS CE. There is a strong main peak and a small side peak as well as minor contributions before and after. The time-period between the beginning of the first peak and the end of the second peak is equivalent to eight half-symbol spaced taps.

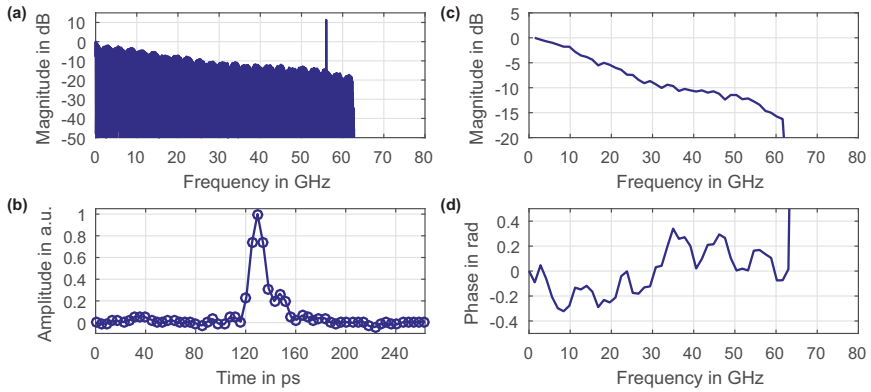


Figure 4.24 Measurement results: received spectrum (a); half-symbol spaced impulse response (b); frequency response for the positive single-ended output: magnitude (c) and phase (d).

In Figs. 4.24(c) and 4.24(d), both the frequency response's magnitude and phase are visualized. The magnitude drops to -13 dB at the 56 GHz clock frequency and then even further. This can be attributed to the oscilloscope's bandwidth limitation mentioned before. Furthermore, a slope change is observed at around 30 GHz, which can be attributed to the mixing operation of the AMUX and the superposition of multiple DAC image replica as shown in Sec. 4.3. The phase response's deviations are small and could be easily mitigated with a pre- or post-equalizer in a further experiment.

4.10.3 Discussion of Results

The presented results from both the S-parameter and the time-domain measurement proof the functioning and the potential of the AMUX IC and the AMUX-DAC concept.

The analog 3 dB bandwidth of the AMUX signal path exceeds 67 GHz and is approximately between 70 and 80 GHz, if the trend of the IS211 and IS231 is extrapolated linearly in the logarithmic scale, which enable the generation of high symbol rate signals.

The time-domain measurement provides eye diagrams for both 56 GBd PAM-4 as well as 112 GBd PAM-2. The PAM-4 eye diagrams at 56 GBd are widely open with average eye openings of ~ 50 and ~ 40 mV for the odd and even symbols, respectively. The 112 GBd PAM-2 eye diagrams are open with an eye opening of ~ 45 and ~ 30 mV for the odd and even symbols, respectively. The difference

between odd and even symbols is primarily caused by a difference in the DAC frequency responses. A strong clock feed-through is present in the 112 GBd signal, which further degrades the quality of the output signal.

In the time-domain measurement, it was not possible to isolate the effects of the AMUX. There are many effects, which contribute to the overall result. Among them are:

- unreliable and unstable on-chip contact
- frequency responses of the DACs, data input probes, clock probe, output probe, cabling, DC blocks, bias tees, oscilloscope, etc.
- unequal DAC frequency responses
- slightly unequal data input probes' frequency responses
- slight amplitude and phase mismatches of the differential input signals
- reflections between components

Moreover, it was not possible to apply pre-distortion due to the utilization of the SHF DACs with both a limited resolution, i.e., two bit, and the dependent BPG data sequences. DSP algorithms for pre-processing and DACs with a higher resolution are expected to improve the results.

4.11 Scaling Model

Jitter has a major influence for the overall AMUX-DAC performance. In this section, a simple noise model is introduced to deduce jitter requirements for scaling the AMUX-DAC concept to both higher frequencies and AMUX orders. Examples are provided for a 2:1 AMUX-DAC and a 4:1 AMUX-DAC.

The following assumptions are made: the DACs are limited by both quantization noise and jitter according to (2.12). The N DACs have equal RMS jitter values τ_{DAC} and the DAC sampling rate is $f_s = 64 \text{ GS/s}$. The DAC's half-rate clock frequency is given as $f_{\text{DAC}} = \frac{f_s}{2}$.

The $N : 1$ AMUX is built from 2:1 AMUXs and consists of $M = \log_2(N)$ stages. The corresponding block diagram is depicted in Fig. 4.25. The individual clock signals for the DACs and the AMUXs are derived from a common clock source. The clock signals are subject to phase noise, but do not have spurious components in the spectrum.

The final AMUX stage operates at the highest frequency; hence, the clock jitter for the final stage is most critical. The AMUXs' clock frequencies are given for a

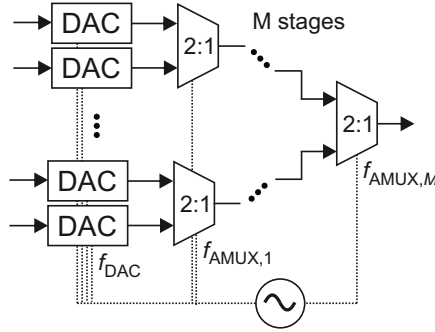


Figure 4.25 AMUX-DAC scaling model for N DACs.

half-clock design for the first and the final stage as $f_{AMUX,1} = f_s$ and $f_{AMUX,M} = M \cdot f_s$, respectively. The RMS jitter for the m th AMUX stage's clock is denoted $\tau_{AMUX,m}$.

The total RMS jitter τ_{tot} is given by

$$\tau_{tot} = \sqrt{\left(\tau_{DAC} \frac{f_{DAC}}{f_{AMUX,M}}\right)^2 + \sum_{m=1}^M \left(\tau_{AMUX,m} \frac{f_{AMUX,m}}{f_{AMUX,M}}\right)^2}, \quad (4.43)$$

whereby the contribution of each stage is weighted with its clock frequency. Based on the half-clock design, (4.43) can be simplified to:

$$\tau_{tot} = \sqrt{\left(\frac{\tau_{DAC}}{2^M}\right)^2 + \sum_{m=1}^M \left(\frac{\tau_{AMUX,m}}{2^{M-m}}\right)^2}. \quad (4.44)$$

Two examples are provided: a 2:1 AMUX-DAC and a 4:1 AMUX-DAC. Both examples are investigated with two types of DACs, i.e., a DAC corresponding to the actual DAC presented in Appendix A and a hypothetical next-generation DAC with a better performance. For the actual and the next-generation DAC a resolution of 6 and 8 bit are assumed in order to limit the SNR close to the measured values. The SNR values are calculated by means of the total jitter and the quantization noise according to (2.12).

In Fig. 4.26, the resulting SNR values are depicted with respect to the frequency. In Fig. 4.26(a), the SNR values for a single DAC are visualized for different RMS jitter values. The SNR values for the actual and the next-generation DAC are indicated with solid and dashed lines, respectively. An upper bound for the SNR is set by the quantization noise at 38 and 50 dB for the actual and the next-generation DAC,

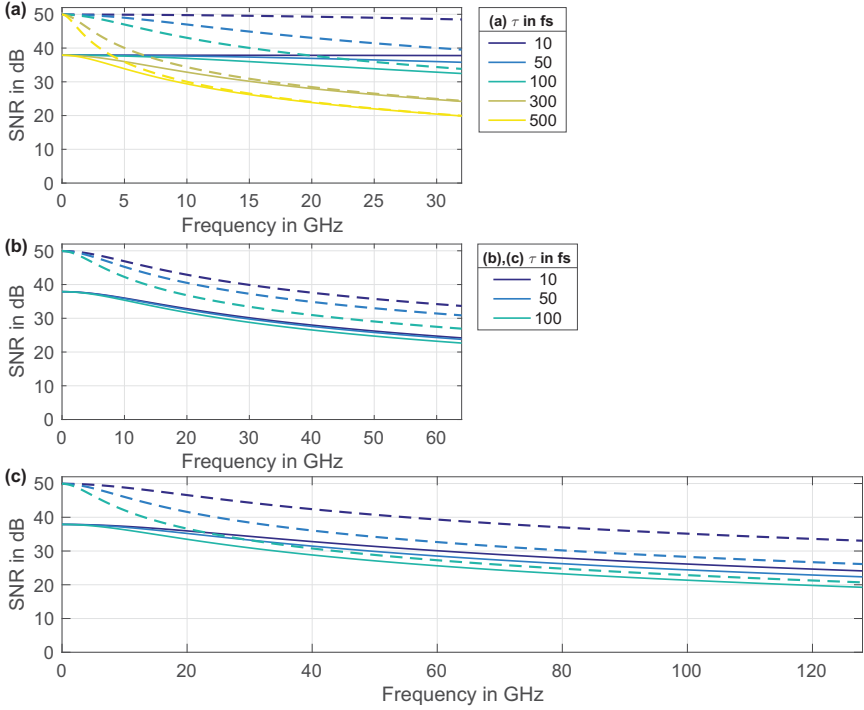


Figure 4.26 SNR values for AMUX-DAC scaling model examples: single DAC (a), 2:1 AMUX-DAC (b), 4:1 AMUX-DAC (c).

respectively. For the actual DAC the quantization noise dominates for RMS jitter values < 10 fs.

In Fig. 4.26(b), the SNR values for the 2:1 AMUX-DAC are depicted. For the actual and the next-generation DAC, a DAC clock RMS jitter of 300 fs and 100 fs is assumed, respectively. Hence, the theoretical SNR curves for the actual DAC roughly match the measured values in Appendix A.

The actual DAC dominates the AMUX-DAC performance for an AMUX clock RMS jitter below 100 fs. The next generation DAC limits the AMUX-DAC performance as well; however, the AMUX clock RMS jitter has a strong impact for all depicted jitter values. There is a 6.7 dB SNR difference between the RMS jitter values of 10 fs and 100 fs at 64 GHz.

In Fig. 4.26(c), the SNR values for the 4:1 AMUX-DAC are visualized. The performance is impaired by both the first and the second stage of the 4:1 AMUX as

well as the DACs. With the actual DAC and an AMUX clock RMS jitter of 50 fs an SNR of 22.3 dB is obtained at 128 GHz. Thereby, an equal RMS jitter is assumed for both AMUX stages.

The best performance, i.e., 33.0 dB SNR at 128 GHz, is achieved by means of the next generation DACs with an AMUX clock RMS jitter of 10 fs. For an AMUX clock RMS jitter of 100 fs, the performances of both the actual and the next generation DAC become almost identical for high frequencies.

Concluding, high-precision clocks are required for AMUX-DAC operation beyond 100 GHz. Moreover, a next generation DAC is desired to enable high SNR values beyond 100 GHz with the AMUX-DAC concept.

4.12 Limitations and Challenges

By using an AMUX in combination with multiple DACs, both the sampling rate and the analog bandwidth are enhanced at the cost of additional noise and jitter. The sampling rate is increased by a factor equal to the number of DACs as for the TI-DAC concept. Additionally, the analog bandwidth is enhanced due to the nonlinear multiplexing operation. The DACs' bandwidth limitations are circumvented and the combined bandwidth of the AMUX-DAC is determined by the analog bandwidths of both the DACs and the AMUX.

Both the AMUX's clock path and the AMUX's data path need to have a high bandwidth to enable the bandwidth extension as shown in Sec. 4.8. This is especially true for NRZ operation, since multiple spectral image replica are superpositioned as shown in Sec. 4.3.

The precise synchronization of the AMUX with the DACs is critical for this concept; Feedback signals are required for the synchronization. The timing deviation of each signal path is measured and the individual clock phases are adjusted accordingly, e.g., in a control loop. In the AMUX-DAC, clock duty cycle mismatches can be tolerated up to 0.5 ± 0.04 for an $\text{EVM} < 10\%$. The EVM rises, if the clock duty cycle mismatch increases further as demonstrated in Sec. 4.8.

A stringent phase noise specification for both the DAC and the AMUX clock are required to enable ultra-broadband signal generation with a high SNR as shown in Secs. 4.9.4 and 4.11. For a 2:1 AMUX-DAC an RMS AMUX clock jitter < 50 fs is required for an $\text{SNR} > 30$ dB at 64 GHz that is only affected by jitter, while assuming an RMS DAC clock jitter of 100 fs.

AMUX-DAC NRZ operation can be supported by DSP to improve the performance. If using DACs in combination with oversampling, DSP is definitely required. By

means of the DSP, the frequency responses of possibly unequal sub-channels can be corrected with the presented MIMO pre-equalizer. However, pre-processing may reduce the ENOB.

Integration of the AMUX with the DACs on a single IC is highly desired to simplify clocking and synchronization. Next to clocking and synchronization issues, a major disadvantage of the setup with discrete components is the change between current and voltage mode. This change is costly in terms of energy and bandwidth due to transimpedance amplifiers, transconductance amplifiers, and resistors. An improved performance could be obtained by designing the AMUX-DAC solely in current mode, which prevents the costly change between the operating modes in terms of power and bandwidth.

A BiCMOS process combining the benefits of both CMOS and bipolar transistors will be a sound choice for the integration. The transistor's $f_T/f_{\max}$ values of the manufacturing process define an upper bound for the maximum performance as explained earlier in Sec. 2.2.

4.13 Summary

In this chapter, the AMUX-DAC was covered. By using an AMUX in combination with multiple DACs, both the sampling rate and the analog bandwidth are enhanced and the DACs' bandwidth limitations are circumvented.

An analytical system model for N DACs was introduced, which had not been available in the literature. Based on this, a 2:1 MIMO system model was derived and the extension to N DACs was outlined. With this, an improved understanding of the AMUX-DAC behavior could be obtained.

Then, signal processing aspects were covered, i.e., clocking, calibration and algorithms. A novel MIMO equalizer was derived from the aforementioned MIMO system model in order to pre-equalize AMUX-DACs with sub-channels having unequal frequency responses. Moreover, algorithmic complexity estimations were presented.

The literature lacked a profound model of both the AMUX by itself and the AMUX-DAC; therefore, a general impairment model for the AMUX-DAC was introduced. Based on this, a generic behavioral 2:1 AMUX model was deduced, which is useful for system-level simulation [43]. It was adapted to reference data from an EDA tool with an NMSE < 20 dB. The model drastically reduces the computation time.

The overall performance for both the AMUX by itself and the AMUX-DAC was investigated by varying multiple parameters, i.e., within the aforementioned behavioral model. The effect of various parameters on the combined frequency response was evaluated. Further, the novel MIMO pre-equalizer notably outperformed a SISO equalizer for unequal signal paths.

Thereafter, experimental results were presented for the AMUX IC with the highest bandwidth > 67 GHz reported at that time [42]. Both an S-parameter measurement and a time-domain measurement were conducted, whereby PAM-4 signals up to 56 GBd and PAM-2 signals up to 112 GBd could be demonstrated without using DSP. The limited resolution of the employed DACs prevented the utilization of pre-compensation algorithms.

In order to complete the treatment of the AMUX-DAC, a noise model for scaling was introduced and analyzed for multiple AMUX-DAC configurations. Finally, conceptual limitations and challenges were discussed.



5 Frequency Interleaving DAC (FI-DAC)

In the previous chapters, time-domain DAC interleaving concepts were covered. In this chapter, a frequency domain concept is proposed. The FI-DAC concept enhances the bandwidth of DACs by using an analog processing system consisting of mixers, filters and a combiner. Multiple frequency bands are aggregated to generate a broadband continuous spectrum. The chapter is structured as follows:

A short introductory literature overview is followed by a detailed explanation of the concept based on a block diagram and a frequency domain illustration [49]. A brief analytical system description lays the foundations for the comprehensive mathematical framework [46, 49], which is required for scaling the FI-DAC towards N DACs. The distribution of data samples among the DACs, the required LO frequencies and the feasible symbol rates are presented within the framework. In order to balance various interdependent FI-DAC parameters two mathematical optimization programs are introduced and FI-DAC parameter examples are provided [49].

The FI-DAC's DSP is covered next. Starting with the DSP block diagram, a MIMO system model for two DACs is presented and extended towards N DACs [44, 45]. Then, both CE and pre-equalization algorithms are described and explained [44, 50]. A novel MIMO pre-equalizer is presented [44, 45] and FI-DAC calibration routines are described [44]. Next, the sub-signals' characteristics are analyzed both with and without pre-equalization [50].

Initial experimental concept verifications with MHz-DACs [45] form the basis for a general impairment model. Thereafter, further experimental results with GHz-DACs are presented. Open high-speed eye diagrams without post-equalization are shown, which were the first at the date of publication [47]. An improved experimental setup further comprises a short-range optical IM/DD transmission [48, 50, 52]. This was shown for the first time at the date of publication [48].

The impact of various parameters on the FI-DAC performance is studied both with experiments and simulations [50, 51]. The investigations include SNR estimations, power and phase mismatches, spectral overlap, clipping, sub-sample shifting, DAC resolution, mixer nonlinearities, phase noise, and jitter.

Moreover, FI-DAC scaling issues are analyzed by means of a simple noise model. Finally, limitations and challenges of the FI-DAC concept are discussed and the chapter is summarized.

5.1 Introduction

The FI-DAC concept employs an analog processing system consisting of mixers, filters and a combiner to generate a broadband signal with a continuous spectrum [68, 260]. Thereby, the digital signal is split by means of DSP into multiple sub-signals, which correspond to different frequency bands. They are individually D/A converted, upconverted with mixers, filtered and seamlessly combined in order to form the desired broadband signal with both a higher overall sampling rate and bandwidth. The corresponding concept for Rx is termed frequency interleaving ADC (FI-ADC) and is commercially realized by oscilloscope vendors [261–263], while the FI-DAC has not been used yet.

Recently, several publications addressed the FI-DAC concept [44–52, 176, 264–267]. In principle, there are two variants of the concept: first, a combined signal is generated with multiple phase-locked frequency bands forming a continuous spectrum [45–47, 176, 264]. Second, a combined signal is generated with multiple independent frequency bands having small guard bands in between [265–267]. With the latter data rates of up to 178 Gbit/s using two frequency bands and an I/Q mixer were transmitted over a short-range optical IM/DD link. The independent frequency bands lower the requirements on both the DSP complexity and the phase stability between the frequency bands. Since, the output spectrum of the second variant is non-continuous, this thesis focuses on the first variant.

The first FI-DAC demonstrations in [45, 264] were published approximately at the same time. In [45], the concept is experimentally verified with two frequency bands that are generated by DACs operating in the sub-GHz range. The cross talk between the frequency bands is removed by a novel MIMO algorithm enabling the DACs to operate at their maximum sampling rate without any oversampling. This way, a higher bandwidth can be achieved in the end.

The experiments with the highest bandwidth were demonstrated in [264]. The FI-DAC operating with three frequency bands generates a total bandwidth of 100 GHz at a sampling rate of 240 GS/s. A 190 GBd 4-level PAM signal was demonstrated for the electrical back-to-back (b2b) case and a 140 GBd binary phase shift keying (BPSK) signal was transmitted in an optical b2b experiment with a high implementation penalty. However, for this FI-DAC, a triplexer and a clock network are employed, that were designed for the application in LeCroy's oscilloscopes and are not well suited for the three CMOS DACs in the FI-DAC. Phase instabilities

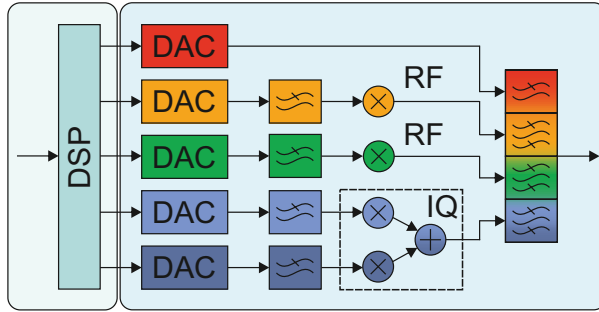


Figure 5.1 Block diagram of the FI-DAC concept with multiple DACs, low-pass filters, RF and I/Q mixers and a frequency multiplexer for filtering and combining the sub-bands. © 2016 IEEE. Reprinted, with permission, from [46].

due to both a 4x and a 32x frequency multiplier require complex Rx DSP, i.e., an adaptive post-equalizer and a phase noise compensation algorithm, in order to recover the received signal and to enable open eye diagrams [176, 268].

Later, the first open eye diagrams without post-equalization at multiple GHz bandwidth due to a better clock network were presented in [47]. Two frequency bands were combined with a diplexer to generate both 2- and 4-level PAM signals with 40 GHz bandwidth and a symbol rate of 80 GBd. Further experiments followed and were published in [48, 50–52].

There are only few other publications covering the frequency interleaving concept: in [269], the effects and the abilities of frequency domain equalization (FDE) for an FI-DAC are presented. In [270, 271], frequency interleaving aspects for Δ - Σ converters are covered.

5.2 Block Diagram and Spectral Illustration

In this section, the FI-DAC concept is explained based on a block diagram and the signal processing is illustrated in the frequency domain.

For frequency interleaving, multiple DACs are used in parallel to perform the D/A conversion for a digital signal, whereby each DAC performs the D/A conversion for a spectral partition of the digital signal. Parts of this section have been previously published in [49].

In Fig. 5.1, a conceptual block diagram of an FI-DAC is shown with five DACs. The digital signal is split by means of a DSP unit into multiple sub-signals, which

correspond to different spectral partitions of the digital signal; they are called sub-bands in the following. After D/A conversion, the sub-bands are converted back to their native frequency positions by means of analog mixers. For the upconversion, different mixer types can be employed, i.e., RF mixers and I/Q mixers. For the RF mixer, either the lower sideband (LSB) or the USB is utilized. Finally, the sub-bands are filtered and combined to generate the analog representation of the combined digital signal, which has a total bandwidth greater than the bandwidth of the DACs. Filtering and combining is achieved either with a single component, i.e., a frequency multiplexer (diplexer, triplexer, quadruplexer, etc.) as depicted in the figure or with multiple components, i.e., with multiple filters and a separate combiner. Besides, an active device is a viable alternative for combining, e.g., a linear active combiner as demonstrated for a TI-DAC in [231].

In Fig. 5.2, both the digital and analog signal processing for the FI-DAC concept is depicted for an exemplary FI-DAC with five DACs corresponding to the block diagram in Fig. 5.1. The figure depicts different exemplary ways to generate the sub-signals rather than an actual implementation. The upper part of Fig. 5.2 illustrates the DSP, i.e., the digital frequency de-interleaving, and the lower part of Fig. 5.2 depicts the analog signal processing, i.e., the analog frequency interleaving.

For the digital frequency de-interleaving, the digital signal in Fig. 5.2(a) is passed through a Fourier transform and the resulting spectrum is shown schematically in Fig. 5.2(b). Next, the spectrum is partitioned into sub-bands as visualized in Fig. 5.2(c). The partitioning of the spectrum is usually done in accordance with the specifications of the analog components, i.e., all of the DACs' sampling rate and bandwidth, the filters' bandwidth, the mixers' bandwidth, etc., which is explained in detail in Secs. 5.4.2-5.5.

After partitioning, the sub-bands are each digitally downconverted to baseband and low-pass-filtered. This can be efficiently achieved by selecting the frequency domain samples of each sub-band and placing them into baseband, whereby digital downconversion and filtering are achieved implicitly. Prior to D/A conversion, guard bands are applied to the sub-bands for both low and high frequencies, indicated with lighter colors in Fig. 5.2(d).

In general, the guard bands increase the spectral distance between desired and undesired spectral components, which are generated by the analog components. The unwanted spectral components encompass DAC images, redundant RF mixer sidebands as well as the LO feedthrough. Due to the guard bands, they can be efficiently suppressed by means of analog filters with finite filter slopes. For the depicted case as a frequency domain implementation, guard bands for both low and high frequencies are easily obtained by inserting additional samples in the sub-bands, i.e., zeros. The generation of guard bands at low and high frequencies is

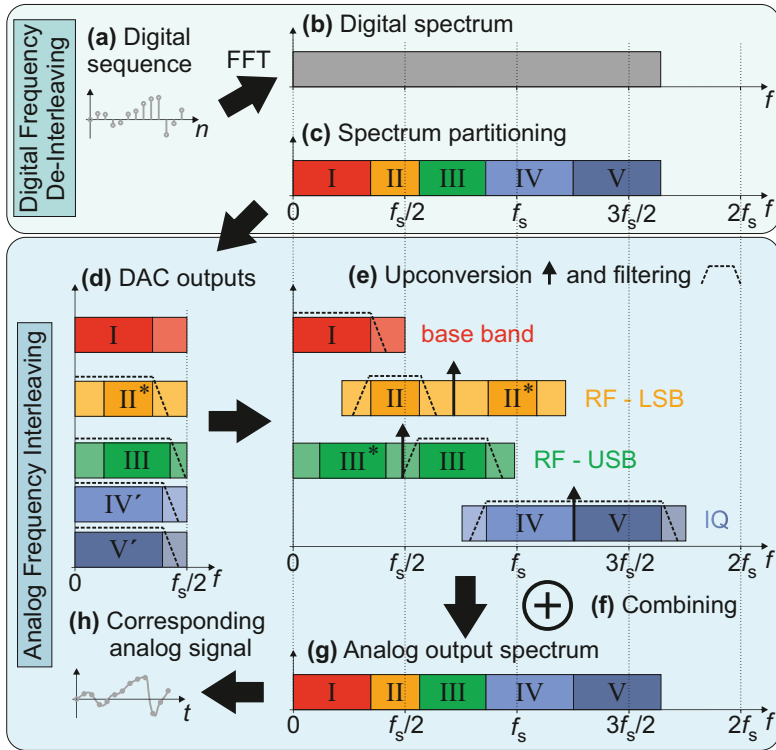


Figure 5.2 Exemplary spectral illustration of the FI-DAC's digital and analog signal processing steps with multiple mixer types for five sub-bands: The digital signal (a,b) is split into multiple sub-signals (c), which are each D/A converted (d). The sub-signals are upconverted, filtered (e) and combined (f) to generate the analog representation (g,h) of (a). © 2016 IEEE. Adapted, with permission, from [46].

equivalent to digital upconversion and oversampling, respectively. The individual sub-signals are further pre-equalized to mitigate both impairments of the analog components and mismatches between the different analog paths [47, 176]. The sub-signals are passed through inverse discrete Fourier transforms (IDFTs) in order to obtain the time domain signals for the DACs.

After D/A conversion, the analog sub-signals are low-pass-filtered in order to suppress DAC hold images. They are further upconverted to their native frequency positions by means of mixers and are band-pass-filtered as depicted in Fig. 5.2(e). The first sub-signal is not upconverted to obtain a combined baseband signal in the end. The BPFs suppress harmonics of the mixers as well as one of the unused

RF mixer sidebands (II* and III*). The LO frequencies depend both on the guard bands and the width of the data sub-bands as shown in Sec. 5.4.3. If the LSB of an RF mixer is used (II), the signal must be generated in the reversed frequency position, which is denoted by the superscript *. This operation is described later in Sec. 5.7. If an I/Q mixer is employed (IV and V), a broader sub-band can be generated, since both mixer sidebands are used; furthermore, less guard bands are required. The sub-bands IV' and V' correspond to the real and the imaginary part of the signal given by the two-sided spectrum [IV, V]. However, analog I/Q mixers are subject to I/Q mismatch, which causes in-band interference and requires additional compensation effort.

All sub-bands are finally combined in Fig. 5.2(f) to form a single continuous spectrum as depicted in Fig. 5.2(g). The corresponding time-domain signal in Fig. 5.2(h) is the analog representation of the digital signal in Fig. 5.2(a). The combined signal has both a higher sampling rate and a higher bandwidth than a signal generated by a single DAC.

With the FI-DAC concept, an arbitrary signal can be generated, either in the time or in the frequency domain. The concept is not limited to time-domain signals, e.g., PAM or quadrature amplitude modulation (QAM), but supports frequency domain modulation formats as well, e.g., DMT or OFDM. If the targeted application tolerates a non-continuous spectrum with gaps, guard bands between the sub-bands simplify the requirements for both DSP and phase stability between the sub-bands [51, 265–267].

The combined DAC's frequency response is composed of the frequency responses of the individual analog paths. The combined frequency response can be both digitally defined and adapted to the user's needs. The pre-processing generates sub-signals, which are either overlapping or non-overlapping in the frequency domain. This depends on the DSP algorithms, the width of the data sub-bands, the DAC sampling rate, etc. The combined sub-bands form a continuous waveform without any spectral gaps for both cases.

Moreover, single sideband (SSB) mixers can be employed, which corresponds to using RF mixers without the need for filtering the unused sideband. Information on the variant with I/Q mixers can be found in Appendix C and in [44].

In the described figure, the unwanted spectral components are removed by means of guard bands and analog filters. Alternatively, these are removed by means of pre-equalization as explained and discussed later in Sec. 5.7. Theoretically, the FI-DAC can be operated without analog filters as well. In turn, the DSP effort reduces the effective resolution.

In wireless communication applications, RF DACs are commonly used. An RF DAC integrates D/A conversion and mixing to improve the analog performance compared to a transmitter realization with an external mixer. RF DACs are well-suited for the implementation in an FI-DAC; however, for ultra-high bandwidth applications with multiple tens of GHz, their current sampling rates and analog bandwidths are insufficient [272].

5.3 Brief Analytical System Description

In this section, a brief analytical description of the FI-DAC's analog processing is provided to improve the understanding of the system. Based on this, the mathematical framework is developed in Sec. 5.4. Further, it is the basis for the DSP presented in Sec. 5.7.

The FI-DAC's analog signal processing steps, i.e., mixing, filtering, and combining, can be applied in reversed order in the digital domain to generate the sub-signals for the N DACs as described in Sec. 5.7.1. The digital input signal $x(k)$ with sampling rate $f_{\text{sym,tot}}$ is split among the N DACs by means of a frequency demultiplexer, multiple mixers and filters. The set of DACs is given by $\Lambda = \{1, \dots, N\}$. The resulting sub-signals are denoted as $x_n(k)$. In Sec. 5.4, a more detailed treatment on the spectral splitting based on integer sample values is provided.

Assuming linear DACs without noise and jitter, the DACs' analog output signals $y_n(t)$ are given by:

$$y_n(t) = \underbrace{\sum_{k \in \mathbb{Z}} x_n(k) \delta(t - kT_s)}_{\text{Ideal D/A Conversion}} * \underbrace{\text{rect}\left(\frac{t}{T_s}\right)}_{\text{ZOH}} * \underbrace{v_n(t)}_{\text{Analog Filter}} \quad . \quad (5.1)$$

The D/A conversion is described by ideal D/A conversion, ZOH operation at the sampling rate f_s as described in Sec. 2.4, and an analog LPF accounting for bandwidth limitations $v_n(t)$. In this description, $v_n(t)$ also accounts for the analog filters following the DACs that are utilized to suppress DAC hold images as well as the low-pass characteristic of the mixer inputs. The low-pass characteristic of the frequency multiplexer for the first sub-signal is not represented by $v_1(t)$. $\delta(t)$, $*$, and $\text{rect}(\cdot)$ denote the Dirac distribution, the convolution, and the rectangular function, respectively. The usage of additional variables for unit consistency, as in (2.3), is omitted to improve clarity and readability.

The corresponding frequency domain representation is given as

$$Y_n(f) = \underbrace{X_n(e^{j2\pi f T_s})}_{\text{Ideal D/A conversion}} \cdot \underbrace{T_s \text{sinc}(T_s f)}_{\text{ZOH}} \cdot \underbrace{V_n(f)}_{\text{Analog Filter}}, \quad (5.2)$$

whereby $X_n(e^{j2\pi f T_s})$, $\text{sinc}(\cdot)$, and $V_n(f)$ denote the DTFT of $x_n(k)$, the sinc-function, and the Fourier transform of $v_n(t)$.

The sub-signals are upconverted to multiple LO frequencies and are further subject to filtering according to

$$\tilde{y}_n(t) = \underbrace{(y_n(t) \cdot \cos(2\pi f_{\text{LO},n} t))}_{\text{Upconversion}} * \underbrace{v_{\text{FM},n}(t)}_{\text{Analog Filter}}, \quad (5.3)$$

whereby $v_{\text{FM},n}(t)$ denotes the combined impulse response of both the mixer's output port and the frequency multiplexer's transfer characteristic. The first sub-signal is not upconverted by setting $f_{\text{LO},1} = 0$.

The upconverted sub-bands, corresponding to the sub-signals $\tilde{y}_n(t)$, are either overlapping or non-overlapping, dependent on the guard bands, the analog filter characteristics and the LO frequencies. For overlapping sub-bands, MIMO DSP algorithms are required, which are covered later in Sec. 5.7.

All sub-bands are finally added in the frequency multiplexer, which results in the combined FI-DAC output signal $y(t)$:

$$y(t) = \sum_{n \in \Lambda} \tilde{y}_n(t). \quad (5.4)$$

5.4 Mathematical Framework

With a FI-DAC, an ultra-broadband waveform can be generated. The concept has been demonstrated so far with two and three DACs, respectively, although scaling to more DACs is theoretically feasible. In order to obtain a continuous spectrum without any spectral gaps, certain conditions need to be fulfilled for the FI-DAC design.

Multiple parameters need to be determined and defined, i.e., DAC sampling rate, LO frequencies, analog frequency responses, number of samples, etc. However, these parameters are interdependent and need to be determined in a joint manner. For a FI-DAC consisting of a few DACs, the complexity is manageable to obtain a feasible and functional parameter set. Nonetheless, scaling the FI-DAC concept to

more DACs requires a profound mathematical framework, which is derived in this section. Parts of this section have been previously published in [46, 49].

The operation of the FI-DAC requires the distribution of the samples of the digital input signal among the DACs, such that each DAC converts a sub-band of the input signal. Two scenarios for the data block length of the system are considered: infinite and finite. The former is present in real-time systems with a continuously operating time-domain signal processing. The latter is present both in real-time systems with a block-based frequency domain signal processing [273] and in “offline” systems with a repetitive data signal, e.g., in an AWG. The mathematical problem consists in splitting the data samples among the DACs, such that each DAC converts a sub-band of the combined digital spectrum. The mathematical problem for the first scenario with an infinite data block length is less complex, since the requirement for an integer number of samples is not present. In this section, the latter scenario with a finite data block length is covered.

This section is structured as follows: starting with several definitions in Sec. 5.4.1, the distribution of data samples among the DACs with respect to the guard bands is addressed in Sec. 5.4.2. The guard bands define the LO frequencies for different mixer types, which is shown in Sec. 5.4.3. Furthermore, the achievable combined bandwidths and symbol rates are discussed in Sec. 5.4.4.

5.4.1 Definitions

In order to develop a mathematical framework, several definitions need to be made. The variables for the FI-DAC are summarized in Table 5.1. Since FI is a frequency-domain concept, this section refers to frequency domain samples, which denote the frequency-discrete sampling points in the frequency domain. The number of frequency domain samples is naturally equal to the number of samples of the corresponding time domain signal.

The FI-DAC consists of N DACs, and the set of DACs is given by $\Lambda = \{1, \dots, N\}$. $K_{D,\text{tot}}$ data samples of the digital signal need to be distributed among the DACs, such the n th DAC converts $K_{D,n}$ data samples. Eventually, all data samples need to be D/A converted in order to generate the correct analog representation of the digital signal. The total number of data samples is given by

$$K_{D,\text{tot}} = \sum_{n \in \Lambda} K_{D,n}. \quad (5.5)$$

Besides, converting $K_{D,n}$ data samples, $K_{O,n}$ additional samples for guard bands are inserted in the frequency domain representation of the digital sub-signals, which are called sub-bands in the following. These guard bands are required to

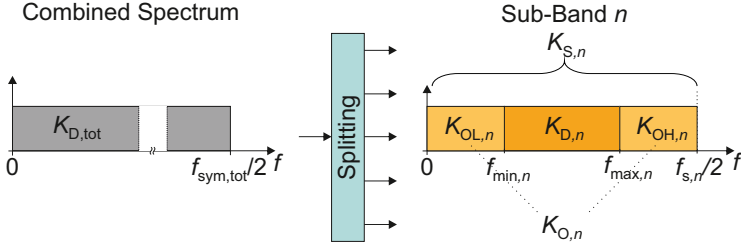


Figure 5.3 The combined spectrum is split into multiple sub-bands: the combined spectrum consists of $K_{D,tot}$ data samples and has a bandwidth equal to half of the total symbol rate $f_{sym,tot}$. The n th sub-band consists of additional samples for guard bands at low frequencies $K_{OL,n}$ and at high frequencies $K_{OH,n}$ as well as data samples $K_{D,n}$.

move undesired image frequency components away from the desired frequency components as depicted in Figs. 5.2(d) and 5.2(e), i.e., DAC hold spectra and mixer sidebands. In Fig. 5.3, both the combined spectrum and the n th sub-band are visualized; the depicted variables are introduced in the following paragraphs.

In general, the oversampling ratio p_n for DAC n , i.e., the relative surplus of additional samples with respect to the data samples, is given by

$$p_n = \frac{K_{S,n}}{K_{D,n}} = \frac{K_{D,n} + K_{O,n}}{K_{D,n}} = 1 + \frac{K_{O,n}}{K_{D,n}}, \quad (5.6)$$

whereby $K_{S,n}$ is the number of total samples for the n th DAC, consisting of $K_{D,n}$ data samples and $K_{O,n}$ additional samples for oversampling.

As depicted in Fig. 5.3, the number of additional samples for oversampling $K_{O,n}$ consists of $K_{OL,n}$ samples at low frequencies and $K_{OH,n}$ at high frequencies:

$$K_{O,n} = K_{OL,n} + K_{OH,n}. \quad (5.7)$$

The number of samples for the n th DAC is thus given by

$$K_{S,n} = K_{D,n} + K_{OL,n} + K_{OH,n}. \quad (5.8)$$

Based on this, two individual oversampling ratios can be specified for the guard bands at low frequencies $p_{L,n}$ and at high frequencies $p_{H,n}$:

$$p_{L,n} = \frac{K_{D,n} + K_{OL,n} + K_{OH,n}}{K_{D,n} + K_{OH,n}} = 1 + \frac{K_{OL,n}}{K_{D,n} + K_{OH,n}}, \quad (5.9)$$

$$p_{H,n} = \frac{K_{D,n} + K_{OL,n} + K_{OH,n}}{K_{D,n} + K_{OL,n}} = 1 + \frac{K_{OH,n}}{K_{D,n} + K_{OL,n}}. \quad (5.10)$$

Table 5.1 Variables for the FI-DAC mathematical framework; source: [49] + additional variables.

Name	Symbol
# of DACs	N
set of DACs	Λ
DAC index	n
sampling rate of DAC n	$f_{s,n}$
total sampling rate of FI-DAC	$f_{s,tot}$
total symbol rate of FI-DAC	$f_{sym,tot}$
# of total data samples	$K_{D,tot}$
# of total samples	$K_{S,tot}$
# of data samples DAC n	$K_{D,n}$
# of additional samples DAC n	$K_{O,n}$
# of additional samples DAC n low	$K_{OL,n}$
# of additional samples DAC n high	$K_{OH,n}$
# of samples DAC n	$K_{S,n}$
oversampling ratio DAC n	p_n
oversampling ratio DAC n low	$p_{L,n}$
oversampling ratio DAC n high	$p_{H,n}$
minimum frequency of DAC n	$f_{min,n}$
maximum frequency of DAC n	$f_{max,n}$
LO frequency for DAC n	$f_{LO,n}$

Based on the oversampling ratios the minimum and the maximum generated frequency of each sub-signal are given by

$$f_{min,n} = \frac{f_{s,n}}{2} \left(1 - \frac{1}{p_{L,n}} \right), \quad (5.11)$$

$$f_{max,n} = \frac{f_{s,n}}{2} \frac{1}{p_{H,n}}, \quad (5.12)$$

whereby $f_{s,n}$ denotes the n th DAC sampling rate. The frequencies are depicted next to the number of samples in Fig. 5.3. The minimum and the maximum

frequency can also be expressed with respect to the number of samples rather than the oversampling ratios according to:

$$f_{\min,n} = \frac{f_{s,n}}{2} \frac{K_{OL,n}}{K_{OL,n} + K_{D,n} + K_{OH,n}} = \frac{f_{s,n}}{2} \frac{K_{OL,n}}{K_{S,n}}, \quad (5.13)$$

$$f_{\max,n} = \frac{f_{s,n}}{2} \frac{K_{OL,n} + K_{D,n}}{K_{OL,n} + K_{D,n} + K_{OH,n}} = \frac{f_{s,n}}{2} \frac{K_{OL,n} + K_{D,n}}{K_{S,n}}. \quad (5.14)$$

The total symbol rate $f_{\text{sym,tot}}$ generated by the FI-DAC is given by twice the sum of the sub-bands' bandwidths

$$f_{\text{sym,tot}} = 2 \cdot \sum_{n \in \Lambda} (f_{\max,n} - f_{\min,n}), \quad (5.15)$$

and the FI-DAC's total sampling rate $f_{s,\text{tot}}$ is given by

$$f_{s,\text{tot}} = \sum_{n \in \Lambda} f_{s,n}. \quad (5.16)$$

For a real-world application, all DAC sampling rates are preferably equal to simplify clocking and synchronization. With $f_{s,n} = f_s \forall n \in \Lambda$ the total sampling rate is given as

$$f_{s,\text{tot}} = N \cdot f_s. \quad (5.17)$$

The theoretically maximum symbol rate $f_{\text{sym,tot}}$ is achieved, if the DACs are operated without oversampling, yielding

$$f_{\text{sym,tot}} \Big|_{p_n=1 \forall n \in \Lambda} = f_{s,\text{tot}}. \quad (5.18)$$

If oversampling is omitted, cross talk between the sub-signals occurs. The cross talk can be mitigated by means of digital signal processing as shown in Sec. 5.7 and in [45].

5.4.2 Distribution of Data Samples

In order to distribute the data samples among the DACs, the number of data samples for each DAC $K_{D,n}$ needs to be calculated. As mentioned in the introduction of Sec. 5.4, the mathematical problem consists in splitting the data samples $K_{D,\text{tot}}$ among the N DACs and taking into account the guard bands, specified by the oversampling ratios $p_{L,n}$ and $p_{H,n}$. For this derivation it is assumed that the oversampling ratios

for both low frequencies $p_{L,n}$ and high frequencies $p_{H,n}$ are defined by the DAC designer for all DACs $n \in \Lambda$ and that the number of total data samples $K_{D,\text{tot}}$ is known.

For the derivation it is further assumed that the DACs operate at the same sampling rate f_s and that all DACs receive their input signals from memories of equal size K_S :

$$f_{s,n} = f_s \quad \forall n \in \Lambda, \quad (5.19)$$

$$K_{S,n} = K_S \quad \forall n \in \Lambda. \quad (5.20)$$

The solution is obtained in two steps: first, a joint oversampling ratio p_n based on $p_{L,n}$ and $p_{H,n}$ is calculated for each DAC. Second, the distribution of the data samples among the DACs is obtained by solving a linear equation system.

Problem Solution Part I: Calculating a Joint Oversampling Ratio for Each DAC

As described before, $p_{L,n}$ and $p_{H,n}$ are given for each DAC and a joint oversampling ratio $p_n(p_{L,n}, p_{H,n})$ for DAC n needs to be determined. In (5.6), $p_n(K_{D,n}, K_{OL,n}, K_{OH,n})$ is given, which depends on samples rather than on oversampling ratios. Therefore, $K_{OL,n}(p_{L,n}, p_{H,n})$ and $K_{OH,n}(p_{L,n}, p_{H,n})$ are obtained by solving the linear equation system given by (5.9) and (5.10) for $K_{OL,n}$ and $K_{OH,n}$.

The solution is trivially given by

$$K_{OL,n} = \frac{p_{H,n}(1 - p_{L,n})}{p_{L,n}p_{H,n} - p_{L,n} - p_{H,n}} K_{D,n}, \quad (5.21)$$

$$K_{OH,n} = \frac{p_{L,n}(1 - p_{H,n})}{p_{L,n}p_{H,n} - p_{L,n} - p_{H,n}} K_{D,n}. \quad (5.22)$$

Depending on the parameters $p_{L,n}$ and $p_{H,n}$, the results of the two previous equations, $K_{OL,n}$ and $K_{OH,n}$, might not be integer-valued. Thus, the input parameters must be chosen accordingly to obtain integer-valued parameters, which is not trivial. Later in Sec. 5.5, a mathematical optimization technique is introduced for parameter balancing.

In order to calculate $p_n(p_{L,n}, p_{H,n})$, (5.21) and (5.22) are inserted into (5.6):

$$p_n = 1 + \frac{K_{OL,n} + K_{OH,n}}{K_{D,n}} = \frac{p_{L,n}p_{H,n}}{p_{L,n} + p_{H,n} - p_{H,n}p_{L,n}}. \quad (5.23)$$

Now, there is a joint oversampling ratio $p_n(p_{L,n}, p_{H,n})$ for each DAC n based on $p_{L,n}$ and $p_{H,n}$. The dependence on $K_{D,n}$, which is still unknown at this point, was canceled out. By inserting (5.11) and (5.12) into (5.23), p_n can also be stated with respect to $f_{\min,n}$ and $f_{\max,n}$ according to:

$$p_n = \frac{f_{\max,n}}{1 - f_{\min,n} f_{\max,n}} \frac{f_{s,n}}{2} . \quad (5.24)$$

Problem Solution Part II: Calculate the Number of Data Samples for Each DAC

In order to distribute the data samples $K_{D,\text{tot}}$ among the N DACs according to the oversampling ratios p_n obtained in the previous section, the following linear equation system is solved:

$$\sum_{n \in \Lambda} K_{D,n} = K_{D,\text{tot}} , \quad (5.25)$$

$$p_n K_{D,n} = p_l K_{D,l} \quad \forall n, l \in \Lambda \wedge n \neq l . \quad (5.26)$$

Reconstructing the digital signal in the analog domain requires all data samples to be D/A converted as in (5.5), which results in (5.25). Furthermore, each DAC uses a memory of equal size as stated in (5.20). Inserting (5.6) in (5.20) yields (5.26).

The linear equation system given by (5.25) and (5.26) is solved for $K_{D,n} \forall n \in \Lambda$. The complete derivation can be found in Appendix D and the solution is given by

$$K_{D,n} = \frac{\prod_{l \in \Lambda_n} p_l}{\sum_{m \in \Lambda} \prod_{i \in \Lambda_m} p_i} K_{D,\text{tot}} \quad \forall n \in \Lambda , \quad (5.27)$$

whereby $\Lambda_k = \Lambda \setminus \{k\}$, $k \in \Lambda$.

With (5.27), the number of data samples for each DAC can be calculated. The number of additional samples for oversampling can now be calculated as well, i.e., $K_{OL,n}$ in (5.21) and $K_{OH,n}$ in (5.22). The resulting symbol rate $f_{\text{sym,tot}}$ is obtained according to (5.15). Thus, all required values are given to generate the sub-signals for each DAC.

However, the system parameters are naturally linked by the equations introduced in the previous sections and are interdependent. In order to generate the sub-signals, integer values are required for $K_{D,n}$, $K_{OL,n}$ and $K_{OH,n}$, which are not given by default in (5.27) as shown in [46]. The parameters $K_{D,n}$, $K_{OL,n}$ and $K_{OH,n}$ are solely integer-valued, if the input parameters $p_{L,n}$ and $p_{H,n}$ are chosen accordingly. The integer

constraint imposes limitations on both the LO frequencies and the feasible symbol rates. In the following sections, the equations for determining the LO frequencies are stated and the symbol rate limitations are further clarified.

5.4.3 Local Oscillator Frequencies

As depicted in Fig. 5.2, the LO frequencies depend on the mixer types and on the oversampling values $p_{L,n}$ and $p_{H,n}$ defining the guards bands for each DAC. In principle, different types of mixers could be employed, i.e., RF mixers using either the LSB or the USB, as well as I/Q mixers generating a sub-band with the doubled RF bandwidth. In the following, formulas for the calculation of the n th LO frequency are provided for different mixer types.

If the RF mixer's USB is used and the LSB is suppressed, the LO frequencies are determined by:

$$f_{\text{LO,USB},n} = \begin{cases} 0, & n = 1 \\ \sum_{i=2}^n f_{\text{max},i-1} - f_{\text{min},i}, & n > 1 \end{cases} \quad (5.28)$$

If the RF mixer's LSB is used and the USB is suppressed, the LO frequencies are given by:

$$f_{\text{LO,LSB},n} = \begin{cases} 0, & n = 1 \\ \sum_{i=1}^n f_{\text{max},i} - f_{\text{min},i-1}, & n > 1 \end{cases} \quad (5.29)$$

whereby $f_{\text{min},1} = f_{\text{min},0} = 0$ for this case.

If I/Q mixers are used, the frequencies of the LOs are given by

$$f_{\text{LO,IQ},n} = \begin{cases} 0, & n = 1 \\ \sum_{i=1}^{n-(n \bmod 2)} f_{\text{max},i}, & n > 1 \end{cases} \quad (5.30)$$

If SSB mixers are employed, these formulas are applicable as well: $f_{\text{LO,LSB},n}$ and $f_{\text{LO,USB},n}$ are used for SSB-LSB and SSB-USB mixers, respectively.

5.4.4 Feasible Symbol Rates

As has been noted before, the combination of the discrete number of samples and the interdependent parameters imposes limitations on the feasible symbol rates. This is analyzed and discussed in the following.

In general, if a single DAC is used with oversampling, the ability to generate a certain symbol rate depends on the number of data samples, the DAC sampling rate, and the oversampling ratio [53]. In the end, an integer number of samples is required for the digital signal. If an FI-DAC with multiple DACs is used, this condition is preserved. However, the ability to generate the targeted symbol rate $f_{\text{sym,tot}}$ now depends on the number of data samples $K_{\text{D,tot}}$, the total sampling rate $f_{\text{s,tot}}$ and the oversampling ratios for all DACs. The first condition for an FI-DAC reads

$$K_{\text{S,tot}} \in \mathbb{N}_{>0}, \quad (5.31)$$

with the total number of samples given by

$$K_{\text{S,tot}} = \sum_{n \in \Lambda} K_{\text{S},n} = \frac{f_{\text{s,tot}}}{f_{\text{sym,tot}}} \cdot K_{\text{D,tot}}. \quad (5.32)$$

This condition is independent of the distribution of the additional samples for oversampling among the DACs. Next to this condition, the total number of samples needs to be distributable among the DACs, yielding a second condition:

$$\frac{K_{\text{S,tot}}}{N} \in \mathbb{N}_{>0}. \quad (5.33)$$

The total symbol rate $f_{\text{sym,tot}}$ reads

$$f_{\text{sym,tot}} = \frac{K_{\text{D,tot}}}{K_{\text{S,tot}}} f_{\text{s,tot}} \left| \frac{K_{\text{S,tot}}}{N} \in \mathbb{N}_{>0} \right|, \quad (5.34)$$

which is equivalent to the formula for the single DAC for $N = 1$ [53].

In Fig. 5.4, two examples are provided. Fig. 5.4(a) covers the standard case of a single DAC and Fig. 5.4(b) covers the case of a FI-DAC consisting of multiple DACs.

In Fig. 5.4(a), the feasible integer symbol rates $f_{\text{sym,tot}}$ in the GBd regime, which are called GBd-integer symbol rates in the following, are depicted for the case of a single DAC as a function of integer sampling rates f_{s} in the GS/s regime. Two

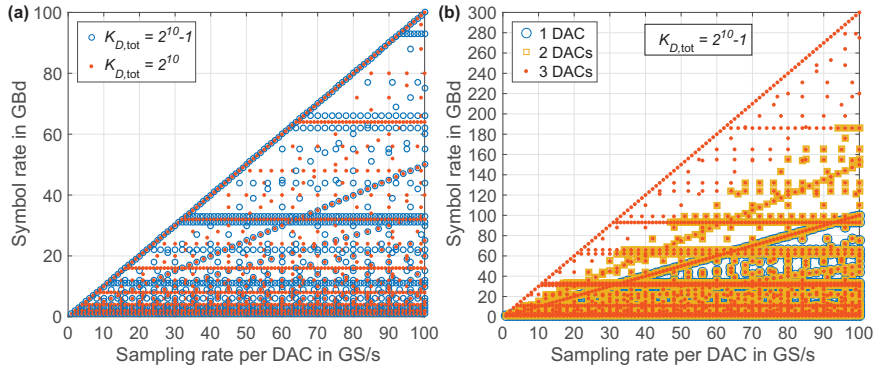


Figure 5.4 Feasible integer GBd-symbol rates for different integer GS/s-sampling rates for: a single DAC for two different numbers of data samples (a); an FI-DAC converting $2^{10} - 1$ data samples consisting of different numbers of DACs (b). © 2018 The Authors of [49]. Adapted, with permission, CC BY 4.0.

different numbers of data samples $K_{D,tot}$ are plotted: $2^{10} - 1 = 1023$ and $2^{10} = 1024$, corresponding to a pseudo-random binary sequence (PRBS) and a two's complement sequence. Naturally, the higher the sampling rate is, the higher the symbol rate will be, as indicated by the general triangular shape of the plotted dots. However, not all GBd-integer symbol rates could be addressed, since not every combination of sampling rate and symbol rate has a dot in the plot. Although, the number of data samples only differs by one, it is obvious, that there are rarely common GBd-integer symbol rates for $K_{D,tot} = 1023$ and $K_{D,tot} = 1024$.

The sampling rate 80 GS/s is analyzed as an example. It is observed, that the GBd-integer symbol rates $\{80, 66, 62, 60, 55, 48, 44, 40 \dots\}$ GBd and $\{80, 40 \dots\}$ GBd could be generated for the number of data samples $K_{D,tot} = 1023$ and $K_{D,tot} = 1024$, respectively. Hence, not every GBd-integer symbol rate in the interval $[0, 80]$ GHz could be addressed and the two common GBd-integer symbol rates between the numbers of data samples are only 80 and 40 GBd. Another noteworthy feature of the plot are the prominent diagonals. The diagonal from the bottom left corner (0 GS/s, 0 GBd) to the upper right corner (100 GS/s, 100 GBd) corresponds to a total oversampling ratio of one sample per symbol. The diagonal from the bottom left corner (0 GS/s, 0 GBd) to the middle of the vertical axis on the right (100 GS/s, 50 GBd) corresponds to a total oversampling ratio of two samples per symbol. On these two diagonals, every GBd-integer symbol rate can be addressed for both numbers of data samples due to the integer oversampling ratio.

In Fig. 5.4(b), the feasible GBd-integer symbol rates for an FI-DAC are depicted for $K_{D,tot} = 2^{10} - 1$ data samples with the number of DACs N being varied between one

and three. By increasing the number of DACs, the total sampling rate $f_{s,\text{tot}}$ increases; thus, the number of addressable GBd-integer symbol rates rises as well. The GBd-integer symbol rates that can be addressed with N DACs could be addressed with $N + 1$ DACs as well, which is indicated by the points for $N \in \{2, 3\}$ plotted on top of the points for $N = 1$. As for the single DAC case, not all GBd-integer symbol rates could be addressed.

As an example, the DAC sampling rate 80 GS/s is analyzed in detail. It can be observed that the GBd-integer symbol rates $\{240, 220, 186, 176, 165\}$ GBd can be addressed solely with three DACs as shown by the red dots in the plot. The following lower GBd-integer symbol rates, $\{155, 132, 124, \dots\}$ GBd could be addressed with a FI-DAC consisting of either two or three DACs as indicated by the red dots framed by the yellow squares. GBd-integer symbol rates ≤ 80 GBd can be addressed by one, two, or three DACs as indicated by the red dots framed by the yellow squares and encircled in blue. The dots ≤ 80 GBd correspond to the dots plotted in Fig. 5.4(a) for $K_{D,\text{tot}} = 2^{10} - 1$. Besides, an odd number of DACs puts an additional constraint on the number of feasible symbol rates, if an even number of data samples is given.

Concluding, the number of data samples and the chosen DAC sampling rate impose constraints on the feasible symbol rates of a FI-DAC. This has to be thoroughly taken into account during FI-DAC design.

5.5 Parameter Balancing

As presented in the previous sections, all system parameters are interdependent, i.e., the number of data samples, the DAC sampling rate, the feasible symbol rates, the oversampling ratios and the LO frequencies. Furthermore, there are integer constraints on the number of samples for each sub-signal. In order to balance these various parameters for a functional FI-DAC design, while taking into account the requirements and the limitations of the analog components, a mathematical optimization technique is introduced in this section. The optimization is performed once during the FI-DAC design phase and it is not required to solve the problem during run-time. Thus, a non-real-time optimization approach is appropriate. Parts of this section have been previously published in [49].

The objective of mathematical optimization consists in minimizing or maximizing a cost function with respect to a set of constraints, which is termed mathematical optimization program [274, 275]. In this particular case, a mixed-integer nonlinear optimization program (MINLP) is used, since the variables of the mathematical program consist of integers and real numbers [276, 277]. Further, nonlinear parameter relations are present in some of the constraints.

Two optimization programs are presented balancing the FI-DAC's parameters: first, a simple optimization program is introduced in Sec. 5.5.1, which is equivalent to the problem solved in Sec. 5.4.2 with additional integer constraints. It is referred to as standard FI-DAC optimization program. Second, an extended optimization program is introduced in Sec. 5.5.2, which contains additional constraints.

The algebraic solutions obtained in Sec. 5.4.2 could be used for the formulation of the optimization program as well. However, the problem is sufficiently described with the original equations; during optimization the interdependencies are implicitly solved.

5.5.1 Standard FI-DAC Optimization Program

The optimization program, which is referred to as standard FI-DAC optimization program, is an MINLP corresponding to the “simple” problem presented and solved in Sec. 5.4.2. Next to the equations used before, the findings of Secs. 5.4.3 and 5.4.4 are taken into account by adding (5.16), (5.32) and (5.34) as constraints to the optimization program. The objective of the optimization function is to maximize the total symbol rate $f_{\text{sym,tot}}$. Parameters for the optimization program are the sampling rate f_s , the number of data samples $K_{D,\text{tot}}$ and the number of DACs N . Further, lower and upper limits for the oversampling ratios $p_{L,n}$ and $p_{H,n}$ are included. Lastly, integer constraints are added. The number of samples $K_{D,n}$, $K_{OL,n}$, $K_{OH,n}$, $K_{O,n}$, $K_{S,n}$ shall be integer-valued.

The optimization program is given as:

$$\begin{array}{ll}
 \max & f_{\text{sym,tot}} \\
 \text{s.t.} & (5.5) - (5.15), (5.16), (5.32), (5.34) \\
 & p_{L,n} \geq p_{L,n}^{\min} \quad (5.35) \\
 & p_{L,n} \leq p_{L,n}^{\max} \quad (5.36) \\
 & p_{H,n} \geq p_{H,n}^{\min} \quad (5.37) \\
 & p_{H,n} \leq p_{H,n}^{\max} \quad (5.38) \\
 & K_{D,n}, K_{OL,n}, K_{OH,n}, \dots \\
 & K_{O,n}, K_{S,n}, K_{S,\text{tot}} \in \mathbb{N}_0 \quad \forall n \in \Lambda \\
 & f_{\text{sym,tot}} \in \mathbb{R}_{>0} \\
 & p_{L,n}, p_{H,n} \in \mathbb{R}_{>0} \quad \forall n \in \Lambda
 \end{array}$$

The output of the optimization program will be an optimal solution set containing the values for the variables $f_{\text{sym,tot}}$, $K_{D,n}$, $K_{OL,n}$, $K_{OH,n}$, $K_{O,n}$, $K_{S,n}$, $p_{L,n}$ and $p_{H,n}$.

5.5.2 Extended FI-DAC Optimization Program

The standard optimization program introduced previously can be extended to cover additional parameters, which shall be balanced with respect to the other parameters.

For the FI-DAC, integer ratios between the LO frequencies $f_{\text{LO},M_n,n}$ introduced in Sec. 5.4.3 and the DAC clock f_{DAC} are desired to efficiently generate the frequencies by dividers and multipliers. This is represented in the optimization program by adding (5.28) to (5.30) for the calculation of the LO frequencies and (5.43) for the integer relation to the DAC clock frequency. In (5.43), the integer-valued variable $\alpha_{\text{LO},M_n,n}$ is introduced. The type of mixer for each DAC is given by the ordered set $\Gamma = \{M_1, \dots, M_N\}$ with $M_n \in \{\{\}, \text{LSB}, \text{USB}, \text{IQ}\}$ and $M_1 = \{\}$ in order to generate a baseband signal.

Furthermore, there is commonly an integer ratio between the DAC clock f_{DAC} and the sampling rate f_s as in (5.44), in which α_{DAC} is the corresponding parameter defined by the DAC architecture.

Moreover, it is desirable to generate a certain symbol rate, instead of just maximizing it. Therefore, the constraint (5.45) is added, which puts an upper bound on the symbol rate. The optimizer attempts to find a solution closest to this target symbol rate $f_{\text{sym,tot,target}}$.

The extended optimization program is given as:

$$\begin{aligned}
 \max \quad & f_{\text{sym,tot}} \\
 \text{s.t.} \quad & (5.5) - (5.15), (5.16), (5.32), (5.34), (5.28) - (5.30) \\
 & p_{\text{L},n} \geq p_{\text{L},n}^{\min} \quad (5.39) \\
 & p_{\text{L},n} \leq p_{\text{L},n}^{\max} \quad (5.40) \\
 & p_{\text{H},n} \geq p_{\text{H},n}^{\min} \quad (5.41) \\
 & p_{\text{H},n} \leq p_{\text{H},n}^{\max} \quad (5.42) \\
 & f_{\text{LO},M_n,n} = \alpha_{\text{LO},M_n,n} \cdot f_{\text{DAC}} \quad (5.43) \\
 & f_s = \alpha_{\text{DAC}} \cdot f_{\text{DAC}} \quad (5.44) \\
 & f_{\text{sym,tot}} \leq f_{\text{sym,tot,target}} \quad (5.45) \\
 & K_{\text{D},n}, K_{\text{OL},n}, K_{\text{OH},n}, K_{\text{O},n}, \dots \\
 & K_{\text{S},n}, \alpha_{\text{LO},M_n,n}, K_{\text{S,tot}} \in \mathbb{N}_0 \quad \forall n \in \Lambda \\
 & f_{\text{sym,tot}} \in \mathbb{N}_{>0} \\
 & p_{\text{L},n}, p_{\text{H},n}, f_{\text{LO},M_n,n} \in \mathbb{R}_{>0} \quad \forall n \in \Lambda
 \end{aligned}$$

5.6 Parameter Balancing Examples

In this section, examples are provided demonstrating the usefulness of mathematical nonlinear optimization to balance the various FI-DAC parameters. To formulate and solve the optimization program, the MINLP is implemented in Zuse Institut Mathematical Programming Language (ZIMPL) [278] and solved with the solver Solving Constraint Integer Programs (SCIP) [279] from Konrad Zuse Center Berlin. The SCIP solver uses a branch-and-bound algorithm based on a linear outer-approximation, whereby primal heuristics are employed during the solving process to speed-up optimization time [280]. Parts of this section have been previously published in [49].

In order to demonstrate the challenges and limitations of the FI-DAC concept regarding the parameter interdependencies an exemplary FI-DAC consisting of $N = 3$ DACs is used, whereby each DAC operates at a sampling rate of $f_s = 80$ GS/s. The FI-DACs exemplary configuration is presented in Fig. 5.2 for the sub-signals I, II and III. The first sub-signal is a baseband signal and both the second and the third sub-signal are upconverted. The LSB of sub-signal II and the USB of sub-signal III are utilized. The triplexer combining the sub-signals is assumed to have cutoff frequencies at 34.5 and 66.5 GHz as in [176]. The number of data samples is chosen to be $K_{D,tot} = 1023$ as in the example in Sec. 5.4.4.

In the following, the parameter balancing is performed with both the standard FI-DAC optimization program and the extended FI-DAC optimization program.

5.6.1 Standard FI-DAC Optimization Program Example

For the standard FI-DAC optimization program, two scenarios are investigated. First, no constraints are imposed on the oversampling ratios in (5.35) - (5.38) and the generated symbol rate is maximized. The result of the optimization is depicted in Figs. 5.5(a) and 5.5(b). Fig. 5.5(a) visualizes the distribution of the samples among the three DACs, whereby each DAC utilizes a block length of $K_S = 341$ samples. Since, $K_{D,tot} = 1023$ is divisible by $N = 3$, all DACs are used to their full extent. The corresponding analog signal is shown in Fig. 5.5(b) with a total symbol rate of $f_{sym,tot} = 240$ GBd. The bandwidth of each sub-band is 40 GHz and the LOs are located both at 80 GHz. Since, no guard bands are utilized, neither DAC image replica nor unused mixer sidebands are sufficiently suppressed with analog filters. Moreover, the parameters do not match the triplexer's cutoff frequencies defined previously.

For the second example, lower and upper bounds for the oversampling ratios $p_{L,n}$ and $p_{H,n}$ are introduced to account for the triplexer's cutoff frequencies at 34.5

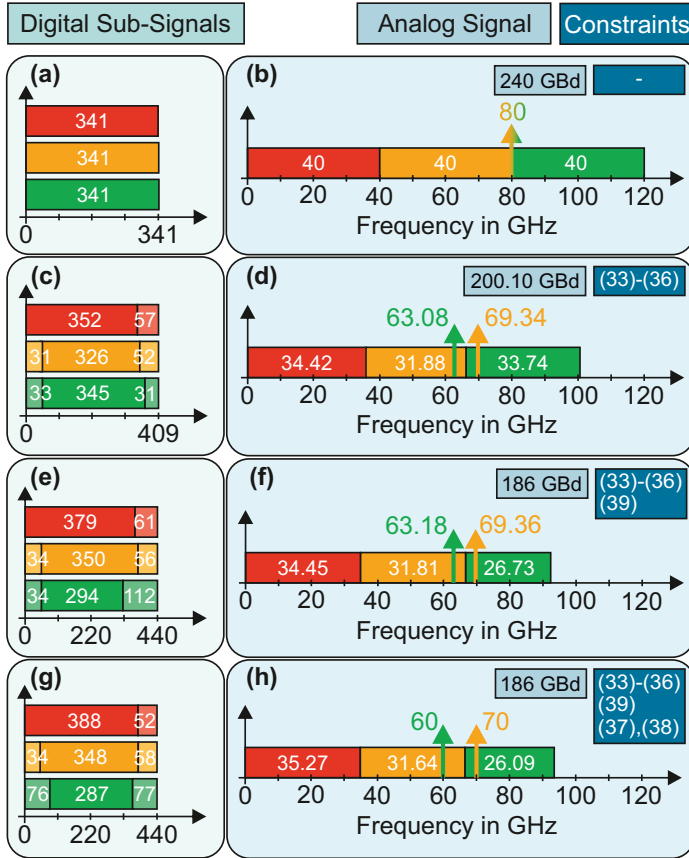


Figure 5.5 Solutions of the FI-DAC optimization program showing the digital input spectrum as a function of samples (left) and the analog output spectrum as a function of frequency (right) for an exemplary FI-DAC consisting of three DACs, each running at $f_s = 80$ GS/s and jointly converting $K_{D,tot} = 1023$ data samples. The equation numbers of the applied constraints are displayed to the right as well as the total symbol rate $f_{sym,tot}$: standard FI-DAC optimization program without constraints (a,b); standard FI-DAC optimization program with oversampling ratio constraints (c,d); extended FI-DAC optimization program with oversampling ratio constraints and GBd-integer symbol rate constraint (e,f); extended FI-DAC optimization program with oversampling ratio constraints, GBd-integer symbol rate constraint and integer ratio constraint between DAC clock and LO frequencies (g,h). © 2018 The Authors of [49]. Adapted, with permission, CC BY 4.0.

and 66.5 GHz, respectively. Matching the first sub-signal to this filter requires a lower bound for the oversampling ratio, i.e., $p_{H,1}^{\min} = 40/34.5$. In order to filter the DAC images of the second and the third sub-signal, the oversampling ratio bound

is chosen to be $p_{H,n}^{\min} = 40/37 \forall n \in \{2, 3\}$. Thereby, it is assumed that a spectral gap of 2×3 GHz is sufficient for a steep roll-off LPF. Furthermore, a spectral gap of 2×3 GHz is assumed sufficient to suppress redundant mixer sidebands. The oversampling ratio for low frequencies is then given as $p_{L,n}^{\min} = 40/(40 - 3) \forall n \in \{2, 3\}$.

In order to match the second sub-signal to the triplexer's characteristic, the value of $p_{H,2}^{\min}$ is further increased to $40/35$. The DACs are now used with oversampling, i.e., guard bands, and the result is visualized in Figs. 5.5(c) and 5.5(d). The sub-bands match the characteristic of the triplexer. The bandwidths of the sub-bands are given by 34.43, 31.88, and 33.74 GHz, respectively. They add up to a total bandwidth of 100.05 GHz corresponding to a symbol rate of 200.10 GBd. The LO frequencies are given by 69.34 and 63.08 GHz for the second and the third sub-signal, respectively.

Generally, it is desirable to have a specific symbol rate for the communication system design. In order to generate a specific symbol rate, the standard FI-DAC optimization program is insufficient. In the next section, the extended FI-DAC optimization program is utilized.

5.6.2 Extended FI-DAC Optimization Program Example

The parameter balancing is rerun with the extended FI-DAC optimization program in this section. At first, only one additional integer constraint is introduced, to achieve a GBd-integer symbol rate: $(f_{\text{sym,tot}} \cdot 10^{-9}) \in \mathbb{N}_{>0}$. The result of the optimization is depicted in Figs. 5.5(e) and 5.5(f). Although, the symbol rate is only required to be GBd-integer-valued, the symbol rate given as the result of the optimization is drastically reduced to $f_{\text{sym,tot}} = 186$ GBd. This can be attributed to the limited set of feasible GBd-integer symbol rates given by $\{240, 220, 186, 176, 165, 155, 132, \dots\}$ as depicted in Fig. 5.4 according to (5.34).

For this reduction, especially the third sub-band is shrunk down to 26.73 GHz. The LO frequencies are 69.36 and 63.18 GHz for the second and the third sub-signal, respectively.

For the DAC system design, an integer ratio between the DAC clock and the LO frequencies is further desirable to design a clock network with a small amount of frequency dividers and multipliers. Thus, the constraints (5.43) and (5.44) are added. Hereby, it is assumed that the DAC clock is given by $f_{\text{DAC}} = f_s/32 = 2.5$ GHz as in [47, 176]. However, with the constraints formulated earlier no solution is feasible. Thus, some of the previous constraints for both the lower and the upper bound for the oversampling ratios are mildly relaxed in order to obtain a feasible solution. The

results of the optimization are depicted in Figs. 5.5(g) and 5.5(h). The symbol rate is still 186 GBd; however, the LO frequencies are now set at integer ratios of the DAC clock at 70 and 60 GHz for the second and the third sub-signal, respectively. The distribution of the data samples among the DACs resembles the distribution in the previous scenario.

In the examples provided above, the upcoming challenges for the FI-DAC design are indicated. On the one hand, the combination of the finite number of data samples, the DAC sampling rate, and the number of DACs determine the achievable symbol rate. On the other hand, further aspects need to be considered for the FI-DAC design, i.e., bandwidth limitations and LO frequencies.

For an implementation, the last configuration would be chosen, since the required LOs are easier to generate. Next to the presented solutions, other optimal solutions with a different distribution of both data samples and additional samples for guard bands are viable as well. They could be obtained by tweaking existing constraints or by adding further constraints. Additional constraints could be used to include further aspects of the FI-DAC design, which have not been covered yet, e.g., equal oversampling ratios for the DACs with $p_{H,1} = p_{H,2}$ or else. The presented optimization program will be very helpful in order to scale the FI-DAC concept to an arbitrary number of DACs while managing the parameter complexity.

5.7 Digital Signal Processing for FI-DAC

The FI-DAC operation requires DSP to mitigate both the impairments of the analog components and the mismatches between the different analog paths. In this section, relevant aspects of the DSP are covered: first, a block diagram of the DSP chain is presented; then, a MIMO system model is derived for the special case of two DACs without guard bands. Algorithms for both the system identification and the pre-equalization are addressed and algorithmic complexity estimations are provided based on analytical formulas. Afterwards, a MIMO pre-equalizer for the special case of two DACs without guard bands is derived. FI-DAC calibration concepts and calibration issues are highlighted next. Finally, the sub-signals' statistical characteristics are evaluated for the case of two DACs and methods for PAPR reduction are discussed. Parts of this section have been previously published in [44, 45, 47, 48, 50].

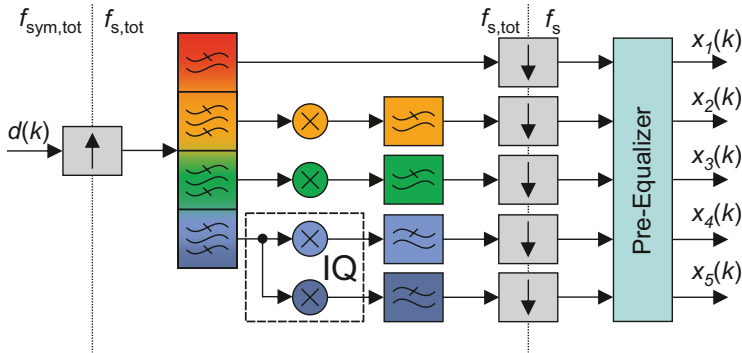


Figure 5.6 FI-DAC DSP block diagram with different mixers: The digital signal $d(k)$ at the sampling rate $f_{\text{sym,tot}}$ is upsampled to the FI-DAC's combined sampling rate $f_{\text{s,tot}}$. It is split in the frequency domain into several sub-signals. Each sub-signal is downconverted to baseband by means of mixers and further low-pass-filtered. After downsampling to the DAC sampling rate f_s the sub-signals are pre-equalized.

5.7.1 Block Diagram

In Fig. 5.6, the block diagram for the FI-DAC DSP is shown with different mixer types. It is virtually a mirror image of the analog processing system depicted in Fig. 5.1.

The digital input signal $d(k)$ is an arbitrary signal at the sampling rate $f_{\text{sym,tot}}$, e.g., a sine signal, a PAM signal with or without pulse shaping, a DMT signal, etc. It is upsampled to the combined FI-DAC sampling rate $f_{\text{s,tot}}$. A digital frequency demultiplexer, i.e., a diplexer, triplexer, etc., splits the combined signal into several sub-signals. The digital frequency demultiplexer can be realized either with ideal rectangular filters allowing for no overlap between the sub-bands, or with other filters allowing for a certain overlap. The sum of the filters' frequency responses need to equal one in the frequency interval $[0, f_{\text{s,tot}}/2]$; this way, the combined signal is reconstructed in the analog domain without undesired spectral dips or peaks resulting from the digital frequency demultiplexing. The digital frequency demultiplexer's realization with raised-cosine characteristics is presented in Appendix E.

Each sub-signal except for the first one, is downconverted to baseband with digital mixers. Either digital RF mixers or digital I/Q mixers are used. The mixers' output signals are low-pass-filtered to suppress unwanted aliases, which have been folded back into the digital spectrum due to the mixing operation. Then, each sub-signal is downsampled to the DAC sampling rate f_s . Eventually, the sub-signals are fed into the pre-equalizer, which is described in Secs. 5.7.3 and 5.7.4. Thereby, either the

sub-signals are either pre-equalized individually or in a joint manner with a MIMO algorithm. The resulting sub-signals are given as $x_n(k)$ with $n \in \Lambda$.

The signal processing operations described above can be implemented either in the time domain or in the frequency domain. For an efficient implementation in the frequency domain, the appropriate frequency domain samples are selected from the combined spectrum and placed into the baseband sub-bands directly, whereby frequency demultiplexing, downconversion and low-pass-filtering are achieved implicitly, as shown later in Secs. 5.7.2 and 5.7.4.

5.7.2 MIMO System Model for two DACs

A special FI-DAC variant is realized by setting all oversampling ratios $p_n = 1 \forall n \in \Lambda$; hence, the FI-DAC is operated without any guard bands to maximize the bandwidth. In this section, a frequency domain MIMO system model is derived for this case. The derivation is performed for two DACs, although it is trivial to extend it to N DACs. Due to the missing guard bands, there is cross talk between the sub-bands. A MIMO pre-equalizer is used to mitigate the cross talk and it is derived in Sec. 5.7.4. Note that parts of this section have been previously published in [45]. For the first time, both a MIMO system model and a MIMO pre-equalizer were presented for the FI-DAC.

Another frequency-domain MIMO model was presented later in [176], whereby the cross talk covers individual frequency bins of the signal rather than frequency bands. This more general model is beneficial, since any cross talk terms can be described including nonlinear distortions; however, this more complex model exacerbates to obtain channel state information (CSI), to calculate the equalizer coefficients and to equalize the distortions.

Block Diagram

In Fig. 5.7, a block diagram of the FI-DAC with two DACs is depicted including insets of the spectra at different points in the diagram.

The DSP is performed as follows: the digital signal $d(k)$ with $K_{D,\text{tot}}$ samples is given in vector notation as $\mathbf{d} = [d(0), d(1), \dots, d(K_{D,\text{tot}} - 1)]^T \in \mathbb{R}$, whereby $[\cdot]^T$ denotes the matrix transpose operator. It is passed through a DFT¹³ and the corresponding

¹³ The DFT is defined for a finite duration discrete-time signal $d(k)$ with K samples as [281]:

$$D(\mu) = \mathfrak{F}_{\text{DFT}} \{d(k)\} = \sum_{k=0}^{K-1} d(k) e^{-j\mu \frac{2\pi}{K} k}, \quad \mu = 0, \dots, K-1.$$

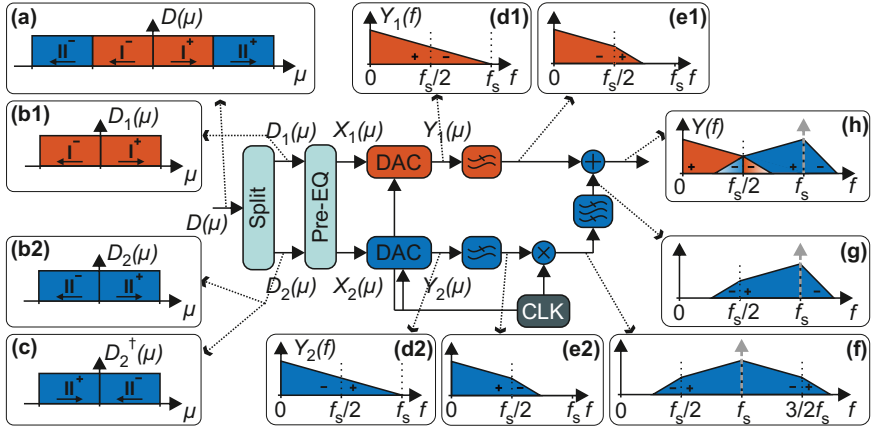


Figure 5.7 The FI-DAC block diagram for two DACs without guard bands: the digital signal (a) is split in the frequency domain into two sub-signals (b1,b2). The second sub-signal (lower signal path) is further reversed in frequency to account for using the mixer's LSB (c). Both sub-signals are pre-equalized in a joint manner. After D/A conversion (d1,d2) the sub-signals are low-pass-filtered (e1,e2). The second sub-signal is upconverted to its native frequency position with an RF mixer (f) and the LSB is selected with a BPF (g). Finally, both sub-signals are combined to form the output signal (h).

two-sided spectrum is given by $\mathbf{D}(\mu) = \mathfrak{F}_{\text{DFT}}\{\mathbf{d}\} = [\mathbf{D}^+(\mu), \mathbf{D}^-(\mu)]$ as depicted in Fig. 5.7(a), whereby μ denotes the discrete frequency. The plus and minus superscript denote the positive and the negative frequency half of the spectrum, respectively.

In Fig. 5.7(a), the direction of frequency is additionally denoted by the arrows. For the initial spectrum split, the vector notation $\mathbf{D}(\mu)$ is beneficial; however, the notation $D(\mu)$ is usually better suited and used instead afterwards. The spectrum $\mathbf{D}(\mu)$ is split into two parts of equal size $\mathbf{D}_1(\mu)$ and $\mathbf{D}_2(\mu)$, which are indicated in Figs. 5.7(b1) and 5.7(b2) as sub-bands I and II:

$$\mathbf{D}(\mu) = [\mathbf{D}^+(\mu), \mathbf{D}^-(\mu)] = [\mathbf{D}_1^+(\mu), \mathbf{D}_2^+(\mu), \mathbf{D}_2^-(\mu), \mathbf{D}_1^-(\mu)] \quad (5.46)$$

$$\mathbf{D}_1(\mu) = [\mathbf{D}_1^+(\mu), \mathbf{D}_1^-(\mu)] \quad (5.47)$$

$$\mathbf{D}_2(\mu) = [\mathbf{D}_2^+(\mu), \mathbf{D}_2^-(\mu)]. \quad (5.48)$$

By selecting the frequency domain sampling points for $\mathbf{D}_1(\mu)$ and $\mathbf{D}_2(\mu)$ from $\mathbf{D}(\mu)$, multiple operations are performed simultaneously. For the first sub-band, the splitting operation corresponds to low-pass filtering with a rectangular filter shape and additional downsampling. For the second sub-band, the splitting operation corresponds to band-pass filtering with a rectangular filter shape, downconversion,

and downsampling. A digital LO is not mandatory for downconversion; however, the more complex architecture depicted in Fig. 5.6 can be applied equivalently.

After the spectrum split, the separated sub-bands are pre-equalized as depicted in Fig. 5.7. For the system model derivation, the equalizer is transparent, such that $X_1(\mu) = D_1(\mu)$ and $X_2(\mu) = D_2^\dagger(\mu)$. The second sub-band is generated in reversed frequency position denoted by the superscript $\dagger$ to account for using the RF mixer's LSB as depicted in Figs. 5.7(c) and 5.7(g). This operation is performed by either shifting the discrete spectrum circularly by half of the sampling points or by multiplying the sub-signal with an alternating sequence of +1 and -1 in the time domain [282]. If the RF mixer's USB is used, this operation is omitted.

The DACs operate at the sampling rate f_s and their output spectra are attenuated by the sinc-function due to the ZOH operation, which is illustrated with the triangular shape in Figs. 5.7(d1) and 5.7(d2). Due to the missing guard bands and the LPFs' finite roll-off, the hold images cannot be suppressed completely as visualized in Figs. 5.7(e1) and 5.7(e2).

The second sub-signal is further upconverted to its native frequency position as depicted in Fig. 5.7(g). Two alternatives exist for the LO frequency: by setting the LO frequency to $f_s/2$ or f_s , the USB or the LSB is used, respectively. The latter alternative is depicted in Figs. 5.7(f) and 5.7(g). A BPF suppresses unused sideband as visualized in Fig. 5.7(g). Finally, both sub-signals are combined to form the analog representation $Y(f)$ of the digital spectrum $D(\mu)$ with a bandwidth of f_s and a sampling rate of $2f_s$ as depicted in Fig. 5.7(h).

Analysis of the Combined Output Signal

The combined signal depicted in Fig. 5.7(h) is affected by cross talk between the sub-bands around $f_s/2$ and image components $> f_s$. The combined signal's spectrum is controlled in the frequency intervals $[0, f_s/2]$ and $[f_s/2, f_s]$ by the first and the second DAC, respectively. Spectral components $> f_s$ cannot be directly assessed. Commonly, they are either shaped with an appropriate analog filter to obtain a certain pulse form or suppressed with an analog filter in combination with oversampling. Thus, the focus of this section is on the cross talk between the sub-bands, which needs to be mitigated rather than spectral components $> f_s$.

In order to derive a system model the combined signal $y(t)$ is ideally sampled with a sampling rate of $2f_s$ yielding the digital signal $y(k)$. The spectral representation $Y(\mu) = \mathfrak{F}_{\text{DFT}}\{y(k)\}$ is depicted at the top of Fig. 5.8(a). Despite $y(k)$ being a real-valued signal, the two-sided spectrum is visualized to ease the understanding of the cross talk.

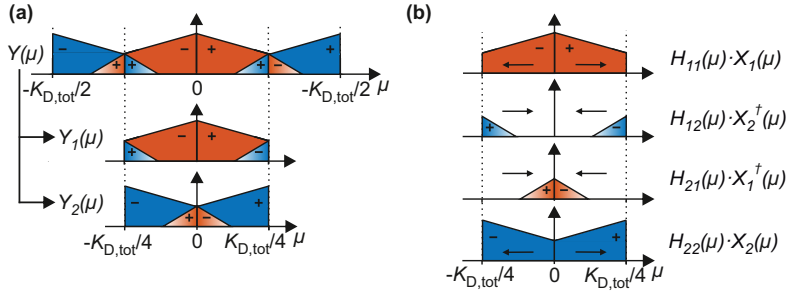


Figure 5.8 Separating the spectrum $Y(\mu)$ into two sub-bands $Y_1(\mu)$ and $Y_2(\mu)$ (a); Splitting the received sub-bands $Y_1(\mu)$ and $Y_2(\mu)$ into desired and cross talk components (b). (a), (b): Adapted, with permission, from [45].

The spectrum $Y(\mu)$ is split into the two sub-bands $Y_1(\mu)$ and $Y_2(\mu)$ as illustrated in Fig. 5.8(a). The splitting operation is performed as before in (5.46) to (5.48):

$$\mathbf{Y}(\mu) = [\mathbf{Y}^+(\mu), \mathbf{Y}^-(\mu)] = [\mathbf{Y}_1^+(\mu), \mathbf{Y}_2^+(\mu), \mathbf{Y}_2^-(\mu), \mathbf{Y}_1^-(\mu)] \quad (5.49)$$

$$\mathbf{Y}_1(\mu) = [\mathbf{Y}_1^+(\mu), \mathbf{Y}_1^-(\mu)] \quad (5.50)$$

$$\mathbf{Y}_2(\mu) = [\mathbf{Y}_2^+(\mu), \mathbf{Y}_2^-(\mu)]. \quad (5.51)$$

Now, there are two sub-bands that are each controlled by one of the DACs: $Y_1(\mu)$ by the first DAC and $Y_2(\mu)$ by the second DAC.

MIMO System Model

The combined output signal $y(t)$ depicted in Fig. 5.7(h) shall be a correct representation of the digital input signal $d(k)$ depicted in Fig. 5.7(a). However, cross talk terms distort the signal.

Another representation of the sub-bands $Y_1(\mu)$ and $Y_2(\mu)$ is visualized in Fig. 5.8(b). The sub-bands $Y_1(\mu)$ and $Y_2(\mu)$ are each separated into the desired components $H_{11}(\mu) \cdot X_1(\mu)$ and $H_{22}(\mu) \cdot X_2(\mu)$ and the undesired cross talk components $H_{12}(\mu) \cdot X_2^\dagger(\mu)$ and $H_{21}(\mu) \cdot X_1^\dagger(\mu)$. The spectral representations of the sub-signals being fed into the DACs are denoted as $X_n(\mu)$. The channel frequency responses characterizing the MIMO system are denoted by $H_{ln}(\mu)$ with $l, n \in \{1, 2\}$.

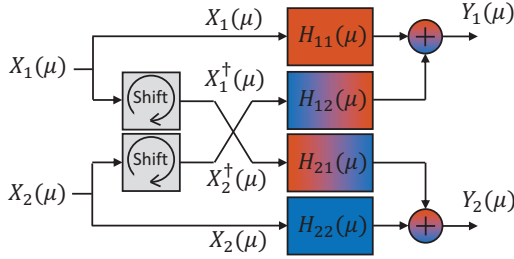


Figure 5.9 MIMO system model block diagram for two DACs with spectrum shift operations. Adapted, with permission, from [45].

The cross talk between the channels can be modeled as a linear frequency domain MIMO system:

$$Y_1(\mu) = H_{11}(\mu) \cdot X_1(\mu) + H_{12}(\mu) \cdot X_2^\dagger(\mu) \quad (5.52)$$

$$Y_2(\mu) = H_{21}(\mu) \cdot X_1^\dagger(\mu) + H_{22}(\mu) \cdot X_2(\mu). \quad (5.53)$$

Contrary to a standard MIMO model the cross talk terms are in reversed frequency position denoted by the superscript $\dagger$. In Fig. 5.9, the block diagram of this MIMO system model is depicted, whereby the reversed frequency position is obtained by shifting the discrete spectrum by half of the sampling points [282].

Following [283], in which a similar mathematical problem was faced for I/Q imbalance, the 2×2 MIMO model given by (5.52) and (5.53) can be extended to a standard 4×4 MIMO model as shown in Appendix F. Thereby, both a compact notation and a straight-forward solution are obtained for the MIMO pre-equalizer by standard matrix inversion.

In Appendix F, a model extension to N frequency bands is presented and multiple pre-equalizer algorithms are described, i.e., ZF, MMSE, and least mean squares (LMS). The pre-equalizer is covered in detail later in Sec. 5.7.4.

5.7.3 Algorithms

Multiple algorithms are necessary to compensate the following impairments of the FI-DAC [44]:

- magnitude and phase distortions in each sub-band,
- magnitude and phase mismatches between the sub-bands,

- cross talk between the sub-bands, and
- nonlinear distortions.

In this section, algorithms compensating these impairments are presented and discussed.

Magnitude and Phase Distortions in Each Sub-Band

Magnitude and phase distortions in each sub-band are compensated by means of linear pre-equalizers, which are applied to each sub-band individually. The pre-equalizers' coefficients are obtained by inverting each sub-channel's impulse response individually [239, 284].

Since, the pre-equalizers operate in baseband, the CE is performed in baseband as well, which requires a digital downconversion of each sub-band prior to estimation.

If a ZF equalizer is utilized, it attempts to enforce a linear phase. However, a linear phase is not favorable for all applications [285]. A minimum phase response avoids pre-shoot of the step response, which can be calculated using Hilbert transforms [286].

Magnitude and Phase Mismatches Between the Sub-Bands

Magnitude and phase mismatches between the sub-bands are mitigated to ensure that the FI-DAC's combined frequency response is free of any steps for both magnitude and phase.

Phase mismatches between the sub-bands can be mitigated by means of dedicated phase estimation and compensation algorithms. However, they are mostly proprietary [287].

Magnitude mismatches between the sub-bands can be estimated by measuring each sub-band's average spectral power density. The spectral power density can be controlled by means of the DACs' peak-to-peak output amplitude. However, the relation between these two parameters is not proportional.

In order to minimize the magnitude differences between the sub-bands, an adaptive algorithm tracking the spectral power density is a straightforward solution, whereby multiple iterations might be required to obtain a sufficient accuracy. For the experiments described in Secs. 5.8 and 5.10, this was done by hand in multiple iterations. For the simulations described in Sec. 5.11, an automatic solution is required compensating the magnitude mismatches. A rather good solution is found by

adapting the DAC peak-to-peak output amplitudes according to the pre-processed digital signals' peak-to-peak amplitude ratios. However, additional corrections were partially required; hence, further investigations are necessary in the future.

Generally, a special algorithm estimating magnitude and phase mismatches between the sub-bands is not required if a MIMO CE is utilized; they are estimated inherently.

Cross Talk Between the Sub-Bands

Cross talk between the sub-bands is compensated with a MIMO pre-equalizer. The MIMO pre-equalizer further corrects both aforementioned aspects of magnitude and phase distortions in each sub-band as well as magnitude and phase mismatches between the sub-bands. However, additional algorithms may improve the accuracy. More information on the MIMO pre-equalizer for the special case of the FI-DAC without guard bands as in [45] is later described in Sec. 5.7.4.

By using digital cross talk compensation, analog filters and guard bands are principally not necessary for the system design. However, analog filters are preferably used to reduce the signal processing effort, which in turn, reduces the ENOB loss due to pre-processing.

Adaptive pre-equalizers may be required to track varying system characteristics for a commercial product [288, 289].

Nonlinear Distortions

Nonlinear distortions are mitigated by means of nonlinear pre-equalizers. Various models exist for nonlinear system identification, e.g. Volterra kernels, memory polynomials, neural networks, etc. A good overview is provided by [290]. The choice of a suitable model for DSP engineering depends on the expected nonlinear properties of the system.

The corresponding nonlinear pre-equalizers are obtained from the identified system by LS inversion, adaptive LMS, indirect learning, etc. [20, 252, 291–293]. Nonlinear MIMO systems are treated, e.g., in [252]. The following work focuses on linear equalization.

Frequency Domain Equalization

Frequency domain equalization (FDE) is based on both FFT and IFFT operations and is performed block-wise in the frequency domain: the signal spectrum is multiplied with the pre-equalizer coefficients. This corresponds to a cyclic convolution in the time domain. For an application with a repetitive signal, e.g. an AWG, this does not impose a constraint; however, for continuous signals a linear convolution is required to prevent interference at block edges.

This issue can be faced with two approaches: first, a cyclic prefix or a unique word can be used at the beginning of each block enabling a one-tap equalizer in the frequency domain [294]. This overhead needs to be removed at the Rx and furthermore, prevents the generation of an arbitrary signal [295].

Second, techniques such as overlap-add or overlap-save [273] can be utilized, whereby the input signal is split into blocks of equal size. Both techniques increase the requirements on the signal processing rate, since the required FFT length is the sum of the equalizer length and the input signal block length.

Nonetheless, FDE is computationally better than the corresponding time-domain equalization, if the system's impulse response has a long memory [295].

Methods for System Identification

Identifying the FI-DAC system requires a CE. The individual signal paths can be estimated either sequentially or jointly.

For this thesis, a time-domain LS CE is utilized, since it is a straight-forward approach: in the time-domain CE, many received samples are used to obtain an accurate estimate of the channel impulse response. It is well suited for the initial experiments with cross talk and is used for the simulations and the experiments with GHz DACs as well. The MIMO LS CE is performed according to [239]:

$$\hat{\mathbf{H}}_{LS} = (\mathbf{S}^H \mathbf{S})^{-1} \mathbf{S}^H \cdot \mathbf{R}, \quad (5.54)$$

whereby $\hat{\mathbf{H}}_{LS}$, $\mathbf{S}$, $\mathbf{R}$, $(\cdot)^H$, and $(\cdot)^{-1}$ denote the estimated channel impulse response matrix, the transmit data matrix, the matrix of received samples, the matrix conjugate

gate transpose and the matrix inverse, respectively.¹⁴ For the CE, the transmitted data samples must be known by the system identification system.

The CE for the FI-DAC without guard bands as in [45] and Sec. 5.8 is performed as follows: the digital transmit signals are upsampled in the time domain by inserting zeros. This way mirror images are generated in the frequency domain to account for the cross talk of the mirror spectra due to D/A conversion. The CE is performed in the baseband without the need for digital downconversion. More information is found in [44].

For the FI-DAC with guard bands as in [47, 48, 50] and Sec. 5.10.2, no cross talk is present, if the guard bands are sufficiently wide. Due to the chosen FI-DAC parameters, the upconversion to $f_s/2$ of the second sub-band in reversed frequency position enables both sub-bands to be fully overlapping from DC to $f_s/2$. The 2×1 MISO CE is performed with two sub-signals, which cover the same frequency interval.

Performing the CE in the time domain is a straightforward approach, since the LS scheme is known to provide a good estimation of any channel. However, there are other techniques, that may improve the CE accuracy, e.g., frequency domain CE or joint time and frequency domain CE [296, 297].

Algorithmic Complexity

In order to provide an estimate on the algorithmic complexity, the number of multiplications for the individual DSP blocks is evaluated in this section. The

¹⁴ The channel impulse response matrix $\hat{\mathbf{H}}_{LS}$ is estimated from the matrix of received samples $\mathbf{R}$ by means of the transmit data matrix $\mathbf{S}$. The channel impulse response matrix $\hat{\mathbf{H}}_{LS}$ has the size $NK_H \times L$, whereby N , K_H , and L denote the number of transmitters, the number of taps of the impulse response, and the number of receivers, respectively. The matrix of received samples $\mathbf{R}$ has the size $K_S \times L$, whereby K_S denotes the number of transmitted (and received) samples. The transmit data matrix $\mathbf{S}$ has the size $K_S \times NK_H$ and is composed of the sub-matrices $\mathbf{S}_n$, whereby the n th sub-matrix corresponds to the n th transmitter:

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_1 & \mathbf{S}_2 & \dots & \mathbf{S}_N \end{bmatrix}.$$

The sub-matrices are circulant Toeplitz matrices of size $K_S \times K_H$ containing the symbols $d_n(k)$ for the n th transmitter:

$$\mathbf{S}_n = \begin{bmatrix} d_n(1) & d_n(K_S) & \dots & d_n(K_S - K_H + 2) \\ d_n(2) & d_n(1) & & \\ \vdots & \vdots & \ddots & \\ d_n(K_S) & d_n(K_S - 1) & \dots & d_n(K_S - K_H + 1) \end{bmatrix}.$$

real-time processing is preferably implemented in the frequency domain to reduce the complexity. The computation of the CE is not required in real-time and can be computed at a significantly reduced rate. A complex multiplication is assumed equivalent to three real-valued multiplications as noted in Sec. 4.4.2. The complexity further depends on the accuracy for the implemented algorithms, which is not covered in this section.

Signal processing operations for the FI-DAC in the frequency domain require both an FFT and multiple IFFTs. The complexity for the split-radix FFT algorithm is stated in Sec. 4.4.2 in (4.32) and (4.33) by $M_{\text{FFT},r}^{\text{sr}}$ and $M_{\text{IFFT},r}^{\text{sr}}$, respectively.

The operations of frequency demultiplexing, mixing, low-pass filtering and down-sampling can be efficiently realized by selecting the appropriate samples from the FFT by rewiring. The cyclic shift in order to obtain one of the sub-band in reversed frequency position is negligible as well.

The number of real-valued multiplications for the $N \times N$ MIMO equalizer is stated by $M_{W,\text{MIMO},f}$ in (4.36) in Sec. 4.4.2. If cross-talk is not present between the frequency bands due to guard bands and analog filters, the complexity is reduced to the diagonal elements and given as

$$M_{W,\text{MIMO},f,\text{diag}} = 3N_P \cdot \left(N \cdot \frac{N_W}{2} \right). \quad (5.55)$$

Since, the system characteristics are expected to vary slowly over time, the coefficients of the MIMO equalizer are calculated in software, i.e., on a dedicated CPU that handles the computation of the CE as well. More details on the complexity are provided in Sec. 4.4.2.

Hence, the total complexity for FI-DAC real-time frequency domain processing is given as

$$M_{\text{FI-DAC}} = M_{\text{FFT},r}^{\text{sr}} + M_{W,\text{MIMO},f} + M_{\text{IFFT},r}^{\text{sr}}. \quad (5.56)$$

As mentioned at the beginning of this section, the processing can be divided into two domains: real-time and non-real-time. The data is processed in real-time in the “data plane”, whereas both the CE and the calculation of the equalizer coefficients are performed in the “control plane” at a significantly reduced rate. The operations in the control plane can be performed in software, whereas the operations in the in the data plane need to be performed in hardware. For an implementation, the complexity in the data plane being stated in (5.56) is far more critical than the complexity for CE and equalizer coefficients calculation. The algorithmic complexity for the real-time domain depends on the available hardware, i.e., if an FPGA or an ASIC is utilized. The maximum DSP clock frequency

sets a lower bound for the parallelization degree N_p and the FFT size depends on latency requirements, the number of available logic gates and the desired frequency resolution.

5.7.4 MIMO Algorithm

Based on the MIMO system model developed in Sec. 5.7.2 a MIMO pre-equalizer with coefficients $W_{in}(\mu)$ is derived in this section. It mitigates the analog components' impairments, such as magnitude and phase distortions as well as cross talk. A similar mathematical is present in MIMO OFDM systems: I/Q imbalance results in interference from the mirror frequency subcarriers [283, 298]. Accordingly, the 2×2 MIMO system from Sec. 5.7.2 is generalized towards an N DAC system. The system with N TxS and L RxS covers the spectrum in both regular and reversed frequency position for both Tx and Rx vectors. Now, there is a MIMO system with N DACs. The full derivation is presented in Appendix F. Parts of this section have been previously published in [44, 45].

The MIMO system is described in the frequency domain as

$$\mathbf{Y}(\mu) = \underline{\mathbf{C}}(\mu)\mathbf{X}(\mu) + \mathbf{V}(\mu) . \quad (5.57)$$

$\mathbf{Y}(\mu)$ is the vector containing the FI-DACs output spectrum separated into L sub-bands with size $2L \times 1$ to cover the sub-bands in both regular and in reversed frequency position. $\mathbf{X}(\mu)$ is the vector of N sub-bands, which are fed into the DACs with size $2N \times 1$ and $\mathbf{V}(\mu)$ is the vector of noise sub-bands with size $2L \times 1$. The channel matrix $\underline{\mathbf{C}}(\mu)$ has the size $2L \times 2N$. $\mathbf{X}(\mu)$ is calculated according to

$$\mathbf{X}(\mu) = \underline{\mathbf{W}}(\mu)\mathbf{D}(\mu), \quad (5.58)$$

whereby $\underline{\mathbf{W}}(\mu)$ is the $2L \times 2N$ pre-equalizer matrix.

There is a plurality of appropriate pre-equalizer algorithms in order to mitigate the impairments described by the channel matrix $\underline{\mathbf{C}}(\mu)$, e.g., ZF pre-equalizers, MMSE pre-equalizers or adaptive pre-equalizers [299]. In Appendix F, multiple algorithms are presented. In this section, the ZF solution is featured with $L = N$, since it is used for the experiments in Sec. 5.8.

The ZF pre-equalizer disregards the noise and thus, the received signal is given by

$$\mathbf{Y}(\mu) = \underline{\mathbf{C}}(\mu)\mathbf{X}(\mu) \Big|_{\mathbf{V}(\mu)=0} = \underline{\mathbf{C}}(\mu)\underline{\mathbf{W}}_{ZF}(\mu)\mathbf{D}(\mu) . \quad (5.59)$$

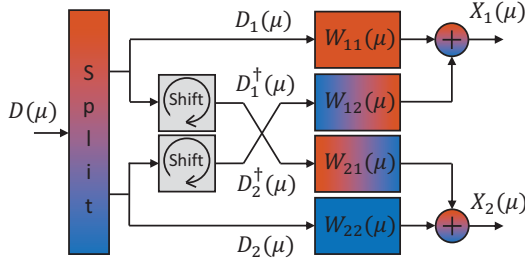


Figure 5.10 FI-DAC MIMO pre-equalizer for two DACs: The input signal is split in the frequency domain and processed by a 2×2 MIMO pre-equalizer with two additional shift operations. Adapted, with permission, from [45].

Accordingly, the ZF solution $\underline{\mathbf{W}}_{ZF}(\mu)$ is obtained for a quadratic channel matrix $\underline{\mathbf{C}}(\mu)$ by

$$\Rightarrow \underline{\mathbf{W}}_{ZF}(\mu) = \underline{\mathbf{C}}^{-1}(\mu). \quad (5.60)$$

The equalizer coefficients $W_{ln}(\mu)$ in $\underline{\mathbf{W}}_{ZF}(\mu)$ for the case of two DACs with $l, n \in \{1, 2\}$ are given by:

$$W_{11}(\mu) = \frac{H_{22}^{\dagger}(\mu)}{H_{11}(\mu)H_{22}^{\dagger}(\mu) - H_{12}(\mu)H_{21}^{\dagger}(\mu)}, \quad (5.61)$$

$$W_{12}(\mu) = -\frac{H_{12}(\mu)}{H_{11}(\mu)H_{22}^{\dagger}(\mu) - H_{12}(\mu)H_{21}^{\dagger}(\mu)}, \quad (5.62)$$

$$W_{21}(\mu) = -\frac{H_{21}(\mu)}{H_{11}^{\dagger}(\mu)H_{22}(\mu) - H_{12}^{\dagger}(\mu)H_{21}(\mu)}, \quad (5.63)$$

$$W_{22}(\mu) = \frac{H_{11}^{\dagger}(\mu)}{H_{11}^{\dagger}(\mu)H_{22}(\mu) - H_{12}^{\dagger}(\mu)H_{21}(\mu)}. \quad (5.64)$$

The FDE steps are depicted for the case of two DACs in Fig. 5.10. The digital spectrum $D(\mu)$ is split into two parts, $D_1(\mu)$ and $D_2(\mu)$, as described in (5.46) to (5.48). A shifted and a non-shifted version of $D_1(\mu)$ and $D_2(\mu)$ are generated. The equalizer coefficients $W_{ln}(\mu)$ are multiplied with the spectra $D_n(\mu)$ and $D_n^{\dagger}(\mu)$ with $l, n \in \{1, 2\}$. The resulting spectra are transformed back to the time domain via multiple parallel IFFT operations and the time domain sub-signals are fed into the DACs. The pre-equalizer is later used in Sec. 5.8.

The operation of shifting the digital spectrum by half of the sampling points in the frequency domain corresponds to a multiplication of the digital signal with an alternating ± 1 sequence in the time domain [282].¹⁵

Alternatively to the presented solution completely without guard bands, it is conceivable to use guard bands partially in combination with analog filters [44]. Hence, only some cross talk terms are present in the channel matrix, which simplifies the pre-equalizer and will generally reduce the ENOB loss due to pre-processing. In [269], partial guard bands are used with two sub-bands with 20 GHz each.

5.7.5 Calibration

In order to compensate the impairments of the analog components, the FI-DAC's frequency response needs to be known. It can be obtained by means of a calibration measurement, whereby three calibration procedures are distinguished: initial calibration, start-up calibration, and run-time calibration. The initial calibration is performed as a factory calibration after manufacturing. The start-up calibration is performed each time the FI-DAC is powered on. The run-time calibration is performed during operation of the FI-DAC to account for changing environment parameters, e.g., temperature. Parts of this section have been previously published in [44].

1. Initial Calibration

The initial calibration is performed as a factory calibration after manufacturing: first, the equivalent baseband sub-channel from each DAC to the combined output is determined and second, the corresponding pre-equalizer coefficients are calculated.

Either the FI-DAC is measured as a whole or the component's parameters are measured individually and are digitally combined afterwards. The calibration may comprise an S-parameter or X-parameter measurement [300]. Alternatively, the frequency response can be estimated by means of CE algorithms as discussed in Sec. 5.7.3.

¹⁵ For the DFT $D(\mu)$ of $d(k)$ with an even number of samples K the following holds true [282]:

$$\mathfrak{F}_{\text{DFT}} \left\{ d(k) \cdot e^{j\pi k} \right\} = D \left(\mu - \frac{K}{2} \right) = D \left(\mu + \frac{K}{2} \right) = D^*(\mu) .$$

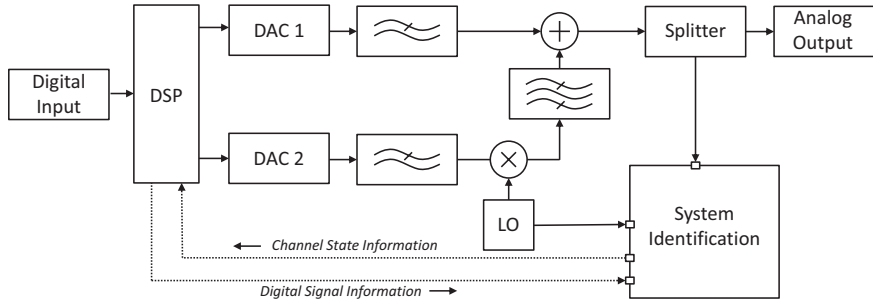


Figure 5.11 Setup for the calibration of the FI-DAC. Adapted from [44].

II. Start-Up Calibration

The FI-DAC may require a calibration during start-up to account for temperature offset and component aging compared to the FI-DAC status at the time of the initial calibration [288]. The start-up calibration is performed each time the FI-DAC is powered on.

The FI-DAC setup is extended by a system identification as depicted in Fig. 5.11. A fraction of the output signal is branched off by a splitter and fed into the system identification. Furthermore, information on the LO, e.g., frequency, phase and amplitude may be fed into the system identification. In order to identify the system properly, the digital signals, that have been fed into the DACs, need to be known by the system identification.

The calibration signals are preferably orthogonal CE sequences to ensure a good system identification result. The sequences can be orthogonal with respect to frequency, time, code, and/or space [301].

In Fig. 5.12, different realizations for the system identification are depicted. In Fig. 5.12(a), the full spectrum of the output signal is acquired and processed by the DSP block. In Fig. 5.12(b), fractions of the output signal are obtained by filtering and downconversion, limiting the oscilloscope requirements in terms of both bandwidth and sampling rate. The filter and the LO are tunable, enabling the iterative identification of the full spectrum. The sequentially obtained information is recombined digitally in order to gain information on the full spectrum. Moreover, an equivalent-time sampling oscilloscope may be used instead.

In Fig. 5.12(c), only information on the LO's frequency and phase is acquired, since the LO is the most critical dynamical element in the setup. It is either obtained by a notch filter in combination with an oscilloscope or by reading the information about

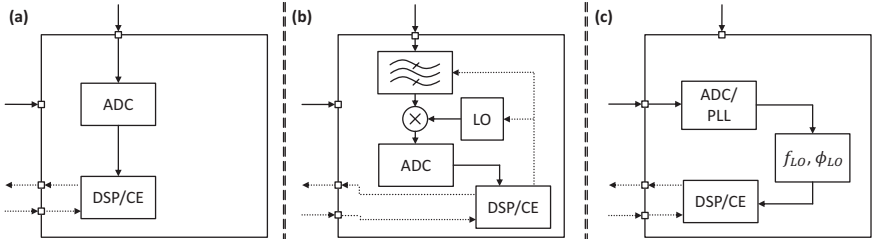


Figure 5.12 Different realizations of the system identification: measuring the full output signal (a), measuring only fractions of the output signal (b), receiving LO information from an oscilloscope or from a PLL (c). Adapted from [44].

the current frequency and phase from an incorporated PLL that may be used for the LO stabilization. For this variant, an initial system identification is required and the component variations with respect to the initial calibration need to be negligible.

The implementation variants are not limited to the realizations depicted in Fig. 5.12, since both hybrid and related variants are feasible.

III. Run-Time Calibration

During FI-DAC operation, it might be necessary to compensate for changing parameters, e.g., the component's temperature variations [288], which affect the frequency responses. Therefore, a re-calibration needs to be performed during run-time in order to continuously adapt the pre-equalizer coefficients. For the adaptivity, the FI-DAC's frequency response need to be tracked. Multiple alternatives are viable as discussed previously.

The re-calibration might comprise measuring distinct frequency lines or frequency ranges. Further, the re-calibration may include tracking the LOs and the compensation of phase and/or frequency deviations.

The CE requires distinguishable sequences for every signal path of the FI-DAC [301]. With a properly designed DSP algorithm, the channel might be estimated with an equivalent-time sampling oscilloscope, while an arbitrary signal is generated by the FI-DAC. Hence, the run-time calibration can be performed without disturbing with the FI-DAC's operation and without the need for special CE sequences.

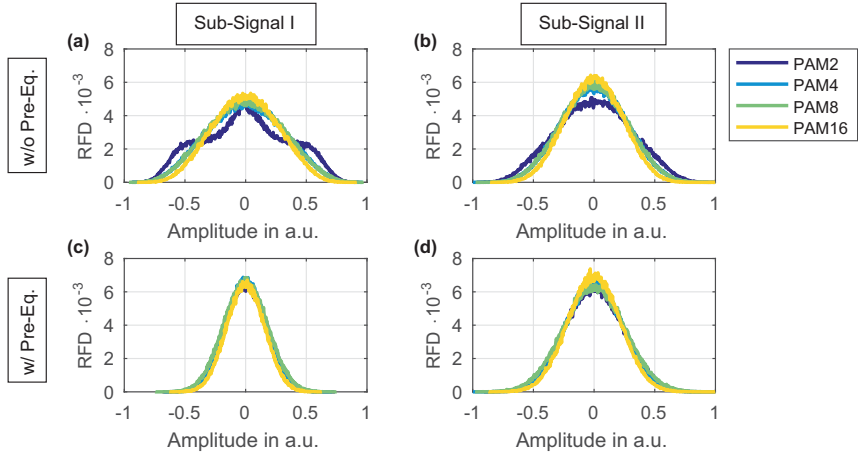


Figure 5.13 Relative frequency distribution (RFD) of the digital sub-signals I and II for different PAM orders: without pre-equalization (a,b); with pre-equalization (c,d). Adapted with permission from [50], © 2018 Optical Society of America.

5.7.6 Sub-Signal Characteristics

In order to improve the understanding of the FI-DAC's behavior, the characteristics of the individual sub-signals are analyzed. Parts of this section have been previously published in [50].

In Fig. 5.13, the relative frequency distribution (RFD) is depicted by means of histograms for the two digital time-domain sub-signals before feeding them into the DACs for different PAM orders. The sub-signals are jointly normalized to a maximum absolute value of one prior to calculating the 500-bin histograms for each PAM order. The analyzed sub-signals correspond to the simulated standard case with nominal parameters in Sec. 5.10.4.

In Figs. 5.13(a) and 5.13(b), the histograms are visualized for the case without pre-equalization for sub-signal one (I) and two (II) on the left and on the right side, respectively. A nearly Gaussian distribution of amplitude values is observed, that becomes more prominent with rising modulation order. For PAM-2, the Gaussian distribution is distorted, i.e., there are still remnants from the PAM-2 distribution visible. The standard deviation of sub-signal one and two is roughly equal and seems to decrease for higher modulation order.

In Figs. 5.13(c) and 5.13(d), the histograms are shown for the case with pre-equalization. Now, the distribution has a Gaussian shape for all PAM orders.

Table 5.2 90 %-percentile of the PAPR values of the FI-DAC sub-signals for different PAM orders: both without and with pre-equalization. Each statistical sample is composed of 100 observations.

PAM Order	PAPR w/o pre-Eq.		PAPR w/ pre-Eq.	
	Sig I	Sig II	Sig I	Sig II
2	7.94	11.55	17.88	18.26
4	10.83	15.29	19.66	20.67
8	12.09	16.17	20.80	21.85
16	12.51	15.79	20.39	21.48

Furthermore, the standard deviation for the second sub-signal is larger relative to the first sub-signal. This can be adjusted either by not utilizing the full-scale range of the first DAC or by controlling the external output amplifiers' gain.

As for DMT signals [302], the Gaussian distribution of amplitude values in the time domain relates to a high peak-to-average power ratio (PAPR). The PAPR is calculated for 100 digital data signals with different seeds of the random number generator for each modulation order. Both, the case with and without pre-equalization is evaluated. Then, the empirical compound distribution function of the PAPRs is calculated. In Tab. 5.2, the 90 %-percentile of the linear PAPR values are listed. For a PAM-4 modulation without pre-equalization, the PAPR is greater than 10.83 and 15.29 in 90 % of the observations in the statistical sample for signal one and two, respectively. For a PAM-4 with pre-equalization, the PAPR is greater than 19.66 and 20.67 in 90 % of the observations in the statistical sample for signal one and two, respectively.

However, high PAPRs are not desired if analog components with nonlinear transfer functions are present, i.e., amplifiers and mixers. Therefore, PAPR mitigation concepts are described in the following.

Mitigation Concepts for high PAPR

The digital signal is split into multiple sub-bands in the frequency domain, which are independently transformed to the time domain. Thus, the sub-signals do not have the same statistical characteristics as the combined digital signal, but a nearly Gaussian distribution as shown previously. In order to mitigate the resulting high PAPR values, multiple techniques are feasible.

For the FI-DAC with guard bands, an artificial overlap between the sub-bands can be digitally generated. This way, the PAPR could principally be reduced as shown in Sec. 5.11.5.

Another approach is clipping. Although, clipping is a nonlinear operation, it can be described linearly in the time domain with a series of $\delta(t)$ -impulses having an amplitude equal to the clipped signal amplitude. The clipping is modeled by subtracting these impulses from the waveform at the time instants, where clipping occurs [303]. In general, high clipping levels can be modeled as noise.

For the optimum clipping level, a trade-off is present: on the one hand, the ENOB is enhanced due to the increased average sub-signals' power. On the other hand, the ENOB is reduced due to the additional nonlinear clipping distortions as will be shown in Sec. 5.11.6.

Alternatively, the sub-signals can be shifted with a common delay value in the sub-sample regime to reduce the PAPR, which is later investigated in Sec. 5.11.7. This technique could be further improved by using analog phase shifters for the LOs in order to obtain an optimum delay for each sub-signal.

For DMT or OFDM signals, multiple other techniques exist, which are not applicable to PAM signals [304–307].

5.8 Initial Experiments with MHz-DACs

In order to verify the FI-DAC concept and the proposed DSP algorithms, initial experiments are conducted with MHz-DACs under laboratory conditions. After describing the experimental setup, the algorithms for CE and pre-equalization are specified. Lastly, the measurement results are presented.

Parts of this section have been previously published in [45]. This was the first demonstration of a MIMO algorithm for the FI-DAC concept.

Setup

The experimental setup is depicted in Fig. 5.14. For the D/A conversion the two-channel arbitrary waveform generator Tektronix AWG5012 with 14 bits nominal vertical resolution is used at a sampling rate of 900 MS/s and is driven by an external clock. The LPFs following the DACs are MiniCircuits SLP-450 with the cutoff frequency $f_{co} = 440$ MHz. The RF mixer MiniCircuit ZFM-150-S is driven by a 900 MHz LO with 10 dBm power provided by the signal generator Rohde und Schwarz SMC 100A. The LO is used for both DAC clock and mixer to achieve

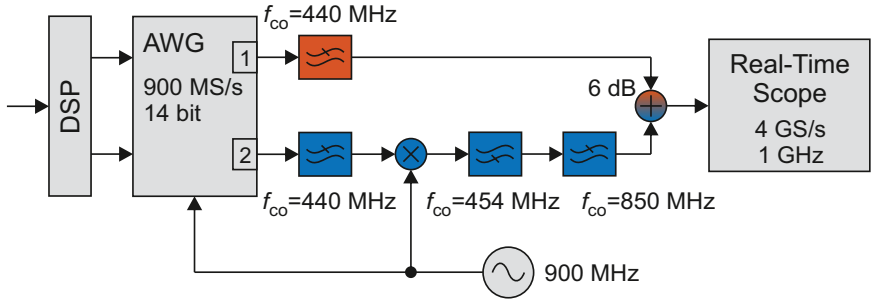


Figure 5.14 Experimental setup: an AWG generates two sub-signals, which are processed by means of filters and a mixer. The sub-signals are combined and captured with an oscilloscope.

phase stability between the sub-bands. In lack of an appropriate BPF, a combination of the LPF MiniCircuits SLP-850 with $f_{co} = 850$ MHz and the HPF MiniCircuits SHP-500 with $f_{co} = 454$ MHz is used to suppress the unused USB after the RF mixer.

The sub-signals are combined by means of the passive 6 dB power combiner MiniCircuit ZX10R-14 and acquired with the real-time oscilloscope Agilent MSO 7104A with 4 GS/s, 1 GHz analog bandwidth and 8 bit vertical resolution. The oscilloscope and the signal generator are synchronized with a 10 MHz reference clock. The sub-bands' spectral power densities are roughly matched by inserting attenuators in the first path, which are not depicted in Fig. 5.14. The amplitude is fine-tuned by manually adjusting the gain of the AWG output amplifiers. In the power combiner, each sub-signal is split and fed both into the output port and back into the other signal path. In order to reduce unwanted coupling effects and reflections at the power combiner, additional attenuators are placed close to the combiner, which reduce the SNR in turn.

DSP & Channel Estimation

The DSP steps are performed as described in Sec. 5.7.2. The digital input signal is Fourier transformed and split into two sub-bands. The sub-bands are jointly pre-equalized with the ZF MIMO algorithm presented in Sec. 5.7.4. The second sub-band is further shifted by half of the number of samples to account for using the RF mixer's LSB as explained in Sec. 5.7. Then, both sub-bands are each passed through an IFFT and fed into the DACs. Thus, two sub-bands are generated each occupying a frequency range of 450 MHz: the first one in the baseband from 0 to 450 MHz and the second one in the passband from 450 to 900 MHz. For the actual data signal, oversampling with an oversampling ratio of 1.125 is applied prior to

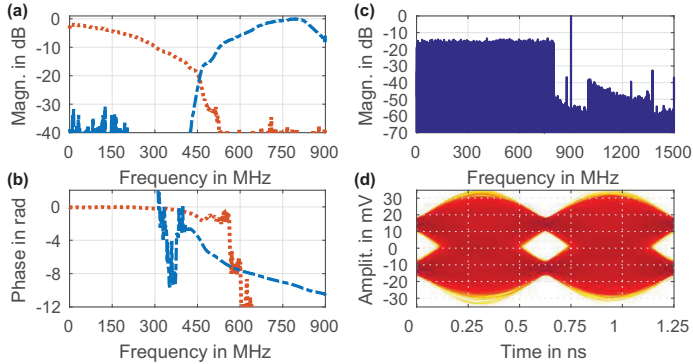


Figure 5.15 FI-DAC experiment with two DACs operating at 900 MS/s to generate a signal with 1.6 GBd and 800 MHz total bandwidth: frequency response's magnitude (a) and phase (b); output spectrum (c); eye diagram (d).

these steps: a data signal with a combined bandwidth of 800 MHz and a symbol rate of 1.6 GBd is generated.

First, a MIMO CE is performed. A De Bruijn binary sequence (DBBS) of length 2^{14} undergoes the processing steps described in the previous paragraph. A 2×1 LS MISO CE with 512 taps yields the result depicted in Figs. 5.15(a) and 5.15(b). There are two channel frequency responses visualized: the lines starting at the upper left and upper right boundary of the plot represent the channel from the first DAC to the combined output and the channel from the second DAC to the combined output, respectively. A strong magnitude roll-off is observed, which can be attributed to the DAC's analog 3 dB bandwidth of 250 MHz. The crossover frequency of the frequency responses is roughly at 450 MHz as intended. The phase is almost linear in the frequency intervals, where substantial magnitude components are present. The slope of the phase for the second sub-band indicates a delay between the first and the second sub-signal.

The analog impairments need to be pre-compensated to obtain the desired output signal. Therefore, the digital data signal, a 2^{17} DBBSs sequence, is upsampled to an oversampling factor of 1.25 to obtain the targeted symbol rate of 1 600 MBd. The MIMO channel frequency response obtained in the previous step is interpolated to match the length of the digital data signal to apply FDE. After pre-equalization, the second sub-band is further shifted by half of the number of samples to generate it in reversed frequency position. This operation corresponds to a multiplication with an alternating ± 1 sequence in the time domain as described in Sec. 5.7.4. Then, the sub-bands are passed through IFFTs and are fed into the DACs.

Results

The spectrum of the combined signal and the corresponding eye diagram with 2^{15} symbols are visualized in Figs. 5.15(c) and 5.15(d), respectively. A clear eye opening is visible, although the eye amplitude is < 40 mV. This amplitude loss can be attributed to the RF mixer, the attenuators, the passive power combiner, and the limited bandwidth of the AWG. A flat spectrum, corresponding to a sinc pulse shape in the time domain, is observed from 0 to 800 MHz. Due to the oversampling, the RF mixer's USB is moved away from the desired spectrum to frequencies ≥ 1 GHz; however, it cannot be fully suppressed by the analog filters. The LO at 900 MHz is fed through to the output and has a spectral level almost 15 dB greater than the magnitude of the signal, due to insufficient suppression by both the mixer and the following BPF.

Concluding, major requirements for the functionality of the FI-DAC are phase stability between the LO and the DAC clock, components with flat frequency responses and high resolution DACs. The successful interleaving of two frequency bands was demonstrated experimentally for two MHz-DACs.

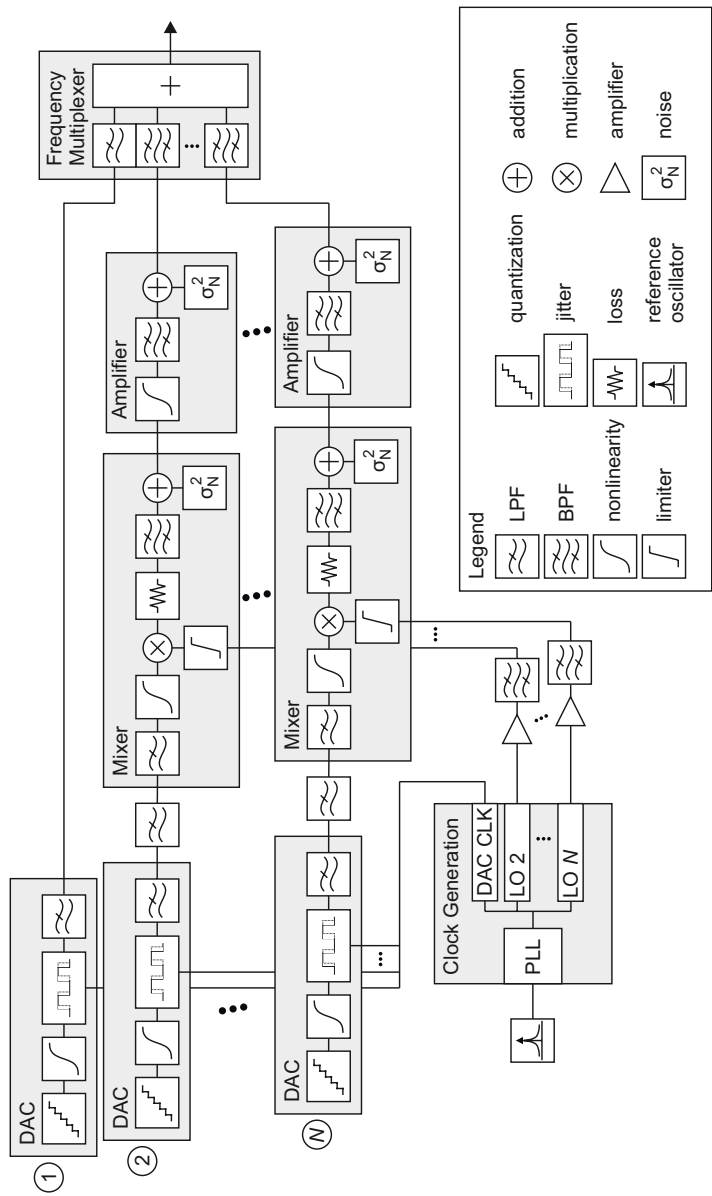
5.9 FI-DAC Impairment Model

From the initial experiments, the major effects in the FI-DAC are deduced to obtain an impairment model for N DACs. The block diagram representing the model is depicted in Fig. 5.16. For this model, the behavior of the following components is taken into account: DACs, passive mixers, amplifiers, frequency multiplexer and clock network. Moreover, there are reflections between all components and thermal noise contributions from active components. In the following, the impairments of the individual components depicted in Fig. 5.16 are described in detail.

The DACs are subject to multiple impairments as described in Sec. 2.5. The DAC introduces nonlinear distortions due to the finite vertical resolution, i.e., quantization noise, as described in Sec. 2.5.3. Further static and dynamic nonlinear distortions are present as described in Secs. 2.5.1 and 2.5.2. Moreover, the DAC is subject to clock phase noise, which translates to the DAC output as jitter, as described in Sec. 2.5.4. An LPF accounts for both the DACs' analog frequency response and the sinc roll-off due to the ZOH operation as described in Sec. 2.4.

Then, an LPF follows, which suppresses undesired hold images. Dependent on the filter design, the LPF either absorbs the signal energy, which is not transmitted to the LPF's output, or reflects it back to the DAC. The LPF for the first path is included in the frequency multiplexer.

Figure 5.16 FI-DAC impairment model block diagram.



The impairments of the passive mixer are described in detail in Sec. 2.7.1. The IF circuitry has a low-pass behavior. Furthermore, non-linear distortions are introduced by the switching operation of the diodes, which generates harmonics both for the IF signal and the LOs. Hence, the IF signal is upconverted not only to the LO frequency, but to multiples of it. This needs to be taken into account for the frequency plan of the FI-DAC. The passive mixer is subject to both an insertion loss and a band-pass characteristic of the RF circuitry. Moreover, the mixer's diodes introduce additional noise that limits the signal quality. Next to the mentioned limitations, both the input signal and the clock signal are fed through to the output.

An amplifier follows, compensating the loss of each mixer. A variable output power is required for each path to match the spectral power density of the frequency bands to each other. Due to the amplifier gain, the mixer can be operated with a low IF power to keep the mixer's nonlinear distortions small. However, the amplifier introduces additional impairments, such as nonlinear distortions, a bandwidth limited frequency response with a band-pass characteristic and noise. The noise is usually described with the noise figure, which is ideally 0 dB, but ranges between 3 and 6 dB for real amplifiers [248].

The signals are combined with a frequency multiplexer [199], which is essentially a filter array with a combined output, consisting of an LPF, multiple BPFs, and either a BPF or a HPF for band N . The frequency multiplexer is preferably realized with multiple diplexers as described in Sec. 2.7.2.

Dependent on the impedance matching, there will be strong reflections for frequency components outside the desired frequency bands of the frequency multiplexer. The amplifiers in Fig. 5.16 will terminate the reflections. However, a setup without amplifiers following the mixers will result in reflections being mixed again, which increases the noise level.

The clock generation is a highly important sub-system, since the LOs' phase noise translates into broadband noise for the upconverted sub-signals. For the clock generation, an internal reference frequency is derived in a PLL from a reference oscillator, e.g., a crystal. This internal reference frequency is subject to phase noise. The phase noise of each LO will be determined by the internal reference frequency and the employed clock architecture consisting of multipliers, dividers or multiple PLLs. Since, all LOs are phase-locked, the LOs' phase noise is correlated. Moreover, the spectral purity of the internal reference frequency needs to be ensured, which has spurs from the crystal's frequency. The LO signals are amplified and further band-pass filtered to drive each passive mixer with the corresponding LO.

Power Consumption and Size

The interleaved architecture depicted in Fig. 5.16 results in an increased power consumption and an increased size compared to a single DAC solution. Parts of this section have been previously published in [50].

For a practical realization, both the power consumption and the size are important parameters. If using N DACs, the total power consumption P_{TOT} can be approximated by

$$P_{\text{TOT}} = N \cdot P_{\text{DAC}} + \sum_{n=2}^N (P_{\text{LO},n} + P_{\text{AMP},n}), \quad (5.65)$$

whereby P_{DAC} , $P_{\text{LO},n}$ and $P_{\text{AMP},n}$ denote the power for a single DAC, the power for the n th LO generation, and the power for the n th amplifier compensating both the passive mixer's losses and the combiner's losses, respectively. Hence, the power consumption of the FI-DAC scales with the number of DACs N and with the number of upconverted bands $N - 1$.

A comparable single DAC running at twice the sampling rate will also consume twice the power. The FI-DAC's additional power consumption stems from the more complex clock generation, the additional amplifiers, and the more complex DSP.

For the size of the FI-DAC, only a qualitative approximation is given at this point, since the size strongly depends on the DACs and the analog components. The size of the FI-DAC naturally scales with the number of DACs. For each additional DAC, both a passive mixer and an amplifier compensating the mixer's loss is required. Furthermore, the size of the frequency multiplexer increases for each additional DAC, since another port for signal combining is required. Moreover, integration of broadband microwave filters with steep roll-offs is difficult; hence, the frequency multiplexer will be most likely realized with multiple diplexers as described in Sec. 2.7.2. The resonators for these filters require a size that scales inversely with the desired frequency range of the filters. A qualitative comparison with the AMUX-DAC is provided in Ch. 6.

5.10 Experiments with GHz-DACs

For the FI-DAC concept, the experiments with the highest bandwidth reported so far in [176] used a triplexer and a clock network, which were not well-suited for the employed DACs. The triplexer's frequency bands far exceed the analog 3 dB bandwidth of the DACs and the clock network comprises frequency multipliers up to factor 32. Hence, complex post processing routines are needed, i.e., an adaptive

post-equalizer as well as a phase tracking and a phase compensation algorithm to recover the signals.

The concept was further developed by the author of this thesis and the first open eye diagrams without using a post-equalizer for the FI-DAC with high-speed DACs were presented in [47]. Based on this, further experiments were conducted with an improved setup and the first transmission over an optical IM/DD link was presented in [48]. Parts of this section have been previously published in [47–52].

In this section, the signal processing for a FI-DAC with guard bands consisting of two DACs is explained in detail, covering both the analog and the digital signal processing. Then, an initial experiment with GHz-DACs is presented, whereby two sub-bands are generated with bandwidths of 22.5 GHz and 17.5 GHz, respectively. They are combined with a diplexer to form a signal with a continuous bandwidth of 40 GHz. Thereafter, an improved version of the experiment is presented, in which the FI-DAC is used for signal generation in an optical IM/DD link.

5.10.1 Overview on Signal Processing for Two DACs

For the experiments with high-speed DACs, two frequency bands are generated and combined with a diplexer. Contrary to Sec. 5.8, the DAC output signals encompass guard bands. Parts of this section have been previously published in [47].

A conceptual block diagram of the FI-DAC with guard bands consisting of two DACs each operating at the sampling rate f_s is presented in Fig. 5.17. The signal is processed in the digital and in the analog domain, which is termed digital frequency de-interleaving and analog frequency interleaving, respectively.

For the digital frequency de-interleaving, the digital signal in Fig. 5.17(a) is passed through a DFT. The spectrum is depicted schematically in Fig. 5.17(b) and is split into two sub-bands as visualized in Fig. 5.17(c). After splitting, the sub-bands are each downconverted to baseband as explained in Sec. 5.7.1.

For the analog frequency interleaving, the first sub-signal is generated in baseband as depicted in Fig. 5.17(d), whereby the guard band is denoted with lighter colors. This way, DAC hold images can be efficiently suppressed with an analog filter. The second sub-signal is generated at an intermediate frequency in Fig. 5.17(e). This way, both DAC hold images and the unwanted mixer sideband can be efficiently suppressed with analog filters. Furthermore, the signal is generated in the reversed frequency position, denoted by the superscript *, since the RF mixer's LSB is utilized. Besides, the mixer's USB could be used as well, which is described in [46] and shown experimentally in [264]. After the LPF, the second sub-signal is upconverted to its native frequency position in Fig. 5.17(f). Both sub-signals

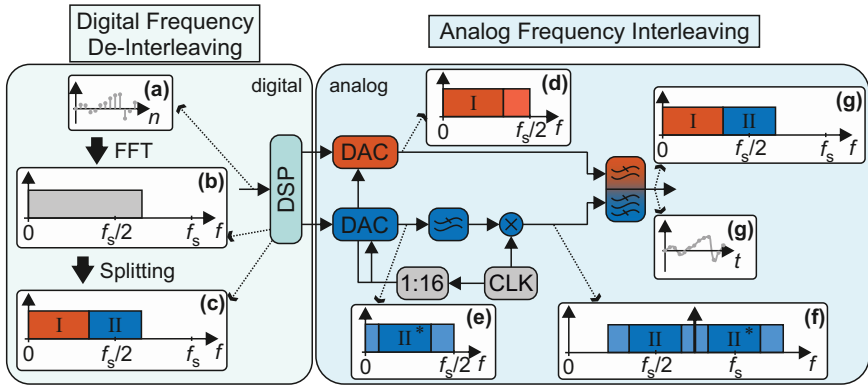


Figure 5.17 Block diagram of the FI-DAC consisting of two DACs: The digital signal (a) is passed through an FFT (b), split in the frequency domain (c) and each sub-band is both individually processed and D/A converted (d,e). The second sub-signal is low-pass filtered and upconverted to its native frequency position (f). A diplexer filters and combines both sub-signals to form the analog representation (g) of the digital signal (a). Adapted from [47], with permission, © 2017 SPIE.

are then combined with a diplexer. The first sub-signal is filtered with a low-pass characteristic to suppress DAC hold images and the second sub-signal is filtered with a band-pass characteristic to suppress the unused mixer sideband. The resulting signal in Fig. 5.17(g) is the analog representation of the digital signal in Fig. 5.17(a).

5.10.2 Digital Signal Processing for Two DACs

The Tx DSP for the FI-DAC with two DACs is visualized as a block diagram in Fig. 5.18. The individual steps are visualized in the frequency domain, though a time domain realization is feasible as well. The processing is applied for the experiments and the simulations. Parts of this section have been previously published in [50].

The 40 GHz input signal is depicted in Fig. 5.18(a). 2^{18} symbols are used in the simulations. For the experiments, 2^{18} and 2^{15} symbols are used in the initial and the improved experiment, respectively. The signal is passed through an FFT and the resulting spectral representation is visualized in Fig. 5.18(b). This signal is pulse-shaped by means of upsampling and filtering with a raised cosine characteristic, as visualized in Fig. 5.18(c). The roll-off-factor is 0.05 for Sec. 5.10.3 and 0.03 for all other sections. Then, the signal is split with a frequency demultiplexer, i.e., a digital diplexer, into two frequency bands as depicted in Figs. 5.18(d) and 5.18(e). The splitting can be performed both with a spectral overlap as shown in Fig. 5.18(g),

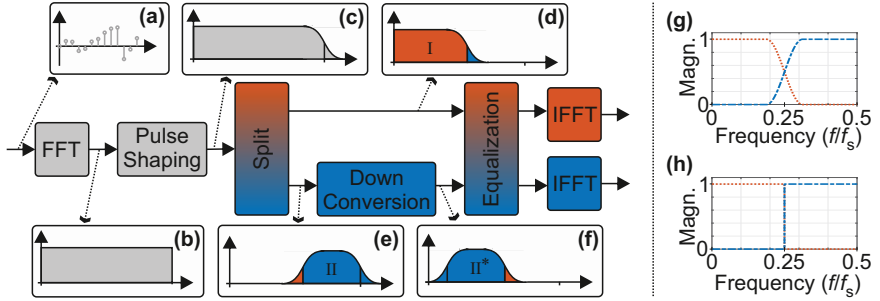


Figure 5.18 FI-DAC DSP block diagram for two DACs: The time domain signal is passed through an FFT (a) and the corresponding frequency domain representation (b) is pulse-shaped (c). Then, the signal is split in the frequency domain with a digital diplexer (d,e). The second signal is further downconverted into baseband with a digital mixer (f). Both signals are pre-equalized and passed through IFFTs, which are then fed to the DACs. The split can be performed both with an overlap (g) and without (h). Adapted with permission from [50], © 2018 Optical Society of America.

e.g., in a raised-cosine manner, or without a spectral overlap by means of ideal rectangular frequency domain filters as visualized in Fig. 5.18(h). The crossover frequency is adapted according to the specifications of the analog components. More information on the digital frequency demultiplexer is found in Appendix E. The high-frequency signal is further down-converted into baseband with a digital RF mixer, as depicted in Fig. 5.18(f). In both experiment and simulation, the RF mixer's LSB is used, which requires the generation of the baseband sub-signal in reversed frequency position, which is denoted by the superscript *.

Then, both sub-signals are jointly pre-equalized to mitigate both magnitude and phase mismatches of the two asymmetrical analog paths. The integer sample delay between the two sub-signals is compensated in a bulk manner by shifting one of the sub-signals according to a delay estimation by means of a cross-correlation. The ZF-FDE coefficients are obtained with a MISO 2x1 LS CE in the time domain. The calculated impulse responses are Fourier transformed and interpolated to the number of sub-signals' samples. Inversion of these frequency domain coefficients result in the desired pre-equalizer coefficients. After clipping both sub-signals to a linear PAPR of 16 (only in the experiments), they are fed through an IFFT and the time-domain signals are fed to the DACs.

At the Rx, the DSP includes synchronization, clock phase recovery, LS channel estimation, a half-symbol spaced MMSE post-equalizer with 220 taps [308], and bit error counting. A sufficient number of taps was chosen for the post-equalizer to prevent a performance degradation due to a limited number of taps. Bit error counting is feasible up to a BER of 1.91×10^{-6} ; it is performed both before and

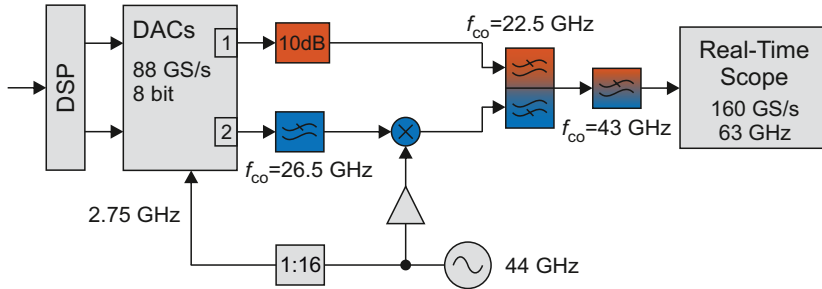


Figure 5.19 FI-DAC setup consisting of two DACs for the initial experiment with GHz-DACs. Adapted from [47], with permission, © 2017 SPIE.

after the post-equalizer. Smaller BER values are obtained by means of a BER estimation based on the EVM [254, 309].¹⁶

5.10.3 Initial Experiment

An initial experiment with two DACs is conducted, in which a combined signal with a sampling rate of 176 GS/s and 40 GHz electrical analog bandwidth is generated. The setup consists of two DACs, each having a 3 dB bandwidth of 20 GHz and operating at a sampling rate of 88 GS/s. Parts of this section have been previously published in [47].

Setup

The experimental setup is depicted in Fig. 5.19 with two DACs operated at 88 GS/s. The first sub-signal contains frequency components from DC to 22.4 GHz and the second sub-signal contains frequency components from 4 to 21.6 GHz. Both sub-signals are pre-processed to realize a 2.5 GHz raised-cosine overlap. The first sub-signal (upper signal path) is attenuated by 10 dB to ensure an almost equal attenuation in both analog paths. The second sub-signal (lower signal path) is filtered with an LPF having a 3 dB bandwidth of 26.5 GHz to suppress DAC hold images. This sub-signal is further upconverted with a passive RF mixer [188] to 44 GHz. The mixer has a conversion loss of up to 9 dB in the frequency range of interest. Both signals are combined in the diplexer, which has a crossover frequency of 22.5 GHz. The low-pass characteristic suppresses the DAC hold images of the first sub-signal. After the diplexer, an LPF with a 3 dB bandwidth

¹⁶ The relation between EVM and BER is given in footnote 12.

of 43 GHz suppresses both the LO, which has been fed through the RF mixer, and the RF mixer's USB. The combined signal is acquired with a 160 GS/s real-time oscilloscope with 63 GHz analog bandwidth.

It is essential for the FI-DAC, that the LO frequency and the DAC clock are phase-locked. Hence, the 44 GHz LO signal of the frequency generator is split and divided by 16 in order to generate the 2.75 GHz clock signal for the DACs. However, the PLL in the DAC chip imposes additional jitter constraints and further limits the phase stability between the sub-bands.

Results

The results of the initial experiments with GHz-DACs are depicted in Figs. 5.20 and 5.21. In Fig. 5.20(a), the flat output spectrum of the FI-DAC is visualized. A strong LO feed-through at 44 GHz is observed, which is not sufficiently suppressed by the LPF. Further, two tones are visible at 22 and 24 GHz. The tone at 22 GHz is the fed-through DAC clock and the tone at 24 GHz is presumably a mixing product of the strong LO with an oscilloscope clock line at 20 GHz.

The residual FI-DAC magnitude response is depicted in Fig. 5.20(b). In the interval from DC to 22 GHz, the magnitude appears fairly flat with deviations of ± 0.4 dB. For frequencies > 22 GHz, there is a ripple of up to ± 1.5 dB in the magnitude. Hence, the pre-equalization does not fully compensate the present effects. The drop-off at 38 GHz can be attributed to the raised cosine pulse shaping. The resulting 80 GBd eye diagrams for both a PAM-2 and a PAM-4 signal are visualized in Fig. 5.20(c). Both eyes are plotted with 1×10^5 symbols. The 18th order DBBS sequences for the PAM signals have at maximum 18 ones or zeros in a row. Two 18th order DBBS sequences are combined to generate the PAM-4 signal. A clear eye opening for the PAM-2 signal and a degraded eye for the PAM-4 signal are observed with an eye amplitude of 26 mV and 34 mV, respectively. The eye height of the PAM-2 eye diagram is 14 mV. For the PAM-4 signal, a BER of 4.7×10^{-4} is achieved. The LO is digitally filtered with a notch filter prior to plotting the eye diagrams. The wideband noise beyond 42 GHz is not filtered to assess the performance of the DAC output signal generally.

In Fig. 5.20(d), a photograph of the laboratory setup is shown. Next to the analog components for FI, the arbitrary waveform generator, the frequency synthesizer and the real-time oscilloscope are visible. Furthermore, the sampling heads of an equivalent-time sampling oscilloscope are present, which were used initially, but later replaced by the real-time oscilloscope. The analog devices and the DSP are controlled and operated from the laptop shown to the bottom right.

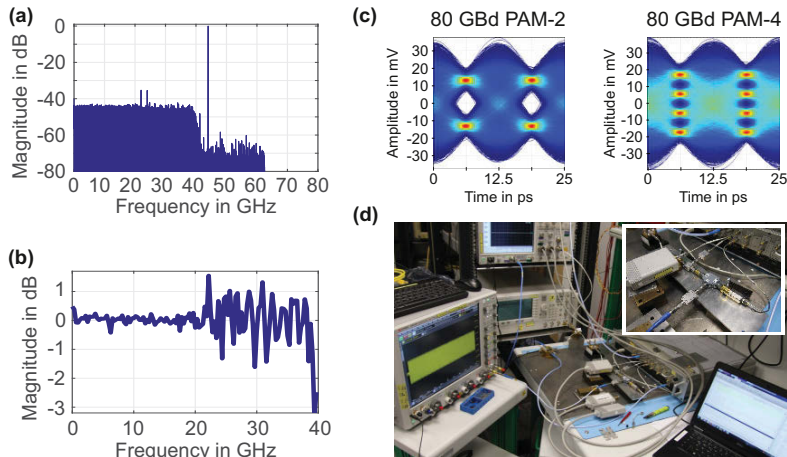


Figure 5.20 FI-DAC experimental results: output spectrum (a); pre-equalized FI-DAC frequency response (b); PAM-2 and PAM-4 eye diagrams at 80 GBd (c); photograph of the laboratory setup (d). (a),(b): Adapted from [47], with permission, © 2017 SPIE.

In Figs. 5.21(a) and 5.21(b), the magnitude and phase of the uncompensated FI-DAC frequency response are depicted. The frequency response is obtained by passing the impulse response through a DFT. The impulse response is estimated with a LS 2×1 MISO CE in the time domain. A 9 dB magnitude drop for the signal path from the first DAC to the output (dotted line) is observed, which is the result of the superposition of the frequency responses of the DAC, the diplexer, the LPF, and the coaxial cables. In the path from the second DAC to the output (dashed line), a ripple of ± 2 dB is observed. The phase has a fairly linear characteristic in the frequency intervals, where substantial magnitude components are present.

In Figs. 5.21(c) and 5.21(d), the SNR is plotted with respect to the frequency for the uncompensated and the pre-equalized FI-DAC. It is obtained from an EVM estimation based on a DMT modulation with 10248 sub-carriers, each modulated with 4-QAM. The sub-carriers from 40 to 45 GHz are not modulated. In Fig. 5.21(c), the SNR for the uncompensated FI-DAC is depicted, which drops from 30 dB at DC to 17 dB at 22 GHz for the first sub-signal and increases from 15 dB at 23 GHz to 22 dB at 40 GHz. The SNR drops at single frequencies are caused by DAC clock feedthrough and oscilloscope clock lines: at 22 GHz, a strong degradation is observed, which is equal to a quarter of the DAC's sampling frequency. Furthermore, there is a strong ripple in the SNR for the second sub-band, which is similar to the findings for the frequency response's magnitude in Fig. 5.21(a). This ripple is caused either by the mixer's nonlinear behavior, the mixer's frequency response,

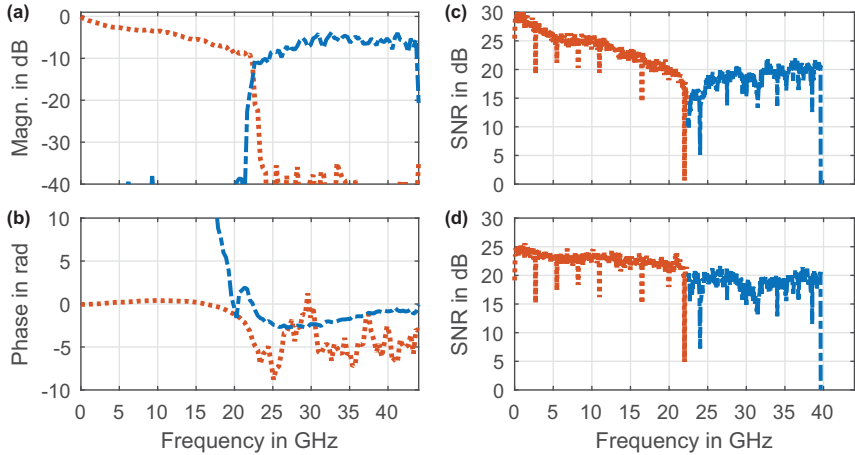


Figure 5.21 FI-DAC experimental results: FI-DAC frequency response (dotted: DAC1→output, dashed: DAC2→output) (a,b); SNR for the uncompensated (c) and the pre-equalized FI-DAC (d). Adapted from [47], with permission, © 2017 SPIE.

by reflections in the setup, or by a limited phase stability between both sub-signals. In Fig. 5.21(d), the SNR for the pre-equalized FI-DAC is depicted. The SNR characteristic is flattened; however, the strong notch at 22 GHz is still visible. The SNR drop at 24 GHz is presumably caused by a mixing product of the strong LO with an oscilloscope clock line at 20 GHz as explained earlier.

In this initial experiment with GHz-DACs, both an electrical 80 GBd PAM-2 and PAM-4 signal with 40 GHz electrical bandwidth were generated. The initial results indicate that serious improvements are necessary for a practical application. The second sub-signal is subject to stronger SNR degradations than the first sub-signal due to the upconversion. The LO's suppression needs to be improved, e.g., by a better suited LPF following the diplexer, or by an enhanced LO feedthrough suppression in the mixer. Furthermore, a wide-band amplifier could be placed after the mixer to compensate the mixer's loss.

5.10.4 Improved Experiment

In this section, an improved version of the experiment in Sec. 5.10.3 is presented. Parts of this section have been previously published in [48–52].

In order to improve the setup, two amplifiers are inserted after the DACs. This way, the mixer is operated with more IF power and the magnitude matching between

both frequency bands can be improved with the variable gain of the amplifier. Due to the increased mixer input power, the LO suppression relative to the signal level is improved. Moreover, the mixer has a better operating point, since the 1 dB compression point is given at 9 dBm input power. The output power of the FI-DAC is enhanced by ~10 dB. Furthermore, the DAC sampling rate is increased from 88 to 90 GS/s to realize an oversampling factor of two, i.e., both sub-bands' upper frequency is 22.5 GHz. Moreover, the FI-DAC's output signal is transmitted over a short-range optical IM/DD link.

Setup

In Fig. 5.22, the experimental setup is depicted. It consists of the FI-DAC and an optical IM/DD link. For the combined 80 GBd signal, the first and the second sub-signal are defined in the frequency intervals from DC to 22.5 GHz and from 5 to 22.5 GHz, respectively. Hence, for both sub-signals two times oversampling is applied to suppress DAC hold images with analog filters. The guard band from DC to 5 GHz for the second sub-signal represents the frequency interval from 40 to 45 GHz after upconversion. It enables the suppression of both the unused RF mixer's USB and the LO with an analog filter. The DSP has been covered in detail in Sec. 5.10.2.

For the D/A conversion, two high-speed CMOS DACs, each with an analog 3 dB bandwidth of ~20 GHz and 8 bit nominal resolution, are operated at a sampling rate of 90 GS/s. The DAC output signals are amplified with the variable gain amplifiers SHF 100BP with an analog 3 dB bandwidth of 29 GHz to adjust the signal amplitudes. In addition to the amplifiers' low-pass characteristic, both sub-signals are passed to LPFs to suppress DAC hold images: the LPF for the first sub-signal is part of the diplexer and the second sub-signal is passed to the LPF Marki Microwave FLP-2650. The passive RF mixer Marki Microwave MML1-1850 is driven by a 45 GHz LO with a power level of 19 dBm. The LO is generated by means of a frequency synthesizer and a driver amplifier. The clock signal for the DACs is obtained with a divide-by-16 clock divider to achieve phase stability between the sub-signals. The sub-signals are combined using a custom-designed diplexer with a crossover frequency of 22.5 GHz to form the combined output signal with 40 GHz bandwidth. The transmission characteristic of the diplexer corresponds to a LPF and a HPF for the first and the second sub-signal, respectively. The diplexer has a roll-off of 25 dB/GHz and an isolation > 53 dB in the stop band. After the diplexer, the LPF Marki Microwave FLP-4300 suppresses both the fed through LO signal and the mixer's unused USB.

The FI-DAC's output signal is further transmitted over a short-range optical IM/DD link. The amplifier SHF 804EA provides the required driving voltage for the

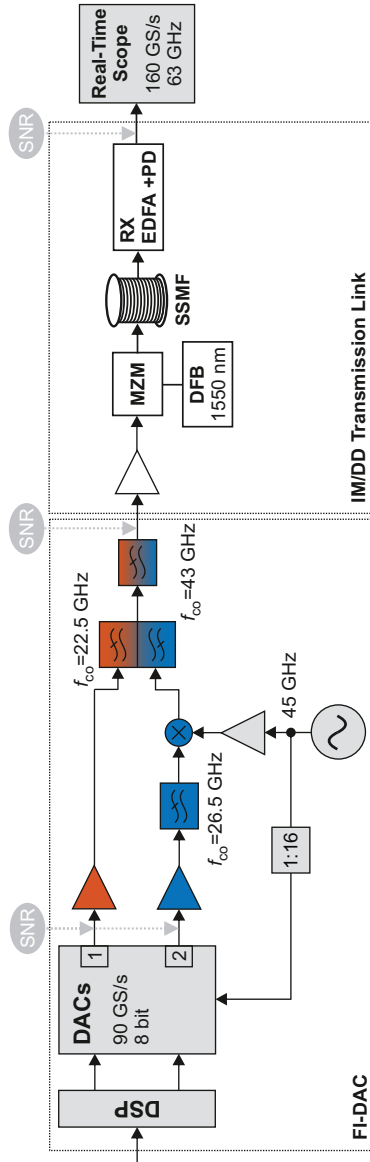


Figure 5.22 Experimental setup consisting of an electrical FI-DAC and an optical IM/DD transmission link. Furthermore, SNR measurement points for DMT transmission are depicted (see Sec. 5.11.2). Adapted with permission from [50], © 2018 Optical Society of America.

electrical-to-optical conversion with a MZM. The MZM is operated at the quadrature point to act as an intensity modulator. A distributed feedback laser provides the optical carrier at 1550 nm. After transmission over 2 km of standard single mode fiber, the signal is optical-to-electrical converted. The conversion is realized with a wideband photodetector having 50 GHz bandwidth, since no wideband photo-receiver (photodiode + transimpedance amplifier) was available in the laboratory. In order to provide the necessary optical power, an erbium doped fiber amplifier and an optical filter with 1 nm bandwidth are used. The signal is acquired with the high-speed real-time sampling oscilloscope Keysight DSAZ634A operating at 160 GS/s with 63 GHz bandwidth.

In order to study several effects of the FI-DAC, an electrical b2b simulation is set up in accordance with the experimental setup. It is later described in Sec. 5.11.1.

Results

In Fig. 5.23(a), both the magnitude and phase of the uncompensated FI-DAC's frequency response are depicted. They are obtained from the impulse response by means of an FFT. In turn, the impulse response is obtained with a MISO 2×1 LS CE for both experiment and simulation. For the experiment, there is drop of 20 dB in the frequency response for the first sub-signal and a 10 dB drop for the second sub-signal to the crossover frequency of 22.5 GHz. The strong drop can be attributed to the multiplication of the individual components' frequency responses, which form the combined frequency response. For signal path one, the following effects contribute to the combined frequency response: DAC analog frequency response, DAC sinc roll-off, amplifier frequency response, attenuators both before and after the amplifier enabling its linear operation, diplexer low-pass characteristic, and coaxial cables. For signal path two, the frequency response of the RF mixer need to be considered extra.

In the simulations, the drop to the crossover frequency is only 8 dB for the first sub-signal and thus, clearly less than in the experiment. The difference is based on the chosen frequency responses of the simulated components. For the simulation only the frequency response of the DAC and the diplexer are considered; the frequency responses of the amplifiers, attenuators and the coaxial cables are neglected. Furthermore, the frequency responses in the simulation are defined by means of Butterworth filters, whereby the 3 dB frequency and the order are approximated with information from the data sheets. Butterworth filters have a relatively flat frequency response up to the cutoff frequency, whereas the frequency responses of real-world components usually have a linear roll-off in the logarithmic domain.

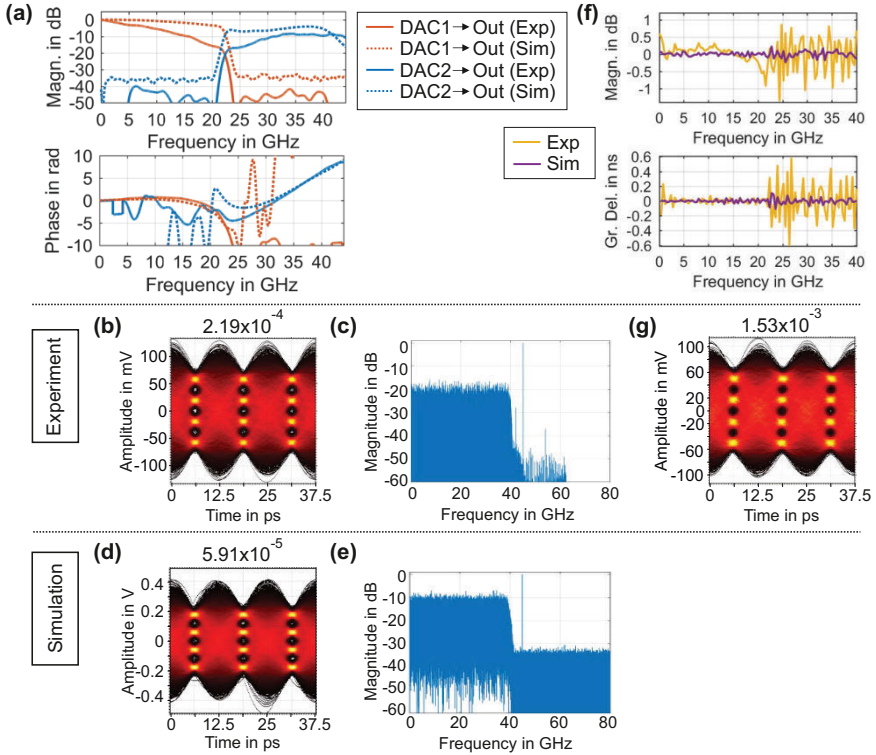


Figure 5.23 Measurement results: frequency response of the uncompensated FI-DAC (a); experimental electrical 80 GBd b2b eye diagram with a BER of 2.19×10^{-4} (b) and corresponding spectrum (c); simulated electrical 80 GBd b2b eye diagram with BER of 5.91×10^{-5} (d) and corresponding spectrum (e); magnitude and phase of the compensated channel's frequency response (f); experimental optical 80 GBd eye diagram for 2 km IM/DD transmission with additional post-equalizer with a BER of 1.5×10^{-3} (g). Adapted with permission from [50], © 2018 Optical Society of America.

A better matching between the experiment and the simulation is obtained for the phase response. The phase is fairly linear in frequency ranges, where substantial magnitude is present.

The frequency responses are corrected by means of ZF pre-equalization, as described in Sec. 5.10.2, enabling an open PAM-4 electrical b2b eye diagram at the output of the FI-DAC system with a BER of 2.19×10^{-4} as depicted in Fig. 5.23(b); the eye diagram is plotted for 5×10^4 symbols and the BER is evaluated for 2^{18} symbols. The corresponding flat spectrum with the raised cosine roll-off is shown in Fig. 5.23(c). Negligible spurious components between 40 and 45 GHz are observed,

which are presumably caused by mixer nonlinearities. Furthermore, broadband oscilloscope noise can be seen for frequencies > 45 GHz. The corresponding simulation results for eye diagram and spectrum are depicted in Figs. 5.23(d) and 5.23(e), respectively. The eye diagram is related to a BER of 5.91×10^{-5} . The eye diagram is plotted with 1×10^5 symbols and the BER is evaluated for 2^{18} symbols. In the spectrum, the noise loading is clearly visible due to the noise floor for frequencies beyond 41 GHz. The difference in BER between simulation and experiment stems from the fact that by using noise loading the SNR values are not matched perfectly, as shown later in Sec. 5.11.2. The LO has been digitally removed with a notch filter prior to plotting the eye diagram for the experimental results due to the lack of an LPF with a lower cutoff frequency than 43 GHz. By using a half-symbol spaced post-equalizer, the BER could be improved for both experiment and simulation to 5.10×10^{-5} and 1.28×10^{-6} , respectively.

Since, the post-equalizer is able to improve the BER for the electrical b2b case, residual distortions are present. In Fig. 5.23(f), the LS channel estimation of the pre-compensated FI-DAC's frequency response is depicted. The experimental results display a fairly flat magnitude for the first sub-band. It shows a ripple of up to 0.9 dB for frequencies < 22.5 GHz. For the second sub-band, there is a stronger ripple of up to 1.5 dB. For the group delay, the findings are similar, i.e., the first sub-band is fairly flat and the second sub-band exhibits stronger ripple. The corresponding simulation results for both sub-bands display a significantly better behavior, although the frequency response in the upper sub-band is distorted stronger than in the lower sub-band as well. These effects can be attributed to the nonlinear transfer function of the mixer in the second path and the integrated PLL in the DAC, which reduces the phase stability between the first and the second sub-band.

In Fig. 5.23(g), the eye diagram is depicted after transmitting the signal over the 2 km SSMF IM/DD link experimentally; the 80 GBd PAM-4 eye diagram is plotted with 5×10^4 symbols. Due to the additional impairments of the optical transmission channel, the received signal is processed with the half-symbol-spaced MMSE post-equalizer to mitigate the frequency response impairments of the optical link. The BER is evaluated for 2^{18} symbols and equals 1.53×10^{-3} for the data rate 160 Gbit/s; hence, it is below the hard decision forward-error-correction threshold of 3.8×10^{-3} for 7 % overhead [310].

5.11 System Parameter Impact on Performance

In order to study the impact of various parameters on the combined output signal, both experimental and simulation results are presented in this section. If possible, experimental and simulation results are directly compared.

Starting with the description of the simulation setup, SNR estimations based on DMT are presented. Then, the analysis of power and phase mismatches between the frequency bands is followed by three methods to mitigate high PAPR values: spectral overlap, clipping, and sub-sample shifting. Thereafter, pure simulation results are presented to study further parameters. Both, the DAC bandwidth and the DAC resolution are varied for different PAM orders. The mixer input signal power is varied and the impact of both jitter and phase noise is investigated.

5.11.1 Simulation Setup

In order to study several effects of the FI-DAC, an electrical b2b simulation is set up in accordance with the experimental setup. The DACs are modeled with 8 bit quantization, a static nonlinear transfer characteristic, clock feedthrough, and an LPF to realize a frequency dependent ENOB. The DAC model is treated in detail in Appendix A. The nonlinear parameters are tuned, such that the resulting ENOB values match the measured parameters of the DACs in the experiment. For the LPF, a 2nd order Bessel filter with a cutoff frequency of 21 GHz is used. The resulting ENOB characteristic is depicted in Fig. A.2: the ENOB equals 5.3 bit at 15 GHz and decreases to 3.0 bit at 40 GHz.

The gain of the amplifiers following the DACs are tuned such that the mixer's IF input power is equal to 2 dBm and that the power spectral density of both sub-signals is equivalent after combining in the diplexer. Both sub-signals are low-pass filtered, whereby the LPF for the first path is included in the following diplexer.

The RF mixer is modeled as a cascade of LPF, static nonlinear transfer characteristic and a band-pass filter. More information on the RF mixer model can be found in Appendix B. Both sub-signals are combined in the passive diplexer, which is modeled as a low-pass and a high-pass Butterworth filter, each having a 3 dB cut-off frequency of 22.5 GHz, and lossless addition. It is followed by a 41 GHz Butterworth LPF to suppress the fed-through mixer LO and the mixer's USB.

Although, several impairments are considered for the simulations, the performance of the simulations does not match the experimental results perfectly. This may result from phase noise and jitter, which are not modeled up to Sec. 5.11.10. In order to close this gap while attempting to obtain similar SNR values as depicted in 5.11.2,

these additional impairments are modeled by white Gaussian noise. At the Rx, white Gaussian noise with a double-sided power spectral density of -137 dBm/Hz is added to the signal. This corresponds to a noise power of -28.0 dBm in a bandwidth of 40 GHz.

For the simulations with phase noise and jitter in Sec. 5.11.10, the phase noise time-domain samples are obtained with an IFFT of white Gaussian noise filtered in the frequency domain, whereby the filter shape is taken from the data sheet of the frequency synthesizer Agilent E8257D [257]. Please see Sec. 5.11.10 for more details.

5.11.2 SNR Estimations Based on DMT

In order to improve the understanding of the signal SNR degradation associated with the FI-DAC concept, both an electrical b2b and an optical transmission experiment is performed using DMT. Parts of this section have been previously published in [50].

The relatively simple SNR estimation at each subcarrier enables a conclusion about the frequency-dependent SNR degradation for both the FI-DAC by itself and the optical transmission. The DMT signal consists of 229/255 modulated subcarriers (512 point FFT/IFFT) with a total bandwidth of 41 GHz. The cyclic prefix has a length of 16 samples and a standard single-tap equalizer for each subcarrier is applied at the Rx. Additionally, an optimized bit loading for a target BER close to 1×10^{-3} is used to maximize the data rate with a practical algorithm [311]. Power loading is not performed. Prior to transmission, the same pre-equalization scheme used for the PAM transmission is applied to mitigate magnitude and phase mismatches between both sub-signals. However, pre-equalization is generally not necessary for a successful DMT transmission over a linear channel due to the independent subcarriers of the DMT signal. For a nonlinear channel, suitable nonlinear pre-equalizers will improve the performance as demonstrated in [312].

In Fig. 5.24, the measured SNR values of the DMT subcarriers are depicted at three points in the transmission system: directly after the DAC, after the diplexer in an electrical b2b scenario and after the optical transmission, as depicted in Fig. 5.22. At first, the behavior for the lower DMT band in the interval from 0 to 22.5 GHz is discussed. Directly after the first DAC, relatively high SNR values around 24 dB are observed (red curve). At this point, only DAC properties like vertical resolution, nonlinearities, jitter and frequency response limit the SNR values. The increase in SNR of 1 dB for higher frequencies can be attributed to the pre-equalization.

At the second measurement point after the diplexer, a clear drop in the SNR of up to 4 dB (yellow curve) is observed. While the SNR degradation at frequencies

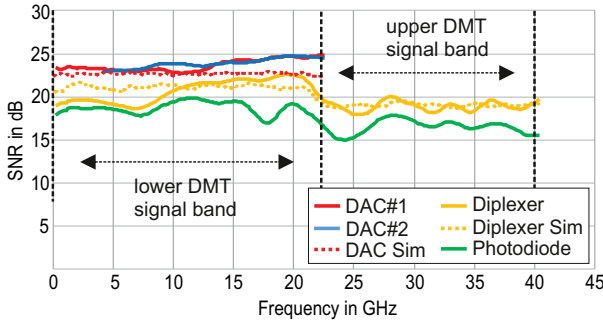


Figure 5.24 Measured SNR values based on a DMT signal at different points in the transmission system: after the DAC1&2, after the diplexer, and after the photodiode. Reprinted with permission from [50], © 2018 Optical Society of America.

between 10 and 22.5 GHz is mainly influenced by the amplifier and the diplexer, it is unclear, where the strong drop at lower frequencies originates from. After optical transmission, a further drop in the SNR is observed, which can be attributed to the limitations of the optical modulator and the driving amplifier (green curve).

For the upper DMT band between 22.5 and 41 GHz, a similar behavior is observed. Directly after the second DAC, similar SNR values compared to the first DAC's output signal are obtained (blue curve). The gap at frequencies from DC to 4 GHz for the blue curve is necessary to limit the total signal bandwidth to 41 GHz, since the LO is located at 45 GHz. After upconversion, significant lower SNR values are observed at the diplexer's output. This results mainly from the mixer's properties, i.e., nonlinearities and bandwidth limitations; additionally, small degradations due to the amplifier and the diplexer contribute to the overall SNR. After the optical transmission, a further SNR drop of ~3 dB occurs as for the lower band.

The noise loading for the simulations is adapted in order to match the measured SNR values in the second band for a mixer input power of 2 dBm. As a result, the experimental SNR values in the first band do not match the simulated SNR values perfectly, since noise loading is performed on the combined signal.

The average drop in SNR for the lower and the upper band are 3 to 5 dB and 5 to 7 dB for the electrical and the optical transmission case, respectively. The degradation of the upper band due to the electrical up-conversion is the primary limitation for the FI-DAC, even with a significant application of DSP. The measured DMT transmission rates for the electrical and the optical case with an optimized bit-loading at a target BER of 1×10^{-3} are 205 Gb/s and 174 Gb/s, respectively. The reduction in data rate can be attributed to the bandwidth limitations of the optical

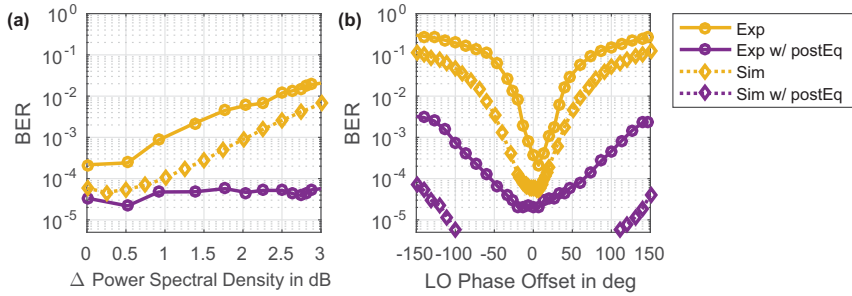


Figure 5.25 FI-DAC results from experiment and simulation: BER versus power spectral density difference between the bands (a); BER versus LO phase offset (b). Adapted with permission from [50], © 2018 Optical Society of America.

modulator. Compared to the PAM transmission, a slightly better performance is observed, which is based on the superior channel adaption properties of the DMT signal.

5.11.3 Power Spectral Density Difference

In order to investigate the required precision for power and phase matching between the two frequency bands, both the spectral power difference and the phase of the LO are varied and the corresponding results are visualized in Fig. 5.25. Parts of this section have been previously published in [50, 51].

In Fig. 5.25(a), the BER is visualized with respect to the mean power spectral density between both bands. The mean power spectral density difference is controlled by increasing the power of the first sub-signal with the gain control of the external amplifier. The mean power spectral density difference relates to a step in the combined frequency response of the FI-DAC. The raw BER without a post-equalizer is rising almost two orders, if the power difference raises to 3 dB. If an additional post-equalizer is utilized, the error is almost fully corrected and the BER is constant. In the simulation, the post-equalizer was able to obtain a roughly constant BER of 2×10^{-6} .

The results indicate that power mismatches < 0.5 dB are tolerable, even if no further post-equalization is applied.

5.11.4 LO Phase Variation

In Fig. 5.25(b), the mixer's LO phase is varied in the interval -150° to 150° . Parts of this section have been previously published in [50].

In the experiment, a manual phase shifter is used before the LO driver amplifier to control the LO phase. The phase shift for the LO frequency is obtained by turn counting and the phase-shift-per-turn value provided in the data sheet of the manual phase shifter. For both the experiment and the simulation, the phase offset has a significant impact on the resulting BER if no post-equalizer is utilized. For the experiment, the phase offset has to be below $\pm 15^\circ$ to ensure a BER below 1×10^{-3} . A phase shift of $\pm 15^\circ$ corresponds to a time shift of ± 0.93 ps for the 45 GHz LO. By using an additional post-equalizer the margin is widened, i.e., $\pm 45^\circ$ and $\pm 90^\circ$ can be tolerated to keep the BER below 1×10^{-4} and 1×10^{-3} , respectively. The BER curve for the experiment without post-equalizer is not centered around zero offset; hence, the pre-equalizer did not match the phases of the frequency bands perfectly, but left a small mismatch.

In general, a good match between the experimental and the simulation-based results is obtained regarding the trend of the curves; however, the simulation results are offset as in Sec. 5.10.4. Phase mismatches are more critical than power mismatches and additional post-equalization is recommended for robustness.

5.11.5 Overlap Variation

In order to reduce the PAPR, a spectral overlap between both frequency bands could be generated by the DSP as depicted in Figs. 5.18(g) and 5.18(h). A raised-cosine overlap translates the original rectangular spectral shape of both sub-signals with commonly high PAPR values into a softer form with lower PAPRs. In Appendix E, more information is provided on the digital frequency demultiplexer based on raised-cosine functions. Parts of this section have been previously published in [50].

The estimated BER and the linear PAPR with respect to the overlap are depicted in Figs. 5.26(a) and 5.26(b), respectively. Regarding the experimental results in Fig. 5.26(a), a small overhead of up to 2 GHz seems to translate into a marginal improvement in terms of BER. An overlap of up to 3 GHz is tolerable without performance degradation. Beyond that, the pre-equalization fails by putting too much effort in the equalization of high frequency components. In the simulation, a wider margin is observed for an overlap of up to 5.7 GHz without performance degradation. This may be based on slower roll-offs and better phase behavior of the standard filter characteristics used in the simulation.

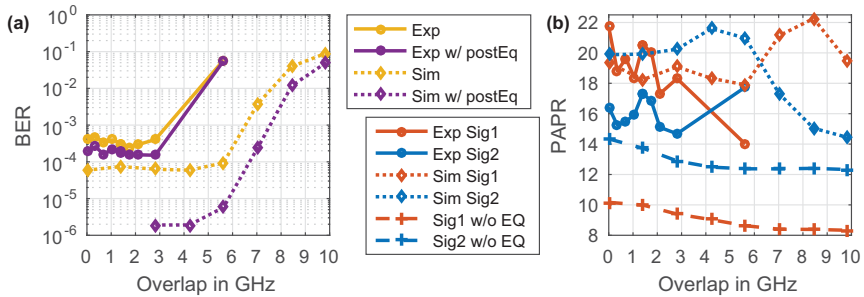


Figure 5.26 FI-DAC results from experiment and simulation: BER versus spectral overlap (a), PAPR versus spectral overlap (b). Adapted with permission from [50], © 2018 Optical Society of America.

In Fig. 5.26(b), the PAPR is depicted with respect to the spectral overlap for sub-signal one (red) and sub-signal two (blue). If no pre-equalization is used (dashed red & blue), the PAPR values decrease monotonically by increasing the overlap. However, the results with pre-equalization do not show a monotonic behavior for neither simulation (dotted lines) nor experiment (solid lines). In the experiment, the PAPR values decrease for a small overlap of up to 1 GHz, but rise again for an overlap of up to 1.5 GHz. They decrease again for an overlap of 2 GHz to 17.3 and 15.1 for sub-signal one and two, respectively. These changes are relatively small, since the linear PAPR is visualized. The simulation results show slightly decreasing values for the first sub-signal, but increasing values for the second sub-signal, whereby the loss of one seems to cancel the benefit of the other. The PAPR values do not decrease, since the values highly depend on the sub-signal characteristics and the frequency response of the pre-equalizer. For more overlap, more pre-equalization is necessary, which in turn worsens the sub-signals' PAPR.

Concluding, for filters with steep roll-offs as used in both experiment and simulation, a small overlap is tolerable. However, larger overlaps cause severe signal quality degradations, since too much effort is spent for the pre-equalization of the frequency responses. A small overlap may be useful for algorithms performing phase matching of the frequency bands.

5.11.6 Clipping

Another alternative to reduce the PAPR is clipping. Clipping is investigated for different PAM orders in this section. Parts of this section have been previously published in [50].

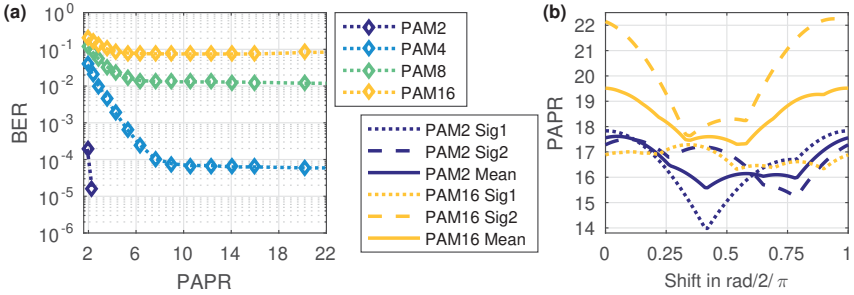


Figure 5.27 FI-DAC results from simulation: BER versus clipping according to a certain PAPR (a); PAPR vs sub-sample shift of the digital sub-signals (b). (a): Adapted with permission from [50], © 2018 Optical Society of America.

The raw PAPR values are summarized in Tab. 5.2 and the 90 % percentile of observations ranges from 17.88 to 21.85 for the pre-equalized sub-signals for different modulation orders. For this investigation, the two sub-signals have a raw PAPR of 17.85/17.28, 19.34/19.88, 17.64/16.22, and 16.87/22.15 for the modulation orders 2, 4, 8, and 16, respectively. Here, the sub-signals are clipped to a certain PAPR. Due to the Gaussian distribution, this corresponds to clipping relative to the sub-signals' power.

In Fig. 5.27(a), the BER is shown with respect to the clipping according to a certain PAPR for different PAM orders. For PAM-2, a BER of $< 2 \times 10^{-5}$ could be achieved with a strong clipping level according to a PAPR of 2.3. For PAM-4, a strong improvement is seen by performing less clipping up to a clipping level according to a PAPR of 12 in order to achieve a BER of 7.63×10^{-5} . This can be further lowered with a PAPR clipping level of 20 to 5.91×10^{-5} , while reaching an error floor. This error floor can be attributed to the noise loading performed for the simulations. For PAM-8 and PAM-16 the error floor is reached at a BER of 1.3×10^{-2} and 7.5×10^{-2} , respectively.

5.11.7 Sub-Sample Shifting

Next to spectral overlap and clipping, both sub-signals can be digitally phase-shifted in order to improve the time-domain behavior. By delaying the sub-signals simultaneously in the interval $[0, 2\pi]$, the phase relation between the sub-signals is kept constant; however, the sub-signals' PAPRs change, since the sub-signals are sinc-interpolated in the time domain on a new time vector. A phase shift of 2π corresponds to a shift of one sample.

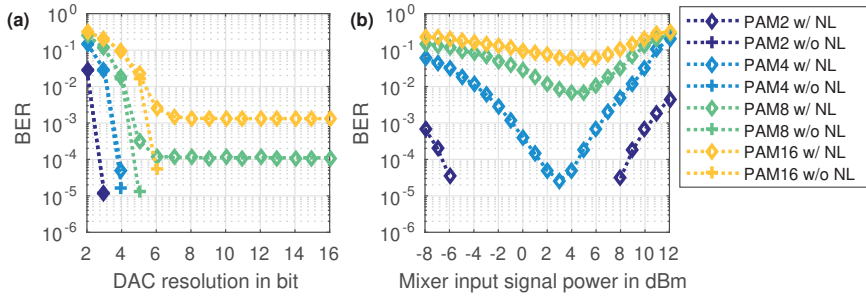


Figure 5.28 FI-DAC results from simulation: BER versus DAC resolution (a); BER versus mixer input signal power (b). (a): Adapted with permission from [50], © 2018 Optical Society of America.

In Fig. 5.27(b), the PAPR is depicted for the pre-equalized sub-signals with PAM-2 and PAM-16 modulation with respect to the sub-sample shift. For each modulation order, the PAPR is visualized for both sub-signals as well as their mean value. The PAPR values for PAM-2 are naturally lower than for PAM-16. The PAPR ranges from 14 to 18 and from 15 to 22.5 for PAM-2 and PAM-16, respectively. The minimum PAPR is not necessarily achieved for the same sub-sample shift for both sub-signals. By using the mean value, a joint optimum for both sub-signals can be found.

Concluding, for a practical realization an additional analog phase shifter is desired, e.g., an LO phase shifter. This way, for both sub-signals the minimum PAPR can be obtained due to the additional degree of freedom.

5.11.8 DAC Resolution Variation

Regarding the results from Sec. 5.7.6, each of the sub-signals has a Gaussian distribution in the time domain, even though a PAM signal is used. Hence, high-resolution DACs are required to generate the analog sub-signals with sufficient accuracy. Parts of this section have been previously published in [50].

In Fig. 5.28(a), the BER is shown with respect to the DAC resolution. For these simulations, the nonlinearities of the DAC are turned off in order to loosen the associated restriction on the maximum ENOB. Hence, the resolution values correspond to effective resolution rather than nominal resolution. Furthermore, noise loading at the Rx is turned off. The simulations are performed with and without nonlinear distortions of the mixer in order to evaluate the fundamental limits of the concept in terms of DAC resolution.

Generally, the more resolution the DACs have, the better the individual sub-signals can be synthesized, which in turn, enables a better performance. For PAM-2, the BER is below 2×10^{-5} for a resolution of 3 bit. For PAM-4, the BER is below 5×10^{-5} for a resolution of 4 bit. For PAM-8 and PAM-16 there is an error floor starting from 6 and 8 bit at a BER of 1.1×10^{-4} and 1.3×10^{-3} , respectively. This can be attributed to the nonlinearities of the mixer, which put a lower bound on the achievable BER.

The simulations without mixer nonlinearities do not show an error floor for BER values $> 1.1 \times 10^{-6}$. Eventually, the mixer nonlinearities are fundamentally limiting the FI-DAC.

5.11.9 Mixer Input Signal Power

In the previous section, the mixer nonlinearities limited the FI-DAC's performance. As described in Sec. 2.7.1 and Appendix B, the nonlinear characteristic of the mixer is dependent on the input power. In this section, the impact of the mixer nonlinearities on the resulting BER values is investigated by varying the mixer input signal power. In Fig. 5.28(b), the results are depicted for different PAM orders.

For all modulation orders, the BER curves display a bath tub shape. The performance is limited for low input power by the FI-DAC's noise floor and for high input power by the mixers' nonlinearities. For PAM-2 the minimum is below the lowest possible simulated BER, as described in Sec. 5.10.2. The minimum for PAM-4, PAM-8, and PAM-16 is at 3, 4, and 5 dBm, respectively. The minimum shifts to higher signal power for higher PAM orders.

Concluding, the mixer impairments are a conceptual limitation in the FI-DAC concept. A trade-off between nonlinearities and noise is observed for the mixer input signal power, which requires optimization. In a FI-DAC implementation, the operating point of the mixer has to be precisely controlled with amplifiers and the nonlinear distortions of both mixer and amplifiers need to be minimized. Further, nonlinear pre-distortion may be employed to compensate the nonlinear impairments of the analog components.

5.11.10 Phase Noise Variation

In order to evaluate the phase noise influence of the DAC clock and the LO on the combined FI-DAC output signal, a simulation is conducted.

The parameters for the phase noise spectrum are obtained from the data sheet of the frequency synthesizer Agilent E8257D [257]. This device has been used for

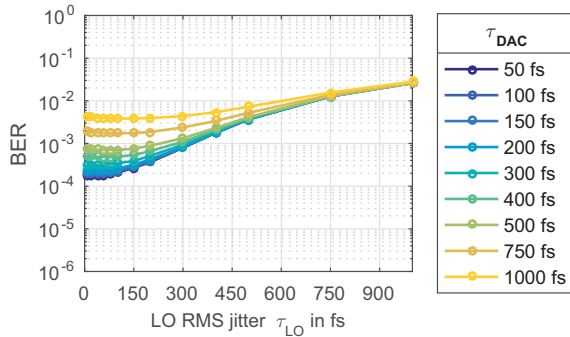


Figure 5.29 FI-DAC BER performance with respect to different LO RMS jitter levels for multiple DAC clock RMS jitter levels.

the experiments in Sec. 5.10. The phase noise time-domain samples are obtained by filtering white Gaussian noise in the frequency domain and a following IFFT. According to the DAC parameter fitting for the Socionext DAC in Appendix A, the DAC's RMS jitter is 150 fs; the DAC's static nonlinear distortions are reduced to retain the SINAD fitting. Please refer to Appendix A for more details. The length of the digital signal is set to 2^{20} samples. An ideal receiver without clock jitter and noise is utilized to highlight the combined DAC's impairments.

In Fig. 5.29, the BER values of the FI-DAC output signal are depicted with respect to the LO RMS jitter for multiple DAC clock RMS jitter values. At low LO RMS jitter values it is observed that the BER is limited to 1.7×10^{-4} and 4.0×10^{-4} for DAC RMS jitter values of 50 and 100 fs by the FI-DAC's nominal parameters, respectively. The DAC jitter dominates for LO RMS jitter values < 100 fs. For LO RMS jitter values > 100 fs, the reverse is true. The difference in EVM between the DAC jitter values decreases if the LO RMS jitter is increased.

Concluding, the overall FI-DAC performance in terms of jitter and phase noise is limited by both the DAC clock's and the LO's RMS jitter. The DAC clock jitter imposes an upper bound on the overall performance, while an increasing LO jitter further limits the FI-DAC's performance.

5.12 Scaling Model

The FI-DAC concept offers ultra-broadband signal generation abilities. In this section, a simple scaling model is introduced to estimate the overall FI-DAC performance with multiple sub-bands.

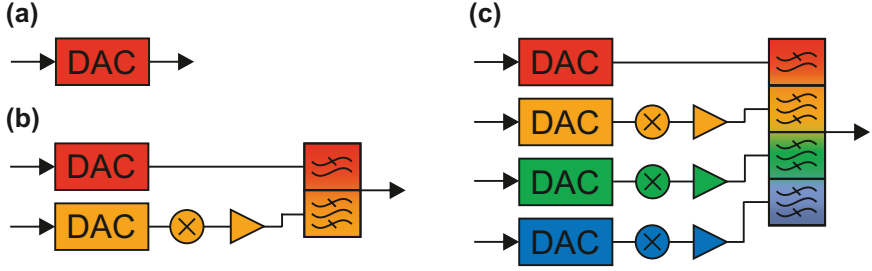


Figure 5.30 Block diagrams for the FI-DAC scaling model examples: single DAC (a), FI-DAC with two DACs (b), FI-DAC with four DACs (c).

During propagation through the FI-DAC's analog processing system, the initial SNR of the DAC output signal, denoted as $\text{SNR}_{\text{in},n}$, is reduced by the analog components. The m th element in the n th signal path reduces the SNR, which is represented by the SNR penalty $\Delta_{\text{SNR},m,n}$. The number of elements in each signal path is $M_{\text{FI},n}$. Hence, the SNR for the n th sub-band at the output of the FI-DAC is given by $\text{SNR}_{\text{out},n}$:

$$\text{SNR}_{\text{out},n} = \text{SNR}_{\text{in},n} - \sum_{m=1}^{M_{\text{FI},n}} \Delta_{\text{SNR},m,n} \quad (5.66)$$

The SNR at the FI-DAC output is composed of the SNRs of each sub-band, whereby it is assumed that there is no overlap between the sub-bands. The capacity is calculated from the SNR according to (1.1) with 1000 subcarriers. The bandwidth of the analog components is assumed high enough for the signals in each path. Generally, nonlinear distortions can be treated as additional noise and could be included in the SNR penalty $\Delta_{\text{SNR},m,n}$.

The performance of three systems is compared to each other: a single DAC, a FI-DAC consisting of two DACs, and a FI-DAC consisting of four DACs. The corresponding block diagrams are depicted in Fig. 5.30. The FI-DAC is composed of DACs, passive mixers, amplifiers and a combiner. The RF mixers' USB is utilized for all upconverted sub-bands.

The DAC SNR characteristic is approximated with a linear trend according to the SNR measurements based on DMT in Sec. 5.11.2. The SNR penalties for the analog components are listed in Tab. 5.3. In the experiment in Sec. 5.11.2, the combined SNR penalty of mixer and amplifier for the upconverted band was 5 dB. In this model, the contribution of the mixer is assumed dominant with respect to the amplifier due to the high conversion loss of the mixer and its nonlinear distortions.

Table 5.3 Parameters for the FI-DAC scaling model examples.

Component	Δ_{SNR} in dB
Mixer 2	4
Amplifier 2	1
Mixer 3	5
Amplifier 3	1
Mixer 4	6
Amplifier 4	1
Combiner	2

* the component numbers
denote the signal path number

Since, only gains of 8 to 10 dB are required to compensate the mixer's losses, the amplifiers are assumed to be linear and to have a low noise figure each. Hence, the mixer and the amplifier for the second signal path introduce SNR penalties of 4 and 1 dB, respectively.

The mixers' SNR penalty is increasing with the mixer number to account for an increased phase noise at higher LO frequencies. The amplifiers compensate the mixers' losses. It is assumed that amplifiers with equal bandwidths can be designed for different frequency bands having similar noise figures and similar nonlinear distortions. Hence, the amplifier in each signal path has the same SNR penalty. The passive combiner has an SNR penalty equivalent to its insertion loss of 2 dB, which is typical for passive broadband microwave filters.

In Fig. 5.31, the results of the FI-DAC scaling model examples are depicted. In Fig. 5.31(a), the SNR of the single DAC is visualized. Two lines are present, which represent the SNR values measured for the uncompensated DAC in the experiments in Sec. 5.10 (solid line) and a hypothetical next-generation DAC, which has an improved SNR by 10 dB (dashed line). The linear decreasing SNR characteristic is a simplification, although it roughly matches the measured values.

In Fig. 5.31(b), the resulting capacity is visualized with respect to the utilized DAC bandwidth. A capacity of 176 and 247 Gbit/s can be achieved by utilizing the actual DAC with a bandwidth of 20 and 40 GHz, respectively. A capacity of 379 Gbit/s is theoretically feasible for the next-generation DAC.

In Fig. 5.31(c), the DAC from Fig. 5.31(a) is used with half the bandwidth, i.e., 20 GHz; hence, the SNR is improved by 3 dB. The corresponding capacity is listed in Tab. 5.4 as "1 DAC".

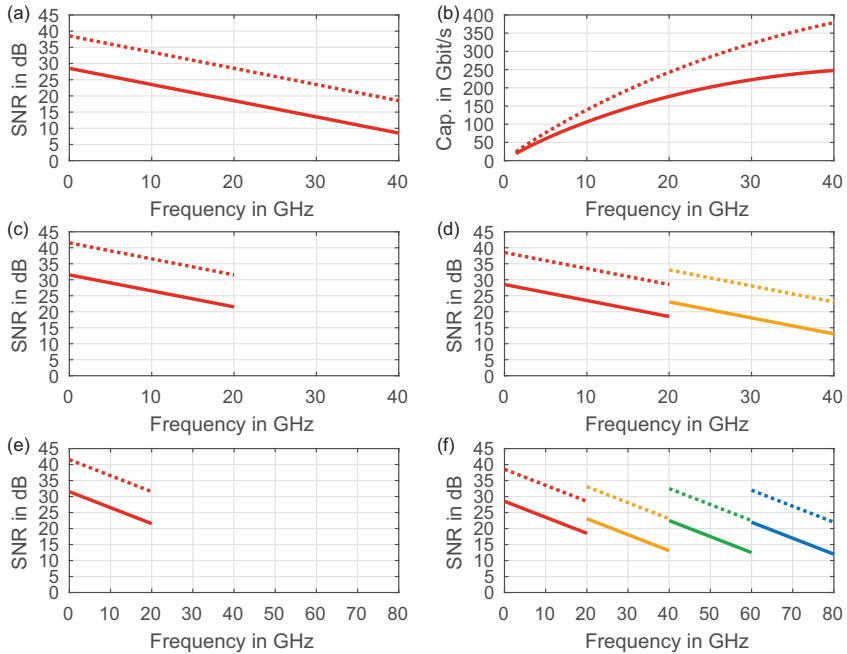


Figure 5.31 FI-DAC scaling model examples results: single DAC SNR (a); single DAC capacity (b); baseband SNR for DAC with 20 GHz bandwidth (c); FI-DAC SNR with two DACs (d); baseband SNR DAC with 20 GHz bandwidth (e); FI-DAC SNR with four DACs (f).

In Fig. 5.31(d), the resulting SNR values are depicted at the FI-DAC's output for the sub-bands one and two in different colors. The SNR of the first and the second sub-band are lowered slightly and clearly, respectively. Both the total capacity and the capacity per frequency band are listed in Tab. 5.4. The total capacity equals to 277 and 410 Gbit/s for the actual and the next-generation DAC, respectively.

In Fig. 5.31(e), the same DAC as in Fig. 5.31(c) is depicted with an extended frequency axes in order to visualize the total FI-DAC bandwidth.

In Fig. 5.31(f), the SNR at the FI-DAC's output is visualized for four sub-bands. The SNR drop between the first and the second sub-band is significant, whereas the subsequent sub-bands have a similar, but decreasing performance. This becomes apparent in Tab. 5.4: the first sub-band provides more than double the capacity compared to the second sub-band. The total capacity is 508 and 772 Gbit/s for the actual and the next-generation DAC, respectively.

Table 5.4 Performance results for the FI-DAC scaling model examples: capacity in Gbit/s.

	#	Regular DAC	Improved DAC
1 DAC	1	176	243
2 DACs	1	156	223
	2	121	187
	Σ	277	410
4 DACs	1	156	223
	2	121	187
	3	117	183
	4	114	179
	Σ	508	772

Concluding, the FI-DAC concept provides a performance gain in terms of capacity, which increases for a higher number of DACs. For the case of a FI-DAC with two actual DACs an almost similar capacity to a single next-generation DAC is obtained.

Please note that the presented model oversimplifies the impairments in the FI-DAC, i.e., noise and nonlinear distortions. However, it enables a fast evaluation of the combined DAC's resulting performance.

5.13 Limitations and Challenges

The DAC bandwidth is enhanced beyond the current technological limits by using the FI-DAC approach. Several conceptual limitations and challenges are discussed in this section. Parts of this section have been previously published in [45, 47, 49, 50].

The interdependencies between multiple FI-DAC parameters must be taken into account for the design, i.e., DAC sampling rate, the analog components' bandwidths, number of data samples, LO frequencies, etc. The fixed analog bandwidths of filters and mixers limit the flexibility of the concept. Due to the fixed clock frequencies, the number of feasible symbol rates is limited.

The pre-equalization requires accurate system information, which can be obtained by means of a CE. Both, power and phase mismatches between the frequency bands distort the signal quality, whereby the phase relation is more critical.

Due to the spectral splitting operation, the resulting time-domain sub-signals have a Gaussian amplitude distribution each with high PAPR values.

The analog components in the present hardware setup are discrete components. The interfaces between the components are costly in terms of energy and bandwidth including the required switching between current- and voltage-mode. Moreover, reflections between the components degrade the signal quality.

Generally, the impairments of the analog components reduce the FI-DAC's ENOB. Usually, high-speed DACs operate in ZOH mode and generate sinc-attenuated frequency images. Since, they appear as cross talk for the neighboring frequency bands, they need to be suppressed. Digital oversampling in combination with an analog LPF is commonly used. Furthermore, the sub-signals' Gaussian amplitude distributions with high PAPR values require high resolution DACs, as for DMT signals. However, the ENOB of high-speed DACs is limited; by using differential signals, the performance is improved.

For the FI-DAC, both RF and I/Q mixers can be employed. High-speed passive broadband mixers are usually limited by a frequency-dependent conversion loss, spurious products due to inter-modulation, a strong LO feed through, LO harmonics as well as spurious products located around the LO harmonics [176, 182, 189, 313]. The influence of the frequency-dependent conversion loss, as well as spurious products due to inter-modulation could be partly mitigated by linear and nonlinear digital pre-processing. The other effects generate frequency components close beside the upconverted frequency band. They can be suppressed by using oversampling in combination with analog filters or by adding inverse signals in the neighboring frequency bands prior to D/A conversion. In the experiments, the mixer reduced the SNR by ~ 3 dB.

The FI-DAC's frequency bands need to be phase-locked to generate a continuous spectrum. Accordingly, the mixers' LOs and the DACs' clocks need to be phase-locked. In the experiments, the integrated PLL in the DACs reduced the phase stability between the frequency bands. Concluding, a common frequency reference is recommended. Further, analog phase shifters for the LOs improve the sub-signals' PAPR in combination with digital sub-sample shifting as shown in Sec. 5.11.7.

The frequency multiplexer both combines the sub-signals and filters undesired frequency components. Analog filters used in the frequency multiplexer can be designed with roll-offs of 4 to 11 dB/GHz [47, 176]. In order to increase the combiner's number of input ports, multiple diplexers are preferably used. However, the integration into a single chip is highly complex, since the required passive components are difficult to integrate on the chip, i.e., coils.

In principle, the concept is scalable to N DACs. For each additional frequency band, a DAC, an LPF, an LO, a mixer, an amplifier, a BPF and an additional port in the combiner are needed. Mixers are available for different frequency bands, even for hundreds of GHz [179, 183], albeit the NF becomes worse for higher frequencies and the loss of the mixer increases.

Finally, the complexity and the size of the combiner will limit the number of DACs that can be interleaved. The parasitic effects in the combiner rise with increasing number of frequency bands, which in turn decreases the analog bandwidth. Furthermore, the phase noise requirements are rising for higher frequencies; hence, the required precision for both clock generation and distribution increases accordingly, whereby the highest LO frequency has the topmost phase noise requirements. If the DAC clock and the LO frequencies are derived from a common reference frequency by means of a frequency divider based clock tree, phase stability between the frequency bands is ensured

For a commercial product, a permanent self-calibration of the analog processing system may be required, which in turn demands an ADC to be integrated in the interleaved DAC.

Concluding, the FI-DAC concept can be applied to high-speed DACs to circumvent the DAC's bandwidth limitations at the expense of an SNR degradation.

5.14 Summary

In this chapter, the FI-DAC was covered. By using multiple parallel DACs with an analog processing system consisting of mixers, filters and a combiner, a broadband continuous spectrum is generated.

First, a short literature overview was given, followed by a detailed explanation of the concept. Based on a block diagram and a frequency domain illustration, the usage of multiple mixer types was pointed out [49]. A brief analytical system description was presented thereafter.

The literature lacked a comprehensive mathematical FI-DAC framework, which is required for scaling the concept to more DACs. In order to manage the parameter complexity, a complete mathematical framework was developed [46, 49]. The mathematical problem of distributing the data samples among the DACs, the calculation of the required LO frequencies, and the limited number of feasible symbol rates were addressed. In order to balance the FI-DAC's various interdependent parameters, both an elementary and an extended MINLP were introduced [49]. Examples for a FI-DAC with three DACs were provided to demonstrate the involved challenges regarding the parameter interdependencies.

Thereafter, the FI-DAC's DSP was covered. A MIMO system model for two DACs was presented for a FI-DAC without guard bands and further extended towards N DACs [44, 45]. Then, both CE and pre-equalization algorithms were described and explained to mitigate the analog component's impairments [44, 50]. Moreover, algorithmic complexity estimations were provided for a frequency domain implementation. Based on the MIMO system model, a novel MIMO pre-equalizer was presented to enable FI-DAC operation without guard bands at maximum bandwidth [44, 45]. Further, FI-DAC calibration routines were described [44] and the sub-signals' characteristics were analyzed [50]. Both with and without pre-equalization the sub-signals are subject to high PAPR values [50].

In initial experiments with MHz-DACs, both the FI-DAC concept in general and the MIMO pre-equalizer were verified [45]. At the time of publication, this was one of the first experimental presentations of the FI-DAC concept ever and the first presentation of a MIMO pre-equalizer for the FI-DAC. Thereafter, a general impairment model for the FI-DAC was derived based on these initial experiments.

Further experiments with GHz-DACs followed. Open high-speed eye diagrams without post-equalization were shown, which were reported for the first time [47]. An improved experimental setup achieved a BER of 2.19×10^{-4} for a PAM-4 signal without post-equalization. Further, the improved setup comprised a short-range optical IM/DD transmission [48, 50, 52], which was demonstrated for the first time for a FI-DAC [48].

The impact of various parameters on the FI-DAC performance was studied with experiments and simulations [50, 51]. SNR estimations based on DMT showed the loss of 3 to 5 dB due to the mixer and the diplexer in the FI-DAC setup. The investigation of power and phase mismatches highlighted the superior importance of phase matching between the frequency bands with an LO phase margin of $\pm 15^\circ$.

In order to reduce the sub-signals' high PAPR, multiple approaches were investigated, i.e., spectral overlap between the frequency bands, digital clipping, and digital sub-sample shifting. A spectral overlap of a few GHz has no major impact on the performance in terms of BER, although it may be useful for phase tracking algorithms. Although, clipping lowered the PAPR, the expected performance gain in terms of BER was not realized. Nevertheless, sub-sample shifting turned out to be beneficial; it is preferably utilized in combination with tunable analog phase shifters for maximum PAPR reduction.

By varying the required effective DAC resolution, the RF mixer was found to limit the FI-DAC performance substantially. If mixer nonlinear distortions are considered, the required number of bits for a performance below the 7 % HD-FEC threshold are ~ 4.5 and ~ 6 bit for PAM-8 and PAM-16, respectively.

There is an optimum operating point for the mixer's input signal power dependent on the noise, the mixer's nonlinearities, and the modulation order. In the end, the FI-DAC is ultimately limited by phase noise and jitter.

FI-DAC scaling issues are analyzed by means of a simple scaling model based on the noise figure. Lastly, conceptual limitations and challenges are discussed for a later commercial application. An integration of all analog components with the DACs is desired; however, the integration of the frequency multiplexer will remain challenging.



6 Summary and Outlook

6.1 Summary

In this thesis, three DAC interleaving concepts enhancing the performance of high-speed DACs were presented: TI-DAC, AMUX-DAC, and FI-DAC.

Both, the AMUX-DAC and the FI-DAC have been covered sparsely in the literature at the beginning of the thesis. Hence, there were plenty of open questions to be answered. The overall objective was to analyze and examine the limitations and the possibilities of both concepts. In the following, the findings and contributions of this thesis are summarized and the corresponding publications are stated.

In the second chapter, an overview on the state of the art high-speed DACs and manufacturing technologies was provided. Current-steering DAC architectures were introduced and the ZOH-DAC operation was described analytically. Furthermore, DAC performance limitations and measures were explained. Thereafter, the three DAC interleaving concepts were shortly introduced including the required analog components for DAC interleaving.

In the third chapter, the TI-DAC concept was briefly covered. An analytical description for both RZ and NRZ pulses was provided. The required pre-processing filter for NRZ pulses was presented in detail and its implications were discussed. The TI-DAC concept enhances the sampling rate, but does not enhance the analog bandwidth.

In the fourth chapter, the AMUX-DAC was presented. An analytical system model for N DACs was introduced, which had not been available in the literature. Based on this, a 2:1 MIMO system model was derived and the extension to N DACs was outlined. An improved understanding of the AMUX-DAC behavior could be obtained with this model. In the signal processing section, a novel MIMO pre-equalizer was derived from the aforementioned MIMO system model to pre-equalize AMUX-DACs with sub-channels having unequal frequency responses.

The literature further lacked a profound behavioral model of the AMUX. Therefore, a general impairment model for the AMUX-DAC was introduced. Based on this, a generic behavioral 2:1 AMUX model was deduced, which is useful for system-level

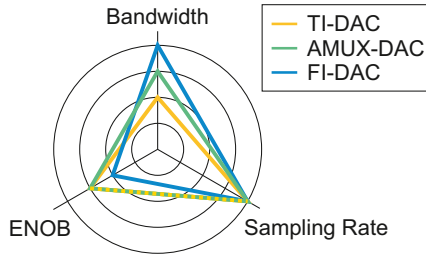


Figure 6.1 Qualitative relations of bandwidth, sampling rate and ENOB for the DAC interleaving concepts.

simulation [43]. The model drastically speeds up computation time. It was adapted to reference data from an EDA tool with an NMSE < -20 dB [43].

The overall performance for both the AMUX by itself and the AMUX-DAC was investigated by varying multiple parameters, i.e., within the aforementioned behavioral model [43]. The effect of various parameters on the combined frequency response was evaluated. Further, the novel MIMO pre-equalizer notably outperformed a SISO equalizer for unequal signal paths.

Thereafter, experimental results were presented for the AMUX IC with the world-wide highest bandwidth > 67 GHz reported at that time [42]. Both an S-parameter measurement and a time-domain measurement were conducted, whereby PAM-4 signals up to 56 Gb/s and PAM-2 signals up to 112 Gb/s could be demonstrated without utilizing DSP [42]. Finally, an AMUX-DAC noise model addressing scaling issues was presented and conceptual limitations and challenges were discussed.

In the fifth chapter, the FI-DAC was presented. The literature lacked a comprehensive mathematical FI-DAC framework, which is required for scaling the concept to more DACs. In order to manage the parameter complexity, a complete mathematical framework was developed [46, 49]. The mathematical problem of distributing the data samples among the DACs, the calculation of the required LO frequencies, and the limited number of feasible symbol rates were addressed [46, 49]. In order to balance the FI-DAC's various interdependent parameters, both an elementary and an extended MINLP were introduced [49]. Examples for a FI-DAC with three DACs were provided to demonstrate the involved challenges regarding the parameter interdependencies [49].

A MIMO system model for two DACs was presented for a FI-DAC without guard bands and further extended towards N DACs [44, 45]. Then, both CE and pre-equalization algorithms were described and explained to mitigate the analog component's impairments [44, 50]. Based on the MIMO system model, a novel MIMO

pre-equalizer was presented to enable FI-DAC operation without guard bands at maximum bandwidth [44, 45]. Besides, the MIMO DSP of the FI-DAC without guard bands and the MIMO DSP of the AMUX-DAC are very similar: for both, the data spectra need to be considered in regular and reversed frequency position.

Further, FI-DAC calibration routines were described [44] and the sub-signals' characteristics were analyzed [50]. Both, with and without pre-equalization the sub-signals are subject to high PAPR values [50].

In initial experiments with MHz-DACs, both the FI-DAC concept in general and the MIMO pre-equalizer were verified [45]. Based on these initial experiments, an impairment model for the FI-DAC was derived. Further experiments with GHz-DACs followed. Open high-speed eye diagrams without post-equalization were shown, which were reported for the first time [47]. An improved experimental setup further comprised a short-range optical IM/DD transmission [48, 50, 52], which was demonstrated for the first time [48].

The impact of various parameters on the FI-DAC performance was studied with both experiments and simulations [50, 51]. SNR estimations based on DMT revealed the loss of 3 to 5 dB due to both the mixer and the diplexer in the FI-DAC setup. Then, multiple approaches were investigated reducing the sub-signals' high PAPR: spectral overlap between the frequency bands, digital clipping, and digital sub-sample shifting. The investigation of power and phase mismatches highlighted the superior importance of phase-matching between the frequency bands. There is an optimum operating point for the mixer's input signal power dependent on the noise, the mixer's nonlinearities, and the modulation order. In the end, the FI-DAC is ultimately limited by phase noise and jitter.

FI-DAC scaling issues are analyzed by means of a simple noise model. Lastly, conceptual limitations and challenges are discussed for a later commercial application.

Concluding, there is a tradeoff between analog bandwidth and ENOB for interleaved high-speed DACs as outlined in the introduction. By extending the bandwidth at the expense of an ENOB loss, the overall capacity is increased, since the capacity relates to the bandwidth linearly and to the SNR logarithmically. However, the capacity gain is traded for both an increased size and increased power consumption of the interleaved DAC.

The results of this thesis are summarized in a qualitative manner by means of a diagram and a table: in Fig. 6.1, a diagram visualizes the relation of the parameters bandwidth, ENOB, and sampling rate for the three interleaving concepts and an overview on the properties of the three interleaving concepts is provided in Tab. 6.1.

Table 6.1 Qualitative comparison of the DAC interleaving concepts.

Parameter	TI-DAC	AMUX-DAC	FI-DAC
Concept	linear	nonlinear	nonlinear
Sampling rate	variable	variable	fixed
Bandwidth	limited by DAC	determined by DACs and AMUX	determined by DACs, mixers and combiner
Nominal resolution	given by DACs	given by DACs	given by DACs
Attenuation	min. -6 dB (passive), 0 dB (active)	0 dB	high due to passive mixers
Scalability	RZ: high attenuation (passive), low attenuation (active) NRZ: pre-equalizer compensates spectral nulls	higher AMUX order	more mixers and filters; combiner with more ports
Variability	given	given	interdependent parameters
Calibration	necessary	necessary	necessary
Pre-distortion	optional	optional	necessary
Integration	good	good	partially good
Power	$N \times$ DAC power	$N \times$ DAC power + AMUX power	$N \times$ DAC power + $(N - 1) \times$ (amplifier power + LO power)

All of the concepts increase the sampling rate. However, for the FI-DAC concept, the variability of the sampling rate is limited due to the interdependent system parameters. An initial timing calibration is required for all of the concepts. A pre-distortion algorithm is preferably used for the TI-DAC and the AMUX-DAC; for the FI-DAC, it is mandatory.

Further, the bandwidth requirements for the AMUX-DAC and the FI-DAC need to be considered when scaling the concepts. The FI-DAC concept uses multiple frequency bands; hence, the analog components in each signal path process bandpass signals. In the AMUX-DAC, the signal paths between the switches and the output port need to support baseband signals from DC to the maximum frequency.

In terms of integration, the AMUX-DAC concept is better suited, since transistors can be well integrated. The frequency multiplexer required for the FI-DAC concept is based on passive filter structures, which can not be integrated well with existing manufacturing technologies. The power consumption is increased by at least the number of DACs N for all interleaving concepts. However, a comparable single DAC operating at $N \times$ the sampling rate, will also consume $N \times$ more power. Thus, in a fair comparison, only the power consumption of the additional analog components increases the total power consumption.

Concluding, applications requiring the maximum bandwidth can use the FI-DAC concept, i.e., test and measurement as well as special high-performance applications. The AMUX-DAC concept is less complex in terms of calibration and synchronization and offers better integration capabilities. For optical communication systems it is an open question, whether the increased electrical transmitter bandwidth at the cost of an enhanced power consumption is beneficial for the total power consumption of the optical communication system.

In summary, the results presented in this thesis significantly contribute to the development of interleaved high-speed DACs, which enable ultra-broadband signal generation for various applications.

6.2 Outlook

The motivation for this thesis is to enhance the DAC bandwidth for optical networks. However, broadband DACs are of great interest for other applications as well, such as radar or mobile communications. For mobile communications, more and more frequency bands are addressed and it may be beneficial to synthesize multiple frequency bands with an interleaved DAC directly.

For both concepts investigated in depth in this thesis, multiple starting points for further research exist. For the AMUX-DAC, an integration of the bipolar AMUX

IC with CMOS DACs and CMOS DSP is desired. Although, an integration of the AMUX with the DACs in a common BiCMOS technology is conceivable, the integration of DSP functionality, e.g., for coherent optical communication systems, is unfavorable in nowadays BiCMOS technologies. Since, the CMOS performance in nowadays BiCMOS technologies is far behind the state-of-the-art in pure CMOS technologies, integration of many DSP features is prohibited by the higher power consumption. Eventually, applications requiring a lot of DSP will favor a hybrid integration, whereas applications with few DSP features might favor an integration in a BiCMOS technology.

Scaling the 2:1 AMUX IC to a 4:1 AMUX is a reasonable opportunity for circuit designers. Moreover, advanced CE and pre-equalization algorithms could be investigated and compared to each other, including nonlinear MIMO pre-equalization [252, 314]. The behavioral 2:1 AMUX model can be extended to N DACs, fitted with measurement data, and further improved by including nonlinear distortions.

Regarding the FI-DAC, an integration of the analog components into a single IC is desired to improve the analog performance. Moreover, different CE and pre-equalizer algorithms could be compared to each other. Especially, nonlinear pre-equalizers will improve the performance by mitigating both the DACs' and the mixers' nonlinear distortions. Calibration concepts could be investigated including the necessity to integrate a sub-sampling oscilloscope into the FI-DAC. Special DSP algorithms will be required to estimate and compensate the frequency response during operation. Moreover, the FI-DAC can be investigated with I/Q mixers by extending the preliminary work in [51].

In the end, both interleaving approaches are limited by the maximum performance of manufacturing technologies for electronic ICs, expressed by $f_T/f_{\max}$. These limits can only be overcome by means of optical concepts. Multiple optical interleaving concepts have been investigated in the literature, which enable ultra-broadband signal generation for various future applications [230, 315–318].

A promising approach for an almost unlimited bandwidth is an optical FI-DAC [319–322]. Multiple frequency bands are seamlessly combined by modulation onto individual lines of an optical frequency comb with I/Q modulators [323, 324]. If an electrical output signal is required, a wideband photodiode will be used for optical-electrical conversion. The mathematical framework presented in this thesis provides a valuable basis for scaling the concept to a large number of wavelengths [49]. Resting upon existing technologies, the optical FI-DAC approach allows ultra-broadband signals spanning multiple THz, which might enable fully flexible grid-less optical networks in the future.

References

- [1] Cisco Systems Inc., “Cisco visual networking index: Forecast and methodology, 2016–2021,” Cisco Systems Inc., Tech. Rep., Sep. 2017. [Online]. Available: <https://www.cisco.com/c/en/us/solutions/collateral/service-provider/visual-networking-index-vni/complete-white-paper-c11-481360.pdf> (Accessed 2018-07-25).
- [2] I. Lee and K. Lee, “The internet of things (IoT): Applications, investments, and challenges for enterprises,” *Business Horizons*, vol. 58, no. 4, pp. 431–440, 2015.
- [3] M. Brettel, N. Friederichsen, M. Keller, and M. Rosenberg, “How virtualization, decentralization and network building change the manufacturing landscape: An industry 4.0 perspective,” *International Journal of Mechanical, Industrial Science and Engineering*, vol. 8, no. 1, pp. 37–44, 2014.
- [4] J. Lee, B. Bagheri, and H.-A. Kao, “A cyber-physical systems architecture for industry 4.0-based manufacturing systems,” *Manufacturing Letters*, vol. 3, pp. 18–23, 2015.
- [5] S. Ben Yoo, “The role of photonics in future computing and data centers,” *IEICE Transactions on Communications*, vol. 97, no. 7, pp. 1272–1280, 2014.
- [6] C. Kachris, K. Kanonakis, and I. Tomkos, “Optical interconnection networks in data centers: recent trends and future challenges,” *IEEE Communications Magazine*, vol. 51, no. 9, pp. 39–45, 2013.
- [7] A. Roy, H. Zeng, J. Bagga, G. Porter, and A. C. Snoeren, “Inside the social network’s (datacenter) network,” in *ACM SIGCOMM Computer Communication Review*, vol. 45, no. 4, 2015, pp. 123–137.
- [8] I. A. T. Hashem, I. Yaqoob, N. B. Anuar, S. Mokhtar, A. Gani, and S. U. Khan, “The rise of “big data” on cloud computing: Review and open research issues,” *Information Systems*, vol. 47, pp. 98–115, 2015.
- [9] NGMN, “NGMN 5G whitepaper,” Next Generation Mobile Networks, Tech. Rep., Feb. 2015. [Online]. Available: https://www.ngmn.org/fileadmin/ngmn/content/downloads/Technical/2015/NGMN_5G_White_Paper_V1_0.pdf (Accessed 2018-07-25).

- [10] S. Gosselin, A. Pizzinat, X. Grall, D. Breuer, E. Bogenfeld, J. T. Gijón, A. Hamidian, and N. Fonseca, “Fixed and mobile convergence: Which role for optical networks?” in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2015.
- [11] E. Agrell, M. Karlsson, A. Chraplyvy, D. J. Richardson, P. M. Krummrich, P. Winzer, K. Roberts, J. K. Fischer, S. J. Savory, B. J. Eggleton *et al.*, “Roadmap of optical communications,” *Journal of Optics*, vol. 18, no. 6, p. 063002, 2016.
- [12] H. Isono, “Recent standardization activities for client and networking optical transceivers and its future directions,” in *Proc. of SPIE OPTO Next-Generation Optical Networks for Data Centers and Short-Reach Links IV*. International Society for Optics and Photonics, 2017.
- [13] D. Kreutz, F. M. Ramos, P. E. Verissimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, “Software-defined networking: A comprehensive survey,” *Proceedings of the IEEE*, vol. 103, no. 1, pp. 14–76, 2015.
- [14] M. Channegowda, R. Nejabati, and D. Simeonidou, “Software-defined optical networks technology and infrastructure: Enabling software-defined optical network operations,” *Journal of Optical Communications and Networking*, vol. 5, no. 10, pp. A274–A282, 2013.
- [15] O. Gerstel, M. Jinno, A. Lord, and S. Yoo, “Elastic optical networking: a new dawn for the optical layer?” *IEEE Communications Magazine*, vol. 50, no. 2, pp. s12–s20, Feb. 2012.
- [16] B. T. Teipen, M. H. Eiselt, K. Grobe, and J.-P. Elbers, “Adaptive data rates for flexible transceivers in optical networks,” *Journal of Networks*, vol. 7, no. 5, p. 776, May 2012.
- [17] C. Rottondi, M. Tornatore, A. Pattavina, and G. Gavioli, “Traffic grooming and spectrum assignment for coherent transceivers in metro-flexible networks,” *IEEE Photonics Technology Letters*, vol. 25, no. 2, pp. 183–186, 2013.
- [18] M. P. Yankov, D. Zibar, K. J. Larsen, L. P. Christensen, and S. Forchhammer, “Constellation shaping for fiber-optic channels with QAM and high spectral efficiency,” *IEEE Photonics Technology Letters*, vol. 26, no. 23, pp. 2407–2410, 2014.
- [19] F. Buchali, F. Steiner, G. Böcherer, L. Schmalen, P. Schulte, and W. Idler, “Rate adaptation and reach increase by probabilistically shaped 64-QAM: An experimental demonstration,” *Journal of Lightwave Technology*, vol. 34, no. 7, pp. 1599–1609, 2016.

- [20] P. W. Berenguer, M. Nölle, L. Molle, T. Raman, A. Napoli, C. Schubert, and J. K. Fischer, "Nonlinear digital pre-distortion of transmitter components," *Journal of Lightwave Technology*, vol. 34, no. 8, pp. 1739–1745, Apr. 2016.
- [21] Y. Bao, Z. Li, J. Li, X. Feng, B.-o. Guan, and G. Li, "Nonlinearity mitigation for high-speed optical OFDM transmitters using digital pre-distortion," *Optics Express*, vol. 21, no. 6, pp. 7354–7361, 2013.
- [22] E. Yamazaki, M. Tomizawa, and Y. Miyamoto, "100-Gb/s optical transport network and beyond employing digital signal processing," *IEEE Communications Magazine*, vol. 50, no. 2, pp. s43–s49, 2012.
- [23] J. C. Rasmussen and T. Drenski, "DSP for short reach optical links," in *Proc. of European Conference on Optical Communication (ECOC)*. VDE, Sep. 2017.
- [24] J. Wei, Q. Cheng, R. V. Pentty, I. H. White, and D. G. Cunningham, "400 Gigabit Ethernet using advanced modulation formats: performance, complexity, and power dissipation," *IEEE Communications Magazine*, vol. 53, no. 2, pp. 182–189, 2015.
- [25] C. Cole, I. Lyubomirsky, A. Ghiasi, and V. Telang, "Higher-order modulation for client optics," *IEEE Communications Magazine*, vol. 51, no. 3, pp. 50–57, 2013.
- [26] F. Lu, B. Zhang, Y. Yue, J. Anderson, and G.-K. Chang, "Investigation of pre-equalization technique for pluggable CFP2-ACO transceivers in beyond 100 Gb/s transmissions," *Journal of Lightwave Technology*, vol. 35, no. 2, pp. 230–237, Jan. 2017.
- [27] K. Kikuchi, K. Tsuzuki, S. Kamei, S. Yamanaka, S. Asakawa, T. Itoh, Y. Nasu, K. Takeda, K. Honda, Y. Kawamura, M. Jizodo, M. Takahashi, M. Usui, H. Mawatari, and T. Saida, "Silicon-photonics-based coherent optical subassembly (COSA) for ultra-compact coherent transceiver," in *Proc. of Compound Semiconductor Integrated Circuit Symposium (CSICS)*, IEEE. IEEE, Oct. 2017.
- [28] T. Drenski and J. C. Rasmussen, "ADC & DAC - technology trends and steps to overcome current limitations," in *Proc. of Optical Fiber Communications Conference and Exposition (OFC)*, Mar. 2018.
- [29] Socionext. (2016) 100G to 400G ADC and DAC for ultra-high-speed optical networks. [Online]. Available: <http://socionextus.com/products/networking-asic/adc-dac/> (Accessed 2018-07-25).
- [30] G. Manganaro, *Advanced Data Converters*. Cambridge University Press, 2012.

- [31] C. E. Shannon, "Communication in the presence of noise," *Proceedings of the IRE*, vol. 37, no. 1, pp. 10–21, 1949.
- [32] T. Sugihara, "Optical precoding technologies with high-speed DAC at 40G or beyond," in *Proc. of Asia Communications and Photonics Conference and Exhibition (ACP)*. OSA, 2011.
- [33] Q.-T. Duong, J. Dąbrowski, and A. Alvandpour, "Design and analysis of high speed capacitive pipeline DACs," *Analog Integrated Circuits and Signal Processing*, vol. 80, no. 3, pp. 359–374, Jul. 2014.
- [34] P. Kinget and M. Steyaert, "Impact of transistor mismatch on the speed-accuracy-power trade-off of analog CMOS circuits," in *Proc. of Custom Integrated Circuits Conference, 1996*, IEEE. IEEE, 1996, pp. 333–336.
- [35] S. Lange, S. Wolf, J. Lutz, L. Altenhain, R. Schmid, R. Kaiser, M. Schell, C. Koos, and S. Randel, "100 GBd intensity modulation and direct detection with an InP-based monolithic DFB laser Mach-Zehnder modulator," *Journal of Lightwave Technology*, vol. 36, no. 1, pp. 97–102, 2018.
- [36] C. Haffner, W. Heni, Y. Fedoryshyn, J. Niegemann, A. Melikyan, D. Elder, B. Baeuerle, Y. Salamin, A. Josten, U. Koch *et al.*, "All-plasmonic Mach-Zehnder modulator enabling optical high-speed communication at the microscale," *Nature Photonics*, vol. 9, no. 8, pp. 525–528, 2015.
- [37] EOSPACE, "Brief product brochure," online, 2016. [Online]. Available: <http://eospace.com/pdf/EOSPACEbriefProductInfo2018.pdf> (Accessed 2018-07-25).
- [38] S. P. Voinigescu, S. Shopov, J. Bateman, H. Farooq, J. Hoffman, and K. Vasilakopoulos, "Silicon millimeter-wave, terahertz, and high-speed fiber-optic device and benchmark circuit scaling through the 2030 ITRS horizon," *Proceedings of the IEEE*, vol. 105, no. 6, pp. 1087–1104, Jun. 2017.
- [39] G. E. Moore, "Cramming more components onto integrated circuits," *Electronics*, vol. 38, no. 8, pp. 114–117, Apr. 1965.
- [40] G. E. Moore, "No exponential is forever: but "forever" can be delayed!" in *Proc. of International Solid-State Circuits Conference (ISSCC)*, Feb. 2003.
- [41] K. Bult, "The effect of technology scaling on power dissipation in analog circuits," in *Analog Circuit Design*. Springer, 2006, ch. 13, pp. 251–294.
- [42] T. Tannert, X.-Q. Du, D. Widmann, M. Grözing, M. Berroth, C. Schmidt, C. Caspar, J. H. Choi, V. Jungnickel, and R. Freund, "A SiGe-HBT 2:1 analog multiplexer with more than 67 GHz bandwidth," in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, Oct. 2017, pp. 146–149.

- [43] C. Schmidt, P. Zielonka, V. Jungnickel, R. Freund, T. Tannert, M. Grözing, M. Berroth, and F. Gerfers, "Behavioral model for a high-speed 2:1 analog multiplexer," in *Proc. of International Midwest Symposium on Circuits and Systems (MWSCAS)*, IEEE. IEEE, Aug. 2018.
- [44] C. Schmidt, C. Kottke, V. Jungnickel, and J. Hilt, "Signal processing systems and signal processing methods," PCT Patent PCT/EP2015/077 000, May 26, 2017. [Online]. Available: <https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2017084705>
- [45] C. Schmidt, C. Kottke, V. Jungnickel, and R. Freund, "Enhancing the bandwidth of DACs by analog bandwidth interleaving," in *Proc. of ITG Symposium on Broadband Coverage in Germany*, VDE. VDE, Apr. 2016, pp. 99–106.
- [46] C. Schmidt, V. H. Tanzil, C. Kottke, R. Freund, and V. Jungnickel, "Digital signal splitting among multiple DACs for analog bandwidth interleaving (ABI)," in *Proc. of International Conference on Electronics, Circuits, & Systems (ICECS)*. IEEE, Dec. 2016, pp. 245–248.
- [47] C. Schmidt, C. Kottke, V. Jungnickel, and R. Freund, "High-speed digital-to-analog converter concepts (invited)," in *Proc. of SPIE Photonics West*. SPIE, Jan. 2017.
- [48] C. Schmidt, C. Kottke, R. Freund, and V. Jungnickel, "Bandwidth enhancement for an optical access link by using a frequency interleaved DAC," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*. OSA, Mar. 2018.
- [49] C. Schmidt, C. Kottke, V. H. Tanzil, R. Freund, V. Jungnickel, and F. Gerfers, "Digital-to-analog converters using frequency interleaving: Mathematical framework and experimental verification," *Circuits, Systems, and Signal Processing*, vol. 37, no. 11, pp. 4929–4954, Nov. 2018.
- [50] C. Schmidt, C. Kottke, R. Freund, F. Gerfers, and V. Jungnickel, "Digital-to-analog converters for high-speed optical communications using frequency interleaving: impairments and characteristics," *Optics Express*, vol. 26, no. 6, pp. 6758–6770, Mar. 2018.
- [51] C. Kottke, C. Schmidt, V. Jungnickel, and R. Freund, "Performance of bandwidth extension techniques for high-speed short-range IM/DD links," *Journal of Lightwave Technology*, vol. 37, no. 2, pp. 665–672, Jan. 2019.
- [52] C. Kottke, C. Schmidt, R. Freund, and V. Jungnickel, "Bandwidth extension techniques for high-speed access networks (invited)," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*. OSA, Mar. 2018.

- [53] A. Oppenheim and R. Schaffer, *Discrete-Time Signal Processing*. Pearson, 2013.
- [54] J. Mitola, "The software radio architecture," *IEEE Communications Magazine*, vol. 33, no. 5, pp. 26–38, 1995.
- [55] J. Mitola and G. Q. Maguire Jr, "Cognitive radio: making software radios more personal," *IEEE Personal Communications*, vol. 6, no. 4, pp. 13–18, 1999.
- [56] S. Balasubramanian, S. Boumaiza, H. Sarbishaie, T. Quach, P. Orlando, J. Volakis, G. Creech, J. Wilson, and W. Khalil, "Ultimate transmission," *IEEE Microwave Magazine*, vol. 13, no. 1, pp. 64–82, 2012.
- [57] Huawei Technologies Corporation, Ltd. (2018) OptiX OSN 8800 product specifications. [Online]. Available: <https://e.huawei.com/en/products/fixed-network/transport/wdm/osn-8800> (Accessed 2018-07-25).
- [58] D. J. Richardson, "Silicon photonics: Beating the electronics bottleneck," *Nature Photonics*, vol. 3, no. 10, pp. 562–564, 2009.
- [59] F. Maloberti, *Data converters*, 1st ed. Springer, 2007.
- [60] J. de la Rosa and R. Rio, *CMOS Sigma-Delta Converters: Practical Design Guide*. Wiley, 2013.
- [61] F. Gerfers and M. Ortmanns, *Continuous-Time Sigma-Delta A/D Conversion: Fundamentals, Performance Limits and Robust Implementations*, ser. Springer Series in Advanced Microelectronics. Springer, 2006.
- [62] M. Gustavsson, J. J. Wikner, and N. Tan, *CMOS data converters for communications*. Springer Science & Business Media, 2000.
- [63] A. Rodríguez-Vázquez, F. Medeiro, and E. Janssens, *CMOS Telecom Data Converters*. Springer Science & Business Media, 2013.
- [64] X. Chen, S. Chandrasekhar, P. Pupaiaikis, and P. Winzer, "Fast DAC solutions for future high symbol rate systems," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2017.
- [65] K. Schuh, F. Buchali, W. Idler, Q. Hu, W. Templ, A. Bielik, L. Altenhain, H. Langenhagen, J. Rupeter, U. Dümmler, T. Ellermeyer, R. Schmid, and M. Möller, "100 GSa/s BiCMOS DAC supporting 400 Gb/s dual channel transmission," in *Proc. of European Conference on Optical Communication (ECOC)*. VDE, Sep. 2016, pp. 37–39.
- [66] T. Ellermeyer, R. Schmid, A. Bielik, J. Rupeter, and M. Möller, "DA and AD converters in SiGe technology: Speed and resolution for ultra high data rate applications," in *Proc. of European Conference and Exhibition on Optical Communication (ECOC)*. IEEE, 2010.

- [67] J. Cao, D. Cui, A. Nazemi, T. He, G. Li, B. Catli, M. Khanpour, K. Hu, T. Ali, H. Zhang, H. Yu, B. Rhew, S. Sheng, Y. Shim, B. Zhang, and A. Momtaz, "A transmitter and receiver for 100Gb/s coherent networks with integrated 4x64GS/s 8b ADCs and DACs in 20nm CMOS," in *Proc. of International Solid-State Circuits Conference (ISSCC)*. IEEE, Feb. 2017.
- [68] C. Laperle and M. OSullivan, "Advances in high-speed DACs, ADCs, and DSP for optical coherent transceivers," *Journal of Lightwave Technology*, vol. 32, no. 4, pp. 629–643, Feb. 2014.
- [69] Keysight Technologies, "M8196A 92 GS/s arbitrary waveform generator data sheet," Oct. 2017. [Online]. Available: <https://literature.cdn.keysight.com/litweb/pdf/5992-0971EN.pdf> (Accessed 2018-07-25).
- [70] Anritsu, "64 Gbaud PAM4 DAC G0374A product introduction," online, Oct. 2016. [Online]. Available: <https://www.anritsu.com/en-GB/test-measurement/support/downloads/product-introductions/dwl18014> (Accessed 2018-07-25).
- [71] SHF AG, "SHF 614B datasheet," Apr. 2016. [Online]. Available: https://www.shf.de/wp-content/uploads/datasheets/datasheet_shf_614_b.pdf (Accessed 2018-07-25).
- [72] Tektronix, "AWG70000A series datasheet," May 2018. [Online]. Available: <http://download.tek.com/datasheet/AWG70000A-Arbitrary-Waveform-Generator-Datasheet-76W283808.pdf> (Accessed 2018-07-25).
- [73] Q. Ye, Y. Zhang, X. Li, and Y. Zhang, "A 12-bit 10GSps ultra high speed DAC in InP HBT technology," in *International Conference on Integrated Circuits and Microsystems (ICICM)*, IEEE. IEEE, Nov. 2017.
- [74] G. Engel, M. Clara, H. Zhu, and P. Wilkins, "A 16-bit 10Gsps current steering RF DAC in 65nm CMOS achieving 65dBc ACLR multi-carrier performance at 4.5GHz Fout," in *Proc. of Symposium on VLSI Circuits*. IEEE, Jun. 2015.
- [75] M. Alavi, *Radio frequency digital to analog converters : implementation in nanoscale CMOS*. Elsevier, 2017.
- [76] S. Balasubramanian and W. Khalil, "Architectural trends in GHz speed DACs," in *Prof. of NORCHIP*. IEEE, 2012, pp. 1–4.
- [77] A. Konczykowska, F. Jorge, J. Dupuy, M. Riet, V. Nodjiadjim, H. Aubry, and A. Adamiecki, "84 GBd (168 Gbit/s) PAM-4 3.7 Vpp power DAC in InP DHBT for short reach and long haul optical networks," *Electronics Letters*, vol. 51, no. 20, pp. 1591–1593, 2015.
- [78] A. Konczykowska, J.-Y. Dupuy, F. Jorge, M. Riet, V. Nodjiadjim, and H. Mardoyan, "Extreme speed power-DAC: Leveraging InP DHBT for ultimate

- capacity single-carrier optical transmissions,” *Journal of Lightwave Technology*, vol. 36, no. 2, pp. 401–407, Jan. 2018.
- [79] H. Huang, J. Heilmeyer, M. Grözing, M. Berroth, J. Leibrich, and W. Rosenkranz, “An 8-bit 100-GS/s distributed DAC in 28-nm CMOS for optical communications,” *IEEE Transactions on Microwave Theory and Techniques*, vol. 63, no. 4, pp. 1211–1218, Apr. 2015.
- [80] M. Nagatani, H. Wakita, H. Nosaka, K. Kurishima, M. Ida, A. Sano, and Y. Miyamoto, “75 GBd InP-HBT MUX-DAC module for high-symbol-rate optical transmission,” *Electronics Letters*, vol. 51, no. 9, pp. 710–712, 2015.
- [81] Y. Arayashiki, M. Ikeda, and Y. Amano, “80 GBd 6-bit DAC in InP DHBT for arbitrary waveform generator,” *Electronics Letters*, vol. 52, no. 23, pp. 1937–1938, Nov. 2016.
- [82] IDRS, “International roadmap for devices and systems,” IDRS, Tech. Rep., 2017. [Online]. Available: <https://irds.ieee.org/roadmap-2017> (Accessed 2018-07-25).
- [83] R. A. Camillo-Castillo, Q. Z. Liu, J. W. Adkisson, M. H. Khater, P. B. Gray, V. Jain, R. K. Leidy, J. J. Pekarik, J. P. Gambino, B. Zetterlund, C. Willets, C. Parrish, S. U. Engelmann, A. M. Pyzyna, P. Cheng, and D. L. Harame, “SiGe HBTs in 90nm BiCMOS technology demonstrating 300GHz/420GHz $fT/fMAX$ through reduced Rb and C cb parasitics,” in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, 2013, pp. 227–230.
- [84] GlobalFoundries, “SiGe 9HP 90nm SiGe BiCMOS technology product brief,” online, 2017. [Online]. Available: <https://www.globalfoundries.com/sites/default/files/product-briefs/sige-9hp-product-brief.pdf> (Accessed 2018-07-25).
- [85] V. Jain, T. Kessler, B. J. Gross, J. J. Pekarik, P. Candra, P. B. Gray, B. Sadhu, A. Valdes-Garcia, P. Cheng, R. A. Camillo-Castillo, K. Newton, A. Natarajan, S. K. Reynolds, and D. L. Harame, “Device and circuit performance of SiGe HBTs in 130nm BiCMOS process with $fT/fMAX$ of 250/330GHz,” in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, 2014, pp. 96–99.
- [86] GlobalFoundries, “SiGe 8XP & SiGe 8HP 130 nm SiGe BiCMOS technologies product brief,” online, 2017. [Online]. Available: <https://www.globalfoundries.com/sites/default/files/product-briefs/130nm-sige-performance-brief.pdf> (Accessed 2018-07-25).
- [87] H. Rücker, B. Heinemann, and A. Fox, “Half-Terahertz SiGe BiCMOS technology,” in *Proc. of Topical Meeting on Silicon Monolithic Integrated Circuits in RF Systems (SiRF)*. IEEE, 2012, pp. 133–136.

- [88] B. Heinemann, H. Rucker, R. Barth, F. Barwolf, J. Drews, G. G. Fischer, A. Fox, O. Fursenko, T. Grabolla, F. Herzel, J. Katzer, J. Korn, A. Kruger, P. Kulse, T. Lenke, M. Lisker, S. Marschmeyer, A. Scheit, D. Schmidt, J. Schmidt, M. A. Schubert, A. Trusch, C. Wipf, and D. Wolansky, "SiGe HBT with $f_t/f_{\max}$ of 505 GHz/720 GHz," in *Proc. of International Electron Devices Meeting (IEDM)*, IEEE. IEEE, Dec. 2016.
- [89] J. Böck, K. Aufinger, S. Boguth, C. Dahl, H. Knapp, W. Liebl, D. Manger, T. Meister, A. Pribil, J. Wursthorn *et al.*, "SiGe HBT and BiCMOS process integration optimization within the DOTSEVEN project," in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, 2015, pp. 121–124.
- [90] P. Chevalier, G. Avenier, G. Ribes, A. Montagné, E. Canderle, D. Céli, N. Derrier, C. Deglise, C. Durand, T. Quémerais *et al.*, "A 55 nm triple gate oxide 9 metal layers SiGe BiCMOS technology featuring 320 GHz f_T /370 GHz $f_{\max}$ HBT and high-Q millimeter-wave passives," in *Proc. of International Electron Devices Meeting (IEDM)*. IEEE, 2014, pp. 3–9.
- [91] P. Chevalier, C. Fellous, L. Rubaldo, F. Pourchon, S. Pruvost, R. Beerkens, F. Saguin, N. Zerounian, B. Barbalat, S. Lepilliet, D. Dutartre, D. Celi, I. Telliez, D. Gloria, F. Aniel, F. Danneville, and A. Chantre, "230-GHz self-aligned SiGeC HBT for optical and millimeter-wave applications," *IEEE Journal of Solid-State Circuits*, vol. 40, no. 10, pp. 2025–2034, Oct. 2005.
- [92] W. Khalil, J. Wilson, B. Dupaix, S. Balasubramanian, and G. Creech, "Toward millimeter-wave DACs: Challenges and opportunities," in *Proc. of Compound Semiconductor Integrated Circuit Symposium (CSICS)*. IEEE, Oct. 2012.
- [93] U. Seckin and C.-K. K. Yang, "A comprehensive delay model for CMOS CML circuits," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 55, no. 9, pp. 2608–2618, 2008.
- [94] N. Deferm and P. Reynaert, *CMOS Front Ends for Millimeter Wave Wireless Communication Systems*. Springer International Publishing, 2015.
- [95] Y. Cao, "Modeling of interconnect parasitics," in *Predictive Technology Model for Robust Nanoelectronic Design*. Springer, 2011, pp. 81–103.
- [96] R. J. Baker, *CMOS: Circuit Design, Layout, and Simulation*, 3rd ed., ser. IEEE Press Series on Microelectronic Systems. Wiley, 2011.
- [97] N. Planes, O. Weber, V. Barral, S. Haendler, D. Noblet, D. Croain, M. Bocat, P.-O. Sassoulas, X. Federspiel, A. Cros, A. Bajolet, E. Richard, B. Dumont, P. Perreau, D. Petit, D. Golanski, C. Fenouillet-Beranger, N. Guillot, M. Rafik, V. Huard, S. Puget, X. Montagner, M.-A. Jaud, O. Rozeau, O. Saxod,

- F. Wacquant, F. Monsieur, D. Barge, L. Pinzelli, M. Mellier, F. Boeuf, F. Arnaud, and M. Haond, "28nm FDSOI technology platform for high-speed low-voltage digital applications," in *Proc. of Symposium on VLSI Technology*. IEEE, Jun. 2012.
- [98] H. J. Lee, S. Rami, S. Ravikumar, V. Neeli, K. Phoa, B. Sell, and Y. Zhang, "Intel 22nm FinFET (22FFL) process technology for RF and mm wave applications and circuit design optimization for FinFET technology," in *Proceedings of International Electron Devices Meeting (IEDM)*, Dec. 2018.
- [99] W. Sansen, "Analog CMOS from 5 micrometer to 5 nanometer," in *Proc. of International Solid-State Circuits Conference (ISSCC)*. IEEE, Feb. 2015.
- [100] V. Teppati, S. Tirelli, R. Lövbom, R. Flückiger, M. Alexandrova, and C. R. Bolognesi, "Accuracy of microwave transistor f_T and f_{MAX} extractions," *IEEE Transactions on Electron Devices*, vol. 61, no. 4, pp. 984–990, 2014.
- [101] O. Esame, Y. Gurbuz, I. Tekin, and A. Bozkurt, "Performance comparison of state-of-the-art heterojunction bipolar devices (HBT) based on Al-GaAs/GaAs, Si/SiGe and InGaAs/InP," *Microelectronics Journal*, vol. 35, no. 11, pp. 901–908, Nov. 2004.
- [102] D. J. Paul, "Si/SiGe heterostructures: from material and physics to devices and circuits," *Semiconductor Science and Technology*, vol. 19, no. 10, pp. R75–R108, Sep. 2004.
- [103] J. A. Del Alamo, "Nanometre-scale electronics with III–V compound semiconductors," *Nature*, vol. 479, no. 7373, pp. 317–323, 2011.
- [104] J. Ajayan and D. Nirmal, "A review of InP/InAlAs/InGaAs based transistors for high frequency applications," *Superlattices and Microstructures*, vol. 86, pp. 1–19, Oct. 2015.
- [105] C. Qu, Y. Liu, X. Liu, and Z. Zhu, "Inductance modeling of interconnections in 3-D stacked-chip packaging," *IEEE Microwave and Wireless Components Letters*, vol. 28, no. 4, pp. 281–283, Apr. 2018.
- [106] M. Lisker, A. Trusch, A. Kruger, M. Fraschke, P. Kulse, Y. Borokhovych, B. Tillack, I. Ostermay, T. Kramer, A. Thies, O. Kruger, F.-J. Schmuckle, V. Krozer, and W. Heinrich, "InP-si BiCMOS heterointegration using a substrate transfer process," *ECS Journal of Solid State Science and Technology*, vol. 3, no. 2, pp. 17–20, nov 2014.
- [107] M. Urteaga, Z. Griffith, M. Seo, J. Hacker, and M. J. Rodwell, "InP HBT technologies for THz integrated circuits," *Proceedings of the IEEE*, vol. 105, no. 6, pp. 1051–1067, 2017.

- [108] J. Godin, V. Nodjiadjim, M. Riet, P. Berdaguer, O. Drisse, E. Derouin, A. Konczykowska, J. Moulu, J.-Y. Dupuy, F. Jorge, J.-L. Gentner, A. Scaennec, T. Johansen, and V. Krozer, "Submicron InP DHBT technology for high-speed high-swing mixed-signal ICs," in *Proc. of IEEE Compound Semiconductor Integrated Circuits Symposium (CSICS)*. IEEE, Oct. 2008.
- [109] A. Konczykowska, "InP HBT technology activities in Europe," in *Proc. of International Conference on Indium Phosphide and Related Materials*, Berlin, Germany, May 2011.
- [110] C. R. Bolognesi, R. Flückiger, M. Alexandrova, W. Quan, R. Lövblom, and O. Ostinelli, "InP/GaAsSb DHBTs for THz applications and improved extraction of their cutoff frequencies," in *Proc. of International Electron Devices Meeting (IEDM)*. IEEE, 2016, pp. 29–5.
- [111] T. Kraemer, M. Rudolph, F. J. Schmueckle, J. Wuerfl, and G. Traenkle, "InP DHBT process in transferred-substrate technology with f_t and f_{max} over 400 GHz," *IEEE Transactions on Electron Devices*, vol. 56, no. 9, pp. 1897–1903, Sep. 2009.
- [112] T. Shivan, N. Weimann, M. Hossain, T. Johansen, D. Stoppel, S. Schulz, O. Ostinelli, R. Doerner, C. Bolognesi, V. Krozer, and W. Heinrich, "A 95 GHz bandwidth 12 dBm output power distributed amplifier in InP-DHBT technology for optoelectronic applications," in *Proc. of German Microwave Conference (GeMiC)*. IEEE, Mar. 2018.
- [113] R. Driad, R. Makon, V. Hurm, F. Benkhelifa, R. Losch, J. Rosenzweig, and M. Schlechtweg, "InP-based DHBT technology for high-speed mixed signal and digital applications," in *Proc. of International Conference on Indium Phosphide & Related Materials*, IEEE. IEEE, May 2009.
- [114] GCS, "InP HBT technologies," online, 2013. [Online]. Available: http://www.gcsincorp.com/dedicated_pure-play_wafer_foundry/InP%20HBT%20Technologies.php (Accessed 2018-07-25).
- [115] Y. Yang, D. Rasbot, C.-H. Chen, B. Lee, W. Yau, D. Hou, and D. Wang, "A foundry-ready ultra high f_t InP/InGaAs DHBT technology," in *Proc. of CS MANTECH Conference*, Portland, Oregon, USA, May 2010, pp. 55–58.
- [116] Northrop Grumman, "Microelectronics products & services foundry services GaAs, InP and GaN technologies flyer," online, 2015. [Online]. Available: <http://www.northropgrumman.com/BusinessVentures/Microelectronics/Documents/pageDocs/FoundryFlysheet.pdf> (Accessed 2018-07-25).
- [117] V. Radisic, D. W. Scott, C. Monier, S. Wang, A. Cavus, A. Gutierrez-Aitken, and W. Deal, "InP HBT transferred substrate amplifiers operating to 600 GHz," in *Proc. of International Microwave Symposium (IMS)*. IEEE, 2015.

- [118] M. Urteaga, R. Pierson, P. Rowell, V. Jain, E. Lobisser, and M. Rodwell, "130nm InP DHBTs with $f_t > 0.52$ THz and $f_{max} > 1.1$ THz," in *Proc. of Device Research Conference (DRC)*. IEEE, 2011, pp. 281–282.
- [119] J. C. Rode, H.-W. Chiang, P. Choudhary, V. Jain, B. J. Thibeault, W. J. Mitchell, M. J. Rodwell, M. Urteaga, D. Loubychev, A. Snyder *et al.*, "Indium phosphide heterobipolar transistor technology beyond 1-THz bandwidth," *IEEE Transactions on Electron Devices*, vol. 62, no. 9, pp. 2779–2785, 2015.
- [120] A. Leuther, A. Tessmann, I. Kallfass, H. Massler, R. Loesch, M. Schlechtweg, M. Mikulla, and O. Ambacher, "Metamorphic HEMT technology for submillimeter-wave MMIC applications," in *Proc. of International Conference on Indium Phosphide and Related Materials (IPRM)*. IEEE, May 2010.
- [121] R. Lai, X. B. Mei, W. R. Deal, W. Yoshida, Y. M. Kim, P. H. Liu, J. Lee, J. Uyeda, V. Radisic, M. Lange, T. Gaier, L. Samoska, and A. Fung, "Sub 50 nm InP HEMT device with f_{max} greater than 1 THz," in *Proc. of International Electron Devices Meeting (IEDM)*, IEEE. IEEE, 2007.
- [122] X. Mei, W. Yoshida, M. Lange, J. Lee, J. Zhou, P.-H. Liu, K. Leong, A. Zamora, J. Padilla, S. Sarkozy *et al.*, "First demonstration of amplification at 1 THz using 25-nm InP high electron mobility transistor process," *IEEE Electron Device Letters*, vol. 36, no. 4, pp. 327–329, 2015.
- [123] S. Shin, H. Jiang, W. Ahn, H. Wu, W. Chung, P. D. Ye, and M. A. Alam, "Performance potential of Ge CMOS technology from a material-device-circuit perspective," *IEEE Transactions on Electron Devices*, vol. 65, no. 5, pp. 1679–1684, May 2018.
- [124] V. Stojanović, R. J. Ram, M. Popović, S. Lin, S. Moazeni, M. Wade, C. Sun, L. Alloatti, A. Atabaki, F. Pavanello, N. Mehta, and P. Bhargava, "Monolithic silicon-photonics platforms in state-of-the-art CMOS SOI processes (invited)," *Optics Express*, vol. 26, no. 10, pp. 13 106–13 121, May 2018.
- [125] D. Thomson, A. Zilkie, J. E. Bowers, T. Komljenovic, G. T. Reed, L. Vivien, D. Marris-Morini, E. Cassan, L. Viot, J.-M. Fédéli *et al.*, "Roadmap on silicon photonics," *Journal of Optics*, vol. 18, no. 7, p. 073003, 2016.
- [126] M. Smit, X. Leijtens, H. Ambrosius, E. Bente, J. van der Tol, B. Smalbrugge, T. de Vries, E.-J. Geluk, J. Bolk, R. van Veldhoven, L. Augustin, P. Thijs, D. D'Agostino, H. Rabbani, K. Lawniczuk, S. Stopinski, S. Tahvili, A. Corradi, E. Kleijn, D. Dzibrou, M. Felicetti, E. Bitincka, V. Moskalenko, J. Zhao, R. Santos, G. Gilardi, W. Yao, K. Williams, P. Stabile, P. Kuindersma, J. Pello, S. Bhat, Y. Jiao, D. Heiss, G. Roelkens, M. Wale, P. Firth, F. Soares, N. Grote, M. Schell, H. Debregeas, M. Achouche, J.-L. Gentner, A. Bakker,

- T. Korthorst, D. Gallagher, A. Dabbs, A. Melloni, F. Morichetti, D. Melati, A. Wonfor, R. Penty, R. Broeke, B. Musk, and D. Robbins, "An introduction to InP-based generic integration technology," *Semiconductor Science and Technology*, vol. 29, no. 8, p. 083001, Jun. 2014.
- [127] K. Doris, A. H. van Roermund, and D. M. Leenaerts, *Wide-bandwidth high dynamic range D/A converters*. Springer, 2006.
- [128] Analog Devices Inc., *Linear Circuit Design Handbook*, 1st ed. Newnes, 2008.
- [129] K. Kammeyer, *Nachrichtenübertragung*, 5th ed. Springer Vieweg, 2011.
- [130] R. F. H. Fischer, *Precoding and Signal Shaping for Digital Transmission*. John Wiley & Sons, Inc., Jul. 2002.
- [131] I. N. Bronshtein and K. A. Semendyayev, *Handbook of mathematics*. Springer Science & Business Media, 2013.
- [132] S. Balasubramanian, G. Creech, J. Wilson, S. Yoder, J. McCue, M. Verhelst, and W. Khalil, "Systematic analysis of interleaved digital-to-analog converters," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 58, no. 12, pp. 882–886, Dec. 2011.
- [133] Maxim Integrated, "Equalizing techniques flatten DAC frequency response," Maxim Integrated, Application Note 3853, Aug. 2012. [Online]. Available: <http://pdfserv.maximintegrated.com/en/an/AN3853.pdf> (Accessed 2018-07-25).
- [134] Keysight Technologies, "The ABC's of arbitrary waveform generation," Keysight Technologies, Application Note, 2014. [Online]. Available: <https://literature.cdn.keysight.com/litweb/pdf/5989-4138EN.pdf> (Accessed 2018-07-25).
- [135] A. Balteanu, P. Schvan, and S. P. Voinigescu, "A 6-bit segmented RZ DAC architecture with up to 50-GHz sampling clock and 4 V_{pp} differential swing," in *Proc. of International Microwave Symposium (IMS)*. IEEE, Jun. 2012.
- [136] S. W. Smith, *The Scientist and Engineer's Guide to Digital Signal Processing*, 1st ed. California Technical Publishing, 1997.
- [137] S. Balasubramanian, "Studies on high-speed digital-to-analog conversion," Ph.D. dissertation, Ohio State University, 2013.
- [138] Texas Instruments, "Understanding data converters," Texas Instruments, Application Report, 1999. [Online]. Available: <https://www.ti.com/lit/an/slaa013/slaa013.pdf> (Accessed 2018-07-25).
- [139] S. Luschas and H. S. Lee, "Output impedance requirements for DACs," in *Proc. of International Symposium on Circuits and Systems (ISCAS)*, May 2003.

- [140] P. Hendriks, "Specifying communications DACs," *IEEE Spectrum*, vol. 34, no. 7, pp. 58–69, 1997.
- [141] W. Kester, "Evaluating high speed DAC performance," Analog Devices, Tutorial MT-013, 2009. [Online]. Available: <http://www.analog.com/media/en/training-seminars/tutorials/MT-013.pdf> (Accessed 2018-07-25).
- [142] A. Van den Bosch, M. Steyaert, and W. Sansen, "Solving static and dynamic performance limitations for high speed D/A converters," in *Analog Circuit Design*. Springer, 2003, pp. 189–210.
- [143] T. Chen and G. G. Gielen, "The analysis and improvement of a current-steering DACs dynamic SFDR-I: The cell-dependent delay differences," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 53, no. 1, pp. 3–15, 2006.
- [144] G. Chen, H. Chen, M. Haurylau, N. A. Nelson, D. H. Albonesi, P. M. Fauchet, and E. G. Friedman, "Predictions of CMOS compatible on-chip optical interconnect," *Integration, the VLSI Journal*, vol. 40, no. 4, pp. 434–446, Jul. 2007.
- [145] F. Jondral, *Software Radio - Adaptivität durch Parametrisierung*. J. Schlembach Fachverlag, 2002.
- [146] R. M. Gray, "Quantization noise spectra," *IEEE Transactions on Information Theory*, vol. 36, no. 6, pp. 1220–1244, 1990.
- [147] A. Suárez, S. Sancho, S. Ver Hoeye, and J. Portilla, "Analytical comparison between time-and frequency-domain techniques for phase-noise analysis," *IEEE Transactions on Microwave Theory and Techniques*, vol. 50, no. 10, pp. 2353–2361, 2002.
- [148] A. Hajimiri and T. H. Lee, "A general theory of phase noise in electrical oscillators," *IEEE Journal of Solid-State Circuits*, vol. 33, no. 2, pp. 179–194, 1998.
- [149] T. H. Lee and A. Hajimiri, "Oscillator phase noise: A tutorial," *IEEE Journal of Solid-State Circuits*, vol. 35, no. 3, pp. 326–336, 2000.
- [150] A. Demir, A. Mehrotra, and J. Roychowdhury, "Phase noise in oscillators: A unifying theory and numerical methods for characterization," *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, vol. 47, no. 5, pp. 655–674, 2000.
- [151] V. Manassewitsch, *Frequency Synthesizers: Theory and Design*. New York, NY, USA: John Wiley & Sons, 1987.
- [152] W. H. Tranter, K. S. Shanmugan, T. S. Rappaport, and K. L. Kosbar, *Principles of communication systems simulation with wireless applications*. Prentice Hall New Jersey, 2004.

- [153] S. Erb and W. Pribyl, "Comparison of jitter decomposition methods for BER analysis of high-speed serial links," in *Proc. of International Symposium on Design and Diagnostics of Electronic Circuits and Systems (DDECS)*. IEEE, 2010, pp. 370–375.
- [154] Q. Dou and J. A. Abraham, "Jitter decomposition in high-speed communication systems," in *Proc. of European Test Symposium*. IEEE, 2008, pp. 157–162.
- [155] Maxim Integrated, "An introduction to jitter in communications systems," Maxim Integrated, Application Note 1916, Mar. 2003. [Online]. Available: <http://pdfserv.maximintegrated.com/en/an/AN1916.pdf> (Accessed 2018-07-25).
- [156] F. M. Gardner, *Phaselock techniques*. John Wiley & Sons, 2005.
- [157] W. Kester, "Converting oscillator phase noise to time jitter," Analog Devices, Tutorial MT-008, 2009. [Online]. Available: <http://www.analog.com/media/en/training-seminars/tutorials/MT-008.pdf> (Accessed 2018-10-28).
- [158] B.-G. Goldberg, "The effects of clock jitter on data conversion devices," *RF Design*, vol. 25, pp. 26–32, Aug. 2002.
- [159] N. Kurosawa, H. Kobayashi, H. Kogure, T. Komuro, and H. Sakayori, "Sampling clock jitter effects in digital-to-analog converters," *Measurement*, vol. 31, no. 3, pp. 187–199, 2002.
- [160] W. Kester, "Aperture time, aperture jitter, aperture delay time - removing the confusion," Analog Devices, Tutorial MT-007, 2005. [Online]. Available: <http://www.analog.com/media/en/training-seminars/tutorials/MT-007.pdf> (Accessed 2018-07-30).
- [161] J. Bergeron, "Analyzing and managing the impact of supply noise and clock jitter on high speed DAC phase noise," *Analog Dialogue*, vol. 51, no. 3, pp. 1–6, Mar. 2017. [Online]. Available: <http://www.analog.com/media/en/analog-dialogue/volume-51/number-1/articles/analyzing-and-managing-the-impact-of-supply-noise-and-clock-jitter-on-high-speed-dac-phase-noise.pdf> (Accessed 2018-07-30).
- [162] V. F. Kroupa, "Jitter and phase noise in frequency dividers," *IEEE Transactions on Instrumentation and Measurement*, vol. 50, no. 5, pp. 1241–1243, 2001.
- [163] S. Levantino, L. Romanò, S. Pellerano, C. Samori, and A. L. Lacaita, "Phase noise in digital frequency dividers," *IEEE Journal of Solid-State Circuits*, vol. 39, no. 5, pp. 775–784, 2004.
- [164] V. Kroupa, "Noise properties of PLL systems," *IEEE Transactions on Communications*, vol. 30, no. 10, pp. 2244–2252, 1982.

- [165] E. Ojefors, B. Heinemann, and U. R. Pfeiffer, "Active 220-and 325-GHz frequency multiplier chains in an SiGe HBT technology," *IEEE Transactions on Microwave Theory and Techniques*, vol. 59, no. 5, pp. 1311–1318, 2011.
- [166] M. Driscoll and T. Merrell, "Spectral performance of frequency multipliers and dividers," in *Proc. of Frequency Control Symposium*. IEEE, 1992, pp. 193–200.
- [167] R. C. van de Beek, E. A. Klumperink, C. S. Vaucher, and B. Nauta, "Low-jitter clock multiplication: A comparison between PLLs and DLLs," *IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing*, vol. 49, no. 8, pp. 555–566, 2002.
- [168] U. Tietze and C. Schenk, *Halbleiter-Schaltungstechnik*. Springer, 1999.
- [169] J. B. Johnson, "Thermal agitation of electricity in conductors," *Physical Review*, vol. 32, no. 1, pp. 97–109, 1928.
- [170] H. Nyquist, "Thermal agitation of electric charge in conductors," *Physical Review*, vol. 32, no. 1, pp. 110–113, 1928.
- [171] Fraunhofer HHI, "70 GS/s arbitrary waveform generator flyer," 2013. [Online]. Available: https://www.hhi.fraunhofer.de/fileadmin/PDF/PN/MAI/flyer_70awg_01.pdf (Accessed 2018-07-30).
- [172] H. Yamazaki, M. Nagatani, S. Kanazawa, H. Nosaka, T. Hashimoto, A. Sano, and Y. Miyamoto, "Digital-preprocessed analog-multiplexed DAC for ultra-wideband multilevel transmitter," *Journal of Lightwave Technology*, vol. 34, no. 7, pp. 1579–1584, Apr. 2016.
- [173] H. Yamazaki, M. Nagatani, F. Hamaoka, S. Kanazawa, H. Nosaka, T. Hashimoto, and Y. Miyamoto, "300-Gbps discrete multi-tone transmission using digital-preprocessed analog-multiplexed DAC with halved clock frequency and suppressed image," in *Proc. of European Conference on Optical Communication (ECOC)*, 2016.
- [174] D. Ferenci, M. Grözing, and M. Berroth, "A 25 GHz analog multiplexer for a 50 GS/s D/A-conversion system in InP DHBT technology," in *Proc. of Compound Semiconductor Integrated Circuit Symposium (CSICS)*, Oct. 2011.
- [175] M. Nagatani, H. Wakita, H. Yamazaki, M. Mutoh, M. Ida, Y. Miyamoto, and H. Nosaka, "An over-110-GHz-bandwidth 2:1 analog multiplexer in 0.25- μm InP DHBT technology," in *Proc. of International Microwave Symposium (IMS)*, June 2018, pp. 655–658.
- [176] X. Chen, S. Chandrasekhar, S. Randel, G. Raybon, A. Adamiecki, P. Pupalais, and P. J. Winzer, "All-electronic 100-GHz bandwidth digital-to-analog

- converter generating PAM signals up to 190 GBaud,” *Journal of Lightwave Technology*, vol. 35, no. 3, pp. 411–417, feb 2017.
- [177] A. Thiede, *Integrierte Hochfrequenzschaltkreise: Grundlagen des computer-gestützten Entwurfs*. Springer, 2013.
- [178] U. Rohde and M. Rudolph, *RF/Microwave Circuit Design for Wireless Applications*. Wiley, 2013.
- [179] Y.-J. Hwang, C.-H. Lien, H. Wang, M. W. Sinclair, R. G. Gough, H. Kano-niuk, and T.-H. Chu, “A 78-114 GHz monolithic subharmonically pumped GaAs-based HEMT diode mixer,” *IEEE Microwave and Wireless Components Letters*, vol. 12, no. 6, pp. 209–211, Jun. 2002.
- [180] M. Gavell, M. Fern Dahl, H. Zirath, M. Urteaga, and R. Pierson, “A fundamental upconverting Gilbert mixer for 100 GHz wireless applications,” in *Proc. of Compound Semiconductor Integrated Circuit Symposium (CSICS)*. IEEE, 2010.
- [181] SAGE Millimeter, “SFU-10-N1 data sheet,” 2015. [Online]. Available: <https://www.sagemillimeter.com/content/datasheets/SFU-10-N1.pdf> (Accessed 2018-07-30).
- [182] SAGE Millimeter, “SFB-65310410-1010KF-N1 data sheet,” 2018. [Online]. Available: <https://www.sagemillimeter.com/content/datasheets/SFB-65310410-1010KF-N1.pdf> (Accessed 2018-07-30).
- [183] B. Thomas, A. Maestrini, and G. Beaudin, “A low-noise fixed-tuned 300-360-GHz sub-harmonic mixer using planar Schottky diodes,” *IEEE Microwave and Wireless Components Letters*, vol. 15, no. 12, pp. 865–867, Dec. 2005.
- [184] I. Kallfass, P. Harati, I. Dan, J. Antes, F. Boes, S. Rey, T. Merkle, S. Wagner, H. Massler, A. Tessmann, and A. Leuther, “MMIC chipset for 300 GHz indoor wireless communication,” in *Proc. of International Conference on Microwaves, Communications, Antennas and Electronic Systems (COMCAS)*. IEEE, Nov. 2015.
- [185] F. Marki and C. Marki, “Mixer basics primer - a tutorial for RF & microwave mixers,” Marki Microwave, Tech. Rep., 2010. [Online]. Available: http://www.markimicrowave.com/assets/appnotes/mixer_basics_primer.pdf (Accessed 2018-07-30).
- [186] F. Ellinger, *Radio Frequency Integrated Circuits and Technologies*. Springer, 2007.
- [187] I. A. Glover, S. Pennock, and P. Shepherd, *Microwave devices, circuits and subsystems for communications engineering*. John Wiley & Sons, 2006.

- [188] Marki Microwave, “ML1-1850 data sheet,” Nov. 2014. [Online]. Available: <http://www.markimicrowave.com/Assets/DataSheets/ml1-1850.pdf> (Accessed 2018-07-30).
- [189] Marki Microwave, “MM1-2567LS data sheet,” Nov. 2016. [Online]. Available: <http://www.markimicrowave.com/Assets/DataSheets/mm1-2567LS.pdf> (Accessed 2018-07-30).
- [190] G. D. Vendelin, A. M. Pavio, and U. L. Rohde, *Microwave circuit design using linear and nonlinear techniques*. John Wiley & Sons, 2005.
- [191] F. Giannini and G. Leuzzi, *Nonlinear microwave circuit design*. John Wiley & Sons, 2004.
- [192] B. C. Henderson, “Mixers in microwave systems (part 2),” *Watkins-Johnson Company Tech-Notes*, vol. 17, no. 2, 1990.
- [193] D. Misra, *Radio-frequency and microwave communication circuits: analysis and design*. John Wiley & Sons, 2004.
- [194] B. C. Henderson, “Predicting intermodulation suppression in double-balanced mixers,” *RF and Microwave Designer’s Handbook*, pp. 495–501, 1993.
- [195] B. C. Henderson, “Mixers in microwave systems (part 1),” *Watkins-Johnson Company Tech-Notes*, vol. 17, no. 1, 1990.
- [196] D. Regev, “Characterization of spurious-response suppression in double-balanced mixers,” *IEEE Transactions on Microwave Theory and Techniques*, vol. 38, no. 2, pp. 123–128, 1990.
- [197] E. Rubiola, “Tutorial on the double balanced mixer,” *ArXiv Physics e-prints arXiv:physics/0608211*, Aug. 2006.
- [198] Marki Microwave, “T3-1040 data sheet,” Dec. 2013. [Online]. Available: <http://www.markimicrowave.com/Assets/datasheets/T3-1040.pdf> (Accessed 2018-07-30).
- [199] IEEE, “IEEE standard for microwave filter definitions,” *IEEE Std 1549-2011*, pp. 1–22, May 2012.
- [200] Teledyne, “TMS filter facts types,” Teledyne, Tech. Rep., 2016. [Online]. Available: <http://www.teledynemicrowave.com/teledyne-microwave-marketing/marcom-cch/teledyne-microwave-brochures/110-tms-filter-facts-types-brochure> (Accessed 2018-07-30).
- [201] R. Rehner, “Ultra-breitbandige Filter, Multiplexer und Mischer für den Aufbau hochintegrierter Millimeterwellen-Empfangssysteme,” Ph.D. dissertation, Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), 2009.

- [202] E. de Assuncao, L. de Menezes, and H. Abdalla, "Microwave multiplexers using complementary triplexer filters," in *Proc. of International Microwave and Optoelectronics Conference*. IEEE, 1999, pp. 169–173.
- [203] J. Malherbe, "Microwave transmission line filters," *Artech House*, 1979.
- [204] H. Meng and K.-L. Wu, "Direct optimal synthesis of a microwave bandpass filter with general loading effect," *IEEE Transactions on Microwave Theory and Techniques*, vol. 61, no. 7, pp. 2566–2573, 2013.
- [205] I. Ashiq and A. Khanna, "Ultra-broadband contiguous planar DC-35–65 GHz diplexer using softboard suspended stripline technology," in *Proc. of International Microwave Symposium (IMS)*. IEEE, 2013.
- [206] I. Ashiq and A. Khanna, "A novel ultra-broadband DC-36-to-66-GHz hybrid diplexer using waveguide and SSL technology," in *Proc. of European Microwave Conference (EuMC)*. IEEE, 2014, pp. 1111–1114.
- [207] I. Ashiq and A. Khanna, "A novel planar contiguous diplexer DC-67-100 GHz using organic liquid crystal polymer (LCP)," in *Proc. of International Microwave Symposium (IMS)*. IEEE, 2015.
- [208] J. Hoffman, J.-R. Martin-Gosse, S. Shopov, J. J. Pekarik, R. Camillo-Castillo, V. Jain, D. Harame, and S. P. Voinigescu, "Analog circuit blocks for 80-GHz bandwidth frequency-interleaved, linear, large-swing front-ends," *IEEE Journal of Solid-State Circuits*, vol. 51, no. 9, pp. 1985–1993, 2016.
- [209] F. Yuan, *CMOS current-mode circuits for data communications*, ser. Analog circuits and signal processing series. Springer Science & Business Media, 2007.
- [210] J. Yu, J. Zhang, Z. Jia, X. Li, H. C. Chien, Y. Cai, F. Li, Y. Wang, and X. Xiao, "Transmission of 8×128.8 GBaud single-carrier PDM-QPSK signal over 2800-km EDFA-only SMF-28 link," in *Proc. of European Conference on Optical Communication (ECOC)*. IEEE, 2015.
- [211] C. Uhl, H. Hettrich, and M. Möller, "A 100 Gbit/s 2 Vpp power multiplexer in SiGe BiCMOS technology for directly driving a monolithically integrated plasmonic MZM in a silicon photonics transmitter," in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, 2017, pp. 106–109.
- [212] G. Raybon, A. Adamiecki, J. Cho, P. Winzer, A. Konczykowska, F. Jorge, J. Dupuy, M. Riet, B. Duval, K. Kim *et al.*, "Single-carrier all-ETDM 1.08-Terabit/s line rate PDM-64-QAM transmitter using a high-speed 3-bit multiplexing DAC," in *Proc. of IEEE Photonics Conference (IPC)*. IEEE, 2015.

- [213] R. Marston, *CMOS Circuits Manual*, 1st ed. London: Heinemann, 1987.
- [214] A. K. Maini, *Digital electronics: principles, devices and applications*. John Wiley & Sons, 2007.
- [215] A. Godse and U. Bakshi, *Digital And Linear Integrated Circuits*. Technical Publications, 2009.
- [216] A. Tanabe, M. Umetani, I. Fujiwara, T. Ogura, K. Kataoka, M. Okihara, H. Sakuraba, T. Endoh, and F. Masuoka, "0.18- μm CMOS 10-Gb/s multiplexer/demultiplexer ICs using current mode logic with tolerance to threshold voltage fluctuation," *IEEE Journal of Solid-State Circuits*, vol. 36, no. 6, pp. 988–996, Jun. 2001.
- [217] M. Nagatani, H. Yamazaki, H. Wakita, H. Nosaka, K. Kurishima, M. Ida, A. Sano, and Y. Miyamoto, "A 50-GHz-bandwidth InP-HBT analog-MUX module for high-symbol-rate optical communications systems," in *Proc. of International Microwave Symposium (IMS)*. IEEE, May 2016.
- [218] M. Nagatani, H. Wakita, H. Yamazaki, H. Nosaka, K. Kurishima, M. Ida, and Y. Miyamoto, "A 128-GS/s 63-GHz-bandwidth InP-HBT-based analog-MUX module for ultra-broadband D/A conversion subsystem," in *Proc. of International Microwave Symposium (IMS)*, IEEE. IEEE, Jun. 2017, pp. 134–136.
- [219] E. Olieman, A.-J. Annema, and B. Nauta, "An interleaved full Nyquist high-speed DAC technique," *IEEE Journal of Solid-State Circuits*, vol. 50, no. 3, pp. 704–713, 2015.
- [220] H. Okawara, "Analysis of pseudo-interleaving AWG," in *Proc. of International Conference on Test*, Austin, TX, USA, 2005.
- [221] C. Krall, C. Vogel, and K. Witrisal, "Time-interleaved digital-to-analog converters for UWB signal generation," in *Proc. of International Conference on Ultra-Wideband*. IEEE, Sep. 2007, pp. 366–371.
- [222] A. Jha and P. R. Kinget, "Wideband signal synthesis using interleaved partial-order hold current-mode digital-to-analog converters," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 55, no. 11, pp. 1109–1113, Nov. 2008.
- [223] C. Vogel, "The impact of combined channel mismatch effects in time-interleaved ADCs," *IEEE Transactions on Instrumentation and Measurement*, vol. 54, no. 1, pp. 415–427, 2005.
- [224] N. Kurosawa, H. Kobayashi, K. Maruyama, H. Sugawara, and K. Kobayashi, "Explicit analysis of channel mismatch effects in time-interleaved ADC systems," *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, vol. 48, no. 3, pp. 261–271, 2001.

- [225] D. Fu, K. C. Dyer, S. H. Lewis, and P. J. Hurst, "A digital background calibration technique for time-interleaved analog-to-digital converters," *IEEE Journal of Solid-State Circuits*, vol. 33, no. 12, pp. 1904–1911, 1998.
- [226] M. Seo, M. J. Rodwell, and U. Madhow, "Comprehensive digital correction of mismatch errors for a 400-Msamples/s 80-dB SFDR time-interleaved analog-to-digital converter," *IEEE Transactions on Microwave Theory and Techniques*, vol. 53, no. 3, pp. 1072–1082, 2005.
- [227] Y. C. Lim, Y.-X. Zou, J. W. Lee, and S.-C. Chan, "Time-interleaved analog-to-digital-converter compensation using multichannel filters," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 56, no. 10, pp. 2234–2247, 2009.
- [228] S. Saleem and C. Vogel, "Adaptive compensation of frequency response mismatches in high-resolution time-interleaved ADCs using a low-resolution ADC and a time-varying filter," in *Proc. of International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2010, pp. 561–564.
- [229] S. Mendel and C. Vogel, "A compensation method for magnitude response mismatches in two-channel time-interleaved analog-to-digital converters," in *International Conference on Electronics, Circuits and Systems (ICECS)*. IEEE, 2006, pp. 712–715.
- [230] M. Khafaji, M. Jazayerifar, K. Jamshidi, and F. Ellinger, "Optically-assisted time-interleaving digital-to-analogue converters," in *Proc. of Photonics Society Summer Topical Meeting Series*. IEEE, 2016, pp. 216–217.
- [231] H. Hettrich, R. Schmid, L. Altenhain, J. Wurtele, and M. Moller, "A linear active combiner enabling an interleaved 200 GS/s DAC with 44 GHz analog bandwidth," in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, Oct. 2017.
- [232] J. Deveugele, P. Palmers, and M. S. J. Steyaert, "Parallel-path digital-to-analog converters for Nyquist signal generation," *IEEE Journal of Solid-State Circuits*, vol. 39, no. 7, pp. 1073–1082, Jul. 2004.
- [233] G. I. Radulov, P. J. Quinn, P. Harpe, H. Hegt, and A. van Roermund, "Parallel current-steering D/A converters for flexibility and smartness," in *Proc. of International Symposium on Circuits and Systems (ISCAS)*, May 2007, pp. 1465–1468.
- [234] P. T. M. van Zeijl and M. Collados, "On the attenuation of DAC aliases through multiphase clocking," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 56, no. 3, pp. 190–194, Mar. 2009.
- [235] S. Balasubramanian and W. Khalil, "Direct digital-to-RF digital-to-analogue converter using image replica and nonlinearity cancelling architecture," *Electronics Letters*, vol. 46, no. 14, pp. 1030–1032, Jul. 2010.

- [236] K. Hsia, "DAC348x device configuration and synchronization," Texas Instruments, Tech. Rep., Apr. 2013. [Online]. Available: <http://www.ti.com/lit/an/slaa584/slaa584.pdf> (Accessed 2018-07-31).
- [237] Maxim Integrated, "Synchronizing multiple high-speed multiplexed DACs for transmit applications," Maxim Integrated, Application Note 3901, Sep. 2006. [Online]. Available: <http://pdfserv.maximintegrated.com/en/an/AN3901.pdf> (Accessed 2018-07-25).
- [238] O. Weikert and U. Zolzer, "Efficient MIMO channel estimation with optimal training sequences," in *Proc. of Workshop on Commercial MIMO-Components and Systems*, 2007.
- [239] N. Benvenuto and G. Cherubini, *Algorithms for Communications Systems and Their Applications*. Wiley, 2002.
- [240] E. W. Weisstein, "Complex multiplication," From MathWorld—A Wolfram Web Resource. [Online]. Available: <http://mathworld.wolfram.com/ComplexMultiplication.html> (Accessed 2018-10-01).
- [241] P. Duhamel, "Algorithms meeting the lower bounds on the multiplicative complexity of length- 2^n DFTs and their connection with practical algorithms," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 38, no. 9, pp. 1504–1511, 1990.
- [242] S. Bouguezal, M. O. Ahmad, and M. S. Swamy, "Arithmetic complexity of the split-radix FFT algorithms," in *Proc. of International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. IEEE, 2005.
- [243] M. Frigo and S. G. Johnson, "The design and implementation of FFTW3," *Proceedings of the IEEE*, vol. 93, no. 2, pp. 216–231, 2005.
- [244] P. Duhamel, "Implementation of "split-radix" FFT algorithms for complex, real, and real-symmetric data," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 34, no. 2, pp. 285–295, 1986.
- [245] P. Duhamel and M. Vetterli, "Fast Fourier transforms: a tutorial review and a state of the art," *Signal processing*, vol. 19, no. 4, pp. 259–299, 1990.
- [246] M. Garrido, J. Grajal, M. Sanchez, and O. Gustafsson, "Pipelined radix- 2^k feedforward FFT architectures," *IEEE Transactions on Very Large Scale Integration Systems*, vol. 21, no. 1, pp. 23–32, 2013.
- [247] T. Haustein, A. Forck, H. Gäbler, V. Jungnickel, and S. Schiffermüller, "Real-time signal processing for multiantenna systems: algorithms, optimization, and implementation on an experimental test-bed," *EURASIP Journal on Advances in Signal Processing*, vol. 2006, no. 1, p. 027573, Jan. 2006.
- [248] M. H. Jones, *A Practical Introduction to Electronic Circuits*, 3rd ed. Cambridge University Press, 1995.

- [249] P. Zielonka, "Analyse des Multiplexing Digital-Analog-Wandlers für optische Kommunikationssysteme," Master's thesis, Technische Universität Berlin, 2017.
- [250] P. Babu, "Spectral analysis of nonuniformly sampled data and applications," Ph.D. dissertation, Uppsala universitet, 2012.
- [251] R. D. Nowak, "Penalized least squares estimation of Volterra filters and higher order statistics," *IEEE Transactions on Signal Processing*, vol. 46, no. 2, pp. 419–428, 1998.
- [252] F. Giri and E.-W. Bai, *Block-oriented nonlinear system identification*, 1st ed. Springer, 2010.
- [253] Marki Microwave, "FLP-4300 data sheet," May 2011. [Online]. Available: <https://www.markimicrowave.com/Assets/datasheets/FLP-4300.pdf> (Accessed 2018-10-08).
- [254] R. Schmogrow, B. Nebendahl, M. Winter, A. Josten, D. Hillerkuss, S. Koenig, J. Meyer, M. Dreschmann, M. Huebner, C. Koos, J. Becker, W. Freude, and J. Leuthold, "Error vector magnitude as a performance measure for advanced modulation formats," *IEEE Photonics Technology Letters*, vol. 24, no. 1, pp. 61–63, Jan. 2012.
- [255] H. Mahmoud and H. Arslan, "Error vector magnitude to SNR conversion for nondata-aided receivers," *IEEE Transactions on Wireless Communications*, vol. 8, no. 5, pp. 2694–2704, May 2009.
- [256] F. Xiong, *Digital Modulation Techniques*, 2nd ed. Norwood, MA, USA: Artech House, Inc., 2006.
- [257] Agilent Technologies, "E8257D PSG microwave analog signal generator data sheet," 2009. [Online]. Available: <https://literature.cdn.keysight.com/litweb/pdf/5989-0698EN.pdf?id=484534> (Accessed 2018-07-31).
- [258] SHF AG, "SHF 613A datasheet," Sep. 2015. [Online]. Available: https://www.shf.de/wp-content/uploads/datasheets/datasheet_shf_613_a.pdf (Accessed 2018-07-31).
- [259] H. Dai, Y. Li, and G. Shen, "Explore maximal potential capacity of WDM optical networks using time domain hybrid modulation technique," *Journal of Lightwave Technology*, vol. 33, no. 18, pp. 3815–3826, 2015.
- [260] P. J. Pupalaikis, "High speed arbitrary waveform generator," US Patent 7,535,394, May 19, 2009.
- [261] G. Ding, C. Dehollain, M. Declercq, and K. Azadet, "Frequency-interleaving technique for high-speed A/D conversion," in *Proc. of International Symposium on Circuits and Systems (ISCAS)*, May 2003.

- [262] P. Pupalaikis, "Digital bandwidth interleaving," LeCroy Corporation, White Paper, Apr. 2010. [Online]. Available: http://cdn.teledynelecroy.com/files/whitepapers/dbi_explained_15april10.pdf (Accessed 2018-07-31).
- [263] LeCroy Corporation, "The interleaving process in digital bandwidth interleaving (DBI) scopes," LeCroy Corporation, White Paper, Dec. 2009. [Online]. Available: http://cdn.teledynelecroy.com/files/whitepapers/interleaving_process_in_dbi_scopes.pdf (Accessed 2018-07-31).
- [264] X. Chen, S. Chandrasekhar, S. Randel, G. Raybon, A. Adamiecki, P. Pupalaikis, and P. Winzer, "All-electronic 100-GHz bandwidth digital-to-analog converter generating PAM signals up to 190-GBaud," in *Proc. of Optical Fiber Communications Conference and Exposition (OFC) Postdeadline Papers*, OSA, OSA, 2016, pp. Th5C-5.
- [265] C. Kottke, C. Schmidt, K. Habel, and V. Jungnickel, "178 Gb/s short-range optical transmission based on OFDM, electrical up-conversion and signal combining," in *Proc. of European Conference on Optical Communication (ECOC)*. VDE, Sep. 2016, pp. 866-868.
- [266] C. Kottke, K. Habel, C. Schmidt, and V. Jungnickel, "154.9 Gb/s OFDM transmission using IM-DD, electrical IQ-mixing and signal combining," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*. OSA, Mar. 2016, p. Th3C.4.
- [267] C. Kottke, C. Schmidt, K. Habel, V. Jungnickel, and R. Freund, "Performance of single-and multi-carrier modulation with additional spectral up-conversion for wideband IM/DD transmission," in *Proc. of ITG Symposium on Photonic Networks*. Leipzig, Germany: VDE, 2017.
- [268] X. Chen, S. Chandrasekhar, P. Winzer, P. Pupalaikis, I. Ashiq, A. Khanna, A. Steffan, and A. Umbach, "180-GBaud Nyquist shaped optical QPSK generation based on a 240-GSa/s 100-GHz analog bandwidth DAC," in *Proc. of Asia Communications and Photonics Conference (APC)*. OSA, 2016, pp. AS4A-1.
- [269] M. Sichma, S. Bieder, and A. Czyliw, "A 40 GHz arbitrary waveform generator by frequency multiplexing," in *Proc. of International Conference on OFDM and Frequency Domain Techniques (ICOF)*. Essen, Germany: VDE, Oct. 2016.
- [270] J. J. McCue, B. Dupaix, L. Duncan, B. Mathieu, S. McDonnell, V. J. Patel, T. Quach, and W. Khalil, "A time-interleaved multimode RF-DAC for direct digital-to-RF synthesis," *IEEE Journal of Solid-State Circuits*, vol. 51, no. 5, pp. 1109-1124, May 2016.

- [271] A. Bhide and A. Alvandpour, "An 11 GS/s 1.1 GHz bandwidth interleaved $\Delta \Sigma$ DAC for 60 GHz radio in 65 nm CMOS," *IEEE Journal of Solid-State Circuits*, vol. 50, no. 10, pp. 2306–2318, Oct. 2015.
- [272] C. Erdmann, E. Cullen, D. Brouard, R. Pelliconi, B. Verbruggen, J. Mcgrath, D. Collins, M. D. L. Torre, P. Gay, P. Lynch, P. Lim, A. Collins, and B. Farley, "A 330mW 14b 6.8GS/s dual-mode RF DAC in 16nm FinFET achieving -70.8dBc ACPR in a 20MHz channel at 5.2GHz," in *Proc. of International Solid-State Circuits Conference (ISSCC)*. IEEE, Feb. 2017.
- [273] F. Pancialdi, G. Vitetta, R. Kalbasi, N. Al-Dhahir, M. Uysal, and H. Mheidat, "Single-carrier frequency domain equalization," *Signal Processing Magazine, IEEE*, vol. 25, no. 5, pp. 37–56, Sep. 2008.
- [274] D. Bertsimas and J. N. Tsitsiklis, *Introduction to linear optimization*. Athena Scientific, 1997.
- [275] D. P. Bertsekas, *Nonlinear programming*. Athena scientific Belmont, 1999.
- [276] P. Belotti, C. Kirches, S. Leyffer, J. Linderoth, J. Luedtke, and A. Mahajan, "Mixed-integer nonlinear optimization," *Acta Numerica*, vol. 22, pp. 1–131, 2013.
- [277] J. Lee and S. Leyffer, *Mixed Integer Nonlinear Programming*, ser. The IMA Volumes in Mathematics and its Applications, J. Lee and S. Leyffer, Eds. Springer Science & Business Media, 2012, vol. 154.
- [278] T. Koch, "Rapid mathematical prototyping," Ph.D. dissertation, Technische Universität Berlin, 2004.
- [279] T. Achterberg, "SCIP: Solving constraint integer programs," *Mathematical Programming Computation*, vol. 1, no. 1, pp. 1–41, Jul. 2009.
- [280] S. Vigerske and A. Gleixner, "SCIP: global optimization of mixed-integer nonlinear programs in a branch-and-cut framework," *Optimization Methods and Software*, vol. 33, no. 3, pp. 563–593, Jun. 2017.
- [281] S. V. Vaseghi, *Advanced Digital Signal Processing and Noise Reduction*, 4th ed. John Wiley & Sons, Ltd, 2008.
- [282] A. Neubauer, *DFT - Diskrete Fouriertransformation*, 1st ed. Vieweg & Teubner Verlag, 2012.
- [283] R. M. Rao and B. Daneshrad, "Analog impairments in MIMO-OFDM systems," *IEEE Transactions on Wireless Communications*, vol. 5, no. 12, pp. 3382–3387, 2006.
- [284] J. C. Santamarina and D. Fratta, *Discrete signals and inverse problems: an introduction for engineers and scientists*. John Wiley & Sons, 2005.

- [285] P. J. Pupalais, "Group delay and its impact on serial data transmission and testing," in *Proc. of DesignCon*, 2006.
- [286] N. Damera-Venkata, B. L. Evans, and S. R. McCaslin, "Design of optimal minimum-phase digital FIR filters using discrete hilbert transforms," *IEEE Transactions on Signal processing*, vol. 48, no. 5, pp. 1491–1495, 2000.
- [287] P. Pupalais, "Method of crossover region phase correction when summing signals in multiple frequency bands," US Patent 7,711,510, May 4, 2010.
- [288] P. J. Pupalais, B. Yamrone, R. Delbue, A. S. Khanna, K. Doshi, B. Bhat, and A. Sureka, "Technologies for very high bandwidth real-time oscilloscopes," in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, 2014, pp. 128–135.
- [289] S. Haykin, *Adaptive filter theory*. Prentice-Hall, Inc., 1986.
- [290] G. B. Giannakis and E. Serpedin, "A bibliography on nonlinear system identification," *Signal Processing*, vol. 81, no. 3, pp. 533–580, Mar. 2001.
- [291] T. Koh and E. Powers, "Second-order volterra filtering and its application to nonlinear system identification," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 33, no. 6, pp. 1445–1455, 1985.
- [292] A. Napoli, P. W. Berenguer, T. Rahman, G. Khanna, M. M. Mezghanni, L. Gardian, E. Riccardi, A. C. Piat, S. Calabrò, S. Dris *et al.*, "Digital pre-compensation techniques enabling high-capacity bandwidth variable transponders," *Optics Communications*, vol. 409, pp. 52–65, 2018.
- [293] W. Liu, J. C. Principe, and S. Haykin, *Kernel adaptive filtering: a comprehensive introduction*. John Wiley & Sons, 2011.
- [294] H. Witschnig, T. Mayer, A. Springer, A. Koppler, L. Maurer, M. Huemer, and R. Weigel, "A different look on cyclic prefix for SC/FDE," in *Proc. of International Symposium on Personal, Indoor and Mobile Radio Communications*. IEEE, Sep. 2002, pp. 824–828.
- [295] D. Falconer, S. Ariyavisitakul, A. Benyamin-Seeyar, and B. Eidson, "Frequency domain equalization for single-carrier broadband wireless systems," *IEEE Communications Magazine*, vol. 40, no. 4, pp. 58–66, Apr 2002.
- [296] G. Kang, M. Weckerle, E. Costa, and P. Zhang, "Time and frequency domain joint channel estimation in multi-branch systems," in *Proc. of International Conference on Applied Electromagnetics, Wireless and Optical Communications*. Corfu, Greece: WSEAS, 2005, pp. 25–30.
- [297] Z. Cheng and D. Dahlhaus, "Time versus frequency domain channel estimation for OFDM systems with antenna arrays," in *Proc. of International Conference on Signal Processing*, Aug. 2002, pp. 1340–1343.

- [298] R. M. Rao and B. Daneshhrad, "I/Q mismatch cancellation for MIMO-OFDM systems," in *Proc. of International Symposium on Personal, Indoor and Mobile Radio Communications*. IEEE, 2004, pp. 2710–2714.
- [299] A. Tarighat, R. Bagheri, and A. Sayed, "Compensation schemes and performance analysis of IQ imbalances in OFDM receivers," *IEEE Transactions on Signal Processing*, vol. 53, no. 8, pp. 3257–3268, Aug 2005.
- [300] D. Root, J. Verspecht, J. Horn, and M. Marcu, *X-Parameters: Characterization, Modeling, and Design of Nonlinear RF and Microwave Components*, ser. Cambridge RF and Microwave Engineering Series. Cambridge University Press, 2013.
- [301] H. Minn and N. Al-Dhahir, "Optimal training signals for MIMO OFDM channel estimation," *IEEE Transactions on Wireless Communications*, vol. 5, no. 5, pp. 1158–1168, May 2006.
- [302] W. Shieh and I. Djordjevic, *OFDM for optical communications*. Amsterdam: Elsevier, 2010.
- [303] A. Gatherer and M. Polley, "Controlling clipping probability in DMT transmission," in *Proc. of Asilomar Conference on Signals, Systems and Computers*. IEEE, 1997.
- [304] X. Wang, T. T. Tjhung, and C. Ng, "Reduction of peak-to-average power ratio of OFDM system using a companding technique," *IEEE Transactions on Broadcasting*, vol. 45, no. 3, pp. 303–307, 1999.
- [305] T. Jiang, Y. Yang, and Y.-H. Song, "Exponential companding technique for PAPR reduction in OFDM systems," *IEEE Transactions on Broadcasting*, vol. 51, no. 2, pp. 244–248, 2005.
- [306] J. Armstrong, "Peak-to-average power reduction for OFDM by repeated clipping and frequency domain filtering," *Electronics letters*, vol. 38, no. 5, pp. 246–247, 2002.
- [307] S.-J. Heo, H.-S. Noh, J.-S. No, and D.-J. Shin, "A modified SLM scheme with low complexity for PAPR reduction of OFDM systems," in *Proc. of International Symposium on Personal, Indoor and Mobile Radio Communications*. IEEE, 2007.
- [308] R. A. Casas, F. L. de Victoria, I. Fijalkow, P. Schniter, and T. J. Endrea, "On MMSE fractionally-spaced equalizer design," in *Proc. of International Conference on Digital Signal Processing*, Jul. 1997, pp. 395–398.
- [309] R. Hassun, M. Flaherty, R. Matreci, and M. Taylor, "Effective evaluation of link quality using error vector magnitude techniques," in *Proc. of Wireless Communications Conference*. IEEE, 1997, pp. 89–94.

- [310] G. Tzimpragos, C. Kachris, I. B. Djordjevic, M. Cvijetic, D. Soudris, and I. Tomkos, "A survey on FEC codes for 100G and beyond optical networks," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 209–221, 2016.
- [311] P. Chow, J. Cioffi, and J. Bingham, "A practical discrete multitone transceiver loading algorithm for data transmission over spectrally shaped channels," *IEEE Transactions on Communications*, vol. 43, no. 2/3/4, pp. 773–775, feb 1995.
- [312] C. Kottke, C. Caspar, V. Jungnickel, R. Freund, M. Agustin, J. R. Kropp, and N. N. Ledentsov, "High-speed DMT and VCSEL-based MMF transmission using pre-distortion," *Journal of Lightwave Technology*, vol. 36, no. 2, pp. 168–174, Jan. 2018. [Online]. Available: <http://jlt.osa.org/abstract.cfm?URI=jlt-36-2-168>
- [313] S. Egami, "Nonlinear, linear analysis and computer-aided design of resistive mixers," *IEEE Transactions on Microwave Theory and Techniques*, vol. 22, no. 3, pp. 270–275, 1974.
- [314] A. Napoli, M. M. Mezghanni, T. Rahman, D. Rafique, R. Palmer, B. Spinnler, S. Calabrò, C. Castro, M. Kuschnerov, and M. Bohn, "Digital compensation of bandwidth limitations for high-speed DACs and ADCs," *Journal of Lightwave Technology*, vol. 34, no. 13, pp. 3053–3064, Jul. 2016.
- [315] S. Cundiff and A. Weiner, "Optical arbitrary waveform generation," *Nature Photonics*, vol. 4, no. 11, pp. 760–766, 2010.
- [316] A. M. Weiner, "Ultrafast optical pulse shaping: A tutorial review," *Optics Communications*, vol. 284, no. 15, pp. 3669–3692, 2011.
- [317] A. Leven, J. Lin, P. Kondratko, U. Koc, J. Lee, K.-Y. Tu, Y. Baeyens, and Y. Chen, "An interferometer-based optoelectronic arbitrary waveform generator," in *Proc. of International Topical Meeting on Microwave Photonics*. IEEE, 2005, pp. 73–76.
- [318] A. Leven, Y. Yang, R. Kopf, A. Tate, T. Hu, J. Frackoviak, R. Reyes, N. Weimann, Y. Chen, R. DeSalvo *et al.*, "High speed arbitrary waveform generation and processing using a photonic digital-to-analog converter," in *Proc. of IEEE LEOS Summer Topicals*. IEEE, Jul. 2007, pp. 174–175.
- [319] D. Geisler, N. Fontaine, R. Scott, and S. Yoo, "Demonstration of a flexible bandwidth optical transmitter/receiver system scalable to terahertz bandwidths," *IEEE Photonics Journal*, vol. 3, no. 6, pp. 1013–1022, Dec. 2011.
- [320] D. J. Geisler, N. Fontaine, R. P. Scott, L. Paraschis, O. Gerstel, and S. J. B. Yoo, "Flexible bandwidth arbitrary modulation format, coherent optical transmission system scalable to terahertz BW," in *Proc. of European Conference and Exposition on Optical Communications (ECOC)*. OSA, 2011.

- [321] N. Fontaine, R. Scott, D. Geisler, and S. Yoo, "Dynamic optical arbitrary waveform generation and measurement to increase the flexibility, fidelity and bandwidth of optical networks," in *Proc. of International Symposium on Communication Systems, Networks Digital Signal Processing (CSNDSP)*, Jul. 2012.
- [322] R. Rios-Müller, J. Renaudier, P. Brindel, H. Mardoyan, P. Jennevé, L. Schmalen, and G. Charlet, "1-Terabit/s net data-rate transceiver based on single-carrier Nyquist-shaped 124 GBaud PDM-32QAM," in *Proc. of Optical Fiber Communication Conference (OFC) Post Deadline Papers*. OSA, 2015.
- [323] S. T. Cundiff, J. Ye, and J. L. Hall, "Optical frequency synthesis based on mode-locked lasers," *Review of Scientific Instruments*, vol. 72, no. 10, pp. 3749–3771, 2001.
- [324] J. Pfeifle, V. Brasch, M. Lauermaun, Y. Yu, D. Wegner, T. Herr, K. Hartinger, P. Schindler, J. Li, D. Hillerkuss *et al.*, "Coherent terabit communications with microresonator kerr frequency combs," *Nature photonics*, 2014.
- [325] J. J. Wikner and N. Tan, "Modeling of CMOS digital-to-analog converters for telecommunication," *IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing*, vol. 46, no. 5, pp. 489–499, 1999.
- [326] I. Myderrizi and A. Zeki, "Current-steering digital-to-analog converters: functional specifications, design basics, and behavioral modeling," *IEEE Antennas and Propagation Magazine*, vol. 52, no. 4, pp. 197–208, 2010.
- [327] K. Andersson and J. J. Wikner, "Characterization of a CMOS current-steering DAC using state-space models," in *Proc. of Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2000, pp. 668–671.
- [328] M. C. Jeruchim, P. Balaban, and K. S. Shanmugan, *Simulation of communication systems: modeling, methodology and techniques*. Springer Science & Business Media, 2006.
- [329] E. Eskinat, S. H. Johnson, and W. L. Luyben, "Use of Hammerstein models in identification of nonlinear systems," *AIChE Journal*, vol. 37, no. 2, pp. 255–268, feb 1991.
- [330] J. He, J. S. Yang, Y. Kim, and A. S. Kim, "System-level time-domain behavioral modeling for a mobile WiMax transceiver," in *Proc. of International Behavioral Modeling and Simulation Workshop*. IEEE, 2006, pp. 138–143.
- [331] T. I. Laakso, V. Valimaki, M. Karjalainen, and U. K. Laine, "Splitting the unit delay - FIR/all pass filters design," *IEEE Signal Processing Magazine*, vol. 13, no. 1, pp. 30–60, Jan. 1996.
- [332] IEEE, "IEEE standard for terminology and test methods of digital-to-analog converter devices," *IEEE Std 1658-2011*, pp. 1–126, Feb. 2012.

- [333] D. Schreurs, M. O'Droma, A. A. Goacher, and M. Gadringer, *RF power amplifier behavioral modeling*. Cambridge University Press New York, NY, USA, 2008.
- [334] N. J. Kasdin, "Discrete simulation of colored noise and stochastic processes and $1/f^\alpha$ power law noise generation," *Proceedings of the IEEE*, vol. 83, no. 5, pp. 802–827, may 1995.
- [335] S. Bittner, S. Krone, and G. Fettweis, "Tutorial on discrete time phase noise modeling for phase locked loops," 2015. [Online]. Available: https://mns.ifn.et.tu-dresden.de/personalSites/stefan.krone/Documents/Bittner_S_PN_Tutorial.pdf (Accessed 2018-08-01).
- [336] K. S. Kundert, "Predicting the phase noise and jitter of PLL-based frequency synthesizers," The Designers Guide, Tech. Rep., Oct. 2015. [Online]. Available: <http://www.designers-guide.org/Analysis/PLLnoise+jitter.pdf> (Accessed 2018-08-01).
- [337] F. Brandonisio and M. P. Kennedy, "Modelling and estimating phase noise with Matlab," in *Noise-Shaping All-Digital Phase-Locked Loops*. Springer, 2014, pp. 143–152.
- [338] M. H. Kouider, "Investigation of electrical bandwidth interleaving for broad-band optical transmission systems," Master's thesis, Technische Universität Berlin, 2017.
- [339] A. Y. Kibangou and G. Favier, "Wiener-Hammerstein systems modeling using diagonal Volterra kernels coefficients," *IEEE Signal Processing Letters*, vol. 13, no. 6, pp. 381–384, 2006.
- [340] D. Faria, L. Dunleavy, and T. Svensen, "The use of intermodulation tables for mixer simulations," *Microwave Journal*, vol. 45, no. 4, pp. 60–69, 2002.
- [341] A. Cidronali, G. Loglio, J. Jargon, and G. Manes, "Behavioral model for reducing the complexity of mixer analysis and design," *International Journal of RF and Microwave Computer-Aided Engineering*, vol. 15, no. 4, pp. 362–370, 2005.
- [342] X. Yang, D. Chaillot, P. Roblin, W.-R. Liou, J. Lee, H.-D. Park, J. Strahler, and M. Ismail, "Poly-harmonic modeling and predistortion linearization for software-defined radio upconverters," *IEEE Transactions on Microwave Theory and Techniques*, vol. 58, no. 8, pp. 2125–2133, 2010.
- [343] J. Xu, M. C. Yagoub, R. Ding, and Q.-J. Zhang, "Neural-based dynamic modeling of nonlinear microwave circuits," *IEEE Transactions on Microwave Theory and Techniques*, vol. 50, no. 12, pp. 2769–2780, 2002.

- [344] N. B. Carvalho, J. C. Pedro, W. Jang, and M. B. Steer, "Nonlinear simulation of mixers for assessing system-level performance," *International Journal of RF and Microwave Computer-Aided Engineering*, vol. 15, no. 4, pp. 350–361, 2005.
- [345] O. Gunther, M. Hartmann, P. Wenig, and R. Weigel, "Behavioral mixer model for rapid system simulation of a 77 GHz FMCW-radar transceiver," in *Proc. of Asia-Pacific Microwave Conference (APMC)*. IEEE, 2007.
- [346] S. Joeres, H.-W. Groh, and S. Heinen, "Event driven analog modeling of RF frontends," in *Proc. of International Behavioral Modeling and Simulation Workshop*. IEEE, 2007, pp. 46–51.
- [347] J. Wood, *Behavioral modeling and linearization of RF power amplifiers*. Artech House, 2014.
- [348] Y. Egashira, Y. Tanabe, and K. Sato, "A novel IQ imbalance compensation method with pilot-signals for OFDM system," in *Proc. of Vehicular Technology Conference*, IEEE. IEEE, Sep. 2006.
- [349] J. Tubbax, B. Come, L. V. der Perre, L. Deneire, S. Donnay, and M. Engels, "Compensation of IQ imbalance in OFDM systems," in *IEEE International Conference on Communications*. IEEE, 2003.
- [350] T. Chow, *Mathematical Methods for Physicists: A Concise Introduction*. Cambridge University Press, 2000.

List of Publications

In this section, the author's publications are listed. The list is divided into two sections: the publications relevant for the PhD thesis and other publications. The publications in each section are categorized and ordered into journals, conferences, and patents. In each category, the publications are ordered chronologically.

Relevant for PhD Thesis

- [A1] C. Kottke, C. Schmidt, V. Jungnickel, and R. Freund, "Performance of bandwidth extension techniques for high-speed short-range IM/DD links," *Journal of Lightwave Technology*, vol. 37, no. 2, pp. 665–672, Jan. 2019.
- [A2] C. Schmidt, C. Kottke, V. H. Tanzil, R. Freund, V. Jungnickel, and F. Gerfers, "Digital-to-analog converters using frequency interleaving: Mathematical framework and experimental verification," *Circuits, Systems, and Signal Processing*, vol. 37, no. 11, pp. 4929–4954, Nov. 2018.
- [A3] C. Schmidt, C. Kottke, R. Freund, F. Gerfers, and V. Jungnickel, "Digital-to-analog converters for high-speed optical communications using frequency interleaving: impairments and characteristics," *Optics Express*, vol. 26, no. 6, pp. 6758–6770, Mar. 2018.
- [A4] C. Schmidt, P. Zielonka, V. Jungnickel, R. Freund, T. Tannert, M. Grözing, M. Berroth, and F. Gerfers, "Behavioral model for a high-speed 2:1 analog multiplexer," in *Proc. of International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, Aug. 2018.
- [A5] C. Schmidt, C. Kottke, R. Freund, and V. Jungnickel, "Bandwidth enhancement for an optical access link by using a frequency interleaved DAC," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*. OSA, Mar. 2018.
- [A6] C. Kottke, C. Schmidt, R. Freund, and V. Jungnickel, "Bandwidth extension techniques for high-speed access networks (invited)," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*. OSA, Mar. 2018.

- [A7] T. Tannert, X.-Q. Du, D. Widmann, M. Grözing, M. Berroth, C. Schmidt, C. Caspar, J. H. Choi, V. Jungnickel, and R. Freund, "A SiGe-HBT 2:1 analog multiplexer with more than 67 GHz bandwidth," in *Proc. of Bipolar/BiCMOS Circuits and Technology Meeting (BCTM)*. IEEE, Oct. 2017, pp. 146–149.
- [A8] C. Schmidt, C. Kottke, V. Jungnickel, and R. Freund, "High-speed digital-to-analog converter concepts (invited)," in *Proc. of SPIE Photonics West*. SPIE, Jan. 2017.
- [A9] C. Schmidt, V. H. Tanzil, C. Kottke, R. Freund, and V. Jungnickel, "Digital signal splitting among multiple DACs for analog bandwidth interleaving (ABI)," in *Proc. of International Conference on Electronics, Circuits, & Systems (ICECS)*. IEEE, Dec. 2016, pp. 245–248.
- [A10] C. Schmidt, C. Kottke, V. Jungnickel, and R. Freund, "Enhancing the bandwidth of DACs by analog bandwidth interleaving," in *Proc. of ITG Symposium on Broadband Coverage in Germany*. Berlin, Germany: VDE, Apr. 2016, pp. 99–106.
- [A11] C. Schmidt, C. Kottke, V. Jungnickel, and J. Hilt, "Signal processing systems and signal processing methods," PCT Patent PCT/EP2015/077 000, May 26, 2017. [Online]. Available: <https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2017084705>

Other

- [B12] D. Schulz, J. Hohmann, P. Hellwig, J. Hilt, C. Schmidt, R. Freund, and V. Jungnickel, "Outdoor measurements using an optical wireless link for fixed-access applications," *Journal of Lightwave Technology*, vol. 37, no. 2, pp. 634–642, Jan. 2019.
- [B13] R. Ryf, M. A. Mestre, S. Randel, C. Schmidt, A. H. Gnauck, R.-J. Essiambre, P. J. Winzer, R. Delbue, P. Pupalakis, A. Sureka, Y. Sun, X. Jiang, D. W. Peckham, A. McCurdy, and R. Lingle, "Mode-multiplexed transmission over a 209-km DGD-compensated hybrid few-mode fiber span," *IEEE Photonics Technology Letters*, vol. 24, no. 21, pp. 1965–1968, Nov. 2012.
- [B14] G. Raybon, A. L. Adamiecki, S. Randel, C. Schmidt, P. J. Winzer, A. Konczykowska, F. Jorge, J.-Y. Dupuy, L. L. Buhl, S. Chandrasekhar, X. Liu, A. H. Gnauck, C. Scholz, and R. Delbue, "All-ETDM 80-GBaud (640-Gb/s) PDM 16-QAM generation and coherent detection," *IEEE Photonics Technology Letters*, vol. 24, no. 15, pp. 1328–1330, Aug. 2012.

- [B15] C. Kottke, C. Schmidt, K. Habel, V. Jungnickel, and R. Freund, "Performance of single-and multi-carrier modulation with additional spectral up-conversion for wideband IM/DD transmission," in *Proc. of ITG Symposium on Photonic Networks*. Leipzig, Germany: VDE, 2017.
- [B16] C. Kottke, C. Schmidt, K. Habel, and V. Jungnickel, "178 Gb/s short-range optical transmission based on OFDM, electrical up-conversion and signal combining," in *Proc. of European Conference on Optical Communication (ECOC)*. VDE, Sep. 2016, pp. 866–868.
- [B17] C. Kottke, K. Habel, C. Schmidt, and V. Jungnickel, "154.9 Gb/s OFDM transmission using IM-DD, electrical IQ-mixing and signal combining," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*. OSA, Mar. 2016, p. Th3C.4.
- [B18] R. Rath, C. Schmidt, and W. Rosenkranz, "Is tomlinson-harashima precoding suitable for fiber-optic communication systems?" in *Proc. of ITG Symposium on Photonic Networks*, VDE. Leipzig, Germany: VDE, May 2013.
- [B19] R. Ryf, S. Randel, M. A. Mestre, C. Schmidt, A. H. Gnauck, R.-J. Essiambre, P. Winzer, R. Delbue, P. Pupalais, A. Sureka, Y. Sun, X. Jiang, A. H. McCurdy, D. W. Peckham, and R. Lingle, "209-km single-span mode- and wavelength-multiplexed transmission over hybrid few-mode fiber," in *Proc. of European Conference and Exhibition on Optical Communication (ECOC)*. OSA, 2012, p. Tu.1.C.1.
- [B20] R. Ryf, R.-J. Essiambre, S. Randel, M. A. Mestre, C. Schmidt, and P. Winzer, "Impulse response analysis of coupled-core 3-core fibers," in *Proc. of European Conference and Exhibition on Optical Communication (ECOC)*. OSA, 2012, p. Mo.1.F.4.
- [B21] S. Randel, C. Schmidt, R. Ryf, R.-J. Essiambre, and P. J. Winzer, "MIMO-based signal processing for mode-multiplexed transmission," in *Proc. of Photonics Society Summer Topical Meeting Series*. IEEE, Jul. 2012, pp. 181–182.
- [B22] R. Ryf, M. A. Mestre, S. Randel, C. Schmidt, A. H. Gnauck, R.-J. Essiambre, P. J. Winzer, R. Delbue, P. Pupalais, A. Sureka, Y. Sun, X. Jiang, D. W. Peckham, A. McCurdy, and R. Lingle, "Mode-multiplexed transmission over a 184-km DGD-compensated few-mode fiber span," in *Proc. of Photonics Society Summer Topical Meeting Series*. IEEE, Jul. 2012, pp. 173–174.
- [B23] S. Randel, R. Ryf, C. Schmidt, M. A. Mestre, P. J. Winzer, and R. J. Essiambre, "MIMO processing for space-division multiplexed transmission," in *Proc. of Advanced Photonics Congress*. OSA, Jun. 2012, p. SpW3B.4.
- [B24] S. Randel, A. Sierra, S. Mumtaz, A. Tulino, R. Ryf, P. Winzer, C. Schmidt, and R. Essiambre, "Adaptive MIMO signal processing for mode-division

- multiplexing,” in *Proc. of Optical Fiber Communication Conference and Exposition (OFC)*. OSA, Mar. 2012, p. OW3D.5.
- [B25] R. Ryf, M. A. Mestre, A. Gnauck, S. Randel, C. Schmidt, R. Essiambre, P. Winzer, R. Delbue, P. Pupaiaikis, A. Sureka, Y. Sun, X. Jiang, D. Peckham, A. H. McCurdy, and R. Lingle, “Low-loss mode coupler for mode-multiplexed transmission in few-mode fiber,” in *Proc. of Optical Fiber Communication Conference and Exposition (OFC)*. OSA, Mar. 2012, p. PDP5B.5.
- [B26] R. Ryf, R. Essiambre, A. Gnauck, S. Randel, M. A. Mestre, C. Schmidt, P. Winzer, R. Delbue, P. Pupaiaikis, A. Sureka, T. Hayashi, T. Taru, and T. Sasaki, “Space-division multiplexed transmission over 4200-km 3-core microstructured fiber,” in *Proc. of Optical Fiber Communication Conference and Exposition (OFC)*. OSA, Mar. 2012, p. PDP5C.2.
- [B27] S. Randel, R. Ryf, A. Gnauck, M. A. Mestre, C. Schmidt, R. Essiambre, P. Winzer, R. Delbue, P. Pupaiaikis, A. Sureka, Y. Sun, X. Jiang, and R. Lingle, “Mode-multiplexed 6×20-GBd QPSK transmission over 1200-km DGD-compensated few-mode fiber,” in *Proc. of Optical Fiber Communication Conference and Exposition (OFC)*. OSA, Mar. 2012, p. PDP5C.5.
- [B28] C. Schmidt and V. Jungnickel, “Optical communication system and method,” PCT Patent PCT/EP 2016/063 069, Dec. 14, 2017. [Online]. Available: <https://patentscope.wipo.int/search/de/detail.jsf?docId=WO2017211413>

A DAC Behavioral Model

A behavioral DAC model is required to study both the FI-DAC's and the AMUX-DAC's performance. The model should have a limited number of parameters that sufficiently depict the major characteristics of a high-speed DAC.

From a system-level perspective, a simple DAC model consists of quantization and low-pass filtering. However, such a simple model does not account for a frequency dependent ENOB. Therefore, nonlinear distortions are implemented alongside with jitter resulting from clock phase noise.

A well-fitting model depicts the actual circuit design, i.e., by simulating the individual current sources [127, 325–327]. However, information on the actual DAC design is required, which is usually not available for commercial DACs. Therefore, a generic approach is taken by implementing a two-box model [328], which consists of a static nonlinearity and a LPF, i.e., a Hammerstein model [329]. It is further extended by quantization, clock feedthrough, hold upsampling, and jitter.

This appendix is structured as follows. First, the model is introduced and explained by means of a block diagram. Second, the model is fitted to measurement data from a current high-speed DAC. Finally, the implementation of jitter and phase noise is outlined.

Block Diagram

The block diagram of the behavioral DAC model is depicted in Fig. A.1. The digital input signal s_{IN} is normalized to the peak-to-peak output amplitude of the DAC V_{pp} and quantized with a resolution of 2^b according to b bits. Furthermore, the attenuated clock signal s_{CLK} is added to the signal to account for clock feedthrough.

Then, a static nonlinear transfer function is applied. In [330], the nonlinear distortions are implemented as deviations from the ideal quantization curve by using random distributions for both DNL and INL. However, this does not provide one of the typical nonlinear transfer function shapes, i.e., bow-shape, s-shape, etc. [140]. In [325], a bow-shape nonlinearity is analytically derived for the code-dependent output impedance.

For this model, a generic shape is desired that can be fitted to various DACs. A nonlinear polynomial is utilized, whose coefficients up to the fifth order are obtained

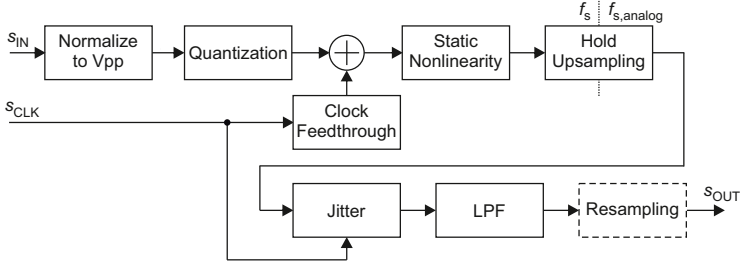


Figure A.1 Behavioral DAC model block diagram.

by first, measuring the harmonics' power levels for a sine input and second, solving the equation system presented in Appendix B.

Thereafter, the actual D/A conversion is performed by hold upsampling with an integer factor, i.e., the digital samples are duplicated according to the factor $\left\lceil \frac{f_{s,\text{analog}}}{f_s} \right\rceil$, whereby f_s , $f_{s,\text{analog}}$, and $\lceil \cdot \rceil$ denote the sampling rates of the DAC, the sampling rate of the analog simulation and the ceiling operator, respectively.

Then, timing jitter is applied to the signal. Hereby, only timing jitter common to all current sources is modeled [127] rather than individual jitter contributions from each current source, as described in Sec. 2.5. Common jitter is applied by shifting each analog sample according to the actual phase deviation of the sine clock signal. This operation is performed in the frequency domain by means of a linear phase [331]. More information on the jitter implementation is provided in the section after the next section.

An LPF accounts for the frequency-dependent DAC output signal. The frequency response can be estimated, e.g., by means of a sine wave frequency sweep [332].

Eventually, the output signal s_{OUT} is optionally resampled, if the ratio of the analog simulation's sampling rate $f_{s,\text{analog}}$ and the DAC sampling rate f_s is not integer-valued.

Model Fitting

The behavioral model is fitted with measurement data for a 28 nm Socionext CMOS DAC on an evaluation board [28, 29], which has been used for the FI-DAC experiments. The DAC has a nominal vertical resolution of 8 bit.

The reference measurement data is obtained with a SINAD measurement at 84 GS/s, which comprises a full spectrum capture for each test vector. Each test vector is a sine wave with a different frequency, which is chosen according to the regulations

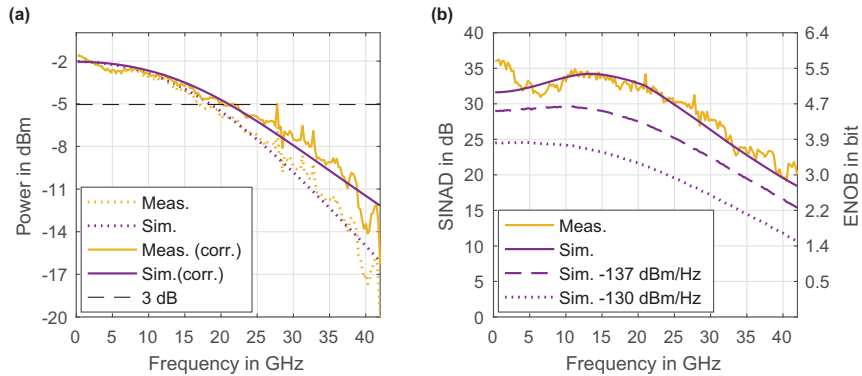


Figure A.2 Behavioral DAC model fitting: frequency response (a); SINAD and ENOB (b).

stated in the IEEE Standard for Terminology and Test Methods for DAC Devices [332]. The SINAD is defined according the definition in this standard [332] as the ratio of the RMS amplitude of the DAC filtered reconstructed output sine wave to the RMS amplitude of the output noise and distortion. The model blocks are fitted in the following order: LPF, clock-feedthrough, and static nonlinearity.

The DAC LPF's frequency response is obtained from the test frequencies' power and is approximated with a 2nd order Bessel LPF with a cutoff frequency of 21 GHz. A standard filter type is chosen to enable a variation of both the cutoff frequency and the filter order in the simulations. The fitting for the magnitude response is depicted in Fig. A.2(a) for the cases with and without sinc correction. The measured frequency response has more ripple than the Bessel filter's frequency response and a dip at around 5 GHz. Overall, a profound frequency response fitting is achieved, which worsens for frequencies > 25 GHz.

From the SINAD measurement data, the power of the fed through clock is calculated to be -37 dBm.

The nonlinear harmonics' power levels in dBc are calculated for each test frequency. The harmonics' power varies with frequency; hence, a mean value is chosen for each harmonic for the calculation of the nonlinear polynomial to achieve a well-fitted SINAD. The harmonics' power levels are given as $[34, 40, 50, 46]$ dBc for the 2nd to the 5th harmonic. The resulting polynomial coefficients are given in ascending order by $[0.00, 1.00, 0.06, -0.96, 1.62, 20.53]$, whereby the first coefficient denotes the DC offset.

In Fig. A.2(b), both the SINAD measurement and the simulation results are depicted. On the right axis, the corresponding ENOB values are displayed. The SINAD has its maximum value of about 36 dB close to DC and decreases to about 20 dB at

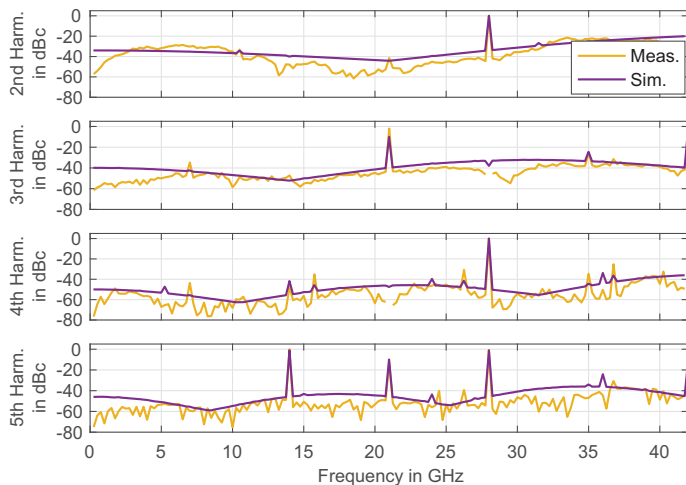


Figure A.3 Behavioral DAC model fitting: power levels of harmonics.

the highest frequency. The simulation data matches the measurement results well, except for some greater deviations between DC and 3 GHz. The dips in the intervals from 2 to 10 GHz and 32 to 40 GHz may result from the internal time-interleaved architecture of the DAC.

In Secs. 5.10.4 and 5.11, noise loading is applied at the receiver to match the experimental and the simulation results. In order to evaluate the implications on the DAC performance, the SINAD is depicted in Fig. A.2(b) with noise loading according to a PSD of -137 and -130 dBm/Hz, respectively. Due to the noise loading, the SINAD is decreased; furthermore, the increase in SINAD for frequencies between 5 and 20 GHz is depressed. An ideal amplifier was used before the noise loading, which has the same gain value as the amplifier in the first signal path in the simulations in Secs. 5.10.4 and 5.11.2.

In Fig. A.3, the harmonics' power levels are depicted. Since, only harmonics up to the 5th order are modeled, the harmonics' power in dBc is overestimated compared to the measurement to achieve a well fitted SINAD. Nonetheless, the spikes resulting from mixing with the fed through clock and from aliasing, match the measurement data very well.

In Fig. A.4, the THD and the SFDR are depicted. As for the harmonics, they do not fit the measurement data perfectly. However, the spikes match very well as before. Furthermore, a symmetrical appearance is observed with stronger degradations at both low and high frequencies. This may be attributed to the internal time-interleaved architecture of the DAC.

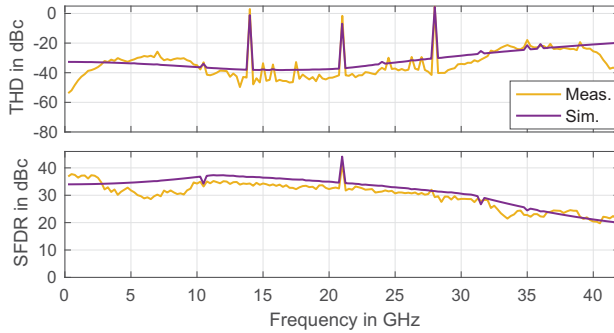


Figure A.4 Behavioral DAC model fitting: THD and SFDR.

For the measurements in this thesis, the DAC is operated single-ended. By using the differential output signal, i.e., with a balun, the output signal's quality can possibly be improved due to the canceling of even order harmonics and clock feedthrough.

Concluding, the derived model depicts sufficiently the SINAD characteristic of the measured reference data. The fitting can be further improved by using the exact frequency response rather than a standard filter's response. In order to enhance the fitting quality even further, more complex models such as memory polynomial, Volterra series, or neural networks can be utilized [328,333].

Jitter and Phase Noise

The previous SINAD fitting was based on a LPF, a static nonlinearity, and clock feedthrough. Phase noise of the DAC clock was not considered, although the behavioral DAC model supports it as depicted in Fig. A.1. In this section, the model fitting is performed again to include phase noise of the DAC clock. The phase noise spectrum relates to a certain RMS jitter according to (2.10). The behavioral model is adapted to a DAC, which has an integrated PLL, whereby its parameters are not known; hence, it is regarded as a black box.

In Fig. A.5(a), the measured SINAD values are depicted alongside with SINAD curves for different RMS jitter values according to (2.11). By assuming that the measured SINAD values between 10 and 30 GHz are mainly determined by the RMS jitter, the RMS jitter is estimated to equal 200 fs. However, since nonlinear distortions and clock feedthrough are also included in the model, the DAC clock RMS jitter is assumed less with 150 fs.

The phase noise spectrum's profile for the DAC clock signal is obtained from the data sheet of the frequency synthesizer Agilent E8257D [257]. Its magnitude is

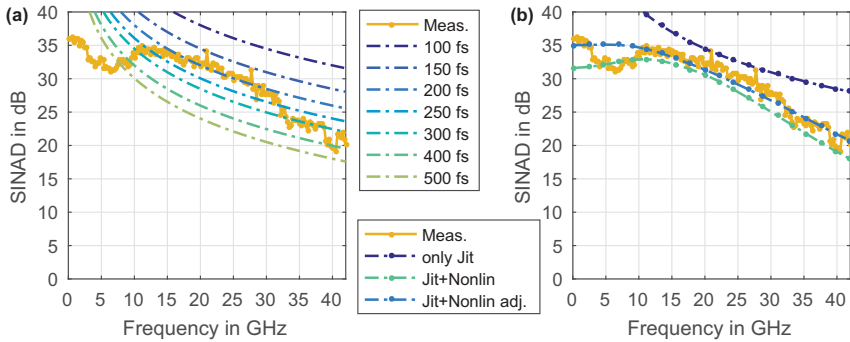


Figure A.5 Behavioral DAC model fitting: jitter fitting for measured SINAD values: theoretical SINAD curves for different RMS jitter levels (a), behavioral DAC model results (b).

shifted in the logarithmic domain to obtain the RMS jitter value of 150 fs. The time domain phase noise samples are generated by filtering white Gaussian noise in the frequency domain with the phase noise spectrum profile and an additional IFFT. The corresponding RMS jitter is calculated by integrating the phase noise spectrum in the interval 100 Hz to 300 MHz according to (2.10). The lower cutoff frequency is further limited by the spectral resolution of the digital signal. The upper frequency is related to the DAC PLL loop filter cutoff frequency. In the experiments, the DAC clock frequency is obtained by dividing the frequency synthesizer's output signal by 16. The DAC PLL is assumed to have a maximum input frequency of 3 GHz and a PLL loop filter cutoff frequency of $0.1 \cdot 3 \text{ GHz} = 300 \text{ MHz}$. Therefore, the phase noise spectrum is further filtered with a rectangular frequency domain filter to account for the PLL loop filter's bandwidth. Information on more complex modeling of oscillator and PLL phase noise can be found in [334–337].

In the experiment, the frequency synthesizer and the DACs operate with a PLL each; thus, the random number generators for the DAC clock phase noise spectrum and the LO phase noise spectrum are initialized with different seeds in the simulations.

In Fig. A.5(b), the simulated SINAD values with the behavioral DAC model are depicted next to the measured SINAD values. If only jitter limitations are active, the simulated SINAD values match the theoretical values for an RMS jitter of 150 fs depicted in Fig. A.5(a). If the LPF, the static nonlinear transfer function and the clock feedthrough are active as well, the curve is below the measured SINAD values. In order to improve the fitting, the static nonlinear characteristic's influence is reduced by setting the harmonics' power levels to [39, 45, 55, 51] dBc for the 2nd to the 5th harmonic. The resulting SINAD values provide a better fit.

B RF Mixer Behavioral Model

For the FI-DAC simulations, a simple behavioral RF mixer model is required, which sufficiently depicts the relevant mixer characteristics. It shall be parametrized based on the data sheet information of a typical high-speed RF upconversion mixer.

A typical data sheet comprises information on the frequency responses of the IF circuitry and the RF balun [188, 189]. Furthermore, port-to-port isolation values are stated and information on the nonlinear behavior is given, e.g., with an SST, in which the IMPs are listed for each LO harmonic relative to the fundamental of the upconverted IF signal for the first LO harmonic.

This appendix begins with a brief mathematical mixer description including static nonlinear distortions based on a nonlinear polynomial. Then, the behavioral model is explained by means of a block diagram. Finally, the relation between the nonlinear polynomial's coefficients and the mixer's SST is derived. The model was developed during the supervision of a master's thesis [338].

Mathematical Description

A simple mathematical description of the mixer considers only static nonlinear distortions, i.e., LO harmonics and spurious products. The output of the mixer is given by the sum of the products of the n th LO harmonic $s_{LO,n}(t)$ with the n th sub-signal $s_{OUT,n}(t)$:

$$s_{OUT}(t) = \sum_{n=0}^{N_{LO}} s_{OUT,n}(t) \cdot s_{LO,n}(t) . \quad (B.1)$$

N_{LO} LO harmonics are considered including the fundamental, whereby $n = 0$ accounts for the IF signal feedthrough.

The sub-signals $s_{OUT,n}(t)$ are obtained by applying a static nonlinear polynomial of M_P th order, which is defined by its coefficients $a_{m,n}$:

$$s_{OUT,n}(t) = \sum_{m=0}^{M_P} a_{m,n} s_{IN}^m(t) = a_{0,n} + a_{1,n} s_{IN}(t) + \dots + a_{M,n} s_{IN}^{M_P}(t) . \quad (B.2)$$

The coefficients $a_{m,n}$ are obtained from the SST; the derivation is presented in the section after the next section. A different polynomial is applied for each LO harmonic. The LO is ideally given by a cosine function according to

$$s_{\text{LO},n}(t) = \cos(2\pi n f_{\text{LO}} t) . \quad (\text{B.3})$$

Block Diagram

Next to the effects described in the previous section, the behavioral mixer model covers additional aspects. In this section, the model is explained in detail by means of a block diagram.

The block diagram is depicted in Fig. B.1; the blocks' colors white and grey denote the processing for the signal and the LO, respectively. Thick lines denote multiple parallel signals. The block diagram represents a classical three-box model consisting of a first filter, a memoryless nonlinearity and a second filter for the main signal path [328, 333]. It is an extended Wiener Hammerstein model [339].

The input signal $s_{\text{IN}}(t)$ is low-pass-filtered to account for the IF circuitry. Then, a nonlinear polynomial is applied. For each LO harmonic a different polynomial is used, which is calculated based on the conversion loss CL, the SST, and the input signal power P_{IN} [196, 340–342].

The sub-signals $s_{\text{OUT},n}(t)$ are each multiplied with the corresponding LO $s_{\text{LO},n}(t)$, and the first upconverted sub-signal $s_{\text{OUT},1}(t) \cdot s_{\text{LO},1}(t)$ is further band-pass-filtered to account for the RF circuitry. Finally, all sub-signals are added to form the output signal. For the IF-to-RF isolation, a nonlinear polynomial is applied to the input signal and the result is added to the output, which corresponds to the case $n = 0$ in (B.1).

The n th LO harmonic is generated by taking the LO input signal to the power of n and applying a rectangular BPF to filter unwanted harmonics resulting from the exponentiation. The signal is further normalized to ensure an amplitude of one for each harmonic. Then, the LO harmonics are scaled to the LO input power. Thereafter, the LO harmonics are attenuated according to the LO-to-RF isolation and are added to the output signal.

This simple model will provide sufficient insights on the impact of the mixer's nonlinear distortions on the combined FI-DAC output signal. The model can be easily modified to represent a downconversion mixer. More sophisticated models for RF mixers, e.g., neural network models, Volterra series models, models with noise sources or event-driven models can be found in [343], [344], [341, 345], or [346], respectively.

A cosine input signal $s_{\text{IN}}(t) = V_0 \cos(\omega_{\text{IN}}t)$ is used, yielding

$$\begin{aligned} s_{\text{OUT},n}(t) = & a_{0,n} + a_{1,n}V_0 \cos(\omega_{\text{IN}}t) + a_{2,n}(V_0 \cos(\omega_{\text{IN}}t))^2 \\ & + a_{3,n}(V_0 \cos(\omega_{\text{IN}}t))^3 + a_{4,n}(V_0 \cos(\omega_{\text{IN}}t))^4 \\ & + a_{5,n}(V_0 \cos(\omega_{\text{IN}}t))^5 \end{aligned} \quad (\text{B.5})$$

$$\begin{aligned} = & a_{0,n} + \frac{1}{2}V_0^2 a_{2,n} + \frac{3}{8}V_0^4 a_{4,n} \\ & + \left(V_0 a_{1,n} + \frac{3}{4}V_0^3 a_{3,n} + \frac{5}{8}V_0^5 a_{5,n} \right) \cos(\omega_{\text{IN}}t) \\ & + \left(\frac{1}{2}V_0^2 a_{2,n} + \frac{1}{2}V_0^4 a_{4,n} \right) \cos(2\omega_{\text{IN}}t) \\ & + \left(\frac{1}{4}V_0^3 a_{3,n} + \frac{5}{16}V_0^5 a_{5,n} \right) \cos(3\omega_{\text{IN}}t) \\ & + \frac{1}{8}V_0^4 a_{4,n} \cos(4\omega_{\text{IN}}t) \\ & + \frac{1}{16}V_0^5 a_{5,n} \cos(5\omega_{\text{IN}}t) . \end{aligned} \quad (\text{B.6})$$

In (B.6), the first and the second line represent the DC term and the desired input signal, respectively. The subsequent lines are the IMPs up to order $M_P = 5$. The nonlinear transfer characteristic causes self-biasing, i.e., a change of the DC component of the output signal, amplitude modulation (AM)-to-AM compression, i.e., a change of the amplitude of the desired signal, and harmonic distortions, i.e., new frequency components at multiples of the input signal frequency [347].

In order to obtain the nonlinear polynomial's coefficients $a_{m,n}$ from the SST, an equation system is solved. The equation system is obtained by calculating the power of each equation line in (B.6). Then, these powers are set equal to the power

of the corresponding spurious components of the SST. By assuming a reference impedance of one, the equation system is given by:

$$10 \frac{P_{0,n}}{10} = \frac{1}{T_{\text{IN}}} \int_0^{T_{\text{IN}}} \left(a_{0,n} + \frac{1}{2} V_0^2 a_{2,n} + \frac{3}{8} V_0^4 a_{4,n} \right)^2 dt, \quad (\text{B.7})$$

$$10 \frac{P_{1,n}}{10} = \frac{1}{T_{\text{IN}}} \int_0^{T_{\text{IN}}} \left(V_0 a_{1,n} \cos(\omega_{\text{IN}} t) + \frac{3}{4} V_0^3 a_{3,n} \cos(\omega_{\text{IN}} t) + \frac{5}{8} V_0^5 a_{5,n} \cos(\omega_{\text{IN}} t) \right)^2 dt, \quad (\text{B.8})$$

$$10 \frac{P_{2,n}}{10} = \frac{1}{T_{\text{IN}}} \int_0^{T_{\text{IN}}} \left(\frac{1}{2} V_0^2 a_{2,n} \cos(2\omega_{\text{IN}} t) + \frac{1}{2} V_0^4 a_{4,n} \cos(2\omega_{\text{IN}} t) \right)^2 dt, \quad (\text{B.9})$$

$$10 \frac{P_{3,n}}{10} = \frac{1}{T_{\text{IN}}} \int_0^{T_{\text{IN}}} \left(\frac{1}{4} V_0^3 a_{3,n} \cos(3\omega_{\text{IN}} t) + \frac{5}{16} V_0^5 a_{5,n} \cos(3\omega_{\text{IN}} t) \right)^2 dt, \quad (\text{B.10})$$

$$10 \frac{P_{4,n}}{10} = \frac{1}{T_{\text{IN}}} \int_0^{T_{\text{IN}}} \left(\frac{1}{8} V_0^4 a_{4,n} \cos(4\omega_{\text{IN}} t) \right)^2 dt, \quad (\text{B.11})$$

$$10 \frac{P_{5,n}}{10} = \frac{1}{T_{\text{IN}}} \int_0^{T_{\text{IN}}} \left(\frac{1}{16} V_0^5 a_{5,n} \cos(5\omega_{\text{IN}} t) \right)^2 dt. \quad (\text{B.12})$$

T_{IN} denotes the time period of the input signal, i.e., $T_{\text{IN}} = \omega_{\text{IN}} / (2\pi)$. The power $P_{m,n}$ for each spurious in dBW is calculated according to

$$P_{m,n} = P_{\text{IN}} - \text{CL} - \text{SST}_{m,n}, \quad (\text{B.13})$$

whereby P_{IN} , CL, and $\text{SST}_{m,n}$ denote the average input signal power in dBW, the mixer's conversion loss in dB, and the SST entry in dBc.

Usually, the SST is given for a certain reference power $P_{\text{IN,ref}}$ denoted by $\text{SST}_{m,n,\text{ref}}$; hence, the required values $\text{SST}_{m,n}$ are calculated by correcting for the power difference of the actual input power P_{IN} and the reference power $P_{\text{IN,ref}}$ according to [185, 188]:

$$\text{SST}_{m,n} = (m-1)(P_{\text{IN,ref}} - P_{\text{IN}}) + \text{SST}_{m,n,\text{ref}}. \quad (\text{B.14})$$

In order to obtain the nonlinear polynomial's coefficients, the integrals are solved first:

$$10^{\frac{p_{0,n}}{10}} = \frac{9V_0^8 a_{4,n}^2 + (24V_0^6 a_{2,n} + 48V_0^4 a_{0,n})a_{4,n} + 16V_0^4 a_{2,n}^2 + 64V_0^2 a_{0,n}a_{2,n} + 64a_{0,n}^2}{64}, \quad (\text{B.15})$$

$$10^{\frac{p_{1,n}}{10}} = \frac{25V_0^{10} a_{5,n}^2 + (60V_0^8 a_{3,n} + 80V_0^6 a_{1,n})a_{5,n} + 36V_0^6 a_{3,n}^2 + 96V_0^4 a_{1,n}a_{3,n} + 64V_0^2 a_{1,n}^2}{128}, \quad (\text{B.16})$$

$$10^{\frac{p_{2,n}}{10}} = \frac{V_0^8 a_{4,n}^2 + 2V_0^6 a_{2,n}a_{4,n} + V_0^4 a_{2,n}^2}{8}, \quad (\text{B.17})$$

$$10^{\frac{p_{3,n}}{10}} = \frac{25V_0^{10} a_{5,n}^2 + 40V_0^8 a_{3,n}a_{5,n} + 16V_0^6 a_{3,n}^2}{512}, \quad (\text{B.18})$$

$$10^{\frac{p_{4,n}}{10}} = \frac{V_0^8 a_{4,n}^2}{128}, \quad (\text{B.19})$$

$$10^{\frac{p_{5,n}}{10}} = \frac{V_0^{10} a_{5,n}^2}{512}. \quad (\text{B.20})$$

The resulting equation system is solved for $a_{m,n}$ second and the polynomial coefficients read:

$$a_{5,n} = \frac{2^{\frac{9}{2}} 10^{\frac{p_{5,n}}{20}}}{V_0^5}, \quad (\text{B.21})$$

$$a_{4,n} = \frac{2^{\frac{7}{2}} 10^{\frac{p_{4,n}}{20}}}{V_0^4}, \quad (\text{B.22})$$

$$a_{3,n} = \frac{-5V_0^5 a_{5,n} + 2^{\frac{9}{2}} 10^{\frac{p_{3,n}}{20}}}{4V_0^3}, \quad (\text{B.23})$$

$$a_{2,n} = \frac{-V_0^4 a_{4,n} + 2^{\frac{3}{2}} 10^{\frac{p_{2,n}}{20}}}{V_0^2}, \quad (\text{B.24})$$

$$a_{1,n} = \frac{-6V_0^3 a_{3,n} - 5V_0^5 a_{5,n} + 2^{\frac{7}{2}} 10^{\frac{p_{1,n}}{20}}}{8V_0}, \quad (\text{B.25})$$

$$a_{0,n} = \frac{-4V_0^2 a_{2,n} - 3V_0^4 a_{4,n} + 8 \cdot 10^{\frac{p_{0,n}}{20}}}{8}. \quad (\text{B.26})$$

C FI-DAC with I/Q Mixer

The FI-DAC concept is presented in the research literature with RF mixers [45, 50, 176]. However, the concept is also conceivable with I/Q mixers [44, 49]. An I/Q mixer upconverts two input signals to a common LO frequency, whereby the input signals are phase shifted by 90° relative to the LO frequency. Thereby, the mixer's LSB and USB can be used simultaneously enabling a broader spectrum at the RF output port.

In this appendix, the block diagram for the FI-DAC with an I/Q mixer is presented and the generation of the corresponding sub-signals is described. Parts of this appendix have been previously published in [44].

Block Diagram

In Fig. C.1, the FI-DAC's block diagram is depicted with an I/Q mixer. Three DACs are utilized to convert the three sub-signals to the analog domain. The sub-signals two and three are fed into the I/Q mixer. The upconverted signal is combined with the first sub-signal to form the combined analog output signal. In the figure, the combiner, i.e., the frequency multiplexer, is decomposed into two filters and the summation operation.

The concept can be extended to include multiple I/Q mixers; for each additional I/Q mixer two DACs are needed. Besides, both I/Q mixers and RF mixers could be used in one FI-DAC together as shown in Sec. 5.2.

The I/Q mixer variant has several advantages and disadvantages. From a system perspective, an I/Q mixer enables an upconverted signal with the doubled bandwidth compared to an RF mixer. Furthermore, only a single LO is required, whereas for the same bandwidth with RF mixers, two LOs are needed. By using both sidebands, the suppression of the unused sideband is not necessary. Hence, the guard band at low frequencies is dispensable, enabling a broader bandwidth. Besides, the I/Q mixer may be operated as a SSB mixer, if only half of the bandwidth is required. One of the sidebands is suppressed by the mixer obviating the need for analog filters with steep roll-offs for suppression.

However, the advantages need to be reconceived regarding the disadvantages. Although, the I/Q mixer is a single mixer, it is internally composed of two RF

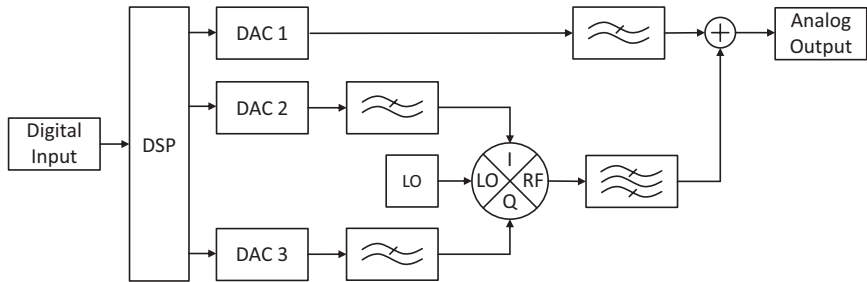


Figure C.1 Block diagram of the FI-DAC with an I/Q mixer.

mixers. An almost equal bandwidth could be generated by means of two individual RF mixers as well. The required 90° phase shift between in-phase and quadrature cannot be maintained for broad bandwidths, which introduces I/Q imbalance; additional estimation and compensation algorithms are required [348, 349]. With an RF mixer, the fed-through LO can be compensated by means of guard bands and analog filters as described in Sec. 5.2. By using an I/Q mixer, the LO is located in the center of the upconverted frequency band rather than at the edge. It can be compensated either by injecting an LO with opposite phase or by tuning the DC offset of the in-phase and the quadrature signal.

Generating the Sub-Signals

The sub-signals for the exemplary FI-DAC depicted in Fig. C.1 are obtained by partitioning the combined digital spectrum into three sub-bands. In Fig. C.2, the partitioning operation is depicted in the frequency domain. First, the data spectrum is partitioned into three sub-bands I, II, and III. The first sub-band corresponds to a real-valued signal, which is fed directly into DAC 1.

Both sub-bands II and III together form the band-pass signal that is generated by the I/Q mixer later. It is downconverted to baseband, low-pass filtered, and downsampled, which is a straightforward operation in the time domain. In the frequency domain, a selection of the respective frequency domain samples is feasible to achieve downconversion, low-pass filtering, and downsampling implicitly.

The baseband spectrum does not have the conjugate symmetry property and the corresponding time-domain signal is complex. The DACs' sub-signals are obtained by taking the real and the imaginary part of the time-domain signal, that is represented by the combined spectrum II and III.

Generating the sub-signals for the I/Q mixer from the combined spectrum, represented by II and III, can be performed in the frequency domain as well. This

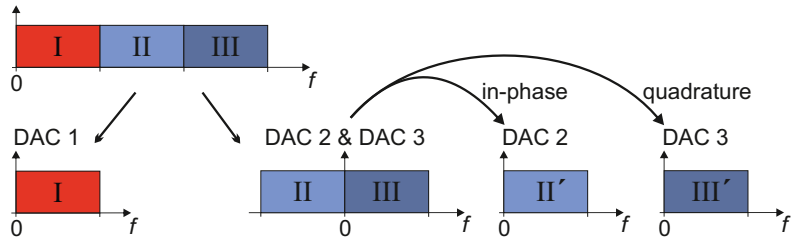


Figure C.2 Spectrum partitioning for the FI-DAC consisting of three DACs and an I/Q mixer.

step requires the exploitation of the general symmetry properties of the Fourier transformation for odd and even functions and spectral components [282].

D FI-DAC Distribution of Data Samples

In Sec. 5.4.2, the solution for an equation system is presented. In this section, the complete derivation is outlined in order to obtain the solution $K_{D,n}$.

The equation system is given as

$$\sum_{n \in \Lambda} K_{D,n} = K_{D,\text{tot}} \quad (\text{D.1})$$

$$p_n K_{D,n} = p_l K_{D,l} \quad \forall n, l \in \Lambda \wedge n \neq l \quad (\text{D.2})$$

whereby $K_{D,n}$, $K_{D,\text{tot}}$, and p_n denote the number of data samples for the n th DAC, the total number of data samples and the oversampling ratio for DAC n . It can be rewritten as

$$\sum_{n \in \Lambda} K_{D,n} = K_{D,\text{tot}} \quad (\text{D.3})$$

$$p_n K_{D,n} = p_1 K_{D,1} \quad \forall n \in \Lambda_1 \quad (\text{D.4})$$

with $\Lambda_n = \Lambda \setminus \{n\}$, $n \in \Lambda$. This equation system can be represented in matrix notation as

$$\underbrace{\begin{pmatrix} 1 & 1 & 1 & \cdots & \cdots & 1 \\ -p_1 & p_2 & 0 & \cdots & \cdots & 0 \\ -p_1 & 0 & p_3 & 0 & \cdots & 0 \\ \vdots & \vdots & 0 & \ddots & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & p_{N-1} & 0 \\ -p_1 & 0 & 0 & \cdots & 0 & p_N \end{pmatrix}}_{\mathbf{P}} \cdot \underbrace{\begin{pmatrix} K_{D,1} \\ K_{D,2} \\ \vdots \\ K_{D,N} \end{pmatrix}}_{\mathbf{K}_D} = \underbrace{\begin{pmatrix} K_{D,\text{tot}} \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\mathbf{K}_{D,\text{tot}}} \quad (\text{D.5})$$

By using capital bold letters for both the matrix and the vectors, (D.5) is simplified to

$$\mathbf{P} \cdot \mathbf{K}_D = \mathbf{K}_{D,\text{tot}} \quad (\text{D.6})$$

This equation system is solved for the number of data samples for the DACs $\mathbf{K}_D$:

$$\Rightarrow \mathbf{K}_D = \underline{\mathbf{P}}^{-1} \cdot \mathbf{K}_{D,\text{tot}} . \quad (\text{D.7})$$

With solely the first element of $\mathbf{K}_{D,\text{tot}}$ being $\neq 0$, only the first column of $\underline{\mathbf{P}}^{-1}$ needs to be calculated. All other entries will equal zero after multiplication with $\mathbf{K}_{D,\text{tot}}$.

Distinct elements of the inverse $\hat{\mathbf{P}} = \underline{\mathbf{P}}^{-1}$ can be calculated by means of the adjugate matrix $\text{adj}(\underline{\mathbf{P}})$ and the cofactor matrix $\tilde{\mathbf{P}} = \text{cof}(\underline{\mathbf{P}})$ [131, 350]:

$$\hat{\mathbf{P}} = \underline{\mathbf{P}}^{-1} = \frac{1}{\det(\underline{\mathbf{P}})} \text{adj}(\underline{\mathbf{P}}) = \frac{1}{\det(\underline{\mathbf{P}})} \text{cof}(\underline{\mathbf{P}})^T = \frac{1}{\det(\underline{\mathbf{P}})} \tilde{\mathbf{P}}^T . \quad (\text{D.8})$$

Thus, the equation system can be efficiently solved:

$$\begin{aligned} \mathbf{K}_D &= K_{D,\text{tot}} \cdot \begin{pmatrix} \hat{p}_{11} \\ \vdots \\ \hat{p}_{N1} \end{pmatrix} = \frac{K_{D,\text{tot}}}{\det(\underline{\mathbf{P}})} \cdot \begin{pmatrix} \tilde{p}_{11} \\ \vdots \\ \tilde{p}_{1N} \end{pmatrix} \\ &= \frac{K_{D,\text{tot}}}{\sum_{m \in \Lambda} \prod_{n \in \Lambda_m} p_n} \cdot \begin{pmatrix} \prod_{n \in \Lambda_1} p_n \\ \vdots \\ \prod_{n \in \Lambda_N} p_n \end{pmatrix} , \end{aligned} \quad (\text{D.9})$$

whereby $\hat{p}_{n1}$ and $\tilde{p}_{1n}$ denote the entries of the first column and the first row of the inverse $\hat{\mathbf{P}}$ and the cofactor matrix $\tilde{\mathbf{P}}$, respectively.

The elements of $\mathbf{K}_D$ in (D.9) are given by

$$K_{D,n} = \frac{\prod_{i \in \Lambda_n} p_i}{\sum_{m \in \Lambda} \prod_{i \in \Lambda_m} p_i} K_{D,\text{tot}} . \quad (\text{D.10})$$

E FI-DAC Digital Frequency Demultiplexer

The split of the combined digital signal is performed with a digital frequency demultiplexer. The individual filter characteristics can be either non-overlapping, i.e., with ideal rectangular frequency domain filters, or overlapping with any other filter characteristic.

Any set of filter functions is reasonable as long as the following condition is fulfilled:

$$\sum_{n \in \Lambda} H_{\text{SF},n}(f) = 1, \quad (\text{E.1})$$

whereby $H_{\text{SF},n}(f)$ and Λ denote the n th filter characteristic and the set of sub-signals, respectively. In this section, the continuous frequency f is used instead of the discrete frequency Ω for convenience reasons.

Usually, the filter characteristics are LPFs, BPFs, or HPFs. For this thesis, shifted raised-cosine functions are used as overlapping filter characteristics. The raised-cosine filter $H_{\text{L},n}^{\text{RC}}(f)$ is commonly defined as a LPF [129]:

$$H_{\text{L},n}^{\text{RC}}(f) = \begin{cases} 1, & |f| \leq (1 - \beta_n) f_{\text{co},n} \\ \frac{1}{2} \left(1 + \cos \left(\frac{2\pi}{\beta_n f_{\text{co},n}} (|f| - (1 - \beta_n) f_{\text{co},n}) \right) \right), & (1 - \beta_n) f_{\text{co},n} < |f| \leq (1 + \beta_n) f_{\text{co},n} \\ 0, & \text{otherwise} \end{cases}, \quad (\text{E.2})$$

whereby β_n denotes the raised-cosine roll-off factor at the crossover frequency $f_{\text{co},n}$.

The required high-pass characteristic is obtained by negating the cosine part of $H_{\text{L},n}^{\text{RC}}(f)$ and exchanging the one- and the zero-level. The high-pass characteristic is then given by

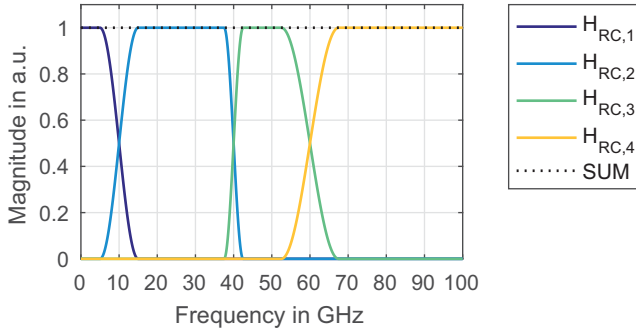


Figure E.1 Exemplary FI-DAC digital frequency demultiplexer frequency response based on raised cosine functions for four sub-signals up to 100 GHz.

$$H_{H,n}^{\text{RC}}(f) = \begin{cases} 0, & |f| \leq (1 - \beta_{n-1})f_{\text{co},n-1} \\ \frac{1}{2} \left(1 - \cos \left(\frac{2\pi}{\beta_{n-1}f_{\text{co},n-1}} (|f| - (1 - \beta_{n-1})f_{\text{co},n-1}) \right) \right), & (1 - \beta_{n-1})f_{\text{co},n-1} < |f| \leq (1 + \beta_{n-1})f_{\text{co},n-1} \\ 1, & \text{otherwise} \end{cases} \quad (\text{E.3})$$

All sub-bands, but the first and the last, are represented by BPFs. The band-pass raised-cosine characteristic is obtained by the multiplication of a LPF and a HPF according to

$$H_{B,n}^{\text{RC}}(f) = H_{L,n}^{\text{RC}}(f) \cdot H_{H,n}^{\text{RC}}(f) . \quad (\text{E.4})$$

In Fig. E.1, an exemplary digital frequency demultiplexer for four sub-signals based on raised-cosine functions is depicted. The frequency responses overlap and the sum of all frequency responses equals one. As a demonstrative example, both different bandwidths for each sub-band and different raised-cosine roll-off factors are chosen for each crossover frequency resulting in asymmetric frequency responses. The raised-cosine roll-off factor is equal for neighboring frequency bands at each crossover frequency.

F FI-DAC MIMO Model and Pre-Equalizer

In this appendix, the derivation for the FI-DAC MIMO algorithm is presented. The system model and the pre-equalizer are presented in Secs. 5.7.2 and 5.7.4. Note that parts of this appendix have been previously published in [44, 45].

In general, the MIMO system is described in the frequency domain with L received frequency bands and N transmitted frequency bands as

$$\mathbf{Y}(\mu) = \underline{\mathbf{C}}(\mu)\mathbf{X}(\mu) + \mathbf{V}(\mu), \quad (\text{F.1})$$

whereby μ denotes the frequency of the DFT (as defined in footnote 13). $\mathbf{Y}(\mu)$ is the vector containing the FI-DACs output spectrum separated into L sub-bands with size $2L \times 1$ to cover the sub-bands in both regular and in reversed frequency position. $\mathbf{X}(\mu)$ is the vector of N sub-bands, which are fed into the DACs with size $2N \times 1$ and $\mathbf{V}(\mu)$ is the vector of noise sub-bands with size $2L \times 1$. The channel matrix $\underline{\mathbf{C}}(\mu)$ has the size $2L \times 2N$. The vectors and the matrix are given by

$$\mathbf{Y}(\mu) = \begin{bmatrix} Y_1(\mu) \\ Y_1^\dagger(\mu) \\ \vdots \\ Y_L(\mu) \\ Y_L^\dagger(\mu) \end{bmatrix}, \quad \mathbf{X}(\mu) = \begin{bmatrix} X_1(\mu) \\ X_1^\dagger(\mu) \\ \vdots \\ X_N(\mu) \\ X_N^\dagger(\mu) \end{bmatrix}, \quad (\text{F.2})$$

$$\mathbf{V}(\mu) = \begin{bmatrix} V_1(\mu) \\ V_1^\dagger(\mu) \\ \vdots \\ V_L(\mu) \\ V_L^\dagger(\mu) \end{bmatrix}. \quad (\text{F.3})$$

The channel matrix $\underline{\mathbf{C}}(\mu)$ is defined as

$$\underline{\mathbf{C}}(\mu) = \begin{bmatrix} C_{1,1}(\mu) & C_{1,2}(\mu) & \cdots & C_{1,2N}(\mu) \\ C_{2,1}(\mu) & C_{2,2}(\mu) & \cdots & C_{2,2N}(\mu) \\ \vdots & \vdots & \ddots & \vdots \\ C_{2L,1}(\mu) & C_{2L,2}(\mu) & \cdots & C_{2L,2N}(\mu) \end{bmatrix}. \quad (\text{F.4})$$

The DSP output sub-bands $\mathbf{X}(\mu)$ are obtained by pre-equalizing the data sub-bands $\mathbf{D}(\mu)$. The pre-equalizer undoing the channel impairments is given by the weight-matrix $\underline{\mathbf{W}}(\mu)$:

$$\mathbf{X}(\mu) = \underline{\mathbf{W}}(\mu)\mathbf{D}(\mu), \quad (\text{F.5})$$

whereby the pre-equalizer $\underline{\mathbf{W}}(\mu)$ and the data sub-bands vector $\mathbf{D}(\mu)$ are given by

$$\underline{\mathbf{W}}(\mu) = \begin{bmatrix} W_{1,1}(\mu) & W_{1,2}(\mu) & \cdots & W_{1,2L}(\mu) \\ W_{2,1}(\mu) & W_{2,2}(\mu) & \cdots & W_{2,2L}(\mu) \\ \vdots & \vdots & \ddots & \vdots \\ W_{2N,1}(\mu) & W_{2N,2}(\mu) & \cdots & W_{2N,2L}(\mu) \end{bmatrix}, \quad (\text{F.6})$$

$$\mathbf{D}(\mu) = \begin{bmatrix} D_1(\mu) \\ D_1^\dagger(\mu) \\ \vdots \\ D_L(\mu) \\ D_L^\dagger(\mu) \end{bmatrix}. \quad (\text{F.7})$$

In principle, the FI-DAC is scalable to any number of DACs. For the scaling to more DACs, it is expected that multiple 2×2 MIMO problems need to be solved individually rather than a higher order joint MIMO problem.

Zero Forcing (ZF) Equalizer

The cost function of a ZF equalizer consists in minimizing the inter-symbol interference (ISI) to zero. Thus, the received sub-bands are given by

$$\mathbf{Y}(\mu) = \underline{\mathbf{C}}(\mu)\mathbf{X}(\mu) \Big|_{\mathbf{V}(\mu)=0} = \underline{\mathbf{C}}(\mu)\underline{\mathbf{W}}_{\text{ZF}}(\mu)\mathbf{D}(\mu). \quad (\text{F.8})$$

Claiming that $\mathbf{Y}(\mu) \stackrel{!}{=} \mathbf{D}(\mu)$ to recover the signal, there are two solutions depending on the channel matrix $\underline{\mathbf{C}}(\mu)$:

1. $\underline{\mathbf{C}}(\mu)$ is quadratic and invertible:

$$\Rightarrow \underline{\mathbf{W}}_{\text{ZF}}(\mu) = \underline{\mathbf{C}}^{-1}(\mu) . \quad (\text{F.9})$$

2. $\underline{\mathbf{C}}(\mu)$ is non-quadratic: the well-known Moore-Penrose pseudo inverse is used [129, 239]:

$$\Rightarrow \underline{\mathbf{W}}_{\text{ZF}}(\mu) = (\underline{\mathbf{C}}^\dagger(\mu)\underline{\mathbf{C}}(\mu))^{-1} \underline{\mathbf{C}}^\dagger(\mu) , \quad (\text{F.10})$$

whereby $(\cdot)^\dagger$ denotes the conjugate transpose and $(\cdot)^{-1}$ the inverse operator, respectively.

Minimum Mean Square Error (MMSE) Equalizer

By considering the noise additionally to the ISI, i.e., $\mathbf{V}(\mu) \neq 0$, a better solution than the ZF pre-equalizer is provided by the MMSE pre-equalizer given by [129, 239]:

$$\underline{\mathbf{W}}_{\text{MMSE}}(\mu) = \left(\underline{\mathbf{C}}^\dagger(\mu)\underline{\mathbf{C}}(\mu) + \frac{\sigma_N^2}{\sigma_S^2} \mathbf{I} \right)^{-1} \underline{\mathbf{C}}^\dagger(\mu) , \quad (\text{F.11})$$

whereby σ_N^2 and σ_S^2 denote the noise and the signal variance, respectively.

Adaptive Equalizer

Adaptive solutions can recalibrate the FI-DAC in case of components drift, temperature variations, etc. The LMS equalizer [239] is chosen as an example for an adaptive equalizer. Other types are feasible as well, e.g., the recursive least squares (RLS) equalizer. The equalizer coefficients for the LMS are updated according to

$$\underline{\mathbf{W}}_{\text{LMS},i+1}(\mu) = \underline{\mathbf{W}}_{\text{LMS},i}(\mu) + \lambda (\mathbf{D}_i(\mu) - \mathbf{Y}_i(\mu)) \mathbf{X}_i^*(\mu) , \quad (\text{F.12})$$

whereby λ denotes the update coefficient of the LMS algorithm.

ZF Equalizer for a FI-DAC with two DACs

As mentioned in Sec. 5.7.2, the FI-DAC without guard bands can be modeled as a MIMO system, whereby $L = N$ is not necessarily true. However, $L = N$ simplifies the solution and enables an appropriate linear formulation.

In the preferred 4×4 MIMO formulation only half of the entries in the channel matrix $\underline{\mathbf{C}}(\mu)$ are $\neq 0$, since the other cross talk terms are not present:

$$\underline{\mathbf{C}}(\mu) = \begin{bmatrix} H_{11}(\mu) & 0 & 0 & H_{12}(\mu) \\ 0 & H_{11}^\dagger(\mu) & H_{12}^\dagger(\mu) & 0 \\ 0 & H_{21}(\mu) & H_{22}(\mu) & 0 \\ H_{21}^\dagger(\mu) & 0 & 0 & H_{22}^\dagger(\mu) \end{bmatrix}. \quad (\text{F.13})$$

Then, the ZF pre-equalizer is given as

$$\underline{\mathbf{W}}_{\text{ZF}}(\mu) = \underline{\mathbf{C}}^{-1}(\mu) \quad (\text{F.14})$$

$$= \text{diag} \left(\begin{bmatrix} \frac{1}{H_{11}(\mu)H_{22}^\dagger(\mu) - H_{12}(\mu)H_{21}^\dagger(\mu)} \\ \frac{1}{H_{11}^\dagger(\mu)H_{22}(\mu) - H_{12}^\dagger(\mu)H_{21}(\mu)} \\ \frac{1}{H_{11}^\dagger(\mu)H_{22}(\mu) - H_{12}^\dagger(\mu)H_{21}(\mu)} \\ \frac{1}{H_{11}(\mu)H_{22}^\dagger(\mu) - H_{12}(\mu)H_{21}^\dagger(\mu)} \end{bmatrix} \right) \cdot \begin{bmatrix} H_{22}^\dagger(\mu) & 0 & 0 & -H_{12}(\mu) \\ 0 & H_{22}(\mu) & -H_{12}^\dagger(\mu) & 0 \\ 0 & -H_{21}(\mu) & H_{11}^\dagger(\mu) & 0 \\ -H_{21}^\dagger(\mu) & 0 & 0 & H_{11}(\mu) \end{bmatrix}. \quad (\text{F.15})$$

Only the first row and the third row are relevant, since the second and the fourth row contain the same information, i.e., they are in reversed frequency position. The result is given as:

$$W_{11}(\mu) = \frac{H_{22}^\dagger(\mu)}{H_{11}(\mu)H_{22}^\dagger(\mu) - H_{12}(\mu)H_{21}^\dagger(\mu)}, \quad (\text{F.16})$$

$$W_{14}(\mu) = -\frac{H_{12}(\mu)}{H_{11}(\mu)H_{22}^\dagger(\mu) - H_{12}(\mu)H_{21}^\dagger(\mu)}, \quad (\text{F.17})$$

$$W_{32}(\mu) = -\frac{H_{21}(\mu)}{H_{11}^\dagger(\mu)H_{22}(\mu) - H_{12}^\dagger(\mu)H_{21}(\mu)}, \quad (\text{F.18})$$

$$W_{33}(\mu) = \frac{H_{11}^\dagger(\mu)}{H_{11}^\dagger(\mu)H_{22}(\mu) - H_{12}^\dagger(\mu)H_{21}(\mu)}. \quad (\text{F.19})$$

Mapping this formulation to the pre-equalizer depicted in Fig. 5.10, the following relations are obtained: $W_{11,\text{fig}}(\mu) = W_{11}(\mu)$, $W_{12,\text{fig}} = W_{14}(\mu)$, $W_{21,\text{fig}} = W_{32}(\mu)$, and $W_{22,\text{fig}} = W_{33}(\mu)$.